Thinking Mathematics Grade 7

Theory – Worked examples – 250 practice and challenge problems

Teacher book – with detailed solutions

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ThinkAlgo Tip.

- Each chapter contains: **theory** (with explanations and proofs), **worked examples** with guided solutions, and **50 problems**: *Core practice* followed by *Extension and challenge*.
- Problems marked ⋆ are hard, ⋆⋆ very hard.
- Try each problem on your own first, then compare with the solution.

1 Algebra and Equations

The big idea. Algebra is *the language of unknown numbers*. Instead of saying at length "some number, add 5, then double it", we simply write $2(x+5)$. The letter x (called a **variable** or an **unknown**) holds the place of a number that we do not know yet or that may change.
In this topic you will practise four core skills:

1. **Translate** words into expressions and formulas ("more than", "times", "sum", "difference"...).
2. **Substitute** numbers to evaluate an expression, or to check whether an equation is true or false.
3. **Solve equations** with the *balance principle*: whatever you do to one side, do exactly the same to the other side.
4. **Spot patterns**: sequences, tables of values, ratios, and the order of numbers.

Every word problem follows the same route: *read carefully* → *choose an unknown* → *write an expression or an equation* → *solve* → *check* → *answer*.

① Variables, algebraic expressions and formulas

1. Variables and expressions. An **algebraic expression** is built from numbers, letters (variables) and the operations $+$, $-$, $\times$, $\div$, powers and brackets. Examples: $3x-1$; $2(n-1)$; $120+30m$; xy.
Short-hand. Between a number and a letter, or between two letters, we leave out the multiplication sign: $3x$ means $3\cdot x$; xy means $x\cdot y$; $2(n-1)$ means $2\cdot(n-1)$. The number is written before the letter.
2. Value of an expression. To evaluate an expression for given values of the variables, we *replace* each letter by its number (best put in brackets), then calculate in the correct order of operations.
Example: for $x=3$, $3x-1=3\cdot 3-1=9-1=8$; $2x^2=2\cdot 3^2=2\cdot 9=18$.
3. Translating words into expressions.

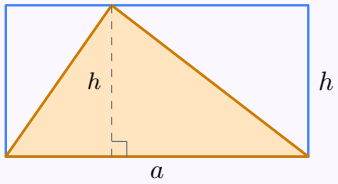
Words	Expression
5 more than x; x increased by 5	$x+5$
5 less than x; x decreased by 5	$x-5$
3 times x; triple x	$3x$
half of x; x divided by 2	$\frac{x}{2}$ or $x\div 2$
start with 120, add 30 each time, m times	$120+30m$
three consecutive whole numbers, the smallest is n	$n,\ n+1,\ n+2$

4. A formula is an equation that tells us how to calculate one quantity from other quantities. Useful geometry formulas (side lengths a, b, c):

- Perimeter of a triangle: $P=a+b+c$ (the sum of the three sides).

- Rectangle with length a and width b: perimeter $P = 2(a + b)$, area $A = a \cdot b$.
- Triangle with base a and height h: area $A = \dfrac{a \cdot h}{2}$.
- Rectangular prism $a \times b \times c$: total surface area $S = 2(ab + bc + ca)$, volume $V = abc$.

Why is it true? Why is the area of a triangle half of (base $\times$ height)? Draw a rectangle with one side equal to the base a of the triangle and the other side equal to the height h. The height splits the triangle into two right triangles; each right triangle is exactly half of a smaller rectangle (in the picture, the shaded and white parts match in pairs). So the triangle covers exactly half of the big rectangle: $A = \dfrac{a \cdot h}{2}$.



Similarly, the perimeter of a rectangle is $a + b + a + b = 2a + 2b = 2(a + b)$ because a rectangle has two equal long sides and two equal short sides. □

ThinkAlgo Tip. When substituting, *always put the number in brackets*, especially a negative number: for $x = -2$, $x^2 = (-2)^2 = 4$, while $3x = 3 \cdot (-2) = -6$. Brackets protect you from sign mistakes.

Common mistake. $3x$ with $x = 4$ is $3 \cdot 4 = 12$, *not* "34". And x^2 (which is $x \cdot x$) is different from $2x$ (which is $x + x$): for $x = 5$, $x^2 = 25$ but $2x = 10$.

ThinkAlgo Tip. Write everything in terms of a few letters. Consider a list in which every number (from the third on) is the sum of the two before it. If the first two numbers are a and b, the next ones are

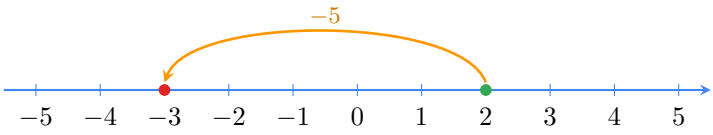
$$a, \quad b, \quad a + b, \quad b + (a + b) = a + 2b, \quad (a + b) + (a + 2b) = 2a + 3b, \ \ldots$$

For instance, with $a = 11$, $b = 20$ the fifth number is $2 \cdot 11 + 3 \cdot 20 = 22 + 60 = 82$, just as in the list $11, 20, 31, 51, 82$. Writing the numbers in terms of a and b turns a question about the list into an equation.

Note for parents. Play the "number machine" game with your child: you say a rule ("multiply by 3, then subtract 1"), your child writes it as the expression $3x - 1$, and then you take turns feeding numbers into the machine. This game makes the letter x feel friendly instead of scary.

② Integers, order of operations and the distributive law

1. Integers. The numbers $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ are called integers. On the number line, a number further to the left is smaller. The opposite of a is $-a$; we have $a + (-a) = 0$.



Example: $2 + (-5) = -3$: starting at 2, move 5 steps to the left and arrive at -3.

2. Adding and subtracting integers.

- Two numbers with the same sign: add the sizes and keep the common sign: $(-11) + (-26) = -37$.
- Two numbers with different signs: subtract the smaller size from the larger one and take the sign of the number with the larger size: $11 + (-37) = -26$.
- Subtracting means adding the opposite: $a - b = a + (-b)$. Example: $-26 - 11 = -26 + (-11) = -37$.

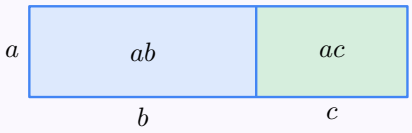
3. Order of operations. Brackets first → powers → multiplication and division (left to right) → addition and subtraction (left to right). Example: $6 + 2 \cdot 3 = 6 + 6 = 12$ (not $8 \cdot 3$).

4. The distributive law. $a(b+c) = ab+ac$ and $a(b-c) = ab-ac$. *Examples:* $3(T+4) = 3T+12$; $20 \cdot 3 - 20 \cdot 2.25 = 20 \cdot (3 - 2.25) = 20 \cdot 0.75 = 15$.

5. Simplifying expressions. Add "like terms": $3x+5x = 8x$; $(a+1)+(b+1)+(c+1)+(d+1) = (a+b+c+d)+4$.

6. Points in the coordinate plane. The point (x, y) has x-coordinate x and y-coordinate y. Moving k units to the right increases x by k; moving k units up increases y by k (moving left or down decreases them).

Why is it true? Why is $a(b+c) = ab+ac$? Take a rectangle with width a and length $b+c$. Its area is $a(b+c)$. But we can also cut it into two smaller rectangles with areas ab and ac. It is the same area, so $a(b+c) = ab+ac$.



□

Common mistake. $3(T+4)$ is *not* equal to $3T+4$: you must multiply 3 by *each* term inside the brackets, $3(T+4) = 3T+3 \cdot 4 = 3T+12$. Likewise $-(a-b) = -a+b$ (change the sign of *every* term in the brackets).

③ Equations and the balance principle

1. Equations. A statement of equality containing an unknown, such as $2(x+8)=38$, is called an **equation**. A number that makes both sides equal when it replaces the unknown is called a **solution**. To *solve an equation* means to find its solutions.
2. The balance principle. Imagine the equation as a pair of scales in balance. The scales stay balanced if we:

- add (or subtract) *the same number* on both sides;
- multiply (or divide) both sides by *the same non-zero number*.

A familiar consequence is the **rule for moving terms**: when a term moves from one side to the other, its *sign changes*.



Example: two identical boxes plus a 3 kg weight balance 11 kg: $2x+3=11$. Remove the 3 kg weight from both pans: $2x=11-3=8$. Halve both pans: $x=8\div 2=4$. Each box weighs 4 kg.
3. Substitution on scales. If A on one pan balances B, then on any other pair of scales we may replace A by B and the scales stay balanced (just like replacing an expression by an equal expression).
4. Always check. After solving, substitute the solution into the original equation to make sure both sides are equal.

Why is it true? Why does a term change sign when it moves across? From $2x+3=11$, subtract 3 from both sides (the scales stay balanced): $2x+3-3=11-3$, that is, $2x=11-3$. Looking at the result, the term $+3$ on the left seems to have "jumped" to the right and become -3. In fact nothing "jumps": we only subtracted the same number from both sides. □

Common mistake. Forgetting to change the sign: from $x+11=-26$, writing $x=-26+11$ is **wrong**. The correct step is $x=-26-11=-37$.

ThinkAlgo Tip. In a multiple-choice question with an equation, you can *substitute each option in turn* into the equation. It is slower but very safe, and it is also the best way to check.

④ Solving word problems with equations

Four steps for a word problem.

1. **Choose an unknown**: call the quantity you want (or a related one) x; write its unit and any condition (for example $x>0$, x is a whole number).
2. **Express** the other quantities in terms of x.
3. **Write an equation** from the conditions of the problem and **solve** it.
4. **Compare** with the conditions, **check**, and **answer** the actual question.

Common types.

- **Consecutive integers:** $n,\ n+1,\ n+2$ (or, more neatly, $n-1,\ n,\ n+1$). Consecutive even

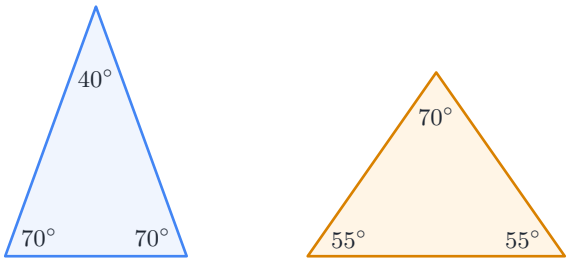
(odd) numbers differ by 2: $n,\ n+2,\ n+4$.

- **Sums in pairs:** three objects a, b, c weighed two at a time give $a+b$, $b+c$, $c+a$. Adding the three results: $(a+b)+(b+c)+(c+a)=2(a+b+c)$, so *the total of the three objects is half of the sum of the results.* To find one object, subtract the pair formed by the other two from the total.
- **Angles in a triangle:** the three angles of a triangle add up to 180°; an isosceles triangle has two equal base angles.
- **Average:** average = sum of the values ÷ number of values; average price = total cost ÷ total mass.

Two small examples.

a) An isosceles triangle with a 70° angle. The 70° angle may be a *base angle* or the *apex angle*, so we must look at two cases:

- If 70° is a base angle: the base angles are 70° and 70°, and the apex angle is $180°-70°-70°=40°$.
- If 70° is the apex angle: the two base angles are equal and add up to $180°-70°=110°$, so each is 55°.



b) Average price. You buy 300 g of type I candy for \$3 and 200 g of type II candy for \$4. The average price of 100 g of candy is $(3+4)\div(300+200)\cdot 100=7\div 500\cdot 100=1.4$, that is, \$1.40. You must *not* average the two prices \$1 and \$2 per 100 g (that gives \$1.50, which is wrong), because different amounts of the two candies were bought.

Why is it true? Three consecutive integers: the middle one equals the sum divided by 3. Call the three numbers $n-1$, n, $n+1$. Their sum is $(n-1)+n+(n+1)=3n$ (the -1 and $+1$ cancel). So the middle number is $n=\text{sum}\div 3$. For example, if the sum is 153, the middle number is 51 and the three numbers are $50, 51, 52$. □

ThinkAlgo Tip. Choosing the *middle number* as the unknown usually gives the neatest equation. For "sums in pairs" problems, do not rush to use three unknowns: *add everything up* first.

Common mistake. Solving the equation is not the end of the problem! Re-read the question: does it ask for the *largest* number, the *lightest* object, or the *total*...? Many students find the correct x but choose the wrong option because they answer a different question.

Note for parents. When your child is stuck, ask: "What don't we know yet? Let's call it x. Now, what does the problem tell us about x?" Let your child say it in words first, and only then write it as an equation.

5 Patterns, tables of values and arithmetic sequences

1. Arithmetic sequences. A list of numbers in which neighbouring numbers always differ by the same number d (called the **common difference**) is called an arithmetic sequence. If the first term is a, then

$$\text{the } n\text{-th term } = a + (n-1) \cdot d.$$

Example: the sequence $7, 10, 13, 16, \ldots$ has $a = 7$, $d = 3$; its n-th term is $7 + 3(n-1) = 3n + 4$.

2. From a table of values to an expression. If each time n increases by 1 the value increases by d, the expression has the form $d \cdot n +$ (some number). Find "some number" by substituting $n = 1$.

n	1	2	3	4	5
value	4	7	10	13	16

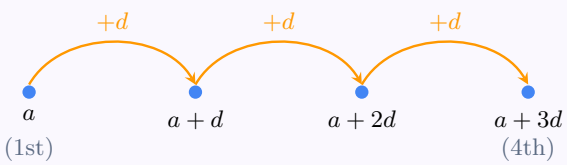
Each time n increases by 1, the value increases by 3, so the expression has the form $3n + b$. Substituting $n = 1$: $3 + b = 4$, so $b = 1$. Hence value $= 3n + 1$.

3. Is a number in the sequence? The number N belongs to the sequence $a,\ a+d,\ a+2d, \ldots$ exactly when $N-a$ is divisible by d (and $N \geq a$). For a sequence counting down, $a,\ a-d,\ a-2d, \ldots$, look at $a - N$ instead.

4. Common terms of two sequences; repeating events. If two arithmetic sequences with differences d_1 and d_2 have a common term, then their common terms are exactly $\text{lcm}(d_1, d_2)$ apart. Similarly, events that repeat every p minutes and every q minutes happen together again every $\text{lcm}(p, q)$ minutes.

ThinkAlgo Tip. *An example of repetition.* The first bus returns to the station every 12 minutes, the second bus every 20 minutes. At 7:00 a.m. both are at the station. The first bus returns after $12, 24, 36, 48, \mathbf{60}, \ldots$ minutes; the second after $20, 40, \mathbf{60}, \ldots$ minutes. The two buses meet at the station again every $\text{lcm}(12, 20) = 60$ minutes, that is, at 8:00 a.m., 9:00 a.m., ... Similarly, if the two sequences $7, 10, 13, \ldots$ (difference 3) and $2026, 2021, 2016, \ldots$ (difference 5) have a common term, then their common terms are $\text{lcm}(3, 5) = 15$ apart.

Why is it true? Why is the n-th term $a + (n-1)d$? Going from the 1st term to the n-th term, we cross $n - 1$ "gaps", and each gap adds d. It is like walking from the 1st lamp post to the 5th: you only pass 4 gaps.



From the 1st term to the 4th term there are 3 steps of $+d$, so the 4th term is $a + 3d$. □

Common mistake. Checking a formula against only *one* column of the table is not enough. For example, for the table $1, 3, 5, 7, 9$, both $3n - 2$ and $2n - 1$ give the value 1 when $n = 1$, but for $n = 2$ we get $3n - 2 = 4 \neq 3$. Test *at least two or three columns.*

⑥ Ratios, units of measurement and uniform motion

1. Directly proportional quantities. If one scoop of food feeds 8 fish, then 2 scoops feed 16 fish and 4 scoops feed 32 fish: when the number of scoops is multiplied by some number, the number of fish is multiplied by the same number. We say y is **directly proportional** to x with constant k if $y = k \cdot x$. The usual method is the **unit method** (work out the value for 1 part first, then multiply).

2. Uniform motion. With s the distance, v the speed and t the time:

$$s = v \cdot t, \qquad t = \frac{s}{v}, \qquad v = \frac{s}{t}.$$

Use matching units: if v is in km/h, then s is in km and t is in hours.

Two people moving *for the same length of time*: whoever is k times as fast covers k times the distance.

Running in opposite directions around a circle: from one meeting to the next, the *total* distance run by the two people is exactly one lap. If the first person is k times as fast as the second, then during each such stretch the first runs $\frac{k}{k+1}$ of a lap and the second runs $\frac{1}{k+1}$ of a lap.

3. Converting units of time. 1 minute = 60 seconds; 1 hour = 60 minutes = $60 \times 60 = 3\,600$ seconds; 1 day = 24 hours; 1 week = 7 days. Therefore 1 week = 7×24 hours = $7 \times 24 \times 60$ minutes = $7 \times 24 \times 60 \times 60$ seconds.

4. Money. Number of bills = total amount ÷ value of one bill. Difference after n times = $n \times$ (higher price − lower price).

ThinkAlgo Tip. Read a product such as $60 \times 60 \times 24 \times 7$ as a *chain of unit conversions*, from right to left: 7 (days in a week) ×24 (hours in a day) ×60 (minutes in an hour) ×60 (seconds in a minute). The result is the number of seconds in one week.

Why is it true? Why can we work out the difference "per time" and then multiply? By the distributive law: $20 \times 3 - 20 \times 2.25 = 20 \times (3 - 2.25) = 20 \times 0.75 = 15$. Both ways always give the same result, but the second is much quicker to do in your head. □

Common mistake. In motion problems, read carefully "how much has been travelled" and "how much is left". After half of a 510 km trip, the distance left is $510 \div 2 = 255$ km, not 510 km.

⑦ Equivalent fractions and a fraction of a quantity

1. The basic property. Multiplying (or dividing) the numerator and the denominator of a fraction by the same non-zero number gives an equal fraction: $\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}$.

2. Recognising equal fractions. For $b \neq 0$, $d \neq 0$: $\frac{a}{b} = \frac{c}{d}$ exactly when $a \cdot d = b \cdot c$ ("cross-multiplying").

3. Fractions with the same numerator. If $\frac{1}{p} = \frac{1}{q}$ (or $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$), then $p = q$. Two equal fractions with equal numerators also have equal denominators.

4. A fraction of a quantity. To find $\frac{m}{n}$ of a number A, calculate $A \cdot \frac{m}{n} = A \div n \cdot m$. *Example:* $\frac{3}{7}$ of 168 km is $168 \div 7 \cdot 3 = 24 \cdot 3 = 72$ km. The rest corresponds to $1 - \frac{3}{7} = \frac{4}{7}$.

Why is it true? Why does $\frac{1}{p} = \frac{1}{q}$ imply $p = q$? Cross-multiply: $1 \cdot q = p \cdot 1$, that is, $q = p$. In other words: if a cake is cut into p pieces and another identical cake into q pieces, and the pieces are the same size, then the numbers of pieces must be the same. □

⑧ Comparing and reasoning about order

1. The transitive property. If $a < b$ and $b < c$, then $a < c$. This lets us line up several numbers: $a < b < c < d$.

2. The "set it equal to k" technique. When several expressions are equal, for example $a - 2 = b + 2 = c + 1 = d - 3$, call the common value k:

$$a = k + 2, \qquad b = k - 2, \qquad c = k - 1, \qquad d = k + 3.$$

Now every number is compared with the same reference point k; ordering the "extras" $-2 < -1 < +2 < +3$ we get $b < c < a < d$.



3. Logical reasoning about order. For sentences like "Anna owns more cats than Ben", translate them into symbols ($A > B$), write down all the conditions and link them with the transitive property. A statement that "*must be true*" has to be true in *every* situation that fits the problem; a single counter-example is enough to reject it.

ThinkAlgo Tip. *A reasoning example.* Anna is taller than Ben; Chloe is shorter than Ben; David is taller than Chloe. Who is the shortest? In symbols (each letter is one child's height): $A > B$, $B > C$, $D > C$. From $A > B > C$ and $D > C$ we see that A, B and D are all greater than C: **Chloe is the shortest**. But we do *not* know whether David is taller or shorter than Anna (for instance, if Anna is 1.6 m, Ben 1.5 m and Chloe 1.2 m, then David being 1.7 m or 1.3 m both fit all the conditions), so the statement "David is the tallest" is *not necessarily* true.

Common mistake. Seeing that $a - 2$ and $b + 2$ are equal, many students conclude too quickly that "a is smaller because something is subtracted". The opposite is true: a must *lose* 2 to equal b *plus* 2, so a is exactly 4 more than b.

Note for parents. Use real objects: "If your bag loses 2 marbles, it has the same number as my bag plus 2 marbles. Which bag has more?" Let your child check with marbles. A concrete picture keeps the child from being fooled by plus and minus signs.

⑨ Worked examples

Example 1 (Translating words and evaluating an expression)

a) Evaluate the expression $A = 2x^2 - 3x + 1$ for $x = -2$.
b) Emma has \$50 in her money box. Every week she puts in another \$20. Write an expression for the amount (in dollars) in the money box after t weeks, then work out that amount after 6 weeks.

Guided solution.

Step 1. Part a: substitute $x = -2$, remembering the brackets

$$A = 2 \cdot (-2)^2 - 3 \cdot (-2) + 1.$$

Step 2. Calculate in order: powers → multiplication → addition and subtraction
$(-2)^2 = (-2) \cdot (-2) = 4$, so $2 \cdot 4 = 8$. $3 \cdot (-2) = -6$, so $-3 \cdot (-2) = +6$. Hence $A = 8 + 6 + 1 = 15$.

Step 3. Part b: find the pattern with a small table

Weeks	0	1	2	t
Amount (dollars)	50	$50 + 20$	$50 + 2 \cdot 20$	$50 + 20t$

The starting amount stays the same; the added part is 20 times the number of weeks. Expression: $50 + 20t$ (dollars).

Step 4. Substitute $t = 6$
$50 + 20 \cdot 6 = 50 + 120 = 170$ (dollars).

Common mistake. Do not write $(50 + 20) \cdot t$: the starting amount is there only *once*; it must not be multiplied by the number of weeks.

Answer: a) $A = 15$; b) $50 + 20t$ (dollars); after 6 weeks there is \$170.

Example 2 (Solving equations with the balance principle)

Solve the equations: **a)** $5x - 7 = 2x + 11$; **b)** $\dfrac{x}{12} = \dfrac{5}{4}$.

Guided solution.

Step 1. Part a: collect the terms with x on the left
Subtract $2x$ from both sides: $5x - 2x - 7 = 11$, that is, $3x - 7 = 11$.

Step 2. Collect the numbers on the right
Add 7 to both sides: $3x = 11 + 7 = 18$.

Step 3. Divide both sides by 3
$x = 18 \div 3 = 6$.

Check: Left side: $5 \cdot 6 - 7 = 30 - 7 = 23$. Right side: $2 \cdot 6 + 11 = 12 + 11 = 23$. Both sides are equal.

Step 4. Part b: use a common denominator (or cross-multiply)

We have $\frac{5}{4} = \frac{5 \cdot 3}{4 \cdot 3} = \frac{15}{12}$. The fractions $\frac{x}{12}$ and $\frac{15}{12}$ have the same denominator and are equal, so $x = 15$. (Cross-multiplying also works: $4x = 12 \cdot 5 = 60$, $x = 60 \div 4 = 15$.)

Recall. $\frac{a}{b} = \frac{c}{d}$ exactly when $ad = bc$ $(b, d \neq 0)$.

Answer: a) $x = 6$; **b)** $x = 15$.

Example 3 (Consecutive even numbers)

The sum of four consecutive even numbers is 116. Find the largest of the four numbers.

Guided solution.

Step 1. Choose an unknown

Let the smallest even number be n (n is even). Consecutive even numbers differ by 2, so the four numbers are $n,\ n+2,\ n+4,\ n+6$.

Step 2. Write an equation

$n + (n+2) + (n+4) + (n+6) = 116$, that is, $4n + 12 = 116$.

Step 3. Solve

$4n = 116 - 12 = 104$, $n = 104 \div 4 = 26$. The four numbers are $26,\ 28,\ 30,\ 32$.

Check: $26 + 28 + 30 + 32 = 116$. Correct.

Step 4. Answer the actual question

The question asks for the *largest* number: it is 32.

ThinkAlgo Tip. Mental shortcut: the four numbers are equally spaced, so their average $116 \div 4 = 29$ lies exactly halfway between the two middle numbers 28 and 30. The four numbers follow at once.

Answer: 32

Example 4 (Weighing in pairs)

Three friends, Anna, Ben and Chloe, step on a scale two at a time. Anna and Ben weigh 72 kg; Anna and Chloe weigh 78 kg; Ben and Chloe weigh 80 kg. How many kilograms does each child weigh?

Guided solution.

Step 1. Choose an unknown and express the other quantities with it

Let Anna's mass be x (kg), $x > 0$. Since Anna and Ben weigh 72 kg, Ben weighs $72 - x$ (kg). Since Anna and Chloe weigh 78 kg, Chloe weighs $78 - x$ (kg).

Step 2. Form an equation from the third weighing

Ben and Chloe weigh 80 kg, so

$$(72 - x) + (78 - x) = 80.$$

Step 3. Solve the equation

Simplify: $72 + 78 - x - x = 150 - 2x$, so $150 - 2x = 80$. Move terms across: $2x = 150 - 80 = 70$,

hence $x = 70 \div 2 = 35$.

Step 4. Find the masses of the other two children

Anna weighs 35 kg; Ben weighs $72 - 35 = 37$ (kg); Chloe weighs $78 - 35 = 43$ (kg).

> **ThinkAlgo Tip. Method 2.** Let the three masses be a, b, c (kg). Add $a + b = 72$, $a + c = 78$, $b + c = 80$ side by side (each letter appears twice): $2(a + b + c) = 230$, so $a + b + c = 115$. Hence $c = 115 - 72 = 43$, $b = 115 - 78 = 37$, $a = 115 - 80 = 35$.

Check: $35 + 37 = 72$, $35 + 43 = 78$, $37 + 43 = 80$. All three weighings match.

Answer: Anna 35 kg, Ben 37 kg, Chloe 43 kg.

Example 5 (Arithmetic sequences)

Consider the sequence 4, 7, 10, 13, ... (each number is 3 more than the one before). **a)** Write an expression for the n-th term. **b)** Find the 50th term. **c)** Is 2026 in the sequence? If so, which term is it?

Guided solution.

Step 1. The n-th term

First term $a = 4$, common difference $d = 3$: the n-th term is $4 + 3(n - 1) = 4 + 3n - 3 = 3n + 1$.

> **Recall.** The n-th term of an arithmetic sequence $= a + (n - 1)d$.

Step 2. The 50th term

$3 \cdot 50 + 1 = 151$.

Step 3. Test the number 2026

We need a whole number n with $3n + 1 = 2026$, that is, $3n = 2025$. Since $2025 \div 3 = 675$ exactly, $n = 675$.

Check: $3 \cdot 675 + 1 = 2025 + 1 = 2026$. Correct.

> **Common mistake.** Do not forget to check that the division is exact. For instance, 2027 is *not* in the sequence because 2026 is not divisible by 3.

Answer: a) $3n + 1$; b) 151; c) yes, it is the 675th term.

Example 6 (Motion, fractions and unit conversion)

A train travels at a constant speed of 72 km/h on a 360 km route. It has already covered $\frac{2}{3}$ of the route. How much longer (in hours and minutes) does the train need to reach its destination?

Guided solution.

Step 1. Distance already covered

$\frac{2}{3}$ of 360 km is $360 \div 3 \cdot 2 = 120 \cdot 2 = 240$ km.

Step 2. Distance left

$360 - 240 = 120$ km. (Or: the remaining part is $1 - \frac{2}{3} = \frac{1}{3}$ of the route, $360 \div 3 = 120$ km.)

Step 3. Time left

Recall. $s = v \cdot t$, so $t = \dfrac{s}{v}$ (time = distance ÷ speed).

Let t (hours) be the time the train still needs. Then $72 \cdot t = 120$, so $t = 120 \div 72 = \dfrac{120}{72} = \dfrac{5}{3}$ hours $= 1\dfrac{2}{3}$ hours.

Step 4. Convert to hours and minutes

$\dfrac{2}{3}$ hour $= \dfrac{2}{3} \cdot 60 = 40$ minutes. So $t = 1$ hour 40 minutes.

Check: In 1 hour 40 minutes $= \dfrac{5}{3}$ hours, the train covers $72 \cdot \dfrac{5}{3} = 24 \cdot 5 = 120$ km, exactly the distance left.

Answer: 1 hour 40 minutes.

Example 7 (A balance-scale problem)

Both scales below are balanced. Identical shapes have equal weights.



How many circles ○ balance two squares □□?

Guided solution.

Step 1. Write each scale as an equation

Let S, T, C be the weights of a square, a triangle and a circle. Scale 1: $S = T + C$. Scale 2: $S + C = 3T$.

Step 2. Substitute scale 1 into scale 2

On scale 2, replace the square by "one triangle and one circle" (allowed by scale 1): $T + C + C = 3T$.

Step 3. Remove identical things from both pans

Remove one triangle from each pan: $C + C = T + T$, that is, $2C = 2T$, so $C = T$: a circle weighs the same as a triangle.

Step 4. Answer

$S = T + C = C + C = 2C$, so $2S = 4C$: two squares balance 4 circles.

Check: Try $C = T = 1$, $S = 2$: scale 1: $2 = 1 + 1$; scale 2: $2 + 1 = 3$. Both are correct.

Answer: 4 circles.

Example 8 (Ordering with the "set it equal to k" technique)

The positive integers a, b, c (all greater than 5) satisfy $\frac{2}{a+3} = \frac{2}{b-1} = \frac{2}{c+5}$. Arrange a, b, c in increasing order.

Guided solution.

Step 1. Equal fractions with the same numerator have equal denominators

The three fractions all have numerator 2 and are equal, so their denominators are equal: $a+3 = b-1 = c+5$.

> **Recall.** If $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$, then $p = q$.

Step 2. Call the common value k

$a+3 = k \Rightarrow a = k-3$; $\quad b-1 = k \Rightarrow b = k+1$; $\quad c+5 = k \Rightarrow c = k-5$.

Step 3. Compare with the reference point k

$k-5 < k-3 < k+1$, so $c < a < b$.

Check: Take $k = 12$: $a = 9$, $b = 13$, $c = 7$. Then $\frac{2}{12} = \frac{2}{12} = \frac{2}{12}$ and indeed $7 < 9 < 13$.

Answer: $c < a < b$.

Core practice

Problem 1

Carrie drives a car at a constant speed of 85 km/h on a 510 km trip. She has now covered exactly half of the distance. How many more hours does she need to finish the trip?

(A) 5 hours (B) 425 hours (C) 12 hours (D) 1.5 hours (E) 3 hours

Detailed solution

Understanding the problem. *Given:* speed 85 km/h (constant), whole trip 510 km, half already covered. *Find:* the time needed for the remaining part.

Step 1. Work out the distance left

Half has been covered, so exactly half is left: $510 \div 2 = 255$ (km).

Step 2. Choose an unknown and form an equation

Recall. Distance = speed × time ($s = v \cdot t$), so time = distance ÷ speed ($t = \frac{s}{v}$).

Let t (hours) be the time Carrie still needs ($t > 0$). In t hours, at 85 km/h, she covers $85t$ km. This must equal the distance left: $85t = 255$.

Step 3. Solve the equation

Divide both sides by 85: $t = 255 \div 85 = 3$ (hours). (Mentally: $85 \times 3 = 255$.)

Common mistake. Calculating $510 \div 85 = 6$ hours gives the time for the *whole* trip, which does not answer the question. And $510 - 85 = 425$ is a meaningless calculation (km minus km/h); that is the trap in option (B).

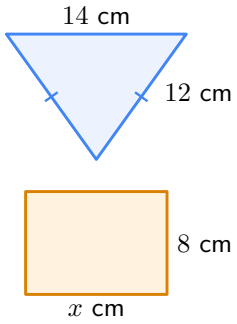
ThinkAlgo Tip. Another way: the whole trip takes $510 \div 85 = 6$ hours; covering half the distance (at a constant speed) means half the time has passed, so $6 \div 2 = 3$ hours are left.

Check: In 3 hours the car covers $85 \times 3 = 255$ km, exactly the distance left.
The result 3 hours is option **(E)**.

Answer: (E) 3 hours

Problem 2

In the diagram, the triangle has one side of length 14 cm and two equal sides of length 12 cm each; the rectangle has width 8 cm and length x cm. The perimeter of the triangle equals the perimeter of the rectangle. Find x.



(A) 8 (B) 10 (C) 11 (D) 14 (E) 15

Detailed solution

Understanding the problem. The triangle has sides 14 cm, 12 cm, 12 cm (the small tick marks show that those two sides are equal). The rectangle measures x cm by 8 cm. The two perimeters are equal. We must find x.

Step 1. Work out the perimeter of the triangle

Recall. The perimeter of a triangle is the sum of its three sides. The perimeter of a rectangle is $2 \times (\text{length} + \text{width})$.

Perimeter of the triangle: $14 + 12 + 12 = 38$ (cm).

Common mistake. The number 12 is written only once in the diagram, but there are *two* sides of 12 cm (marked as equal). Forgetting the third side gives 26 cm and a wrong answer.

Step 2. Write an equation

The perimeter of the rectangle is $2(x + 8)$. Setting the two perimeters equal: $2(x + 8) = 38$.

Step 3. Solve the equation

Divide both sides by 2: $x + 8 = 19$. Subtract 8: $x = 19 - 8 = 11$, which is option **(C)**.

Check: An 11 cm $\times$ 8 cm rectangle has perimeter $11 + 8 + 11 + 8 = 38$ cm, the same as the triangle.

Note for parents. Use a 38 cm piece of string: bend it into a 14 – 12 – 12 triangle, then into a rectangle with an 8 cm side. The "perimeter" is just the length of the string, which does not change when the shape changes.

Answer: (C) 11

Problem 3

Let $x = 3$. Which of the following equations is true?

(A) $2x = 5$ (B) $3x - 1 = 8$ (C) $x + 5 = 3$ (D) $7 - x = 2$ (E) $6 + 2x = 14$

Detailed solution

Understanding the problem. Substitute $x = 3$ into the left side of each equation, work it out, and compare with the right side. Only one equation is true.

Recall. To evaluate an expression, replace the letter by the number and calculate in order: multiplication and division first, then addition and subtraction.

Step 1. Check each option in turn

Option	Left side with $x = 3$	Right side	True/False
(A)	$2x = 2 \cdot 3 = 6$	5	False
(B)	$3x - 1 = 3 \cdot 3 - 1 = 9 - 1 = 8$	8	**True**
(C)	$x + 5 = 3 + 5 = 8$	3	False
(D)	$7 - x = 7 - 3 = 4$	2	False
(E)	$6 + 2x = 6 + 2 \cdot 3 = 6 + 6 = 12$	14	False

Common mistake. In (E), working in the wrong order, $(6 + 2) \cdot 3 = 24$, or reading $2x$ as "23" leads to mistakes. Multiply first, then add: $6 + 2 \cdot 3 = 12$.

Step 2. Conclusion

Only the equation in (B) is true: $3 \cdot 3 - 1 = 8$. So the answer is **(B)**.

Answer: (B) $3x - 1 = 8$

Problem 4

The sum of three consecutive integers is 153. The largest of the three integers is

(A) 52 (B) 50 (C) 53 (D) 54 (E) 51

Detailed solution

Understanding the problem. Three consecutive integers stand next to each other on the number line and differ by 1 (for example $7, 8, 9$). Their sum is 153. We want the *largest* one.

Step 1. Choose an unknown: let the middle number be n

The three numbers are $n - 1$, n, $n + 1$.

Step 2. Write an equation

$(n - 1) + n + (n + 1) = 153$. On the left, -1 and $+1$ cancel, leaving $3n = 153$.

Step 3. Solve

$n = 153 \div 3 = 51$. The three numbers are 50, 51, 52.

Recall. For three consecutive integers, the middle number equals the sum divided by 3 (the middle number is the average).

Why is it true? Another way, letting the smallest number be m: $m+(m+1)+(m+2) = 3m+3 = 153$, so $3m = 150$, $m = 50$. The same three numbers $50, 51, 52$. □

ThinkAlgo Tip. Multiple-choice shortcut: the largest number must be a little more than the average $153 \div 3 = 51$; try $50 + 51 + 52 = 153$ straight away.

Check: $50 + 51 + 52 = 153$. Correct.

Step 4. Answer the actual question

The largest number is 52, which is option **(A)**.

Common mistake. Option (E) 51 is the *middle* number and option (B) 50 is the *smallest* number. Read the question carefully so you do not choose the wrong one.

Answer: (A) 52

Problem 5

One scoop of fish food can feed 8 goldfish. How many goldfish can 4 such scoops of fish food feed?

(A) 12 (B) 16 (C) 8 (D) 64 (E) 32

Detailed solution

Understanding the problem. 1 scoop → 8 fish. Asked: 4 scoops → ? fish. When the number of scoops is multiplied by some number, the number of fish fed is multiplied by the same number (the quantities are directly proportional).

Recall. If y is directly proportional to x with constant k, then $y = k \cdot x$. Here $k = 8$ (fish per scoop), so number of fish = 8× number of scoops.

Step 1. Make a table of the two directly proportional quantities

Let y be the number of fish fed with x scoops of food. Then $y = 8x$.

Scoops x	1	2	3	4
Fish $y = 8x$	8	16	24	?

Step 2. Substitute $x = 4$

$y = 8 \cdot 4 = 32$ (fish).

Common mistake. Option (A) 12 is $8 + 4$ (adding instead of multiplying), and option (D) 64 is 8×8. Each extra scoop feeds 8 *more fish*, so multiply 8 by the number of scoops.

Check: $32 \div 4 = 8$: exactly 8 fish per scoop.
So the answer is **(E)**.

Answer: (E) 32

Problem 6

If $x = 4$ and $y = 3x$, the value of y is

(A) 12 (B) 24 (C) 7 (D) 81 (E) 4

Detailed solution

Understanding the problem. We know $x = 4$. The expression $y = 3x$ means "y equals 3 times x". We must find y.

Recall. $3x$ is short for $3 \cdot x$ (between a number and a letter we leave out the multiplication sign).

Step 1. Substitute $x = 4$
$y = 3 \cdot x = 3 \cdot 4 = 12$.

Common mistake. Do not read $3x$ as "34" (joining the digits); it is not $3+4 = 7$ (option C) or $3^4 = 81$ (option D) either.

ThinkAlgo Tip. Read $y = 3x$ in words: "y is three times x". Three times 4 is 12.

Check: $12 \div 3 = 4 = x$. Correct.
So the answer is **(A)**.

Answer: (A) 12

Problem 7

Each bus ride costs Alex \$2.25, while Sam pays \$3.00 per ride (they use different routes). If each of them rides the bus 20 times, how much less does Alex pay than Sam in total?

(A) \$25 (B) \$10 (C) \$15 (D) \$45 (E) \$60

Detailed solution

Understanding the problem. Per ride: Alex pays \$2.25, Sam pays \$3.00. Each rides 20 times. We want the *difference* after 20 rides.

Step 1. Method 1 – Work out each total, then subtract
Alex pays: $20 \times 2.25 = 45$ (dollars). Sam pays: $20 \times 3 = 60$ (dollars).
Difference: $60 - 45 = 15$ (dollars).

Step 2. Method 2 – Find the difference per ride, then multiply (quicker)
Each ride, Alex pays less than Sam by: $3.00 - 2.25 = 0.75$ (dollars).

After 20 rides: $20 \times 0.75 = 15$ (dollars).

Recall. The distributive law: $20 \times 3 - 20 \times 2.25 = 20 \times (3 - 2.25)$. That is why both methods always agree.

Common mistake. Option (D) \$45 is *Alex's total* and option (E) \$60 is *Sam's total*. The question asks how much *less*, which is the difference.

Check: $45 + 15 = 60$. Correct.
So the answer is **(C)**.

Answer: (C) \$15

Problem 8

Which expression produces the values in the second row of the table, given the values of n in the first row?

n	1	2	3	4	5
Value	1	3	5	7	9

(A) $3n - 2$ (B) $2(n - 1)$ (C) $n + 4$ (D) $2n$ (E) $2n - 1$

Detailed solution

Understanding the problem. We need an expression in n which gives $1, 3, 5, 7, 9$ when $n = 1, 2, 3, 4, 5$.

Step 1. Find the pattern in the row of values
Each time n increases by 1, the value increases by 2 ($1 \to 3 \to 5 \to 7 \to 9$). So the expression has the form $2n + b$.

Recall. If the value always increases by d when n increases by 1, the expression has the form $d \cdot n + b$; find b by substituting $n = 1$.

Step 2. Find b
Substitute $n = 1$: $2 \cdot 1 + b = 1$, so $b = 1 - 2 = -1$. The expression is $2n - 1$.

Step 3. Check the whole table and rule out the other options

n	1	2	3	4	5	Verdict
(A) $3n - 2$	1	4				fails at $n = 2$
(B) $2(n - 1)$	0					fails at $n = 1$
(C) $n + 4$	5					fails at $n = 1$
(D) $2n$	2					fails at $n = 1$
(E) $2n - 1$	1	3	5	7	9	**fits the whole table**

Common mistake. Option (A) $3n-2$ also gives 1 when $n=1$. If you only test the first column, you cannot decide between (A) and (E). Test other columns too.

Check: $2n-1$ for $n=1,2,3,4,5$: $1,3,5,7,9$. It matches the whole table.
So the answer is **(E)**.

Answer: (E) $2n-1$

Problem 9

When two numbers are added, the result is -26. One of the numbers is 11. The other number is

(A) -37 (B) 37 (C) -15 (D) 15 (E) -48

Detailed solution

Understanding the problem. Call the unknown number x. We are told that $x+11=-26$. We must find x.

Step 1. Write an equation
$x+11=-26$.

Step 2. Solve with the balance principle
Subtract 11 from both sides: $x=-26-11$.

Recall. Subtracting a number means adding its opposite: $-26-11=-26+(-11)$. Adding two negative numbers: add the sizes and keep the "$-$" sign: $-(26+11)=-37$.

So $x=-37$.



On the number line: starting from -37 and moving 11 steps to the right, we arrive at -26.

Common mistake. Forgetting to change the sign when moving a term across, and writing $x=-26+11=-15$, is wrong (option C). Check: $-15+11=-4\neq-26$.

Note for parents. You can explain it with a story about owing money: "After someone gives you \$11 to pay back part of a debt, you still owe \$26. How much did you owe at first?" At first you owed more: $26+11=37$ dollars, that is, -37.

Check: $(-37)+11=-26$. Correct.
So the answer is **(A)**.

Answer: (A) -37

Problem 10

Mikhail has \$10 000, all in \$50 bills. How many \$50 bills does he have?

(A) 1 000 (B) 200 (C) 1 250 (D) 500 (E) 2 000

Detailed solution

Understanding the problem. The total amount is \$10 000 (ten thousand dollars). Each bill is worth \$50. We must find the number of bills.

Recall. Number of bills = total amount ÷ value of one bill.

Step 1. Choose an unknown and form an equation
Let n be the number of \$50 bills (n is a whole number). The n bills are worth $50n$ dollars, so $50n = 10\,000$.

Step 2. Solve: divide both sides by 50, simplifying by removing zeros
$n = 10\,000 \div 50$. Remove one zero from both numbers (that is, divide both by 10; the quotient does not change): $10\,000 \div 50 = 1\,000 \div 5 = 200$.

Step 3. Another way of thinking
2 bills of \$50 make \$100. Ten thousand dollars is 100 times \$100, so there are $100 \times 2 = 200$ bills.

Check: $200 \times 50 = 10\,000$ dollars. Correct.

Common mistake. Miscounting the zeros is a very common mistake: $10\,000 \div 50$ is *not* 2 000 (option E) or 1 000 (option A). Write numbers with a space between groups of three digits (10 000) to count accurately.

So the answer is **(B)**.

Note for parents. You can use play money: a bundle of 10 bills of \$50 is \$500. Ask your child: how many such bundles make \$10 000? (20 bundles, that is, 200 bills.)

Answer: (B) 200

Problem 11

The product $60 \times 60 \times 24 \times 7$ equals

(A) the number of minutes in seven weeks

(B) the number of hours in sixty days

(C) the number of seconds in seven hours

(D) the number of seconds in one week

(E) the number of minutes in twenty-four weeks

Detailed solution

Understanding the problem. There is no need to multiply the product out. Read each factor as one step of a time-unit conversion.

Recall. 1 week = 7 days; 1 day = 24 hours; 1 hour = 60 minutes; 1 minute = 60 seconds.

Step 1. Read the product from right to left

- 7 is the number of days in **one week**;
- 24×7 is the number of hours in one week;
- $60 \times 24 \times 7$ is the number of minutes in one week;
- $60 \times 60 \times 24 \times 7$ is the number of **seconds** in one week.

Step 2. Check the other options by writing their products

Option	Expression	Value
(A) minutes in 7 weeks	$60 \times 24 \times 7 \times 7$	70 560
(B) hours in 60 days	24×60	1 440
(C) seconds in 7 hours	$60 \times 60 \times 7$	25 200
(D) seconds in 1 week	$60 \times 60 \times 24 \times 7$	604 800
(E) minutes in 24 weeks	$60 \times 24 \times 7 \times 24$	241 920

Only option (D) has exactly the product $60 \times 60 \times 24 \times 7 = 604\,800$.

Common mistake. Seeing the numbers 60, 24, 7 in the wording, many students rush to choose (A) or (E). Match *every* conversion step: (A) is missing one factor 60 and has an extra factor 7.

So the answer is **(D)**.

Answer: (D) the number of seconds in one week

Problem 12 ⋆

Each of a, b, c, d is a positive integer greater than 3, and

$$\frac{1}{a-2} = \frac{1}{b+2} = \frac{1}{c+1} = \frac{1}{d-3}.$$

Which ordering of these four numbers is correct?

(A) $a < b < c < d$ (B) $c < b < a < d$ (C) $b < a < c < d$

(D) $d < a < c < b$ (E) $b < c < a < d$

Detailed solution

Understanding the problem. Four fractions with the same numerator 1 are equal. We must compare a, b, c, d. (The condition "greater than 3" makes sure all the denominators are non-zero, so the fractions make sense.)

Step 1. Equal fractions with the same numerator have equal denominators

> **Recall.** If $\frac{1}{p} = \frac{1}{q}$, then $p = q$ (cross-multiply: $1 \cdot q = 1 \cdot p$).

Therefore: $a - 2 = b + 2 = c + 1 = d - 3$.

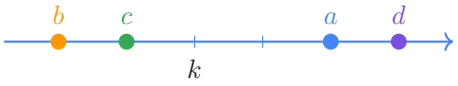
Step 2. Call the common value k and write each number in terms of k

$a - 2 = k \Rightarrow a = k + 2$; $\quad b + 2 = k \Rightarrow b = k - 2$; $\quad c + 1 = k \Rightarrow c = k - 1$; $\quad d - 3 = k \Rightarrow d = k + 3$.

Step 3. Compare with the same reference point k

The "extras" are -2 (for b), -1 (for c), $+2$ (for a), $+3$ (for d). Since $-2 < -1 < 2 < 3$,

$$k - 2 < k - 1 < k + 2 < k + 3, \quad \text{that is,} \quad b < c < a < d.$$



> **Common mistake.** Seeing the minus sign in "$a - 2$", many students think a is small. But a has to *lose* 2 to equal k, so a is bigger than k. On the other hand, b has to *gain* 2 to equal k, so b is smaller than k.

Check: Try $k = 6$ (so that all four numbers are greater than 3): $a = 8$, $b = 4$, $c = 5$, $d = 9$. Then $\frac{1}{8-2} = \frac{1}{4+2} = \frac{1}{5+1} = \frac{1}{9-3} = \frac{1}{6}$ and indeed $4 < 5 < 8 < 9$, that is, $b < c < a < d$. The ordering $b < c < a < d$ is option **(E)**.

Answer: (E) $b < c < a < d$

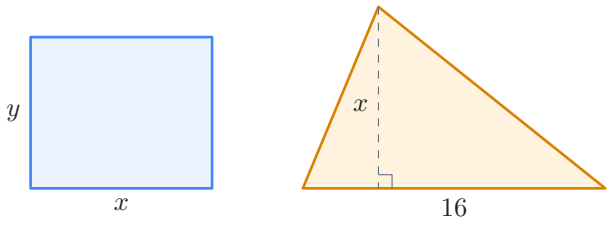
Problem 13

A rectangle has length x and width y. A triangle has base 16 and corresponding height x. If the area of the rectangle equals the area of the triangle, then the value of y is

(A) 16 (B) 4 (C) 8 (D) 12 (E) 32

Detailed solution

Understanding the problem. Rectangle: length x, width y. Triangle: base 16, height x. The two areas are equal. Find y (note: x is not given, but we will see that x "disappears").



Step 1. Write the two area formulas

Recall. Area of a rectangle = length × width. Area of a triangle = $\frac{\text{base} \times \text{height}}{2}$.

Rectangle: $A_1 = x \cdot y = xy$. Triangle: $A_2 = \frac{16 \cdot x}{2} = 8x$.

Step 2. Write an equation and solve

$xy = 8x$. Since x is a length, $x > 0$; dividing both sides by x: $y = 8$.

Why is it true? You can also see it without dividing: $x \cdot y$ and $x \cdot 8$ are "x groups of y" and "x groups of 8". If the products are equal with the same number of groups $x \neq 0$, each group must be the same: $y = 8$. □

Common mistake. Forgetting to halve when finding the area of the triangle leads to $y = 16$ (option A).

Check: Try $x = 5$: rectangle $5 \times 8 = 40$; triangle $\frac{16 \cdot 5}{2} = 40$. Equal. (It works for any x.)
So the answer is **(C)**.

Answer: (C) 8

Problem 14

The symbol $\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 5 & 6 \\ \hline \end{array}$ is evaluated as $3 \times 6 + 4 \times 5 = 38$. If $\begin{array}{|c|c|}\hline 2 & 6 \\ \hline 1 & \\ \hline \end{array}$ is evaluated as 16, then the number that should be placed in the empty space is

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Detailed solution

Understanding the problem. First we must understand the rule from the given example, then apply it to the box with the unknown number.

Step 1. Find the rule

In $\begin{array}{|c|c|}\hline 3 & 4 \\ \hline 5 & 6 \\ \hline \end{array}$: 3 (top left) times 6 (bottom right), plus 4 (top right) times 5 (bottom left): $3{\times}6{+}4{\times}5 = 18 + 20 = 38$. That is: *multiply the two numbers on each diagonal, then add.* In general, $\begin{array}{|c|c|}\hline a & b \\ \hline c & d \\ \hline \end{array} = a \times d + b \times c$.

Step 2. Name the empty space and write an equation

Let the number in the empty space (bottom right) be x. By the rule: $\begin{array}{|c|c|}\hline 2 & 6 \\ \hline 1 & x \\ \hline \end{array} = 2\times x+6\times 1 = 2x+6$. We are told the value is 16: $2x + 6 = 16$.

Step 3. Solve

Subtract 6 from both sides: $2x = 10$. Divide both sides by 2: $x = 5$.

Check: $2 \times 5 + 6 \times 1 = 10 + 6 = 16$. Correct.

Common mistake. Do not pair the wrong numbers: $2 \times 6 + 1 \times x$ (multiplying along rows) breaks the rule; that way you would get $x = 4$ (option D). Check the rule against the example 38 before using it.

So the answer is **(E)**.

Note for parents. Together with your child, "invent" a new symbol (for example, multiply along the rows and then subtract) and quiz each other. The important habit is always to test a rule on the given example before using it.

Answer: (E) 5

Problem 15

Three pumpkins are weighed two at a time in all possible ways. The weights of the pairs are 12 kg, 13 kg and 15 kg. How much does the lightest pumpkin weigh?

(A) 4 kg (B) 5 kg (C) 6 kg (D) 7 kg (E) 8 kg

Detailed solution

Understanding the problem. Three pumpkins can be paired in exactly 3 ways. Let their weights from lightest to heaviest be $a \le b \le c$ (kg). The lightest pair is the two lightest pumpkins, and the heaviest pair is the two heaviest pumpkins: $a + b = 12$, $a + c = 13$, $b + c = 15$.

Step 1. Add all three pairs

Each pumpkin appears in exactly *two* pairs: $(a + b) + (a + c) + (b + c) = 12 + 13 + 15 = 40$, that is, $2(a + b + c) = 40$.

Recall. Three objects weighed in pairs: the total of the three objects is half of the sum of the results.

Step 2. Total weight of the three pumpkins

$a + b + c = 40 \div 2 = 20$ (kg).

Step 3. Find the lightest pumpkin

The lightest pumpkin a is the one that is *not* in the heaviest pair ($b + c = 15$): $a = (a + b + c) - (b + c) = 20 - 15 = 5$ (kg). (Similarly: $b = 20 - 13 = 7$ kg, $c = 20 - 12 = 8$ kg.)

ThinkAlgo Tip. Another way (substitution). From $a+b=12$ we get $b=12-a$; from $a+c=13$ we get $c=13-a$. Substitute into $b+c=15$: $(12-a)+(13-a)=15$, that is, $25-2a=15$, so $2a=10$ and $a=5$.

Check: $5+7=12$, $5+8=13$, $7+8=15$. All three pairs are correct.

Common mistake. Do not take $12\div 2=6$ kg and assume each pumpkin in the lightest pair weighs 6 kg: the two pumpkins in a pair need not weigh the same. Option (E) 8 kg is the *heaviest* pumpkin.

The lightest pumpkin weighs 5 kg, which is option **(B)**.

Note for parents. Let your child take three objects at home and a scale: weigh them in pairs, add the three results and halve, then compare with putting all three on the scale together. Your child will see for themselves why we divide by 2.

Answer: (B) 5 kg

Problem 16

Rosie is learning to save money. Today she has \$120 in her savings account. From now on, every time she makes a deposit she will add exactly \$30. If Rosie makes m such deposits, which expression correctly gives the amount of money (in dollars) in her account?

(A) $120+m$ (B) $30m$ (C) $30+120m$ (D) $150m$ (E) $120+30m$

Detailed solution

Analysis. Given: at the start there is \$120; each deposit adds \$30; there are m deposits. Find: an algebraic expression for the total after m deposits.

Step 1. Try a few concrete values of m to see the pattern

Deposits	Money added	Total (dollars)
0	0	120
1	30	$120+30=150$
2	$30+30=30\cdot 2=60$	$120+60=180$
3	$30\cdot 3=90$	$120+90=210$
m	$30\cdot m=30m$	$120+30m$

Step 2. Write the expression

The money *added* after m deposits is m times 30, that is $30m$. Adding the 120 already in the account gives

$$\text{Total}=120+30m \text{ (dollars)}.$$

Recall. $30m$ means $30 \cdot m$ (we may leave out the multiplication sign between a number and a letter). Adding a number a to itself m times gives $a \cdot m$.

Step 3. Rule out the wrong options

- (A) $120 + m$: adds only \$1 each time – wrong.
- (B) $30m$: forgets the \$120 already there. For $m = 0$ it gives 0 – wrong.
- (C) $30 + 120m$: swaps the roles of the two numbers. For $m = 1$ it gives 150 (the same!), but for $m = 2$ it gives $270 \neq 180$ – wrong.
- (D) $150m$: for $m = 2$ it gives $300 \neq 180$ – wrong.

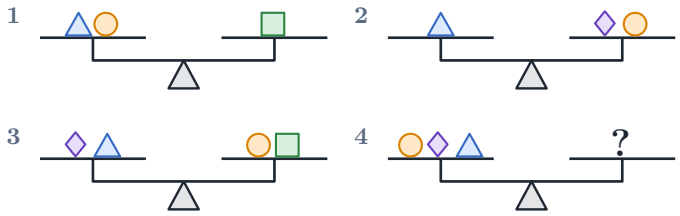
Common mistake. Testing only *one* value, $m = 1$, is not enough: (C), (D) and (E) all give 150. Also try $m = 0$ or $m = 2$ to tell them apart.

Check: For $m = 3$: $120 + 30 \cdot 3 = 210$, exactly as in the table of Step 1.
So the answer is **(E)**.

Answer: (E) $120 + 30m$

Problem 17

All four scales below are balanced. Identical shapes have equal masses. One possible replacement for the "?" that keeps the fourth scale balanced is



(A) △□ (B) ◇△ (C) ○□ (D) □◇ (E) △○

Detailed solution

Analysis. Each balanced scale gives an *equation*: the total mass on the left pan equals the total mass on the right pan. We write the shapes straight into the equations (each shape stands for its mass):

Scale 1: △ + ○ = □, Scale 2: △ = ◇ + ○, Scale 3: ◇ + △ = ○ + □.

Find: the right pan of scale 4 must weigh the same as ○ + ◇ + △. Idea: convert every shape into one common "unit", namely ○.

Recall. If a scale is balanced, then *replacing* an object by objects of the same mass, or *removing* identical objects from both pans, keeps it balanced. This is exactly the property of equations: $a = b \Rightarrow a + c = b + c$ and $a - c = b - c$.

Step 1. Use scale 2 to replace △ in scale 1

By scale 2, $\triangle = \diamond + \bigcirc$. Substituting into scale 1:

$$\square = \triangle + \bigcirc = (\diamond + \bigcirc) + \bigcirc = \diamond + 2\bigcirc.$$

Step 2. Substitute into scale 3 to find ◇

Left pan of scale 3: $\diamond + \triangle = \diamond + (\diamond + \bigcirc) = 2\diamond + \bigcirc$.

Right pan of scale 3: $\bigcirc + \square = \bigcirc + (\diamond + 2\bigcirc) = \diamond + 3\bigcirc$.

So $2\diamond + \bigcirc = \diamond + 3\bigcirc$. Remove one ◇ and one ○ from both pans:

$$\diamond = 2\bigcirc.$$

Step 3. Express every shape in units of ○

$$\diamond = 2\bigcirc, \qquad \triangle = \diamond + \bigcirc = 3\bigcirc, \qquad \square = \diamond + 2\bigcirc = 4\bigcirc.$$

Left pan of scale 4: $\bigcirc + \diamond + \triangle = 1 + 2 + 3 = 6$ (○).

Step 4. Work out each option (in units of ○)

	(A) △□	(B) ◇△	(C) ○□	(D) □◇	(E) △○
Mass	$3 + 4 = 7$	$2 + 3 = 5$	$1 + 4 = 5$	$4 + 2 = \mathbf{6}$	$3 + 1 = 4$

Only (D) weighs exactly 6 ○.

Check: Let ○ weigh 1 kg; then ◇, △, □ weigh 2, 3, 4 kg. Scale 1: $3 + 1 = 4$ ✓; scale 2: $3 = 2 + 1$ ✓; scale 3: $2 + 3 = 1 + 4$ ✓; scale 4: $1 + 2 + 3 = 6 = 4 + 2$ ✓.

ThinkAlgo Tip. When there are several unknowns, choose *one* shape as the unit of measurement and express all the others in terms of it – just like changing all amounts of money into the same kind of banknote.

So the answer is **(D)**.

Answer: (D) □◇

Problem 18

The number that goes in the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Detailed solution

Analysis. Two fractions are equal and the denominator of the second one is 63. Find its numerator.

> **Recall. Basic property of fractions:** multiplying both the numerator and the denominator of a fraction by the same non-zero number gives an equal fraction: $\frac{a}{b} = \frac{a \cdot k}{b \cdot k}$ $(k \neq 0)$.

Step 1. What was the denominator multiplied by?
From 7 to 63: $63 \div 7 = 9$. So the denominator was multiplied by 9.

Step 2. Multiply the numerator by the same number
To keep the fraction unchanged, the numerator must also be multiplied by 9: $3 \cdot 9 = 27$.

$$\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}.$$

Step 3. Another way: "cross-multiplication"
$\frac{a}{b} = \frac{c}{d}$ exactly when $a \cdot d = b \cdot c$. Call the missing number x: $3 \cdot 63 = 7 \cdot x$, so $189 = 7x$ and $x = 189 \div 7 = 27$.

> **Common mistake.** Some students see $63 - 7 = 56$ and then compute $3 + 56 = 59$ (option C). A fraction does *not* stay the same when we *add* the same number to its numerator and denominator! We must *multiply* (or divide).

Check: Dividing 27 and 63 by 9 gives 3 and 7, so $\frac{27}{63} = \frac{3}{7}$ ✓.
So the answer is **(A)**.

Answer: (A) 27

Problem 19

The sum of four numbers is T. Each of the four numbers is increased by 1. The four new numbers are added together, and the sum is then multiplied by 3. The final result is

(A) $3T + 3$ (B) $3T + 4$ (C) $3T + 12$ (D) $T + 12$ (E) $12T$

Detailed solution

Analysis. There are four unknown numbers and we only know their sum is T. We do not need each number – only how the *sum* changes.

Step 1. Name the numbers
Call the four numbers a, b, c, d. We are told $a + b + c + d = T$.

Step 2. Increase each number by 1 and add
The new numbers are $a + 1$, $b + 1$, $c + 1$, $d + 1$. Their sum is

$$(a + 1) + (b + 1) + (c + 1) + (d + 1) = (a + b + c + d) + (1 + 1 + 1 + 1) = T + 4.$$

Recall. Addition is commutative and associative, so we may group the letters together and the numbers together.

Step 3. Multiply by 3

$$3 \cdot (T + 4) = 3 \cdot T + 3 \cdot 4 = 3T + 12.$$

Recall. Distributive property: $k \cdot (x + y) = k \cdot x + k \cdot y$. The 3 must multiply *each* term in the brackets.

Common mistake. Two common mistakes: (1) adding only 1 to the sum (instead of 4, since there are four numbers), which gives $3T + 3$; (2) forgetting to multiply the 4 by 3, which gives $3T + 4$.

Check: Take the numbers $1, 2, 3, 4$: $T = 10$. The new numbers $2, 3, 4, 5$ add to 14; times 3 gives 42. And $3T + 12 = 30 + 12 = 42$ ✓. The other options give 33, 34, 22, 120 – all wrong. So the answer is **(C)**.

Answer: (C) $3T + 12$

Problem 20

Two isosceles triangles each have at least one angle measuring 70°. In the first triangle, the measure (in degrees) of each of the other two angles is even. In the second triangle, the measure of each of the other two angles is odd. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) 280° (B) 250° (C) 220° (D) 200° (E) 300°

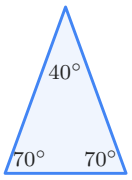
Detailed solution

Analysis. An isosceles triangle has two equal base angles. The 70° angle could be the *apex angle* or a *base angle*. We look at both possibilities and see which one gives "the other two angles even" and which one gives "the other two angles odd".

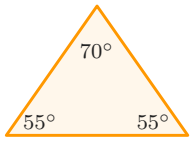
Recall. The three angles of a triangle add up to 180°. In an isosceles triangle the two base angles are equal.

Step 1. Possibility 1: the 70° angle is a base angle
Then the other base angle is also 70°. The apex angle is $180° - 70° - 70° = 40°$.
The angles are 70°, 70°, 40°. The other two angles (apart from one 70° angle) are 70° and 40° – **both even**.

Step 2. Possibility 2: the 70° angle is the apex angle
The two base angles are equal and add up to $180° - 70° = 110°$, so each is $110° \div 2 = 55°$.
The angles are 70°, 55°, 55°. The other two angles are 55° and 55° – **both odd**.



First triangle



Second triangle

Step 3. Find S and T
The first triangle (other two angles even) is possibility 1. Its two *equal* angles are the two 70° angles: $S = 70° + 70° = 140°$.
The second triangle (other two angles odd) is possibility 2. Its two equal angles are the 55° angles: $T = 55° + 55° = 110°$.

Step 4. Compute $S + T$
$S + T = 140° + 110° = 250°$.

Common mistake. "The sum of the equal angles" is the sum of the two *base* angles, not the sum of the two "other" angles. In the first triangle the other two angles are 70° and 40° (sum 110°), but the equal angles are 70° and 70° (sum 140°). Mixing these up gives $110° + 110° = 220°$ (option C).

Check: $70 + 70 + 40 = 180$ and $70 + 55 + 55 = 180$ ✓.
So the answer is **(B)**.

Answer: (B) 250°

Problem 21

Andrea is on a 6-day kayak trip with a total distance of 168 km. At the end of the third day she has covered $\frac{3}{7}$ of the total distance. How many kilometres per day must she average for the rest of the trip?

(A) 29 (B) 24 (C) 27 (D) 32 (E) 26

Detailed solution

Analysis. Given: total distance 168 km; the trip lasts 6 days; after 3 days $\frac{3}{7}$ of the distance is done. Find: the average number of km per day for the remaining days.

Recall. To find $\frac{m}{n}$ of a number, multiply the number by $\frac{m}{n}$ (divide by n, then multiply by m).

Step 1. Distance already covered
$\frac{3}{7}$ of 168 km is $168 \cdot \frac{3}{7} = 168 \div 7 \cdot 3 = 24 \cdot 3 = 72$ km.

Step 2. Days left
The trip lasts 6 days and 3 are over, so $6 - 3 = 3$ days remain.

Step 3. Choose an unknown and form an equation
Let x (km) be the average distance Andrea must paddle per day over the remaining 3 days ($x > 0$). In those 3 days she paddles $3x$ km. The distance done plus the distance still to go equals the whole trip:

$$72 + 3x = 168.$$

Step 4. Solve the equation
Move 72 across: $3x = 168 - 72 = 96$. Divide both sides by 3: $x = 96 \div 3 = 32$ km per day.

ThinkAlgo Tip. You can also think: the part left is $1 - \frac{3}{7} = \frac{4}{7}$ of the distance, that is $168 \div 7 \cdot 4 = 96$ km, shared equally over 3 days gives 32 km per day.

Common mistake. Do not confuse "the fraction of the distance left" ($\frac{4}{7}$) with "the number of days left" (3 days). Option (B), 24 km, is just the length of *one seventh* of the distance.

Check: $72 + 3 \cdot 32 = 72 + 96 = 168$ km ✓.
So the answer is **(D)**.

Answer: (D) 32 km per day

Problem 22 *

A dot starts at the point $(20, 19)$ of the coordinate plane. On the first move it travels 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On the second move it travels 2 units (in either direction) *perpendicular* to the first move. From then on, each move is 1 unit longer than the previous one (3 units, 4 units, 5 units, ...) and is always perpendicular to the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Detailed solution

Analysis. The moves have lengths $1, 2, 3, \ldots, 10$. "Perpendicular to the previous move" means the moves *alternate*: horizontal – vertical – horizontal – vertical... We must find the point the dot can never reach.

Recall. In the coordinate plane, moving right (left) k units increases (decreases) the first coordinate by k; moving up (down) k units increases (decreases) the second coordinate by k.

Step 1. Split the moves into two groups
If move 1 is horizontal, then the horizontal moves are moves $1, 3, 5, 7, 9$ (odd lengths) and the vertical moves are moves $2, 4, 6, 8, 10$ (even lengths). If move 1 is vertical, it is the other way round.
Each move can go in the positive direction (sign +) or the negative direction (sign −). So the

displacement in one direction is

$$\pm 1 \pm 3 \pm 5 \pm 7 \pm 9 \quad \text{(odd group)}, \qquad \pm 2 \pm 4 \pm 6 \pm 8 \pm 10 \quad \text{(even group)}.$$

Step 2. Which numbers can the odd group give?
If every sign is $+$: $1+3+5+7+9=25$. Changing the sign of a move of length a from $+$ to $-$ lowers the total by $2a$ (we lose a and then subtract another a). Hence

$$\text{displacement} = 25 - 2 \times (\text{sum of the lengths whose sign is changed}).$$

To get 21, the lengths whose sign is changed must add up to $(25-21) \div 2 = 2$. But no choice of numbers from $\{1,3,5,7,9\}$ adds up to 2 (choosing 1 gives 1; choosing any further number already gives at least 4). So **the odd group cannot give a displacement of** 21 (and likewise not -21).

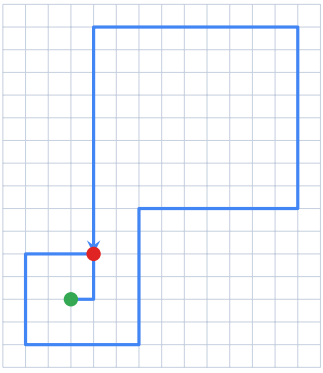
Why is it true? The possible sums of numbers chosen from $\{1,3,5,7,9\}$ are $0,1,3,4,5,6,7,8,9,\ldots$ but *not* 2: the smallest number is 1 and the next smallest is 3, so choosing two or more numbers gives at least $1+3=4$. □

Step 3. Check option (B) $(30,40)$
Displacement: horizontal $30-20=10$, vertical $40-19=21$.
A signed sum of 5 odd numbers is always *odd*; a signed sum of even numbers is always *even*. Since 21 is odd, the vertical direction must be the odd group, so we would need $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9 = 21$ – impossible by Step 2. So the dot **cannot** reach $(30,40)$.

Step 4. All other options can be reached (show the moves)

Point	Displacement (horiz., vert.)	Odd group	Even group
(A) $(27,33)$	$(7,14)$	$1+3+5+7-9=7$	$2+4+6-8+10=14$
(C) $(21,21)$	$(1,2)$	$1-3+5+7-9=1$	$2-4+6+8-10=2$
(D) $(42,44)$	$(22,25)$	$1+3+5+7+9=25$	$2-4+6+8+10=22$
(E) $(37,37)$	$(17,18)$	$-1-3+5+7+9=17$	$2+4-6+8+10=18$

(In (D) the horizontal displacement 22 is even, so the first move must be vertical.) For example, a path to $(21,21)$: right 1, up 2, left 3, down 4, right 5, up 6, right 7, up 8, left 9, down 10:



● starting point $(20,19)$ ● final point $(21,21)$

Common mistake. Looking only at even and odd is not enough: both 10 (even) and 21 (odd) are "allowed" as far as parity goes, yet 21 still cannot be reached. You must really check whether a choice of signs exists.

So the answer is **(B)**.

Answer: (B) $(30, 40)$

Problem 23

Gail works as a server in a restaurant. On Saturday she gets up at 6:30 a.m., starts work at x a.m. and finishes at x p.m. (the same day). How long does Gail work on Saturday?

(A) $24 - 2x$ hours (B) $12 - x$ hours (C) $2x$ hours

(D) 0 hours (E) 12 hours

Detailed solution

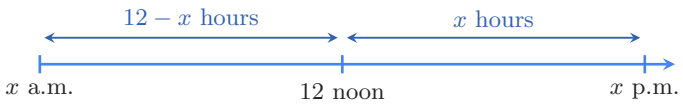
Analysis. She starts at x a.m. and finishes at x p.m. Find how long she works. The wake-up time 6:30 a.m. is *extra* information that we do not need.

Recall. x p.m. (12-hour clock) is the same as $(x + 12)$ o'clock on the 24-hour clock. For example, 3 p.m. is 15:00.

Step 1. Change to the 24-hour clock
Start: x o'clock. Finish: $(x + 12)$ o'clock.

Step 2. Compute the working time
Time = finishing time − starting time = $(x + 12) - x = 12$ hours.

Step 3. Another way to think (two pieces)



From x a.m. to 12 noon is $12-x$ hours; from 12 noon to x p.m. is x hours. In total: $(12-x)+x = 12$ hours. The result does not depend on x!

Common mistake. Option (B) $12 - x$ is only the morning part. Option (D) 0 hours comes from computing $x - x$ and forgetting that one time is "a.m." and the other is "p.m.".

Check: If $x = 8$: she works from 8 a.m. to 8 p.m. (20:00), that is $20 - 8 = 12$ hours ✓.
So the answer is **(E)**.

Answer: (E) 12 hours

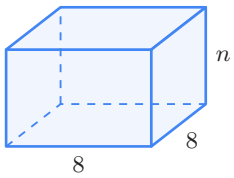
Problem 24

An $8 \times 8 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ cubes. Let A be the surface area of the box and B the combined surface area of all the small cubes that make it up. The sum of all values of n for which $\dfrac{B}{A}$ is an integer is

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Detailed solution

Analysis. We write A and B in terms of n, form the ratio $\frac{B}{A}$, find every positive integer n making it an integer, and finally add those values of n.



Recall. A rectangular box with dimensions a, b, c has surface area (total area of its 6 faces) $2(ab + bc + ca)$. A unit cube has 6 faces of area 1, so its surface area is 6. An $a \times b \times c$ box is made of $a \cdot b \cdot c$ unit cubes.

Step 1. Compute A

$A = 2(8 \cdot 8 + 8 \cdot n + 8 \cdot n) = 2(64 + 16n) = 128 + 32n$.

Step 2. Compute B

Number of small cubes: $8 \cdot 8 \cdot n = 64n$. Each has surface area 6, so $B = 6 \cdot 64n = 384n$.

Step 3. Simplify the ratio

$$\frac{B}{A} = \frac{384n}{128 + 32n} = \frac{32 \cdot 12n}{32 \cdot (4 + n)} = \frac{12n}{n + 4}.$$

Step 4. Split off the whole part

$$\frac{12n}{n+4} = \frac{12(n+4) - 48}{n+4} = 12 - \frac{48}{n+4}.$$

Why is it true? $12(n + 4) - 48 = 12n + 48 - 48 = 12n$, so the numerator is unchanged. Therefore $\frac{12n}{n+4}$ is an integer exactly when $\frac{48}{n+4}$ is an integer, that is, when $n + 4$ is a divisor of 48. □

Step 5. Find the values of n

The positive divisors of 48 are $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$. Since $n \geq 1$, we need $n + 4 \geq 5$; keep only $6, 8, 12, 16, 24, 48$:

$n+4$	6	8	12	16	24	48
n	2	4	8	12	20	44
$\frac{B}{A} = 12 - \frac{48}{n+4}$	4	6	8	9	10	11

Step 6. Add the values
$2+4+8+12+20+44=90$.

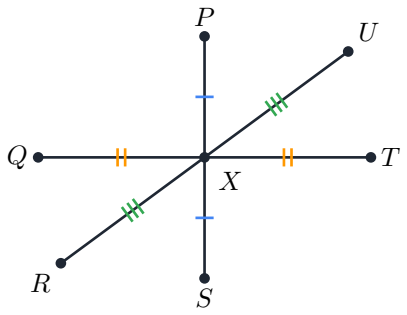
Common mistake. Do not forget the condition $n \geq 1$: the divisors $1, 2, 3, 4$ of 48 give $n+4 \leq 4$, that is $n \leq 0$ – rejected.

Check: For $n=2$: $A = 128+64 = 192$, $B = 768$, $\frac{B}{A} = 4$ ✓. For $n=44$: $A = 128+1408 = 1536$, $B = 16\,896$, $\frac{B}{A} = 11$ ✓.
So the answer is **(B)**.

Answer: (B) 90

Problem 25 *

Three bus routes cross at the stop X as shown (segments with the same number of tick marks have equal length). Every 12 minutes, bus A completes a trip $P \to X \to S \to X \to P$. Every 20 minutes, bus B completes a trip $Q \to X \to T \to X \to Q$. Every 28 minutes, bus C completes a trip $R \to X \to U \to X \to R$. At 1:00 p.m., buses A, B and C leave P, Q and R respectively; each bus drives at a constant speed and turns around instantly at the end of its route. The buses run continuously until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., how many times are two or more buses at X at the same moment?



(A) 18 (B) 19 (C) 20
(D) 21 (E) 22

Detailed solution

Analysis. Each bus runs steadily along a route with two arms of equal length. We find *the times* at which each bus is at X, then count the times that coincide. Measure time t as *the number of minutes after 1:00 p.m.*: then 5:00 p.m. corresponds to $t = 240$ and 10:00 p.m. to $t = 540$.

Step 1. When is bus A at X?
A trip $P \to X \to S \to X \to P$ consists of 4 equal segments ($PX = XS$) and takes 12 minutes, so each segment takes $12 \div 4 = 3$ minutes. Bus A reaches X after 3 minutes, then goes $X \to S \to X$ in 6 minutes, so it is at X again at minute 9; after that at minutes $15, 21, 27, \ldots$ That is,

$$t = 3,\ 9,\ 15,\ 21, \ldots \quad = 3\times 1,\ 3\times 3,\ 3\times 5,\ 3\times 7, \ldots$$

Bus A is at X exactly at the times $t = 3 \times$ (odd number).

Step 2. Similarly for buses B and C

Bus B: each segment takes $20 \div 4 = 5$ minutes; it is at X when $t = 5, 15, 25, \cdots = 5 \times (\text{odd})$.
Bus C: each segment takes $28 \div 4 = 7$ minutes; it is at X when $t = 7, 21, 35, \cdots = 7 \times (\text{odd})$.

Step 3. When are two buses at X together?

Why is it true? Buses A and B are both at $X \iff t = 3 \times (\text{odd}) = 5 \times (\text{odd})$. Then t is odd (a product of odd numbers is odd), and t is divisible by both 3 and 5, hence by 15; so t is an *odd multiple of* 15. Conversely, if $t = 15 \times (\text{odd})$, then $t = 3 \times (5 \times \text{odd}) = 5 \times (3 \times \text{odd})$, where the brackets are odd, so both buses are at X. □

By exactly the same argument: A and C are at X together $\iff$ t is an odd multiple of 21; B and C $\iff$ t is an odd multiple of 35; all three buses $\iff$ t is an odd multiple of 105.

Recall. If a number is divisible by two coprime numbers a and b, then it is divisible by $a \cdot b$. For example: divisible by 3 and by 5 implies divisible by 15.

Step 4. Count in the range $240 \le t \le 540$

- Odd multiples of 15: $15 \times 17 = 255, \ldots, 15 \times 35 = 525$ (for $17, 19, \ldots, 35$): **10** times.
- Odd multiples of 21: $21 \times 13 = 273, \ldots, 21 \times 25 = 525$ (for $13, 15, \ldots, 25$): **7** times.
- Odd multiples of 35: $35 \times 7 = 245, \ldots, 35 \times 15 = 525$ (for $7, 9, \ldots, 15$): **5** times.
- Odd multiples of 105: $105 \times 3 = 315$ and $105 \times 5 = 525$: **2** times (all three buses).

Step 5. Remove the double counting

At $t = 315$ and $t = 525$ all three buses are at X, so each of these times was counted *three times* in $10 + 7 + 5 = 22$ (once in each list), although it should be counted once. We subtract 2 for each of them:

$$22 - 2 \times 2 = 18.$$

The full list of the 18 times (converted to clock times):

Time	Buses	Time	Buses	Time	Buses
5:05 p.m.	B, C	6:57 p.m.	A, C	8:21 p.m.	A, C
5:15 p.m.	A, B	7:15 p.m.	A, B	8:35 p.m.	B, C
5:33 p.m.	A, C	7:25 p.m.	B, C	8:45 p.m.	A, B
5:45 p.m.	A, B	7:39 p.m.	A, C	9:03 p.m.	A, C
6:15 p.m.	**A, B, C**	7:45 p.m.	A, B	9:15 p.m.	A, B
6:45 p.m.	A, B	8:15 p.m.	A, B	**9:45 p.m.**	**A, B, C**

Common mistake. Bus A is at X *twice* in every 12-minute trip (on the way out and on the way back), so the gap between visits is 6 minutes, not 12 minutes. Using only "multiples of 12, 20, 28" gives a wrong answer.

Check: Checking minute by minute from 240 to 540 (for example with a computer) also gives exactly 18 times.
So the answer is **(A)**.

Answer: (A) 18

Problem 26

Steve starts at 7 and counts forward by 3, getting the list $7, 10, 13, \ldots$ Dave starts at 2026 and counts backward by 5, getting the list $2026, 2021, 2016, \ldots$ Which of the following numbers appears in both lists?

(A) 1009 (B) 1006 (C) 1003 (D) 1001 (E) 1011

Detailed solution

Analysis. Each list consists of equally spaced numbers. We find a "test" for a number to belong to each list, then check each option.

Step 1. The test for Steve's list

$7 = 2 \cdot 3 + 1$ leaves remainder 1 when divided by 3. Each time we add 3, the remainder on division by 3 does not change. So every number in Steve's list **leaves remainder** 1 **on division by** 3 (and is ≥ 7). Conversely, every number ≥ 7 with remainder 1 appears, since it equals $7 + 3k$.

Step 2. The test for Dave's list

$2026 = 405 \cdot 5 + 1$ leaves remainder 1 on division by 5. Subtracting 5 does not change the remainder on division by 5. So every number in Dave's list **leaves remainder** 1 **on division by** 5 (and is ≤ 2026), that is, it **ends in the digit** 1 **or** 6.

> **Recall.** Adding or subtracting a multiple of m does not change the remainder on division by m. A number leaves remainder 1 on division by 5 exactly when its last digit is 1 or 6. A number and the sum of its digits leave the same remainder on division by 3.

Step 3. Check each option

Number	Digit sum; remainder mod 3	Last digit	In both?
1009	10; rem. 1 ✓	9 ×	no
1006	7; rem. 1 ✓	6 ✓	**yes**
1003	4; rem. 1 ✓	3 ×	no
1001	2; rem. 2 ×	1 ✓	no
1011	3; rem. 0 ×	1 ✓	no

Check: $1006 = 7 + 3 \cdot 333$ (since $1006 - 7 = 999 = 3 \cdot 333$), so 1006 is the 334th number in Steve's list. $2026 - 1006 = 1020 = 5 \cdot 204$, so 1006 is in Dave's list (after 204 steps of 5) ✓.

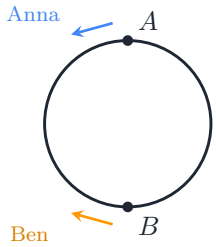
> **ThinkAlgo Tip.** Instead of writing out hundreds of numbers from each list, we only need the remainder – a common "signature" of every number in the list.

So the answer is **(B)**.

Answer: (B) 1006

Problem 27

On a circular track, Anna stands at point A and Ben stands at point B directly opposite A (segment AB is a diameter). Anna runs counterclockwise and Ben runs clockwise. They run at constant but different speeds. After a while they notice that every time they pass each other, it is always at the same three places on the track. The ratio of their speeds is



(A) $3:2$ (B) $3:1$ (C) $4:1$
(D) $2:1$ (E) $5:2$

Detailed solution

Analysis. The two friends run *in opposite directions* around the circle. We find what fraction of a lap each one runs between two consecutive meetings, and deduce how many different meeting points there are.

Step 1. Between two consecutive meetings, together they run exactly 1 lap

Why is it true? Just after meeting, the two separate in opposite directions. The distance between them (measured along the track) grows, and they meet again when the total distance the two have run just covers the whole circle. So between two consecutive meetings, the *total* distance they run is exactly 1 lap. (For the first meeting only: they start half a lap apart, so the total is $\frac{1}{2}$ lap.) □

Step 2. What fraction of a lap does Anna run each time?

In the same amount of time, distance is proportional to speed. If the ratio of speeds Anna : Ben is $p:q$, then during 1 shared lap Anna runs $\dfrac{p}{p+q}$ of a lap and Ben runs $\dfrac{q}{p+q}$ of a lap.

Recall. Sharing a quantity in the ratio $p:q$: the first share is $\frac{p}{p+q}$ and the second share is $\frac{q}{p+q}$ of the total.

So each meeting point is an arc of $\dfrac{p}{p+q}$ lap away from the previous one (in Anna's direction).

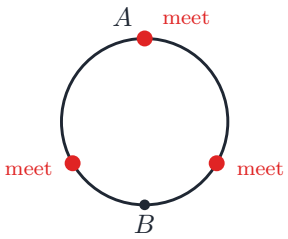
Step 3. Try the ratio $2:1$

Each time Anna runs $\frac{2}{3}$ of a lap. Let M_1 be the first meeting point. The next meeting points are $\frac{2}{3}$, $\frac{4}{3}$, $\frac{6}{3}=2$ laps, ... away from M_1. Since a full lap brings you back to the same place, $\frac{4}{3}$ lap is the same as $\frac{1}{3}$ lap, and 2 laps is M_1 itself. There are only **three** meeting points: M_1 and the two points that divide the circle into three equal parts with it. (If Ben is faster, ratio $1:2$, the result is the same.)

Step 4. How many points do the other ratios give?

Ratio	Anna runs each time	Meeting points (measured from M_1)	Points
(A) 3 : 2	$\frac{3}{5}$ lap	$0, \frac{3}{5}, \frac{1}{5}, \frac{4}{5}, \frac{2}{5}$ then repeats	5
(B) 3 : 1	$\frac{3}{4}$ lap	$0, \frac{3}{4}, \frac{2}{4}, \frac{1}{4}$ then repeats	4
(C) 4 : 1	$\frac{4}{5}$ lap	$0, \frac{4}{5}, \frac{3}{5}, \frac{2}{5}, \frac{1}{5}$	5
(D) 2 : 1	$\frac{2}{3}$ lap	$0, \frac{2}{3}, \frac{1}{3}$	**3**
(E) 5 : 2	$\frac{5}{7}$ lap	7 different places	7

(For example, row (A): $\frac{3}{5} + \frac{3}{5} = \frac{6}{5}$, remove 1 lap to get $\frac{1}{5}$; add $\frac{3}{5}$ to get $\frac{4}{5}$; add $\frac{3}{5}$ to get $\frac{7}{5}$, remove 1 lap to get $\frac{2}{5}$; add $\frac{3}{5}$ to get 1 lap – back at M_1.)



Check: Let the track be 600 m long, Anna run 200 m per minute and Ben 100 m per minute. They start 300 m apart and first meet after $300 \div 300 = 1$ minute, at the point 200 m from A (in Anna's direction). After that they meet every $600 \div 300 = 2$ minutes, and Anna runs another 400 m each time: positions 200 → 600 (that is, A) → 1000 (that is, 400 m) → 1400 (that is, 200 m) … Exactly three points: 0 m, 200 m, 400 m, dividing the circle into three equal parts – as in the picture above.
So the answer is **(D)**.

Answer: (D) 2 : 1

Problem 28

The list $11, 20, 31, 51, 82$ is an example of an increasing list of five positive integers in which the first and second numbers add to the third, the second and third add to the fourth, and the third and fourth add to the fifth. How many such lists have 124 as the fifth number?

(A) 10 (B) 7 (C) 9 (D) 6 (E) 8

Detailed solution

Analysis. The whole list is completely determined by its first two numbers. We express the fifth number in terms of the first two, then count the pairs that work.

Step 1. Express the whole list in terms of the first two numbers a and b

1st	2nd	3rd	4th	5th
a	b	$a+b$	$b+(a+b) = a+2b$	$(a+b)+(a+2b) = 2a+3b$

Check with the example: $a = 11$, $b = 20$ gives the fifth number $2 \cdot 11 + 3 \cdot 20 = 22 + 60 = 82$ ✓.

Step 2. Conditions

- The fifth number is 124: $2a + 3b = 124$.
- a, b are positive integers.
- The list is increasing: we need $a < b$. (The later steps increase automatically: $a + b > b$, $a + 2b > a + b$, ... since $a, b > 0$.)

Step 3. b must be even

$2a$ and 124 are both even, so $3b = 124 - 2a$ is even, hence b is even. Write $b = 2t$ (t a positive integer). Then $2a = 124 - 6t$, that is,

$$a = 62 - 3t.$$

Step 4. Find the range of t

- $a \geq 1$: $62 - 3t \geq 1 \Rightarrow 3t \leq 61 \Rightarrow t \leq 20$.
- $a < b$: $62 - 3t < 2t \Rightarrow 62 < 5t \Rightarrow t > 12.4 \Rightarrow t \geq 13$.

So $t \in \{13, 14, 15, 16, 17, 18, 19, 20\}$: 8 values.

Step 5. List them to check

t	13	14	15	16	17	18	19	20
$a = 62 - 3t$	23	20	17	14	11	8	5	2
$b = 2t$	26	28	30	32	34	36	38	40

For example $t = 13$: the list $23, 26, 49, 75, 124$; $t = 20$: the list $2, 40, 42, 82, 124$.

Common mistake. If we forget the condition "increasing" ($a < b$), we overcount pairs such as $a = 26$, $b = 24$ (the list $26, 24, 50, 74, 124$ – it decreases at the first step).

Check: $t = 12$ gives $a = 26$, $b = 24$: $a > b$, rejected. $t = 21$ gives $a = -1$: not positive, rejected. So there are exactly 8 lists.
So the answer is **(E)**.

Answer: (E) 8

Problem 29

Kathy owns more cats than Alice and more dogs than Bruce. Alice owns more dogs than Kathy but fewer cats than Bruce. Which of the following statements **must** be true?

(A) Bruce owns the fewest cats.

(B) Bruce owns the most cats.

(C) Kathy owns the most cats.

(D) Alice owns the most dogs.

(E) Kathy owns the fewest dogs.

Detailed solution

Analysis. There are two separate kinds of comparison: comparing numbers of *cats* and comparing numbers of *dogs*. We split the information into two groups and put each in order.

Step 1. Information about cats

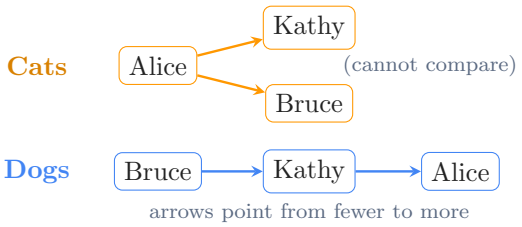
"Kathy owns more cats than Alice": Alice < Kathy. "Alice owns fewer cats than Bruce": Alice < Bruce.

So Alice has the fewest cats. But between Kathy and Bruce, *we do not know who has more.*

Step 2. Information about dogs

"Kathy owns more dogs than Bruce": Bruce < Kathy. "Alice owns more dogs than Kathy": Kathy < Alice.

Together: Bruce < Kathy < Alice. So **Alice has the most dogs**, and Bruce the fewest dogs.



Step 3. Examine each statement

- (A) "Bruce owns the fewest cats": **false**, because Alice has fewer cats than Bruce.
- (B) "Bruce owns the most cats": *may* be true, but not necessarily. For example, Alice 1, Kathy 3, Bruce 2 cats satisfies every condition, yet Kathy has the most cats.
- (C) "Kathy owns the most cats": not necessarily either. For example, with Alice 1, Kathy 2, Bruce 3 cats, Bruce has the most.
- (D) "Alice owns the most dogs": **must be true** (Step 2).
- (E) "Kathy owns the fewest dogs": **false**, because Bruce has fewer dogs than Kathy.

Common mistake. "Must be true" means true in *every* situation that fits the problem. Finding a single situation in which a statement is false is enough to reject it.

So the answer is **(D)**.

Answer: (D) Alice owns the most dogs

Problem 30

Chloe bought 300 g of candy A for \$5.00 and x g of candy B for \$7.00. She worked out that the average price of all the candy she bought was \$1.50 per 100 g. What is the value of x?

(A) 525 (B) 600 (C) 500 (D) 450 (E) 900

Detailed solution

Analysis. Given: the amount paid for each kind of candy, the mass of candy A and the average price. Find: the mass of candy B (in grams).

Recall. Average price = total cost ÷ total mass. Therefore total cost = average price × total mass.

Step 1. Total cost

$5 + 7 = 12$ (dollars).

Step 2. Express the total mass and form an equation

The total mass of candy is $300 + x$ (g), that is, $\dfrac{300 + x}{100}$ "portions of 100 g". Each 100 g portion costs \$1.50 on average, so the total cost is

$$1.5 \cdot \frac{300 + x}{100} = 12.$$

Step 3. Solve the equation

Divide both sides by 1.5: $\dfrac{300 + x}{100} = 12 \div 1.5 = 8$.

Multiply both sides by 100: $300 + x = 800$.

Move 300 across: $x = 800 - 300 = 500$ (g).

	Candy A	Candy B	Both
Mass (g)	300	$x = 500$	800
Cost (dollars)	5	7	12
Price per 100 g (dollars)	≈ 1.67	1.40	1.50

Common mistake. You cannot simply take the average of the two prices (\$1.67 and \$1.40), because the two masses are different. The average price must be computed as *total cost divided by total mass*.

Check: \$12 for 800 g means $12 \div 8 = 1.5$ dollars per 100 g ✓.

So the answer is **(C)**.

Answer: (C) 500

Extension and challenge

Problem 31

The road from Mr Henry's home to the city is 180 km long. He rides a motorbike at a constant speed, and after 3 hours he has covered $\frac{2}{3}$ of the distance.

a) Find the speed of the motorbike.

b) If he keeps the same speed, how much longer will it take him to reach the city?

c) If he wants to arrive 30 minutes earlier than planned in part b), at what speed must he ride for the rest of the way?

Detailed solution

Understanding the problem. *Given:* the whole distance 180 km; in 3 hours he covered $\frac{2}{3}$ of it. *Find:* the speed; the remaining time; the new speed needed to save 30 minutes.

Recall. $s = v \cdot t$; $v = \frac{s}{t}$; $t = \frac{s}{v}$. $\frac{m}{n}$ of A equals $A \div n \cdot m$.

Step 1. Distance covered and distance left

Covered: $\frac{2}{3}$ of 180 km is $180 \div 3 \cdot 2 = 60 \cdot 2 = 120$ (km). Left: $180 - 120 = 60$ (km).

Step 2. Part a: choose an unknown and write an equation

Let the speed of the motorbike be v (km/h), $v > 0$. In 3 hours he covers $3v$ (km). By Step 1 this distance is 120 km, so $3v = 120$, which gives $v = 120 \div 3 = 40$ (km/h).

Step 3. Part b: the time left

Let t (hours) be the time needed for the remaining 60 km at 40 km/h. Then $40t = 60$, so $t = 60 \div 40 = 1.5$ (hours) $= 1$ hour 30 minutes.

ThinkAlgo Tip. Another way, using direct proportion: at a constant speed, time is directly proportional to distance. 120 km takes 3 hours; 60 km is half of 120 km, so it takes half of 3 hours, that is 1.5 hours.

Step 4. Part c: the new speed

New time: 1 hour 30 minutes $-$ 30 minutes $=$ 1 hour. Let the new speed be u (km/h). Then $u \cdot 1 = 60$, so $u = 60$ (km/h).

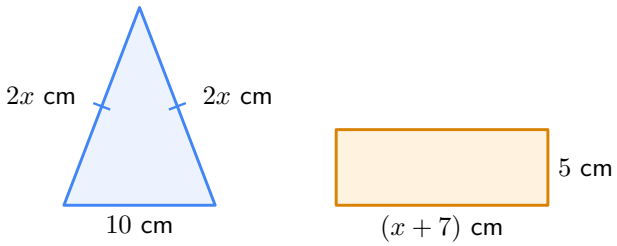
Common mistake. Do not just add to the speed, as in "30 minutes earlier means 30 km/h faster". Recalculate: speed = distance ÷ new time.

Check: $40 \cdot 3 = 120$ km; $40 \cdot 1.5 = 60$ km; $60 \cdot 1 = 60$ km. Total $120 + 60 = 180$ km.

Answer: a) 40 km/h; b) 1 hour 30 minutes; c) 60 km/h.

Problem 32

An isosceles triangle has a base of 10 cm and two equal sides of $2x$ cm each. A rectangle has length $(x + 7)$ cm and width 5 cm. The perimeter of the triangle equals the perimeter of the rectangle.



a) Write an expression in x for the perimeter of each shape.

b) Find x, then find the side lengths of the triangle and the area of the rectangle.

Detailed solution

Understanding the problem. The triangle has sides $2x$, $2x$, 10 (cm). The rectangle measures $x + 7$ by 5 (cm). The two perimeters are equal.

> **Recall.** Perimeter of a triangle = sum of the three sides. Perimeter of a rectangle $= 2 \times$ (length + width). Area of a rectangle = length $\times$ width.

Step 1. Part a: expressions for the perimeters

Triangle: $2x + 2x + 10 = 4x + 10$ (cm). Rectangle: $2 \cdot \big((x + 7) + 5\big) = 2(x + 12) = 2x + 24$ (cm).

> **Common mistake.** When multiplying the bracket by 2, multiply both terms: $2(x+12) = 2x + 24$, not $2x + 12$.

Step 2. Part b: write an equation

$4x + 10 = 2x + 24$.

Step 3. Solve the equation

Subtract $2x$ from both sides: $2x + 10 = 24$. Subtract 10 from both sides: $2x = 14$. Divide by 2: $x = 7$.

Step 4. Work out what is asked

Equal sides of the triangle: $2x = 14$ cm, so the triangle has sides 14 cm, 14 cm, 10 cm.
Rectangle: length $x + 7 = 14$ cm, width 5 cm, area $14 \cdot 5 = 70$ (cm^2).

Check: Perimeter of the triangle $14+14+10 = 38$ cm; perimeter of the rectangle $2(14+5) = 38$ cm. Equal.

Answer: a) $4x + 10$ and $2x + 24$; b) $x = 7$; triangle 14 cm, 14 cm, 10 cm; area of the rectangle 70 cm^2.

Problem 33

We use the rule of Problem 14 again: $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot d + b \cdot c$.

a) Evaluate $\begin{vmatrix} -3 & 4 \\ 5 & 2 \end{vmatrix}$.

b) Find x if $\begin{vmatrix} x & 2 \\ 3 & 4 \end{vmatrix} = \begin{vmatrix} 1 & x \\ 5 & 2 \end{vmatrix}$.

c) Find all integers x such that $\begin{vmatrix} x & x \\ x & x \end{vmatrix} = 50$.

Detailed solution

Understanding the problem. The rule: multiply the two numbers on each diagonal, then add the two products. Part a is an evaluation (with a negative number); parts b and c are equations.

Step 1. Part a

$\begin{vmatrix} -3 & 4 \\ 5 & 2 \end{vmatrix} = (-3) \cdot 2 + 4 \cdot 5 = -6 + 20 = 14.$

Recall. The product of a negative number and a positive number is negative: $(-3) \cdot 2 = -6$. Adding numbers with different signs: $-6 + 20 = 20 - 6 = 14$.

Step 2. Part b: write both sides using the rule

Left side: $x \cdot 4 + 2 \cdot 3 = 4x + 6$. Right side: $1 \cdot 2 + x \cdot 5 = 2 + 5x$. Equation: $4x + 6 = 5x + 2$.

Step 3. Solve part b

Subtract $4x$ from both sides: $6 = x + 2$. Subtract 2 from both sides: $x = 4$.

Check: Left side $4 \cdot 4 + 6 = 22$; right side $2 + 5 \cdot 4 = 22$. Equal.

Step 4. Part c

$\begin{vmatrix} x & x \\ x & x \end{vmatrix} = x \cdot x + x \cdot x = 2x^2$. The equation is $2x^2 = 50$, that is, $x^2 = 25$. Which numbers squared give 25? There is $5 \cdot 5 = 25$, but also $(-5) \cdot (-5) = 25$. So $x = 5$ or $x = -5$.

Common mistake. Many students find only $x = 5$ and forget $x = -5$. The product of two negative numbers is positive, so $(-5)^2 = 25$. The question asks for "all integers", which hints that there may be more than one answer.

Check: $x = -5$: $(-5)(-5) + (-5)(-5) = 25 + 25 = 50$. Correct.

Answer: a) 14; **b)** $x = 4$; **c)** $x = 5$ **or** $x = -5$.

Problem 34

a) The sum of five consecutive odd numbers is 155. Find the five numbers.

b) Leo says: "I have found four consecutive odd numbers whose sum is 150." Is Leo right or wrong? Why?

Detailed solution

Understanding the problem. Consecutive odd numbers differ by 2 (for example 29 and 31). Part a asks us to find numbers; part b asks for an *explanation*, not just a guess.

Step 1. Part a: let the middle number be the unknown
Call the middle number n. The five numbers are $n-4,\ n-2,\ n,\ n+2,\ n+4$.

Step 2. Write and solve the equation
$(n-4)+(n-2)+n+(n+2)+(n+4)=155$. The -4 and $+4$, and the -2 and $+2$, cancel: $5n=155$, so $n=31$.
The five numbers are $27,\ 29,\ 31,\ 33,\ 35$.

Check: $27+29+31+33+35=155$. Correct, and they are consecutive odd numbers.

Step 3. Part b: try to write an equation
Let the smallest odd number be m. The four numbers are $m,\ m+2,\ m+4,\ m+6$. Sum: $4m+12=150$, so $4m=138$, $m=138\div 4=34.5$. The number 34.5 is not an integer, let alone an odd number. So there are *no* four consecutive odd numbers with sum 150.

Why is it true? Another explanation using even and odd: the sum of two odd numbers is even, so the sum of four odd numbers is even – this is not yet a contradiction. But we can go further: $4m+12=4(m+3)$ is always *divisible by* 4, while 150 is not divisible by 4 ($150=4\cdot 37+2$). So the sum of four consecutive odd numbers can never be 150. □

Common mistake. Trying a few sets of numbers (such as $35+37+39+41=152$) and then concluding is not rigorous enough. We need a reason that works in *every* case.

Answer: a) 27, 29, 31, 33, 35; b) Leo is wrong, because the sum of four consecutive odd numbers is always divisible by 4, but 150 is not.

Problem 35

Every 2 scoops of fish food are enough to feed 12 fish for one day.

a) How many fish can 5 scoops of food feed for one day?

b) Mia's fish tank has 18 fish. A box of food contains 60 scoops. For how many days will the box feed the whole tank?

Detailed solution

Understanding the problem. For one day, more scoops feed more fish, and doubling the scoops doubles the fish: the number of fish fed for one day is *directly proportional* to the number of scoops.

Recall. Two quantities x and y are directly proportional when $y=k\cdot x$ for some constant $k\neq 0$. Then the ratio $\frac{y}{x}$ is always k, so we can write the proportion $\frac{y_1}{x_1}=\frac{y_2}{x_2}$.

Step 1. Find the constant of proportionality
Let y be the number of fish fed for one day and x the number of scoops. When $x=2$, $y=12$, so

$k = \dfrac{12}{2} = 6$, that is $y = 6x$: each scoop feeds 6 fish for one day.

Step 2. Part a

For $x = 5$: $y = 6 \cdot 5 = 30$ (fish).

(Or write the proportion $\dfrac{12}{2} = \dfrac{y}{5}$, so $y = \dfrac{12 \cdot 5}{2} = 30$.)

Step 3. Part b: how many scoops does the tank need each day?

Let z be the number of scoops needed for 18 fish for one day. Then $6z = 18$, so $z = 18 \div 6 = 3$ (scoops).

Scoops x	2	5	$z = 3$
Fish $y = 6x$	12	30	18

Step 4. Number of days

The box holds 60 scoops and 3 scoops are used each day. Let the number of days be d: $3d = 60$, so $d = 60 \div 3 = 20$ (days).

Common mistake. A common mistake is to calculate $60 \div 18$ or $60 \cdot 6 \div 18$ without understanding what it means. Always ask yourself: "How many scoops are used each day?" and only then divide.

Check: 20 days $\times$ 3 scoops $= 60$ scoops, exactly one box; and 3 scoops $\times 6 = 18$ fish each day.

Note for parents. The unit "fish-day" can help: each scoop provides 6 "fish-days"; the whole box provides $60 \cdot 6 = 360$ fish-days; dividing by 18 fish gives 20 days.

Answer: a) 30 fish; b) 20 days.

Problem 36

Each bus ride costs Lucy \$2.50. The bus company also sells a monthly pass for \$72 with unlimited rides.

a) Write an expression for the amount (in dollars) Lucy pays if she makes n rides in a month without a monthly pass.

b) What is the smallest number of rides in a month for which the monthly pass is cheaper than paying per ride?

Detailed solution

Understanding the problem. Paying per ride: the cost depends on the number of rides n. Monthly pass: always \$72. We want the smallest number of rides for which paying per ride is *more expensive* than the pass.

Step 1. Part a

Each ride costs \$2.50, so n rides cost $2.5 \cdot n = 2.5n$ (dollars).

Step 2. Part b: write an inequality

The pass is cheaper when the cost of single rides is more than \$72, that is $2.5n > 72$. Divide both

sides by the positive number 2.5: $n > 72 \div 2.5 = 28.8$.

Step 3. Choose the smallest whole number and check

The smallest whole number greater than 28.8 is 29. Check with a table:

Rides n	Per ride $2.5n$ (dollars)	Compared with the \$72 pass
28	70	paying per ride is cheaper
29	72.5	**the pass is cheaper**

So she needs at least 29 rides.

Common mistake. The number of rides is a whole number, so do not answer "28.8 rides". Do not "round" to 28 either: 28 rides cost only \$70, which is still cheaper than the pass. We need the smallest whole number *greater than* 28.8, which is 29.

Note for parents. This is a great "real life" problem to talk about with your child: phone plans, monthly swimming passes... can all be compared in this way.

Answer: a) $2.5n$ dollars; b) at least 29 rides.

Problem 37

Look at the table below (the values increase steadily as n increases):

n	1	2	3	4	5
Value	2	7	12	17	22

a) Write an expression in n for the row of values.

b) Find the value for $n = 40$.

c) For which n is the value 302?

d) Is any value in the table (continued forever) equal to 2026?

Detailed solution

Understanding the problem. Each time n increases by 1, the value increases by 5. It is an arithmetic sequence with first term 2 and common difference 5.

Recall. If the value always increases by d when n increases by 1, the expression has the form $dn + b$; the n-th term of an arithmetic sequence is $a + (n-1)d$.

Step 1. Part a

The expression has the form $5n + b$. Substitute $n = 1$: $5 + b = 2$, so $b = -3$. Expression: $5n - 3$. (Or: $2 + 5(n-1) = 2 + 5n - 5 = 5n - 3$.)

Check: $n = 4$: $5 \cdot 4 - 3 = 17$; $n = 5$: $5 \cdot 5 - 3 = 22$. It matches the table.

Step 2. Part b
$5 \cdot 40 - 3 = 200 - 3 = 197$.

Step 3. Part c: solve an equation
$5n - 3 = 302 \;\Rightarrow\; 5n = 305 \;\Rightarrow\; n = 61$.

Step 4. Part d
$5n - 3 = 2026 \;\Rightarrow\; 5n = 2029$. But 2029 ends in 9, so it is not divisible by 5, and there is no whole number n that works.

Why is it true? Every value $5n - 3$ ends in 2 or 7 (because $5n$ ends in 0 or 5). The number 2026 ends in 6, so it cannot be in the table. □

Answer: a) $5n - 3$; **b)** 197; **c)** $n = 61$; **d) no.**

Problem 38

a) The product $60 \times 24 \times 365$ is the number of which time unit in a non-leap year (with 365 days)? Work out the product.

b) How many days, hours, minutes and seconds are there in one million seconds?

Detailed solution

Understanding the problem. Part a: read the product as a chain of unit conversions. Part b: convert back from the small unit (seconds) to larger units, using division with remainder.

Recall. 1 minute = 60 seconds; 1 hour = 60 minutes = 3 600 seconds; 1 day = 24 hours = $24 \cdot 3\,600 = 86\,400$ seconds.

Step 1. Part a
365 is the number of days in the year; 24×365 is the number of hours in the year; $60 \times 24 \times 365$ is the number of **minutes** in a non-leap year. Calculation: $24 \times 365 = 8\,760$; $60 \times 8\,760 = 525\,600$ (minutes).

Step 2. Part b: convert to days
$1\,000\,000 \div 86\,400$ gives quotient 11, remainder $1\,000\,000 - 11 \cdot 86\,400 = 1\,000\,000 - 950\,400 = 49\,600$ (seconds).

Step 3. Convert the remainder to hours
$49\,600 \div 3\,600$ gives quotient 13, remainder $49\,600 - 13 \cdot 3\,600 = 49\,600 - 46\,800 = 2\,800$ (seconds).

Step 4. Convert the remainder to minutes and seconds
$2\,800 \div 60$ gives quotient 46, remainder $2\,800 - 46 \cdot 60 = 2\,800 - 2\,760 = 40$ (seconds).

Check: $11 \cdot 86\,400 + 13 \cdot 3\,600 + 46 \cdot 60 + 40 = 950\,400 + 46\,800 + 2\,760 + 40 = 1\,000\,000$.

Common mistake. Do not write the result as a decimal such as "11.57 days" and then read it as "11 days 57 hours": the decimal part of a day is not a number of hours. Use division with remainder as above.

Answer: a) The number of minutes in a non-leap year, 525 600; **b)** 11 **days** 13 **hours** 46 **minutes** 40 **seconds.**

Problem 39

Four positive integers a, b, c, d satisfy

$$\frac{3}{a-5} = \frac{3}{b+1} = \frac{3}{c-2} = \frac{3}{d+4} \qquad \text{and} \qquad a+b+c+d = 50.$$

a) Arrange a, b, c, d in increasing order.

b) Find the value of each number.

Detailed solution

Understanding the problem. Four fractions with the same numerator 3 are equal. Use the "set it equal to k" technique as in Problem 12, then use the condition on the sum to find k.

Step 1. The denominators are equal

Recall. If $\frac{m}{p} = \frac{m}{q}$ with $m \neq 0$, then $p = q$.

$a-5 = b+1 = c-2 = d+4$. Call the common value k ($k \neq 0$ because a denominator cannot be 0).

Step 2. Write each number in terms of k

$a = k+5$, $b = k-1$, $c = k+2$, $d = k-4$.

Step 3. Part a: compare

$k-4 < k-1 < k+2 < k+5$, so $d < b < c < a$.

Step 4. Part b: use the sum to find k

$a+b+c+d = (k+5)+(k-1)+(k+2)+(k-4) = 4k+2$. From $4k+2 = 50$ we get $4k = 48$, $k = 12$.

So $a = 17$, $b = 11$, $c = 14$, $d = 8$.

Check: $a-5 = 12$, $b+1 = 12$, $c-2 = 12$, $d+4 = 12$: all four fractions equal $\frac{3}{12}$. The sum is $17+11+14+8 = 50$. And $8 < 11 < 14 < 17$.

Common mistake. When adding $(+5)+(-1)+(+2)+(-4)$, keep the signs: the result is $+2$, not 12.

Answer: a) $d < b < c < a$; **b)** $a = 17$, $b = 11$, $c = 14$, $d = 8$.

Problem 40 ★★

There are four bags of rice with different weights. They are weighed two at a time in all possible ways, giving six results:

$$7 \text{ kg}, \quad 8 \text{ kg}, \quad 9 \text{ kg}, \quad 10 \text{ kg}, \quad 11 \text{ kg}, \quad 12 \text{ kg}.$$

a) Find the total weight of the four bags.

b) Each bag weighs a whole number of kilograms. Find the weight of each bag.

Detailed solution

Understanding the problem. Let the weights of the four bags from lightest to heaviest be $a < b < c < d$ (kg). Four bags can be paired in 6 ways: $a+b$, $a+c$, $a+d$, $b+c$, $b+d$, $c+d$. This problem extends Problem 15 (the three pumpkins).

Step 1. Part a: add all six results

Each bag appears in exactly 3 pairs (paired with each of the 3 other bags). Therefore $3(a+b+c+d) = 7+8+9+10+11+12 = 57$, so $a+b+c+d = 57 \div 3 = 19$ (kg).

> **Recall.** With three objects, each object appears in 2 pairs, so the total = (sum of the pairs)÷2. With four objects, each appears in 3 pairs, so the total = (sum of the pairs)÷3.

Step 2. Which pairs certainly match which results?

- The lightest pair is the two lightest bags: $a+b = 7$.
- The second lightest pair is $a+c = 8$: every other pair (except $a+b$) is heavier than $a+c$, because $a+d > a+c$, $b+c > a+c$, $b+d > a+c$, $c+d > a+c$.
- Similarly, the heaviest pair is $c+d = 12$, and the second heaviest is $b+d = 11$.
- The two middle results (9 and 10) are $a+d$ and $b+c$, but we do *not yet know* which is 9 and which is 10.

> **Why is it true?** Subtract $a+b = 7$ from $a+c = 8$: $c-b = 1$, that is, $c = b+1$. Then $b+c = b+(b+1) = 2b+1$ is an *odd number* (because b is a whole number). Of the two numbers 9 and 10, only 9 is odd. So $b+c = 9$ and $a+d = 10$. □

Step 3. Find each bag

$2b+1 = 9 \Rightarrow b = 4$; $c = b+1 = 5$; $a = 7-b = 3$; $d = 12-c = 7$.

Pair	$a+b$	$a+c$	$b+c$	$a+d$	$b+d$	$c+d$
Sum (kg)	$3+4=7$	$3+5=8$	$4+5=9$	$3+7=10$	$4+7=11$	$5+7=12$

Check: $3+4=7$, $3+5=8$, $3+7=10$, $4+5=9$, $4+7=11$, $5+7=12$: exactly the six results $7, 8, 9, 10, 11, 12$. Total $3+4+5+7 = 19$.

> **Common mistake.** Do not simply assume that the third result (9 kg) is $a+d$ or $b+c$. You need a reason! Without the "whole number" condition, the case $b+c = 10$ giving $a = 2.5$, $b = 4.5$, $c = 5.5$, $d = 6.5$ would also work.

Note for parents. You can experiment with four slips of paper showing $3, 4, 5, 7$: let your child pair them in all possible ways and add, and they will see for themselves that each number is used three times.

Answer: a) 19 kg; b) 3 kg, 4 kg, 5 kg, 7 kg.

Problem 41 *

The three scales below are balanced; identical shapes have equal masses.



Scale 1 Scale 2 Scale 3

a) How many □ shapes balance one ○?

b) How many □ shapes balance one ◇?

c) One ◇ and one ○ are placed on the left pan. If only △ shapes may be used on the right pan, how many △ are needed?

d) The four objects △, ○, □, ◇ (one of each kind) have a total mass of 50 g. Find the mass of each object.

Detailed solution

Analysis. This extends Problem 17. Let T, C, S, D (grams) be the masses of one △, one ○, one □ and one ◇. Each balanced scale gives an equation (mass on the left pan = mass on the right pan):

$$(1)\ 2T = 3C, \qquad (2)\ C + S = T, \qquad (3)\ D = T + S.$$

Idea: express every mass in terms of the same unknown, S.

Recall. Substitution: if we know $T = C + S$, we may replace T by $C + S$ in any equation. Adding or subtracting the same expression on both sides of an equation gives a true equation.

Step 1. Part a – Substitute (2) into (1)
Replace T by $C + S$ in (1): $2(C + S) = 3C$, that is $2C + 2S = 3C$.
Subtract $2C$ from both sides: $2S = 3C - 2C = C$. So $C = 2S$.
One ○ weighs the same as two □: we need 2 **shapes** □.

ThinkAlgo Tip. Why substitute (2) into (1)? Because (1) contains only T and C; after replacing $T = C + S$, the equation contains only C and S, so we can compare C with S at once.

Step 2. Express T in terms of S
From (2): $T = C + S = 2S + S = 3S$.

(Check with (1): $2T = 6S$ and $3C = 3 \cdot 2S = 6S$ ✓.)

Step 3. Part b

From (3): $D = T + S = 3S + S = 4S$. We need 4 **shapes** □.

Step 4. Part c

Left pan: $D + C = 4S + 2S = 6S$. Each △ weighs $T = 3S$, so the number of △ needed is $6S \div 3S = 2$ **shapes** △. (Indeed, $2T = 6S = D + C$.)

Step 5. Part d – An equation in one unknown

$T + C + S + D = 50$; in terms of S: $3S + 2S + S + 4S = 50$, that is $10S = 50$, so $S = 5$. Hence:

Object	□	○	△	◇
In terms of S	S	$C = 2S$	$T = 3S$	$D = 4S$
Mass	5 g	10 g	15 g	20 g

Check: Scale 1: $2 \cdot 15 = 30 = 3 \cdot 10$ ✓. Scale 2: $10 + 5 = 15$ ✓. Scale 3: $20 = 15 + 5$ ✓. Total: $15 + 10 + 5 + 20 = 50$ ✓.

Note for parents. Ask your child to read each scale as an equation with letters ($2T = 3C$, ...) before calculating. "Removing the same object from both pans" is exactly subtracting the same number from both sides – the foundation of solving equations.

Answer: a) 2 shapes □; b) 4 shapes □; c) 2 shapes △; d) □ 5 g, ○ 10 g, △ 15 g, ◇ 20 g.

Problem 42 ⋆

In this problem we only consider triangles in which every angle measures (in degrees) a *whole number*. Two isosceles triangles are said to be of the *same type* if they have the same three angle measures.

a) Prove that there is no triangle whose three angles all have odd measures.

b) How many types of isosceles triangle have all three angles with even measures?

c) How many types of isosceles triangle have exactly two angles with odd measures? Is there an isosceles triangle with exactly one angle of odd measure?

d) How many types of isosceles triangle are there altogether?

Detailed solution

Analysis. This extends Problem 20: instead of looking only at triangles with a 70° angle, we look at *all* isosceles triangles with whole-number angles and sort them by whether their angles are even or odd.

Recall. The three angles of a triangle add up to 180°. An isosceles triangle has two equal base angles. Even and odd: odd + odd = even; even + odd = odd; the sum of three odd numbers is odd; 2×(any number) is even.

Step 1. Part a

Suppose all three angles are odd. The sum of the first two odd numbers is even; adding one more odd number gives an *odd* number. But the sum of the three angles is 180 – an even number. Contradiction. So no triangle has three odd angles.

Step 2. Describe every isosceles triangle by one number x

Let each base angle measure x (degrees). Then the apex angle is $180 - 2x$. For a triangle, the apex angle must be positive, $180 - 2x \geq 1$, that is, $x \leq 89$. So x can be $1, 2, 3, \ldots, 89$, and each value of x gives exactly one type of isosceles triangle.

Why is it true? Two different values of x give two different sets of angles (the base angles differ), and every isosceles triangle has the form $(x, x, 180 - 2x)$. The equilateral triangle $(60, 60, 60)$ corresponds to $x = 60$ – it is counted only once. Note: we can always take the two equal angles as the base angles; for the equilateral triangle any angle will do. □

Step 3. The apex angle is always even

$180 - 2x = 2 \cdot (90 - x)$ is an even number for every x. So the parity pattern of the triangle depends only on x:

- x even: two even base angles, even apex angle ⇒ all three angles even.
- x odd: two odd base angles, even apex angle ⇒ exactly two angles odd.

Step 4. Part b

Count the even numbers from 1 to 89: $2, 4, \ldots, 88$, that is $88 \div 2 = 44$ numbers. So there are 44 **types**.

Step 5. Part c

Count the odd numbers from 1 to 89: $1, 3, \ldots, 89$, that is $(89 + 1) \div 2 = 45$ numbers. So 45 **types** have exactly two odd angles.

An isosceles triangle with exactly one odd angle does **not exist**: by Step 3, the number of odd angles can only be 0 or 2.

Step 6. Part d

Total number of types: $44 + 45 = 89$ (exactly the number of values of x).

x (base angle)	The three angles	Odd angles
1	1°, 1°, 178°	2
2	2°, 2°, 176°	0
55	55°, 55°, 70°	2
70	70°, 70°, 40°	0
89	89°, 89°, 2°	2

Common mistake. Do not forget the "very flat" triangles such as $(1°, 1°, 178°)$ – they are still genuine triangles. And do not allow $x = 90$: then the apex angle would be 0°, which is no longer a triangle.

Check: Listing $x = 1, \ldots, 89$ with a computer gives exactly 44 all-even triples and 45 triples with two odd angles. The rows $x = 55$ and $x = 70$ are exactly the two triangles of Problem 20.

Answer: a) It does not exist; b) 44; c) 45 types, and no isosceles triangle has exactly one odd angle; d) 89.

Problem 43 ★★

A dot starts at the origin $O(0,0)$ and moves by the rule of Problem 22: the first move is 1 unit horizontally or vertically; each later move is 1 unit longer than the previous one and perpendicular to the move just before it (either direction allowed).

a) Explain why after exactly 10 moves the dot cannot be back at O.

b) Find the smallest n ($n \geq 1$) such that after exactly n moves the dot can be back at O. Show one way to do it.

c) Can the dot be back at O after exactly 8 moves?

Detailed solution

Analysis. As in Problem 22, the moves alternate horizontal – vertical. One direction gets the moves of *odd* length $1, 3, 5, \ldots$; the other gets the moves of *even* length $2, 4, 6, \ldots$ The dot returns to O exactly when the signed sum of *each* group is 0.

> **Recall.** Being back at O means the total horizontal displacement is 0 *and* the total vertical displacement is 0. Changing the sign of a term a in a sum changes the sum by $2a$ – an even number – so it **does not change whether the sum is even or odd**.

Step 1. Part a

After 10 moves, the odd group consists of the lengths $1, 3, 5, 7, 9$ (whether the first move is horizontal or vertical). The signed sum $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$ has the same parity as $1 + 3 + 5 + 7 + 9 = 25$, so it is always *odd* and can never be 0. So in that direction the dot is not back where it started, hence the dot is not at O.

Step 2. Part b – Rule out $n \leq 6$

Look at the odd group (in one direction) for each n:

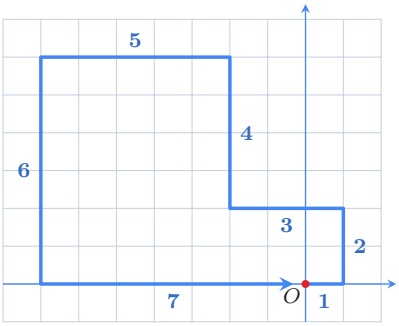
n	Odd lengths	Why the signed sum cannot be 0
1, 2	$\{1\}$	$\pm 1 \neq 0$
3, 4	$\{1, 3\}$	$\pm 1 \pm 3$ can only be $4, 2, -2, -4$
5, 6	$\{1, 3, 5\}$	the signed sum has the parity of $1 + 3 + 5 = 9$ (odd)

So for $n \leq 6$ the dot cannot return to O, and $n \geq 7$.

Step 3. Part b – $n = 7$ works

Odd group: $\{1, 3, 5, 7\}$, choose $+1 - 3 - 5 + 7 = 0$. Even group: $\{2, 4, 6\}$, choose $+2 + 4 - 6 = 0$. Let the first move be horizontal. The moves: right 1, up 2, left 3, up 4, left 5, down 6, right 7. The positions are

$$(0,0) \to (1,0) \to (1,2) \to (-2,2) \to (-2,6) \to (-7,6) \to (-7,0) \to (0,0).$$



So the smallest n is 7.

Step 4. Part c – $n = 8$
Odd group: $\{1, 3, 5, 7\}$, again $+1-3-5+7 = 0$. Even group: $\{2, 4, 6, 8\}$, choose $+2-4-6+8 = 0$. So **yes**: right 1, up 2, left 3, down 4, left 5, down 6, right 7, up 8; the positions are $(1, 0)$, $(1, 2)$, $(-2, 2)$, $(-2, -2)$, $(-7, -2)$, $(-7, -8)$, $(0, -8)$, $(0, 0)$.

Why is it true? A signed sum $\pm a_1 \pm a_2 \pm \dots$ equals 0 exactly when the numbers $a_1, a_2, \dots$ can be split into two groups with equal sums (the group with sign $+$ and the group with sign $-$). For example $\{1, 3, 5, 7\}$: $1 + 7 = 3 + 5 = 8$. So the total of all the numbers must be *even*. □

Common mistake. Do not forget that *both* groups must sum to 0. For $n = 5$: the even group $\{2, 4\}$ cannot give 0, and neither can the odd group $\{1, 3, 5\}$ – one failing group is already enough to rule n out.

Check: Checking every choice of directions with a computer for $n = 1, 2, \dots, 15$: the dot can return to O for $n = 7$, $n = 8$ and $n = 15$; it cannot for any $n \le 6$.

Answer: a) The signed sum of $1, 3, 5, 7, 9$ is always odd, so it is not 0; b) $n = 7$ (for example: right 1, up 2, left 3, up 4, left 5, down 6, right 7); c) Yes.

Problem 44 **

A $4 \times 6 \times n$ rectangular box (n a positive integer) is built from $1 \times 1 \times 1$ unit cubes. Let A be the surface area of the box and B the combined surface area of all the unit cubes.

a) Write A, B and the ratio $\dfrac{B}{A}$ as expressions in n.

b) Prove that $\dfrac{B}{A} < 7.2$ for every n.

c) Find all values of n for which $\dfrac{B}{A}$ is an integer.

Detailed solution

Analysis. This extends Problem 24, but this time the trick of "splitting off the whole part" is not as neat. We use a different method: bound the ratio from above, then test each possible integer value.

Recall. An $a \times b \times c$ box: surface area $2(ab+bc+ca)$, made of $a \cdot b \cdot c$ unit cubes; each unit cube has surface area 6.

Step 1. Part a

$A = 2(4 \cdot 6 + 4 \cdot n + 6 \cdot n) = 2(24 + 10n) = 48 + 20n$.

Number of unit cubes: $4 \cdot 6 \cdot n = 24n$; therefore $B = 6 \cdot 24n = 144n$.

$$\frac{B}{A} = \frac{144n}{48+20n} = \frac{4 \cdot 36n}{4 \cdot (12+5n)} = \frac{36n}{5n+12}.$$

Step 2. Part b

Since $5n+12 > 5n$, we are dividing $36n$ by a number *larger* than $5n$, so the result is *smaller*:

$$\frac{36n}{5n+12} < \frac{36n}{5n} = \frac{36}{5} = 7.2.$$

Recall. For positive numbers: with the same numerator, the fraction with the larger denominator is smaller.

Step 3. Part c – Call the integer value k

If $\frac{B}{A} = k$ is an integer, then k is positive and $k < 7.2$, so $k \in \{1,2,3,4,5,6,7\}$. We have

$$\frac{36n}{5n+12} = k \iff 36n = k(5n+12) = 5kn+12k \iff 36n-5kn = 12k \iff n(36-5k) = 12k.$$

So $n = \dfrac{12k}{36-5k}$, and we only need to test 7 values of k.

Step 4. Test each value of k

k	1	2	3	4	5	6	7
$12k$	12	24	36	48	60	72	84
$36-5k$	31	26	21	16	11	6	1
$n = \frac{12k}{36-5k}$	$\frac{12}{31}$ ×	$\frac{24}{26}$ ×	$\frac{36}{21}$ ×	**3**	$\frac{60}{11}$ ×	**12**	**84**

(The mark × means: not an integer.) So $n \in \{3, 12, 84\}$.

Why is it true? Why have we missed nothing? Every n making $\frac{B}{A}$ an integer gives an integer k from 1 to 7 (part b), and for each k there is exactly one n with $n(36-5k) = 12k$. We have tested all 7 possibilities. □

Check: $n = 3$: $A = 48+60 = 108$, $B = 432$, $\frac{B}{A} = 4$ ✓. $n = 12$: $A = 48+240 = 288$, $B = 1728$, $\frac{B}{A} = 6$ ✓. $n = 84$: $A = 48+1680 = 1728$, $B = 12\,096$, $\frac{B}{A} = 7$ ✓.

ThinkAlgo Tip. When a fraction containing n cannot be "split" neatly, find a number it always stays below. If only a few integer values remain possible, test them all!

Answer: a) $A = 48+20n$, $B = 144n$, $\dfrac{B}{A} = \dfrac{36n}{5n+12}$; **b) see the solution; c)** $n = 3$, $n = 12$, $n = 84$.

Problem 45

Two bus routes cross at the stop X. Bus A runs back and forth along the route P–X–S; each arm (PX or XS) takes exactly 4 minutes. Bus B runs along the route Q–X–T; each arm takes exactly 6 minutes. At 1:00 p.m., bus A leaves P and bus B leaves Q; the buses turn around instantly at the ends of their routes and run continuously.

a) Prove that buses A and B are never at X at the same time.

b) A bus C is added on the route R–X–U; each arm takes 10 minutes, and it leaves R at 1:00 p.m. Between 1:00 p.m. and 7:00 p.m., at which times are two buses at X together?

c) Suppose each arm of bus C's route takes m minutes (m any positive integer). Prove that bus C cannot both meet bus A at X (at some moment) and meet bus B at X (at some moment).

Detailed solution

Analysis. This extends Problem 25. Measure time t in minutes after 1:00 p.m. As in Problem 25, a bus whose arm takes a minutes first reaches X at minute a, and then returns to X every $2a$ minutes (out to the end and back):

$$t = a,\ 3a,\ 5a,\ 7a, \ldots \quad \text{that is } t = a \times (\text{odd number}).$$

So bus A is at X when $t = 4 \times (\text{odd})$; bus B when $t = 6 \times (\text{odd})$; bus C when $t = m \times (\text{odd})$.

> **Recall.** The product of two odd numbers is odd. A number divisible by 4 gives an even number when divided by 2.

Step 1. Part a

Times when bus A is at X: $t = 4 \times (\text{odd})$ – always **divisible by** 4.
Times when bus B is at X: $t = 6 \times (\text{odd}) = 2 \times (3 \times \text{odd}) = 2 \times (\text{odd})$ – dividing by 2 gives an odd number, so it is **not divisible by** 4.
A number cannot be both divisible and not divisible by 4, so A and B are never at X together.

Step 2. Part b – bus C with $m = 10$

Bus C is at X when $t = 10 \times (\text{odd}) = 2 \times (5 \times \text{odd})$ – not divisible by 4, so C **never** meets A at X (same reasoning as part a).
C meets B when $t = 6 \times (\text{odd}) = 10 \times (\text{odd})$. Then $\frac{t}{2} = 3 \times (\text{odd}) = 5 \times (\text{odd})$ is an odd number divisible by 3 and 5, i.e. an odd multiple of 15. So $t = 30 \times (\text{odd})$. Conversely, $t = 30 \times (\text{odd}) = 6 \times (5 \times \text{odd}) = 10 \times (3 \times \text{odd})$ is indeed a time when both buses are at X.
From 1:00 p.m. to 7:00 p.m. means $0 \le t \le 360$: $t = 30,\ 90,\ 150,\ 210,\ 270,\ 330$.

t (min)	30	90	150	210	270	330
Time	1:30 p.m.	2:30 p.m.	3:30 p.m.	4:30 p.m.	5:30 p.m.	6:30 p.m.

At all 6 of these times the two buses at X are B and C.

Step 3. Part c – "the number of halvings"

For each positive integer t, keep dividing t by 2 until the result is odd. Let $h(t)$ be the number of divisions. For example: $12 \to 6 \to 3$, so $h(12) = 2$; $6 \to 3$, so $h(6) = 1$; 5 is already odd, so $h(5) = 0$.

Why is it true? Multiplying by an odd number does not change h. Write $m = 2 \cdot 2 \cdots 2 \times r$ (with $h(m)$ factors 2 and r odd). Then $m \times (\text{odd}) = 2 \cdot 2 \cdots 2 \times (r \times \text{odd})$, where $r \times$ odd is still odd. So $m \times (\text{odd})$ can also be halved exactly $h(m)$ times. $\square$

Applying this:

- Every time bus A is at X: $h(4 \times \text{odd}) = h(4) = 2$.
- Every time bus B is at X: $h(6 \times \text{odd}) = h(6) = 1$.
- Every time bus C is at X: $h(m \times \text{odd}) = h(m)$.

If C meets A then $h(m) = 2$; if C meets B then $h(m) = 1$. A number m cannot have both $h(m) = 2$ and $h(m) = 1$. So C cannot meet both A and B.

Common mistake. Part c does not say that C *never* meets anyone: with $m = 4$, C meets A (at $t = 4$); with $m = 2$, C meets B (at $t = 6$). What we proved is that C cannot meet *both*.

Check: Checking with a computer for $m = 1, 2, \ldots, 12$: no value of m lets C meet both A and B; $m = 4$ and $m = 12$ meet A; $m = 2, 6, 10$ meet B.

Answer: a) Times when A is at X are divisible by 4, times when B is at X are not; b) 1:30, 2:30, 3:30, 4:30, 5:30 and 6:30 p.m. (always buses B and C); c) see the solution.

Problem 46 *

Sam counts forward $7, 10, 13, \ldots$ (adding 3 each time) and stops before going past 2026. Tom counts backward $2026, 2021, 2016, \ldots$ (subtracting 5 each time) and stops before going below 7.

a) How many numbers appear in both lists?

b) Find the sum of all the numbers in part a.

c) Zoe writes the list of positive multiples of 4: $4, 8, 12, \ldots$ How many numbers appear in all three lists (Sam's, Tom's and Zoe's)? What is the largest of them?

Detailed solution

Analysis. This extends Problem 26. As in Problem 26: a number N is in Sam's list $\iff$ it leaves remainder 1 on division by 3 and $7 \le N \le 2026$ (because $7 = 3 \cdot 2 + 1$, and adding 3 does not change the remainder on division by 3); a number N is in Tom's list $\iff$ it leaves remainder 1 on division by 5 and $7 \le N \le 2026$ (because $2026 = 5 \cdot 405 + 1$). Now we find *all* the common numbers.

Step 1. Part a – The test for a common number

N is common $\iff$ N leaves remainder 1 on division by 3 and by 5 $\iff$ $N - 1$ is divisible by 3 and by 5 $\iff$ $N - 1$ is divisible by 15. So $N = 15j + 1$.

Recall. A number divisible by 3 and by 5 (two coprime numbers) is divisible by $3 \cdot 5 = 15$.

Step 2. Part a – Count

We need $7 \le 15j + 1 \le 2026$. Smallest: $j = 1$ gives 16 ($j = 0$ gives $1 < 7$, rejected). Largest: $15j + 1 \le 2026 \Rightarrow 15j \le 2025 \Rightarrow j \le 135$; $j = 135$ gives 2026.

So the common numbers are $16, 31, 46, \ldots, 2026$, for $j = 1, 2, \ldots, 135$: 135 **numbers**.

Step 3. Part b – The sum (Gauss's pairing method)

Recall. Sum of an equally spaced list = (first + last) × (number of terms) ÷2. Reason: write the list forwards and backwards and add in pairs; every pair equals first + last.

$$16 + 31 + \cdots + 2026 = \frac{(16 + 2026) \cdot 135}{2} = \frac{2042 \cdot 135}{2} = 1021 \cdot 135 = 137\,835.$$

Step 4. Part c – Add the condition "divisible by 4"

Look at the common numbers $16, 31, 46, 61, 76, 91, 106, 121, 136, \ldots$ Which are divisible by 4?

$$16\ \checkmark,\ 31,\ 46,\ 61,\ 76\ \checkmark,\ 91,\ 106,\ 121,\ 136\ \checkmark, \ldots$$

In every 4 consecutive terms exactly 1 is divisible by 4, and these numbers are $4 \cdot 15 = 60$ apart: $16, 76, 136, \ldots$

Why is it true? Each step along the list adds 15; adding 15 increases the remainder on division by 4 by 3 (because $15 = 12 + 3$). The remainders of $16, 31, 46, 61$ on division by 4 are $0, 3, 2, 1$, and then they repeat. So exactly one term in every 4 is divisible by 4, and two consecutive such terms are $4 \times 15 = 60$ apart. □

The numbers common to all three lists: $16 + 60i$ with $16 + 60i \le 2026$, i.e. $60i \le 2010$, $i \le 33.5$, so $i = 0, 1, \ldots, 33$: 34 **numbers**. The largest is $16 + 60 \cdot 33 = 16 + 1980 = 1996$.

Check: $1996 = 4 \cdot 499$ ✓; $1996 - 7 = 1989 = 3 \cdot 663$ ✓; $2026 - 1996 = 30 = 5 \cdot 6$ ✓. The next one, $1996 + 60 = 2056 > 2026$, is rejected.

Answer: a) 135 numbers; b) 137 835; c) 34 numbers, the largest is 1996.

Problem 47 ★★

Emma and Noah both start at point A on a circular track; each runs at a constant speed.

a) They run in opposite directions, with the ratio of speeds Emma : Noah equal to 5 : 3. At how many different points of the track do they meet?

b) If they run in the *same direction* (still with ratio 5 : 3), at how many different points does Emma overtake Noah?

c) Find all speed ratios $p : q$ (with $p > q$, where p and q are coprime positive integers) such that, running in opposite directions, they meet at exactly 10 different points.

Detailed solution

Analysis. This extends Problem 27. We reuse two ideas: (1) running in opposite directions, between two consecutive meetings the two together cover 1 lap; (2) distance is proportional to

speed.

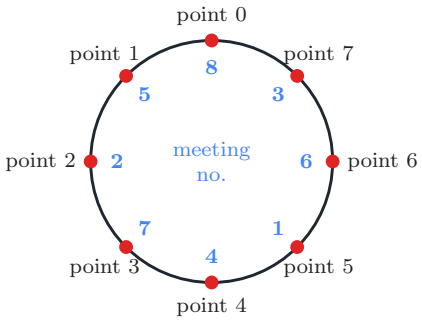
Step 1. Part a – Divide the lap into 8 segments

The ratio is $5:3$, so we divide the lap into $5+3=8$ equal segments and number the division points $0, 1, 2, \ldots, 7$ in Emma's running direction (point 0 is A). Between two meetings the two together run 8 segments, of which Emma runs 5 and Noah 3.

The k-th meeting is at the point $5k$ segments from A; every 8 segments is a full lap, so we remove multiples of 8:

Meeting k	1	2	3	4	5	6	7	8	9
$5k$	5	10	15	20	25	30	35	40	45
Point (remove multiples of 8)	5	2	7	4	1	6	3	0	5

The first eight meetings fall on all 8 different points $0, 1, \ldots, 7$; the 9th meeting is back at point 5 and everything repeats. So there are 8 **meeting points**.



(Blue numbers inside: which meeting falls on that point; the points are numbered in Emma's running direction.)

Step 2. Part b – Same direction

Running in the same direction, Emma overtakes Noah each time Emma has run exactly 1 lap *more* than Noah. In the same time Emma runs 5 parts while Noah runs 3, so Emma gains 2 parts. Thus "1 lap more" corresponds to 2 parts = 1 lap, so each part is $\frac{1}{2}$ lap, and Emma has run $5 \cdot \frac{1}{2} = 2\frac{1}{2}$ laps.

First overtaking: Emma has run $2\frac{1}{2}$ laps, at the point opposite A. Second: 5 laps, exactly at A. Then it repeats. So there are 2 **points** (point A and the point opposite A).

Step 3. Part c – The general rule

Do the same as in part a with ratio $p:q$: divide the lap into $p+q$ segments; the k-th meeting is at the point $k \cdot p$ segments from A (removing multiples of $p+q$).

Why is it true? If p and q are coprime, there are exactly $p+q$ meeting points.
Meeting number $p+q$ is back at A: because $(p+q) \cdot p$ is a multiple of $p+q$.
Meetings $1, 2, \ldots, p+q$ are at different points: suppose meetings i and j $(i < j \le p+q)$ are at the same point. Then $j \cdot p - i \cdot p = (j-i) \cdot p$ is a multiple of $p+q$. Any common divisor of p and $p+q$ also divides $(p+q)-p=q$; since p, q are coprime, p and $p+q$ have no common prime factor. Hence every prime factor of $p+q$ must be in $j-i$, so $j-i$ is divisible by $p+q$. But $0 < j-i < p+q$ – impossible.
So there are exactly $p+q$ meeting points, equally spaced around the circle. □

We need $p+q=10$ with $p > q \ge 1$ coprime:

$p : q$	$9 : 1$	$8 : 2$	$7 : 3$	$6 : 4$
Coprime?	yes	no ($= 4 : 1$)	yes	no ($= 3 : 2$)
Meeting points	10	5	10	5

($5 : 5$ is excluded because $p > q$.) So there are two ratios: $9 : 1$ **and** $7 : 3$.

Common mistake. The ratio $8 : 2$ has $8 + 2 = 10$ but is really $4 : 1$, which gives only $4 + 1 = 5$ meeting points. Always reduce the ratio before applying the "$p + q$ points" rule.

Check: Check Problem 27 with the rule: the ratio $2 : 1$ gives $2 + 1 = 3$ points ✓. A computer simulation of all reduced ratios $p : q$ with $p \le 19$: only $9 : 1$ and $7 : 3$ give 10 points.

Answer: a) 8 points; b) 2 points; c) $9 : 1$ and $7 : 3$.

Problem 48 ★★

We call an *increasing* list of positive integers a *Fibo list* if, from the third number on, each number is the sum of the two numbers just before it (for example $11, 20, 31, 51, 82$ as in Problem 28).

a) Find all Fibo lists of 6 numbers whose last number is 100.

b) How many Fibo lists of 5 numbers have last number 200?

c) Prove that there is no Fibo list of 6 numbers whose last number is 30.

Detailed solution

Analysis. As in Problem 28, the whole list is determined by its first two numbers a and b; the list is increasing exactly when $a < b$.

Step 1. Express the numbers in terms of a and b

1st	2nd	3rd	4th	5th	6th
a	b	$a + b$	$a + 2b$	$2a + 3b$	$(a + 2b) + (2a + 3b) = 3a + 5b$

ThinkAlgo Tip. The coefficients $1, 1, 2, 3, 5, 8, \ldots$ appearing in the table are the famous Fibonacci numbers!

Step 2. Part a – The equation $3a + 5b = 100$
Conditions: $a \ge 1$, $a < b$, $3a + 5b = 100$.

- From $a \ge 1$: $5b = 100 - 3a \le 97$, so $b \le 19$.
- From $a < b$: $100 = 3a + 5b < 3b + 5b = 8b$, so $b > 12.5$, that is $b \ge 13$.
- $3a = 100 - 5b$ must be divisible by 3.

Test b from 13 to 19:

b	13	14	15	16	17	18	19
$100 - 5b$	35	30	25	20	15	10	5
$a = (100 - 5b) \div 3$	×	**10**	×	×	**5**	×	×

There are two lists: **10, 14, 24, 38, 62, 100** and **5, 17, 22, 39, 61, 100**.

Step 3. Part b – The equation $2a + 3b = 200$
Since $2a$ and 200 are even, $3b$ is even, so b is even: $b = 2t$. Then $2a = 200 - 6t$, that is $a = 100 - 3t$.

- $a \geq 1$: $100 - 3t \geq 1 \Rightarrow t \leq 33$.
- $a < b$: $100 - 3t < 2t \Rightarrow 100 < 5t \Rightarrow t > 20 \Rightarrow t \geq 21$.

So $t = 21, 22, \ldots, 33$: there are $33 - 21 + 1 = 13$ lists. (For example $t = 21$: $37, 42, 79, 121, 200$; $t = 33$: $1, 66, 67, 133, 200$.)

Common mistake. For $t = 20$ we get $a = b = 40$: the list $40, 40, 80, 120, 200$ is *not increasing* at the first step, so it is rejected. The number of integers from 21 to 33 is $33 - 21 + 1 = 13$, not 12.

Step 4. Part c – No list ends in 30
We need $3a + 5b = 30$ with $1 \leq a < b$.

- $3a = 30 - 5b \geq 3$, so $5b \leq 27$, that is $b \leq 5$.
- $5b = 30 - 3a$ is divisible by 3, so b is divisible by 3; with $b \leq 5$ only $b = 3$ is possible.
- $b = 3$ gives $3a = 15$, $a = 5$. But $a = 5 > 3 = b$ – the list $5, 3, 8, 11, 19, 30$ is not increasing.

So no Fibo list of 6 numbers ends in 30.

Why is it true? A shorter argument: since $a < b$, the sixth number satisfies $3a + 5b < 3b + 5b = 8b$. To get $3a + 5b = 30$ we would need $b > 30 \div 8$, that is $b \geq 4$; together with $b \leq 5$ and b divisible by 3, no value is left. □

Check: Testing every pair $1 \leq a < b$ with a computer: exactly 2 lists in part a, 13 lists in part b and 0 lists in part c.

Answer: a) $10, 14, 24, 38, 62, 100$ **and** $5, 17, 22, 39, 61, 100$**; b) 13 lists; c) see the solution.**

Problem 49 ★

Four friends Anna, Ben, Chloe and David each have at least one marble. We know that:

(1) Anna and Ben together have as many marbles as Chloe and David together.
(2) Anna and David together have more marbles than Ben and Chloe together.
(3) Ben has more marbles than Anna and Chloe together.

a) Put the four friends in order from fewest to most marbles.
b) Altogether the four friends have 30 marbles. What is the greatest number of marbles Chloe can have? In that case, how many marbles does each friend have?

Detailed solution

Analysis. This extends Problem 29: instead of comparing pairs directly, the information here compares *sums*. Let Anna, Ben, Chloe, David have a, b, c, d marbles (positive integers). We have:

$$(1)\ a+b=c+d, \qquad (2)\ a+d>b+c, \qquad (3)\ b>a+c.$$

Recall. Adding (or subtracting) the two sides of an equation to (from) the corresponding sides of an inequality gives an inequality in the same direction: if $X>Y$ and $M=N$, then $X+M>Y+N$ and $X-M>Y-N$.

Step 1. Compare Anna and Chloe: add (2) and (1)
$(a+d)+(a+b)>(b+c)+(c+d)$, that is $2a+b+d>b+d+2c$. Subtract $b+d$ from both sides: $2a>2c$, so $a>c$: **Anna has more than Chloe**.

Step 2. Compare Ben and David: subtract (1) from (2)
$(a+d)-(a+b)>(b+c)-(c+d)$, that is $d-b>b-d$. Add $b+d$ to both sides: $2d>2b$, so $d>b$: **David has more than Ben**.

Step 3. Compare Anna and Ben: use (3)
$b>a+c\geq a+1>a$ (since $c\geq 1$). So **Ben has more than Anna**.

Step 4. Part a – Conclusion
Putting the three results together: $c<a$, $a<b$, $b<d$, so $c<a<b<d$.
From fewest to most: **Chloe, Anna, Ben, David**.

Step 5. Part b – Use the total of 30
From (1), $a+b=c+d$, and $(a+b)+(c+d)=30$, so $a+b=c+d=15$. Hence $b=15-a$.
Substitute into (3): $15-a>a+c$, that is $2a+c<15$; since $2a+c$ is a whole number, $2a+c\leq 14$.
By part a, $a>c$, so $a\geq c+1$. Therefore $2(c+1)+c\leq 2a+c\leq 14$, that is $3c+2\leq 14$, $3c\leq 12$, $c\leq 4$.

Step 6. Part b – Can $c=4$ actually happen?
With $c=4$: we need $a\geq 5$ and $2a+4\leq 14$, that is $a\leq 5$. So $a=5$, $b=15-5=10$, $d=15-4=11$.
Check all three conditions: (1) $5+10=15=4+11$ ✓; (2) $5+11=16>14=10+4$ ✓; (3) $10>5+4=9$ ✓.

Common mistake. Proving "$c\leq 4$" is only half the job: we must also show a concrete case with $c=4$ before we can claim that 4 is the greatest value.

Check: Listing with a computer every (a,b,c,d) with total 30 satisfying (1), (2), (3): there are 12 of them, the largest c is 4, occurring only in $(5,10,4,11)$; and in every one of them $c<a<b<d$.

Answer: a) Chloe, Anna, Ben, David; b) Chloe has at most 4 marbles; then Anna has 5, Ben 10, Chloe 4, David 11.

Problem 50

A shop sells candy A at \$12 per kilogram and candy B at \$20 per kilogram.

a) How many grams of each kind must be mixed to get 1 kg of mixture costing \$15 per kilogram (the mixture is worth exactly the total value of the candy put in)?

b) How many grams of candy C, costing \$30 per kilogram, must be added to the 1 kg mixture of part a to get a new mixture costing \$18 per kilogram?

c) Prove that if candy A and candy B (both present) are mixed in any proportion, the price of the mixture is always more than \$12 and less than \$20 per kilogram.

Detailed solution

Analysis. This extends Problem 30. The principle: *the total cost of the mixture = the total cost of the parts mixed.* We measure mass in kilograms and money in dollars.

Recall. Cost = price (per 1 kg) × mass (kg). Average price of a mixture = total cost ÷ total mass.

Step 1. Part a – Choose an unknown and write an equation

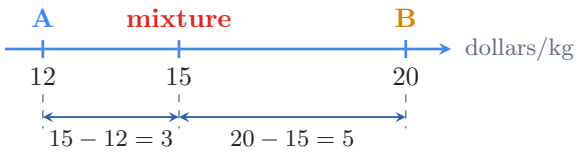
Let the mass of candy A be x (kg), with $0 < x < 1$; then the mass of candy B is $1 - x$ (kg). Cost of A: $12x$; cost of B: $20(1-x) = 20 - 20x$; cost of 1 kg of mixture: $15 \cdot 1 = 15$ (dollars). So

$$12x + 20 - 20x = 15 \iff 20 - 8x = 15 \iff 8x = 5 \iff x = \frac{5}{8} = 0.625.$$

So we need 0.625 kg = 625 g of candy A and $1000 - 625 = 375$ g of candy B.

Check: $12 \cdot 0.625 + 20 \cdot 0.375 = 7.5 + 7.5 = 15$ (dollars) for 1 kg, exactly \$15 per kilogram ✓.

Step 2. Part a – Another way: compare with the price \$15 on a number line



Each kilogram of candy A is \$3 cheaper than \$15; each kilogram of candy B is \$5 dearer than \$15. The "cheaper" part must exactly balance the "dearer" part: $3x = 5(1-x)$. Solving: $3x = 5 - 5x$, $8x = 5$, $x = 0.625$ – the same as Step 1. From $3x = 5y$ (with $y = 1 - x$) we get $\frac{x}{y} = \frac{5}{3}$: the masses of A and B are in the ratio 5 : 3, the two distances 3 and 5 *reversed*.

Step 3. Part b

Let the mass of candy C added be y (kg), $y > 0$. Total cost: $15 + 30y$; total mass: $1 + y$ (kg). For a price of \$18 per kilogram we need

$$15 + 30y = 18(1+y) = 18 + 18y \iff 30y - 18y = 18 - 15 \iff 12y = 3 \iff y = 0.25.$$

So we must add 0.25 kg = 250 g of candy C.

Check: Total cost $15 + 30 \cdot 0.25 = 15 + 7.5 = 22.5$ (dollars) for 1.25 kg: $22.5 \div 1.25 = 18$ ✓. (As in Step 2: 18 is 3 away from 15 and 12 away from 30, so $3 \cdot 1 = 12y$, $y = 0.25$ ✓.)

Step 4. Part c

Let the masses of A and B be x and y (kg), with $x > 0$, $y > 0$. The price of the mixture is $G = \dfrac{12x + 20y}{x + y}$ (dollars per kg).

- Since $y > 0$, $20y > 12y$, so $12x + 20y > 12x + 12y = 12(x + y)$. Dividing both sides by the positive number $x + y$: $G > 12$.
- Since $x > 0$, $12x < 20x$, so $12x + 20y < 20x + 20y = 20(x + y)$. Dividing both sides by $x + y$: $G < 20$.

So $12 < G < 20$: the price of the mixture always lies *between* the prices of the two candies.

Common mistake. You cannot mix A and B to get a price of \$25 or of \$10 per kilogram: an average never goes outside the range from the lowest to the highest price.

Note for parents. You can illustrate part c by mixing water: mixing water at 20°C with water at 60°C gives warm water in between, never hotter than 60°C. "An average always lies in between" is a very useful idea.

Answer: a) 625 g of candy A and 375 g of candy B; b) 250 g of candy C; c) see the solution.

2 Geometry and Measurement

The big idea. Geometry is the study of *shapes*; measurement attaches *numbers* to those shapes: lengths, angle measures, perimeters, areas and volumes. Almost every problem in this topic is solved with a handful of very simple ideas that are used over and over again:

- **Angles have a fixed "budget":** two supplementary angles always add up to 180°, the three angles of a triangle always add up to 180°, and a full turn is 360°. Knowing some of the angles, we can work out the rest.
- **Areas can be cut, rearranged and subtracted:** a complicated shape is split into familiar pieces (rectangles, triangles, trapezoids), or computed as "big shape minus the extra bits".
- **Ratios instead of numbers:** "what percentage is shaded" only requires comparing the shaded part with the whole shape – often without knowing any real sizes.
- **Shape does not change when turned or flipped:** rotating, reflecting or rolling a cube only changes *position*, never *size* or *shape*.
- **A box is built in layers:** volume = base area × height.

Whenever you meet a geometry problem, always **draw the figure and write the given numbers on it**, then ask yourself: "What do I know? What is missing? Which formula connects the two?".

① The coordinate plane

The coordinate system Oxy consists of two number lines that are perpendicular at the origin O:

- the horizontal axis Ox is called the **x-axis**; its positive direction points to the *right*;
- the vertical axis Oy is called the **y-axis**; its positive direction points *up*.

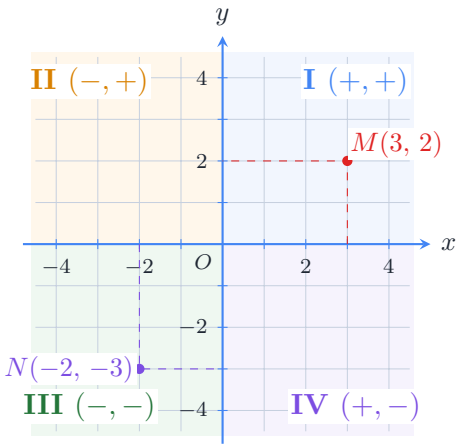
Each point M of the plane corresponds to exactly one pair of numbers (x_0, y_0), written $M(x_0, y_0)$:

- x_0 is the **x-coordinate**: drop a perpendicular from M to the x-axis; its foot is at the number x_0;
- y_0 is the **y-coordinate**: draw a perpendicular from M to the y-axis; its foot is at the number y_0.

The two axes divide the plane into four **quadrants**, numbered counterclockwise:

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
Signs of (x, y)	$(+, +)$	$(-, +)$	$(-, -)$	$(+, -)$

Distances along horizontal and vertical lines. Two points with the same y-coordinate (on the same horizontal line) are a distance apart equal to the difference of their x-coordinates (larger minus smaller). Similarly, two points with the same x-coordinate are a distance apart equal to the difference of their y-coordinates.



ThinkAlgo Tip. Read coordinates in the order "**across first, then up or down**": start at O, move right (positive) or left (negative) to get the x-coordinate; then move up (positive) or down (negative) to get the y-coordinate. For example, to reach $N(-2, -3)$: go 2 squares left, then 3 squares down.

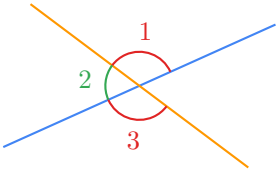
Common mistake. Swapping the two numbers is a very common mistake: $(3, -1)$ and $(-1, 3)$ are two *different* points (one in quadrant IV, the other in quadrant II). The x-coordinate is *always* written first.

Mini example. Let $A(1, 1)$, $B(5, 1)$, $C(5, 4)$. Since A and B have the same y-coordinate 1, $AB = 5 - 1 = 4$. Since B and C have the same x-coordinate 5, $BC = 4 - 1 = 3$. Angle B is a right angle (one side horizontal, one side vertical), so the area of triangle ABC is $\dfrac{4 \cdot 3}{2} = 6$ (square units).

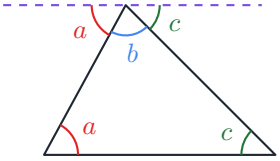
Note for parents. You can turn the coordinate plane into a "treasure hunt" on squared paper: you say "left 2, up 3", your child marks the point and writes its coordinates $(-2, 3)$. Then swap roles: your child gives directions and you draw. After a few rounds, the order "x first, y second" becomes automatic.

② Angles: supplementary, vertically opposite, angle sum of a triangle

- A **right angle** is 90°, a **straight angle** (half a turn) is 180°, a **full turn** is 360°.
- **Two adjacent supplementary angles** (sharing one arm, with the other two arms forming a straight line) add up to 180°.
- **Two vertically opposite angles** (each arm of one is the extension of an arm of the other) are **equal**.
- **The three angles of a triangle add up to** 180°**.** Hence: third angle = 180° − (sum of the two known angles).
- An **exterior angle of a triangle** (supplementary to an interior angle) equals **the sum of the two interior angles not adjacent to it**.
- In a right triangle, the two acute angles add up to 90°.



$\widehat{1}+\widehat{2}=180°$ (supplementary)
$\widehat{1}=\widehat{3}$ (vertically opposite)



$a+b+c=180°$

Why is it true? **Vertically opposite angles are equal.** In the left figure, $\widehat{1}+\widehat{2}=180°$ (supplementary) and $\widehat{3}+\widehat{2}=180°$ (also supplementary). Angles $\widehat{1}$ and $\widehat{3}$ are both supplementary to the same angle $\widehat{2}$, so $\widehat{1}=180°-\widehat{2}=\widehat{3}$.

The angles of a triangle add up to $180°$**.** Through the top vertex of the triangle, draw the line parallel to the base (dashed). For parallel lines, *alternate angles* are equal: the angle on the left at the top vertex equals a, and the angle on the right equals c. The three angles a, b, c at the top vertex together form a straight angle, so $a+b+c=180°$.

Exterior angle. The exterior angle at a vertex is supplementary to the interior angle there: exterior angle $=180°-$ interior angle. But $180°-$ interior angle is exactly the sum of the other two interior angles. So the exterior angle equals the sum of the two interior angles not adjacent to it. □

ThinkAlgo Tip. In "find the angle" problems, write the measure of every angle you find **on the figure** immediately. An angle you have just found is often the key to the next one (supplementary, vertically opposite, angle sum, ...).

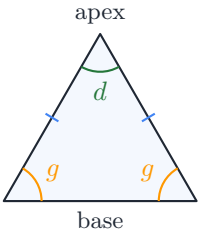
Common mistake. Do not confuse *vertically opposite* angles (equal) with *supplementary* angles (sum $180°$). Vertically opposite angles sit "across from each other" at the intersection point and share no arm; adjacent supplementary angles sit "next to each other" and share an arm.

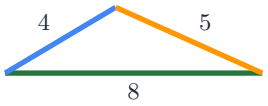
③ Isosceles triangles, equilateral triangles and the triangle inequality

- An **isosceles triangle** has two equal sides (the *legs*); the third side is the *base*, and the angle between the two legs is the *apex angle*.
- In an isosceles triangle, **the two base angles are equal**. So if the apex angle is d and each base angle is g, then

$$d+2g=180°, \qquad d=180°-2g, \qquad g=\frac{180°-d}{2}.$$

- A base angle is always **acute** (less than $90°$), because the two base angles are equal and their sum must be less than $180°$.
- An **equilateral triangle** has three equal sides and three equal angles, each $180°\div 3=60°$.
- **Triangle inequality:** in a triangle, each side is always *less* than the sum of the other two. To check whether three lengths form a triangle, it is enough to check: *longest side* $<$ *sum of the other two.*





$8 < 4 + 5$: a triangle exists

Why is it true? **Why are the base angles equal?** Let M be the midpoint of the base BC of the isosceles triangle ABC ($AB = AC$). Triangles ABM and ACM have $AB = AC$, $BM = CM$ and the common side AM, so they are congruent (side – side – side). Hence the corresponding angles $\widehat{B}$ and $\widehat{C}$ are equal.

Why does the triangle inequality hold? The shortest path between two points is the straight segment. Going from B to C "via" A (that is, travelling $BA + AC$) is always longer than going straight along BC. If the longest side were equal to or greater than the sum of the other two, the two short sides could not "reach each other": they would lie flat along the long side or fail to meet, and no triangle would be formed. □

Common mistake. When a problem only says "an isosceles triangle has *an angle* of 50°", we do **not yet know** whether it is the apex angle or a base angle. We must consider **both cases**:

- 50° is the apex angle: the base angles are $(180° - 50°) \div 2 = 65°$ and 65°;
- 50° is a base angle: the other base angle is also 50°, and the apex angle is $180° - 2 \cdot 50° = 80°$.

Only when the given angle is *greater than or equal to* 90° must it be the apex angle.

④ Rotations, reflections and rolling cubes

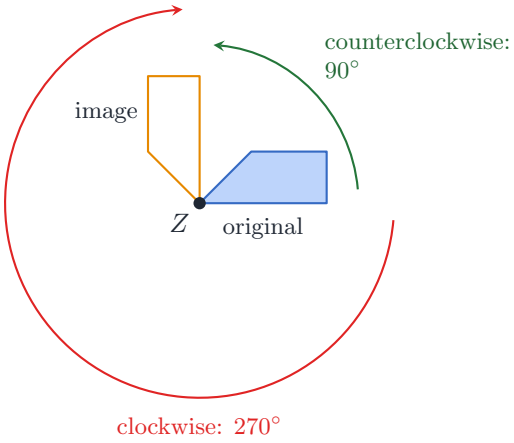
A **rotation** about a point Z (the *centre of rotation*) through an angle α in a given direction (clockwise or counterclockwise) sends each point M to a point M' such that $ZM' = ZM$ and $\widehat{MZM'} = \alpha$.

- Every point of the figure turns through **the same angle**, in the same direction; distances to the centre do not change.
- A rotation keeps the shape and size of the figure; the figure is only "turned", never "flipped over".
- Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise. A 360° rotation brings the figure back to where it started.
- **How to measure the angle of rotation:** choose a segment starting at the centre Z on the original figure, find the matching segment on the new figure, and measure the angle that segment has swept through *in the given direction*.

A **reflection in a line** (flipping over the line, like a mirror) also keeps sizes, but **reverses the orientation** of the figure: the letter P seen in a mirror becomes a backwards P.

In the coordinate plane (centre of rotation at the origin O):

Rotation 90° counterclockwise	$(x, y) \to (-y, x)$
Rotation 180°	$(x, y) \to (-x, -y)$
Rotation 90° clockwise	$(x, y) \to (y, -x)$
Reflection in the x-axis / in the y-axis	$(x, y) \to (x, -y)$ / $(x, y) \to (-x, y)$



Why is it true? Why is a 270° clockwise rotation the same as a 90° counterclockwise one? A full turn is 360°. After going 270° one way, only $360° - 270° = 90°$ more would bring us back to the start; so the end position is exactly where we would arrive by going 90° the other way. In the figure: to turn the horizontal side (pointing right) into the vertical side (pointing up), we can rotate 90° counterclockwise, or rotate 270° clockwise (through pointing down, pointing left, and only then up). □

Rolling a cube. When a cube is rolled, each face is carried along unchanged. So a face with a painted pattern:

- may appear in another position (front, top, side, ...);
- may be *turned* by 90°, 180° or 270° within its own plane;
- **but never** becomes a mirror image, changes size, or "gains" extra paint.

ThinkAlgo Tip. For cube problems, "mark" one special feature of the pattern (for example: the right angle of the black triangle sits at *a corner* of the face) and check that feature in every option. Cutting out a paper square, drawing the pattern on it and turning it is also a very effective check.

5 Units of measurement and the number line

Units of length. Each unit is 10 times the next one:

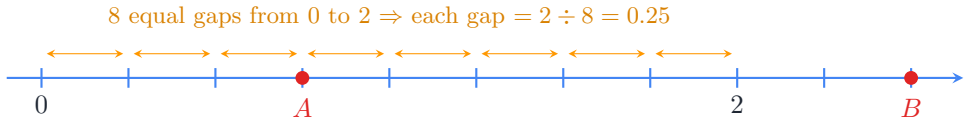
$$1 \text{ km} = 10 \text{ hm}, \quad 1 \text{ hm} = 10 \text{ dam}, \quad 1 \text{ dam} = 10 \text{ m},$$

$$1\text{ m} = 10\text{ dm}, \quad 1\text{ dm} = 10\text{ cm}, \quad 1\text{ cm} = 10\text{ mm}.$$

The most used relations: $1\text{ m} = 100\text{ cm} = 1000\text{ mm}$; $1\text{ km} = 1000\text{ m}$.
Units of area go up by a factor of 100: $1\text{ m}^2 = 100\text{ dm}^2 = 10\,000\text{ cm}^2$.
Units of volume go up by a factor of 1000: $1\text{ m}^3 = 1000\text{ dm}^3$, $1\text{ dm}^3 = 1000\text{ cm}^3 = 1$ litre.
Conversion rule: from a *larger* unit to a *smaller* unit, **multiply**; from a *smaller* unit to a *larger* unit, **divide**.
Number line with equally spaced ticks. If we know the numbers at two ticks and count k *gaps* (not k ticks) between them, then

$$\text{value of one gap} = \frac{\text{difference of the two numbers}}{k}.$$

Why is it true? Why is $1\text{ m}^2 = 10\,000\text{ cm}^2$ and not 100 cm^2? One square metre is the area of a square with side 1 m, i.e. side 100 cm. That square can be divided into 100 rows of 100 squares of side 1 cm, so it contains $100 \times 100 = 10\,000$ squares of 1 cm^2. Likewise, a cube of side 1 dm = 10 cm consists of $10 \times 10 \times 10 = 1000$ small cubes of 1 cm^3. □



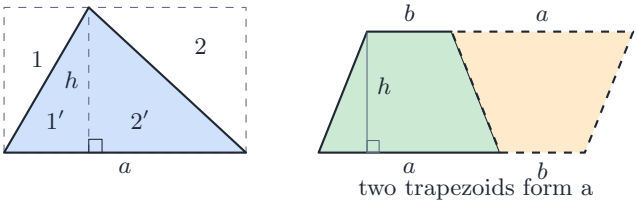
Mini example. On the number line above, A is three gaps from 0, so $A = 3 \cdot 0.25 = 0.75$; B is ten gaps from 0, so $B = 10 \cdot 0.25 = 2.5$.

Common mistake. Counting *ticks* instead of *gaps*: from tick 0 to tick 2 in the figure above there are 9 ticks but only 8 gaps. Between two numbers there is always one gap fewer than ticks (like trees in a row: 9 trees have 8 spaces between them).

⑥ Perimeter and area of plane figures

Figure	Perimeter	Area
Rectangle (length a, width b)	$P = 2 \cdot (a + b)$	$S = a \cdot b$
Square (side a)	$P = 4 \cdot a$	$S = a \cdot a = a^2$
Triangle (base a, height h to that base)	sum of the three sides	$S = \frac{a \cdot h}{2}$
Parallelogram (base a, height h)	2·(sum of two adjacent sides)	$S = a \cdot h$
Trapezoid (parallel sides a, b; height h)	sum of the four sides	$S = \frac{(a+b) \cdot h}{2}$

The **height** of a triangle (parallelogram, trapezoid) is the segment drawn *perpendicular* from the vertex (or from the opposite side) to the line containing the base.
Working backwards: from $S = \frac{a \cdot h}{2}$ we get $h = \frac{2 \cdot S}{a}$ and $a = \frac{2 \cdot S}{h}$; from the perimeter of a square, side $= P \div 4$.



two trapezoids form a parallelogram with base $a + b$

Why is it true? Area of a triangle. Place the triangle inside the rectangle with the same base a and the same height h (left figure). The height splits the rectangle into two smaller rectangles; each smaller rectangle is cut by a diagonal into two equal triangles: the part inside the triangle ($1'$, $2'$) equals the part outside it (1, 2). So the triangle is exactly half of the rectangle: $S = \dfrac{a \cdot h}{2}$.
Area of a parallelogram. Cut off a right triangle at one end and slide it to the other end: we get a rectangle with the same base a and the same height h: $S = a \cdot h$.
Area of a trapezoid. Two identical trapezoids, one turned upside down, fit together into a parallelogram with base $a+b$ and height h (right figure). One trapezoid is half of it: $S = \dfrac{(a+b) \cdot h}{2}$. □

ThinkAlgo Tip. Two triangles with **the same base** (or equal bases) and **the same height** have equal areas, even if they look very different. Consequence: the segment joining a vertex to the midpoint of the opposite side divides a triangle into two parts of equal area.

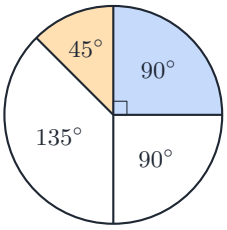
Common mistake.

- The height must be **perpendicular** to the base; a slanted side is *not* the height.
- Forgetting to divide by 2 in the area formulas of the triangle and the trapezoid.
- Mixing up units: perimeter is measured in cm, area in cm^2. A square with perimeter 36 cm has side 9 cm and area 81 cm^2 – not 36 cm^2.

⑦ Parts of an area: fractions, percentages, circles and rearranging

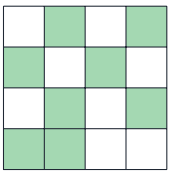
- If a figure is divided into n **equal** parts and m parts are shaded, the shaded part is $\dfrac{m}{n}$ of the figure, i.e. $\dfrac{m}{n} \cdot 100\%$.
- **Circles and sectors.** One turn around the centre is 360°. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the area of the circle. For example: a semicircle corresponds to 180°, a quarter circle to 90° (a right angle at the centre).
- **Spinners:** the pointer stops at random, all directions being equally likely, so the chance of stopping in a region equals the fraction of the area (i.e. of the angle) of that region.
- **Cut and rearrange:** split the figure into familiar pieces and add their areas. **Complement:** required area = area of the big figure − area of the extra pieces.
- **Systematic counting of figures:** when asked how many figures (triangles, parallelograms, ...) appear in a diagram, sort them by *size* (made of 1 piece, 2 pieces, 3 pieces, ...) and count

each type so that nothing is missed or counted twice.



blue: $\frac{90}{360} = \frac{1}{4} = 25\%$

orange: $\frac{45}{360} = \frac{1}{8} = 12.5\%$



8 of 16 squares: $\frac{8}{16} = 50\%$

Why is it true? Why is a sector with angle n° equal to $\frac{n}{360}$ of the circle? Imagine cutting the circle into 360 tiny sectors, each with a central angle of 1°. These tiny sectors are identical (turning the circle by 1° about its centre makes one fit exactly onto the next), so they have equal areas, each $\frac{1}{360}$ of the circle. A sector with angle n° is made of n such tiny sectors. □

Common mistake. You may only count "m parts out of n" when the parts are **equal**. If the pieces have different sizes, reduce them to one kind of piece (for example by cutting them further into equal pieces) or compute the area of each piece.

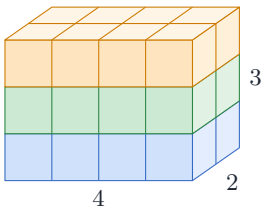
Note for parents. Cut a cake (or a paper circle) into 4, then into 8 equal parts so your child can see: 1 quarter = 90°, 1 eighth = 45°, and half of a quarter is an eighth. The idea "halving the angle halves the area" becomes very concrete.

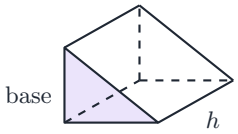
⑧ Solids: volume and surface area

- A **rectangular box** (rectangular prism) with dimensions a, b, c:

$$V = a \cdot b \cdot c, \qquad S_{\text{lateral}} = 2 \cdot (a + b) \cdot c, \qquad S_{\text{total}} = 2 \cdot (ab + bc + ca).$$

- A **cube** with side a: $V = a^3$, $S_{\text{total}} = 6 \cdot a^2$.
- A **right prism** (with a triangular, quadrilateral, ... base): $V = S_{\text{base}} \cdot h$, where h is the height of the prism (the length of a lateral edge).
- The number of 1 cm × 1 cm × 1 cm cubes needed to fill a box exactly equals the volume of the box in cm^3.





each layer $4 \cdot 2 = 8$ cubes, 3 layers: $8 \cdot 3 = 24$ cubes $V = S_{\text{base}} \cdot h$

Why is it true? **Why is $V = a \cdot b \cdot c$?** Stack 1 cm^3 cubes in layers. The bottom layer is a rectangle of a rows $\times$ b columns, containing $a \cdot b$ cubes (exactly the base area). Stack c such layers and we have $a \cdot b \cdot c$ cubes. The argument "each layer = base area, number of layers = height" works for every right prism, so $V = S_{\text{base}} \cdot h$. In particular, for a prism whose base is a right triangle with legs a, b: two such prisms fit together into an $a \times b \times h$ box, so each prism has volume $\dfrac{a \cdot b}{2} \cdot h$. $\square$

ThinkAlgo Tip. The surface area of a box is the total area of its 6 faces, which come in 3 pairs of equal opposite faces. When small blocks are joined into a big block, the faces that are "pressed together" are no longer on the surface, so the surface area of the big block is *less* than the total surface area of the small blocks.

Example 1

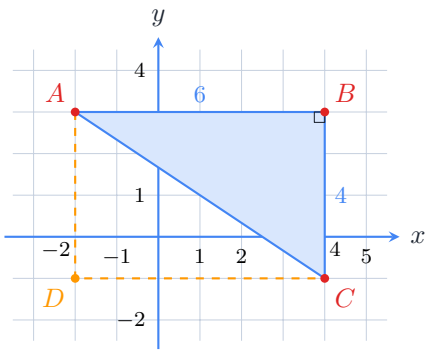
In the coordinate plane Oxy, consider the points $A(-2, 3)$, $B(4, 3)$ and $C(4, -1)$.

a) In which quadrant does each of the points A, B, C lie?

b) Find the lengths AB, BC and the area of triangle ABC.

c) Find the coordinates of the point D such that $ABCD$ is a rectangle.

Guided solution.



Step 1. Use the signs of the coordinates to find the quadrant

$A(-2, 3)$: negative x, positive y $\Rightarrow$ quadrant II. $B(4, 3)$: both positive $\Rightarrow$ quadrant I. $C(4, -1)$: positive x, negative y $\Rightarrow$ quadrant IV.

Step 2. Find the lengths of the horizontal and vertical sides

A and B have the same y-coordinate 3, so AB is horizontal and $AB = 4 - (-2) = 4 + 2 = 6$ (units). B and C have the same x-coordinate 4, so BC is vertical and $BC = 3 - (-1) = 3 + 1 = 4$.

Step 3. Find the area of the triangle

Side AB is horizontal and side BC is vertical, so angle B is a right angle. The area of a right triangle is half the product of the two legs:

$$S_{ABC} = \frac{AB \cdot BC}{2} = \frac{6 \cdot 4}{2} = 12 \text{ (square units)}.$$

Step 4. Find the point D

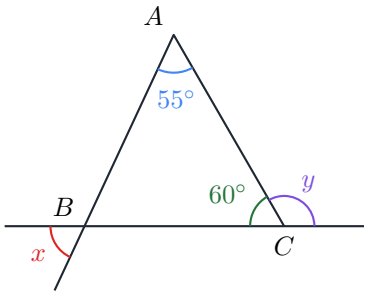
In rectangle $ABCD$, D is "opposite" B: D must be vertically in line with A (same x-coordinate -2) and horizontally in line with C (same y-coordinate -1). So $D(-2, -1)$, in quadrant III.

Check: Rectangle $ABCD$ has area $6 \cdot 4 = 24$, exactly twice the area of triangle ABC (the diagonal AC cuts the rectangle into two equal halves).

Answer: **a)** A: **II**, B: **I**, C: **IV**. **b)** $AB = 6$, $BC = 4$, $S = 12$. **c)** $D(-2, -1)$.

Example 2

In the figure below, triangle ABC has $\widehat{A} = 55°$, $\widehat{C} = 60°$. Sides AB and CB are extended beyond B; side BC is extended beyond C. Find the measures x and y.



Guided solution.

Analysis. Angle x lies *outside* the triangle, but it is *vertically opposite* to the angle $\widehat{ABC}$ of the triangle (the arms of x are the extensions of BC and of BA). So we find $\widehat{ABC}$ first.

Step 1. **Find angle B of the triangle**

Recall. The three angles of a triangle add up to 180°.

$\widehat{ABC} = 180° - (55° + 60°) = 180° - 115° = 65°$.

Step 2. **Find x**

Recall. Vertically opposite angles are equal.

Angle x and angle $\widehat{ABC}$ are vertically opposite, so $x = 65°$.

Step 3. **Find y**

Angle y is supplementary to angle $\widehat{ACB}$, so $y = 180° - 60° = 120°$.

Check: By the exterior angle property, y must equal the sum of the two non-adjacent interior angles: $55° + 65° = 120°$. It matches!

Answer: $x = 65°$, $y = 120°$.

Example 3

a) An isosceles triangle has an angle of 40°. Find the other two angles (consider all possibilities).

b) An isosceles triangle has an angle of 100°. Find the other two angles.

c) Which of the triples of lengths (3, 4, 8) and (3, 5, 7) forms a triangle? An isosceles triangle has two sides of lengths 4 cm and 9 cm. Find its perimeter.

Guided solution.

Step 1. Part a – two cases

Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°.

Case 1: 40° *is the apex angle.* The base angles are equal and add up to $180° - 40° = 140°$, so each is $140° \div 2 = 70°$. The triangle has angles 40°, 70°, 70°.
Case 2: 40° *is a base angle.* The other base angle is also 40°; the apex angle is $180° - 40° - 40° = 100°$. The triangle has angles 40°, 40°, 100°.

Step 2. Part b – only one case

100° cannot be a base angle: otherwise the two base angles alone would add up to $200° > 180°$. So 100° is the apex angle, and each base angle is $(180° - 100°) \div 2 = 40°$.

Step 3. Part c – use the triangle inequality

Recall. Three lengths form a triangle exactly when the longest is less than the sum of the other two.

(3, 4, 8): $8 > 3 + 4 = 7$, *no* triangle. (3, 5, 7): $7 < 3 + 5 = 8$, a triangle exists.
An isosceles triangle with two sides 4 and 9 has its third side equal to 4 or to 9:

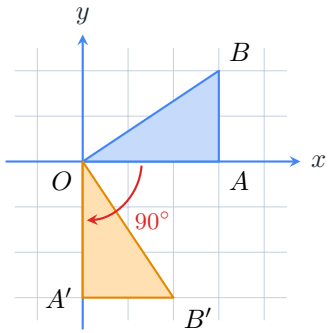
- (4, 4, 9): $9 > 4 + 4 = 8$, rejected;
- (4, 9, 9): $9 < 4 + 9 = 13$, fine. Perimeter $= 4 + 9 + 9 = 22$ cm.

Answer: a) 70° and 70°, or 40° and 100°. b) 40° and 40°. c) only (3, 5, 7); perimeter 22 cm.

Example 4

Triangle OAB with $O(0, 0)$, $A(3, 0)$, $B(3, 2)$ is rotated about the origin O *clockwise* and becomes triangle $OA'B'$ with $A'(0, -3)$, $B'(2, -3)$. What is the angle of rotation? If we rotate counterclockwise instead, through what angle must we turn to get the same result?

Guided solution.



Step 1. Choose a segment starting at the centre of rotation

Segment OA lies along the ray Ox, pointing *right*. Its image is OA' with $A'(0, -3)$, lying on the y-axis and pointing *down*.

Step 2. Measure the angle swept clockwise

A clock hand going from the "3 o'clock" direction (right) to the "6 o'clock" direction (down) sweeps

a right angle. So the clockwise angle of rotation is 90°.

Step 3. Convert to the other direction

Recall. Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise.

Counterclockwise: $360° - 90° = 270°$.

Check: By the table in section 4, a 90° clockwise rotation sends (x, y) to $(y, -x)$: $A(3, 0) \to (0, -3) = A'$ and $B(3, 2) \to (2, -3) = B'$. Correct.

Answer: 90° clockwise, equivalent to 270° counterclockwise.

Example 5

On a number line, the ticks are equally spaced. The first tick is at 0, the seventh tick is at 3 (that is, six gaps away from 0). Point M is at the tick three gaps from 0, point N at the tick nine gaps from 0.

a) Find the numbers at M and N, and compute $M \div N$.

b) If the unit on the line is the metre, convert M and N to centimetres.

Guided solution.



Step 1. Find the value of one gap

From 0 to 3 there are 6 equal gaps, so each gap is worth $3 \div 6 = 0.5$.

Step 2. Find M and N

M is three gaps from 0: $M = 3 \cdot 0.5 = 1.5$. N is nine gaps from 0: $N = 9 \cdot 0.5 = 4.5$.

Step 3. Compute the quotient

$M \div N = 1.5 \div 4.5 = \dfrac{15}{45} = \dfrac{1}{3} \approx 0.33$. (Quick view: N is 3 times as far from 0 as M, so $M \div N = \frac{1}{3}$.)

Step 4. Convert units

Recall. 1 m = 100 cm; from a larger unit to a smaller unit, multiply.

1.5 m $= 1.5 \cdot 100$ cm $= 150$ cm; 4.5 m $= 450$ cm.

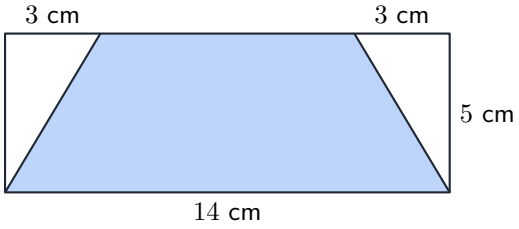
Answer: a) $M = 1.5$; $N = 4.5$; $M \div N = \frac{1}{3}$. b) 150 cm and 450 cm.

Example 6

A rectangle is 14 cm long and 5 cm wide. On its top side, two points are marked, each 3 cm from a top corner; they are joined to the two bottom corners to form the shaded trapezoid (see figure).

a) Find the area of the trapezoid in two ways.

b) A triangle has area 24 cm^2 and a base of length 8 cm. Find the height to that base.



Guided solution.

Step 1. **Method 1 – the trapezoid formula**

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$, where a, b are the parallel sides and h is the height.

Longer parallel side $a = 14$ cm. Shorter parallel side $b = 14 - 3 - 3 = 8$ cm. The height equals the width of the rectangle: $h = 5$ cm.

$$S = \frac{(14+8)\cdot 5}{2} = \frac{22\cdot 5}{2} = \frac{110}{2} = 55 \text{ (cm}^2\text{)}.$$

Step 2. **Method 2 – big shape minus the extra pieces**

Area of the rectangle: $14\cdot 5 = 70$ (cm^2). The two unshaded parts are right triangles, each with legs 3 cm and 5 cm and area $\dfrac{3\cdot 5}{2} = 7.5$ (cm^2). Area of the trapezoid: $70 - 2\cdot 7.5 = 70 - 15 = 55$ (cm^2).

Step 3. **Part b – find the height by working backwards**

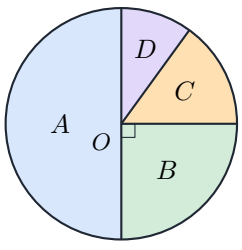
From $S = \dfrac{a\cdot h}{2}$ we get $h = \dfrac{2\cdot S}{a} = \dfrac{2\cdot 24}{8} = \dfrac{48}{8} = 6$ (cm).

Check: $\dfrac{8\cdot 6}{2} = 24$. Correct.

Answer: a) 55 cm^2. b) 6 cm.

Example 7

A circular spinner with centre O is divided into four regions A, B, C, D: region A is a semicircle, region B is a quarter circle, and the rest is split into regions C and D so that the central angle of C is one and a half times the central angle of D. What percentage of the spinner's area does each region take up? About what percentage of spins will stop in region C?



Guided solution.

Step 1. **Central angles of regions A and B**

The whole circle corresponds to 360°. Region A (a semicircle) corresponds to 180°, region B (a

quarter) to 90°.

Step 2. Central angles of regions C and D

Regions C and D share what is left: $360° - 180° - 90° = 90°$.

Let the central angle of region D be x (degrees). "The angle of C is one and a half times the angle of D" means the angle of C is $1.5x$. The two angles add up to 90°, so we get the equation

$$x + 1.5x = 90.$$

Collecting like terms: $2.5x = 90$, so $x = 90 \div 2.5 = 36$. Hence the angle of D is 36° and the angle of C is $1.5 \cdot 36° = 54°$.

Step 3. Convert to percentages

Recall. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the circle.

A: $\dfrac{180}{360} = 50\%$; B: $\dfrac{90}{360} = 25\%$; C: $\dfrac{54}{360} = 0.15 = 15\%$; D: $\dfrac{36}{360} = 0.1 = 10\%$.

Check: $50\% + 25\% + 15\% + 10\% = 100\%$, and $54° = 1.5 \cdot 36°$. Correct.

Answer: A: 50%, B: 25%, C: 15%, D: 10%; about 15% of spins stop in region C.

Example 8

a) How many cubes with edge 1 cm are needed to fill a 5 cm × 3 cm × 2 cm box? Find the total surface area of the box.

b) A right prism has a base that is a right triangle with legs 3 cm and 4 cm, and the prism is 10 cm high. Find its volume. If a prism with the same base has volume 90 cm^3, what is its height?

Guided solution.

Step 1. Part a – count layer by layer

Recall. $V_{\text{box}} = a \cdot b \cdot c$; $S_{\text{total}} = 2 \cdot (ab + bc + ca)$.

The bottom layer has $5 \cdot 3 = 15$ cubes; there are 2 layers, so we need $15 \cdot 2 = 30$ cubes (volume 30 cm^3).

The three pairs of opposite faces have areas $5 \cdot 3 = 15$, $3 \cdot 2 = 6$, $5 \cdot 2 = 10$ (cm^2), so $S_{\text{total}} = 2 \cdot (15 + 6 + 10) = 2 \cdot 31 = 62$ (cm^2).

Step 2. Part b – base area times height

Recall. $V_{\text{prism}} = S_{\text{base}} \cdot h$.

The base is a right triangle: $S_{\text{base}} = \dfrac{3 \cdot 4}{2} = 6$ (cm^2). Volume: $V = 6 \cdot 10 = 60$ (cm^3).

With volume 90 cm^3: $h = V \div S_{\text{base}} = 90 \div 6 = 15$ (cm).

Check: $6 \cdot 15 = 90$. Correct.

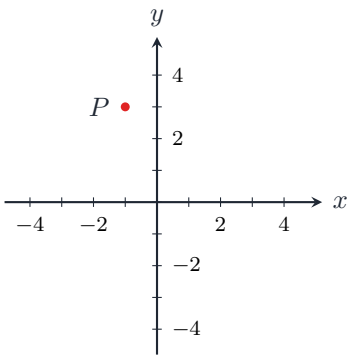
Answer: a) 30 cubes; 62 cm^2. b) 60 cm^3; 15 cm.

Note for parents. After each example, ask your child to *say in words* the formula just used and *why* it is true (for example: "a triangle is half of the rectangle with the same base and height"). Understanding the "why" means less rote memorisation and fewer mistakes on unfamiliar figures.

Core practice

Problem 1

In the coordinate plane below, each tick on the two axes represents 1 unit. Which pair of numbers could be the coordinates of point P?



(A) (1, 3) (B) (1, −3) (C) (−3, 1) (D) (3, −1) (E) (−1, 3)

Detailed solution

Understanding the problem. We read the position of P relative to the two axes to find the *sign* and the *size* of its coordinates, then compare with the five options.

> **Recall.** For a point $M(x_0, y_0)$: x_0 tells whether the point is to the right ($x_0 > 0$) or to the left ($x_0 < 0$) of the y-axis; y_0 tells whether it is above ($y_0 > 0$) or below ($y_0 < 0$) the x-axis. The x-coordinate is always written first.

Step 1. Find the signs of the coordinates

P is to the *left* of the y-axis, so its x-coordinate is **negative**. P is *above* the x-axis, so its y-coordinate is **positive**. So P is in quadrant II, of the form $(-, +)$.

Step 2. Eliminate options

- (A) (1, 3): form $(+, +)$, quadrant I – rejected.
- (B) (1, −3): form $(+, -)$, quadrant IV – rejected.
- (D) (3, −1): form $(+, -)$ – rejected.
- Remaining: (C) (−3, 1) and (E) (−1, 3), both of the form $(-, +)$.

Step 3. Compare the sizes

Drop a perpendicular from P to the x-axis: its foot lies between 0 and −2, exactly at tick −1. Draw a perpendicular to the y-axis: its foot lies between 2 and 4, exactly at tick 3. So P is very close to the y-axis (1 unit away) and fairly high up (3 units above the x-axis): $P(-1, 3)$.
The point (−3, 1) in option (C) is the opposite: far from the y-axis (3 units) and low (1 unit), which does not match the figure.

Common mistake. Option (C) is the "trap" for those who read the y-coordinate first. Remember: *across first, then up or down.*

Check: From O go 1 tick left, then 3 ticks up: we arrive exactly at P.
So the answer is **(E)**.

Answer: (E) $(-1, 3)$

Problem 2

Two angles of a triangle measure 25° and 70°. The third angle of the triangle measures

(A) 85° (B) 105° (C) 65° (D) 95° (E) 75°

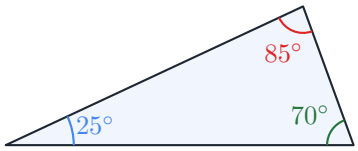
Detailed solution

Understanding the problem. Two angles of the triangle are known; we need the remaining one. We use the angle sum theorem.

Recall. The three angles of a triangle add up to 180°.

Step 1. Add the two known angles
$25° + 70° = 95°$.

Step 2. Subtract that sum from 180°
Third angle $= 180° - 95° = 85°$.



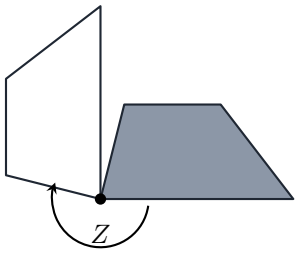
Common mistake. Option (D) 95° is the sum of the two given angles, not the third angle. Do not stop at step 1!

Check: $25° + 70° + 85° = 180°$. Correct.
So the answer is **(A)**.

Answer: (A) 85°

Problem 3

Rotating the shaded quadrilateral about point Z *clockwise* (in the direction of the arrow in the figure) produces the unshaded quadrilateral. The angle of rotation is approximately



(A) 180° (B) 270° (C) 360° (D) 45° (E) 135°

Detailed solution

Understanding the problem. The figure is rotated about Z clockwise. To find the angle, we follow *one side starting at* Z and see what angle it sweeps through.

> **Recall.** In a rotation, every point turns through the same angle about the centre; side lengths do not change. A full turn is 360°; rotating α clockwise is the same as rotating $360° - \alpha$ counterclockwise.

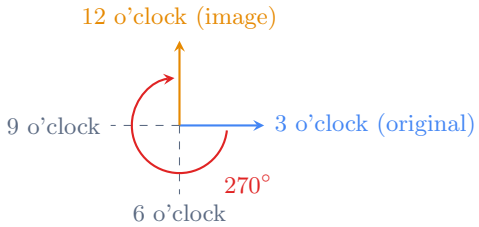
Step 1. Find the matching side

The shaded quadrilateral has its longest side starting at Z, horizontal and pointing *right* (the "3 o'clock" direction on a clock face). In the unshaded quadrilateral, the longest side starting at Z is the *vertical* side pointing *up* (the "12 o'clock" direction). Since a rotation keeps lengths, the horizontal side has become this vertical side.

Step 2. Measure the angle swept clockwise

Going clockwise from 3 o'clock to 12 o'clock, we pass 6 o'clock (down) and 9 o'clock (left) before reaching 12 o'clock: that is 3 right angles.

$$3 \cdot 90° = 270°.$$



Step 3. Check with a second side

The short slanted side of the shaded quadrilateral goes up from Z, leaning slightly right. After a further 270° clockwise turn (that is, 90° counterclockwise) it must point left, leaning slightly up – exactly the bottom side of the unshaded quadrilateral.

Common mistake. Rotating *counterclockwise* would need only 90°, but the problem asks for clockwise (the direction of the arrow), and 90° is not among the options. Do not mix up the two directions.

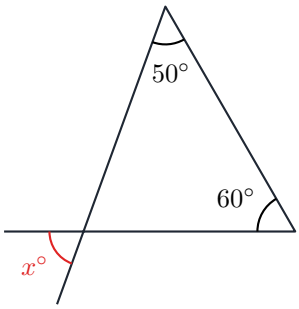
Check: $270° + 90° = 360°$: rotating 270° clockwise and rotating 90° counterclockwise give the same result, and a 90° counterclockwise turn clearly takes a side pointing "right" to a side pointing "up".
So the answer is **(B)**.

Answer: (B) 270°

Note for parents. Take a clock (or draw a clock face) and let your child turn the minute hand clockwise from the 3 to the 12: the hand passes 9 "five-minute steps" = 45 minutes, i.e. $\frac{3}{4}$ of a turn $= \frac{3}{4} \cdot 360° = 270°$. This activity helps your child feel the difference between the two directions of rotation.

Problem 4

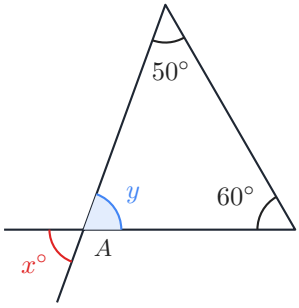
In the figure, two sides of the triangle are extended as shown. The value of x is



(A) 110 (B) 50 (C) 10 (D) 60 (E) 70

Detailed solution

Understanding the problem. The figure shows a triangle with two angles of 50° and 60°. At the third vertex (bottom left), the base and a side are extended outwards, forming the angle $x°$. We first find the interior angle at that vertex, then connect it to x.



Step 1. Find the interior angle y at vertex A

Recall. The three angles of a triangle add up to 180°.

Let y be the interior angle of the triangle at vertex A:

$$y = 180° - (50° + 60°) = 180° - 110° = 70°.$$

Step 2. Spot the pair of vertically opposite angles

Angle $x°$ is formed by the extension of the base (to the left) and the extension of the side (downwards). These are exactly the extensions of the two arms of angle y. So $x°$ and y are **vertically opposite angles**.

Recall. Vertically opposite angles are equal.

Therefore $x° = y = 70°$, i.e. $x = 70$.

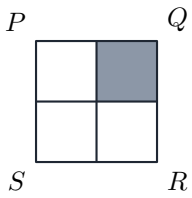
Common mistake. Option (A) 110 is what you get by confusing x with the angle *supplementary* to y (the angle right next to it, above the extended base). Look carefully: angle x sits *opposite* the interior angle across vertex A.

Check: The angle supplementary to y is $180° - 70° = 110°$; angle x is in turn supplementary to that 110° angle, so $x° = 180° - 110° = 70°$. Both ways give the same result.
So the answer is **(E)**.

Answer: (E) 70

Problem 5

Square $PQRS$ is divided into four equal smaller squares as shown. What percentage of the area of square $PQRS$ is shaded?



(A) 33% (B) 40% (C) 25% (D) 50% (E) 15%

Detailed solution

Understanding the problem. The big square is divided into 4 *equal* parts, and 1 part is shaded. We convert this ratio to a percentage.

Recall. If a figure is divided into n equal parts and m parts are shaded, the shaded part is $\frac{m}{n}$ of the figure, i.e. $\frac{m}{n} \cdot 100\%$.

Step 1. Write the fraction

The shaded part is 1 of 4 equal small squares: it is $\frac{1}{4}$ of square $PQRS$.

Step 2. Convert to a percentage

$\frac{1}{4} = \frac{25}{100} = 25\%$.

Check: If $PQRS$ has side 2 cm, the big area is 4 cm^2, and the shaded square has side 1 cm and area 1 cm^2; $1 \div 4 = 0.25 = 25\%$.

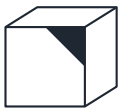
Common mistake. Option (A) 33% comes from comparing the shaded part with the *unshaded* part (1 to 3). A percentage must compare with the *whole* figure.

So the answer is **(C)**.

Answer: (C) 25%

Problem 6

A cube has part of *exactly one face* painted black as shown (the paint is a right triangle at a corner of the face, whose two legs are half the edge of the cube). The other five faces are not painted. After the cube is rolled, which of the following could be the same cube?



(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Detailed solution

Understanding the problem. The cube has only *one* painted face, and that face carries only *one* black triangle. We describe that triangle very precisely and compare it with each option.

Recall. When a cube is rolled, each face is carried along unchanged: the pattern on a face may move or be turned by 90°, 180°, 270°, but it does *not* change size, is *not* mirrored and does *not* gain extra paint.

Step 1. Describe the original pattern

Three features of the pattern:

- there is only **one** black triangle;

- the triangle has its **right angle exactly at a corner** of the square face;
- its two legs are **half the edge** of the face (the triangle covers $\frac{1}{8}$ of the face).

Step 2. Examine each option

- (A): the black triangle covers *half* the face (its legs are whole edges) – too big, rejected.
- (B): one black triangle, right angle at the bottom-right corner of the face, legs half an edge – matches all three features.
- (C): one triangle of the right size, but its right angle is at the *midpoint* of the bottom edge, not at a corner of the face – rejected.
- (D) and (E): *two* black triangles, more paint than the original cube – rejected.

Step 3. Check that (B) can really be obtained

Pattern (B) is the original pattern *turned* 90° *clockwise* within the plane of that face: the top-right corner becomes the bottom-right corner. Place the cube on the table with the painted face towards you and turn the cube a quarter turn about the axis perpendicular to that face (a few rolls back and forth achieve this): we obtain exactly picture (B).

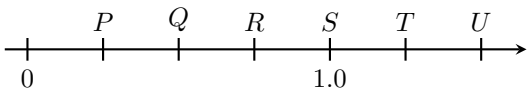
ThinkAlgo Tip. To check, cut out a paper square, colour a triangle in one corner and turn the paper: the triangle can only "travel" around the four corners, never sit in the middle of an edge as in (C).

So the answer is **(B)**.

Answer: (B)

Problem 7

On a number line the ticks are equally spaced. The numbers P, Q, R, S, T, U are marked as shown. Which of the following is closest to the value of $R \div U$?



(A) 0.25 (B) 0.50 (C) 0.75 (D) 1.25 (E) 1.50

Detailed solution

Understanding the problem. The two ticks with known values are 0 and 1.0 (tick S). From them we find the value of each gap, and then R and U.

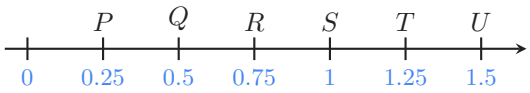
Recall. On a number line with equal spacing: value of one gap = (difference of the two known numbers) ÷ (number of *gaps* between them).

Step 1. Count the gaps from 0 to 1.0

From tick 0 to tick S (at 1.0) the gaps are: $0 \to P$, $P \to Q$, $Q \to R$, $R \to S$, i.e. 4 gaps.

Step 2. Find the value of one gap
$1.0 \div 4 = 0.25$.

Step 3. Find R and U
R is three gaps from 0: $R = 3 \cdot 0.25 = 0.75$. U is six gaps from 0: $U = 6 \cdot 0.25 = 1.5$.



Step 4. Compute the quotient
$R \div U = 0.75 \div 1.5 = \dfrac{75}{150} = \dfrac{1}{2} = 0.5$.
Quick view: R is three gaps from 0 and U is six gaps from 0, so U is twice R and $R \div U = \frac{1}{2}$ – no need to know the size of a gap!

Common mistake. Option (C) 0.75 is the value of R, before dividing by U. Also, counting 5 *ticks* from 0 to S instead of 4 *gaps* gives a wrong value for each gap.

Check: $1.5 \cdot 0.5 = 0.75 = R$. Correct.
So the answer is **(B)**.

Answer: (B) 0.50

Problem 8

One angle of an isosceles triangle measures 50°. The other two angles of this triangle could be

(A) 50° and 90° (B) 40° and 50° (C) 50° and 80°

(D) 30° and 100° (E) 60° and 70°

Detailed solution

Understanding the problem. The problem does not say whether the 50° angle is the apex angle or a base angle, so we consider both cases and then compare with the options.

Recall. An isosceles triangle has two equal base angles; the three angles of a triangle add up to 180°.

Step 1. Case: 50° is the apex angle
The base angles are equal and add up to $180° - 50° = 130°$, so each is $130° \div 2 = 65°$. The other two angles: 65° and 65° (not among the options).

Step 2. Case: 50° is a base angle
The other base angle is also 50°. Apex angle: $180° - 50° - 50° = 80°$. The other two angles: 50° and 80° – this is option (C).

Step 3. Check why the other options are wrong
Each triple of angles must add up to 180° *and* contain two equal angles:

Option	Three angles	Sum	Two equal angles?
(A)	50°, 50°, 90°	190°	yes, but wrong sum
(B)	50°, 40°, 50°	140°	yes, but wrong sum
(C)	50°, 50°, 80°	180°	yes
(D)	50°, 30°, 100°	180°	no
(E)	50°, 60°, 70°	180°	no

Only (C) satisfies both conditions.

Check: $50° + 50° + 80° = 180°$ and two angles are equal (50°).
So the answer is **(C)**.

Answer: (C) 50° and 80°

Problem 9

Ben measures a rope and finds that it is 3.5 m long. How many centimetres is that?

(A) 350 (B) 30.5 (C) 3.05 (D) 3.5 (E) 305

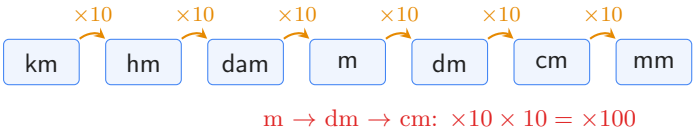
Detailed solution

Understanding the problem. We convert from metres (a larger unit) to centimetres (a smaller unit), so we *multiply*.

Recall. 1 m = 10 dm = 100 cm. From a larger unit to a smaller unit, multiply; from a smaller unit to a larger unit, divide.

Step 1. Write the multiplication
$3.5 \text{ m} = 3.5 \cdot 100 \text{ cm}$.

Step 2. Multiply by 100
Multiplying a decimal by 100 moves the decimal point 2 places to the right: $3.5 \to 35 \to 350$. So 3.5 m = 350 cm.



m → dm → cm: $\times 10 \times 10 = \times 100$

Another way to see it: 3.5 m = 3 m + 0.5 m = 300 cm + 50 cm = 350 cm (half a metre is 50 cm).

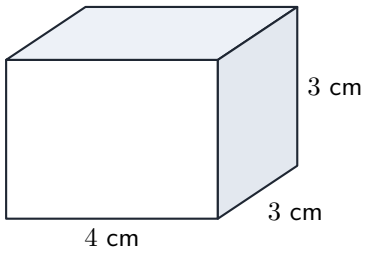
Common mistake. Option (E) 305 comes from reading it as "3 metres 5 centimetres". But 0.5 m is *half a metre* = 50 cm, not 5 cm.

Check: $350 \div 100 = 3.5$. Correct. So the answer is **(A)**.

Answer: (A) 350

Problem 10

Anna wants to build the solid rectangular box shown from small cubes measuring 1 cm×1 cm×1 cm. How many small cubes does Anna need?



(A) 10 (B) 12 (C) 33 (D) 66 (E) 36

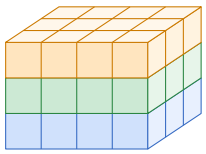
Detailed solution

Understanding the problem. The box has three dimensions: length 4 cm, height 3 cm, depth (width) 3 cm. The number of 1 cm^3 cubes needed is exactly the volume of the box in cm^3.

Recall. Volume of a rectangular box: $V = a \cdot b \cdot c$ (length × width × height).

Step 1. Count the cubes in one layer

We build horizontal layers. The bottom layer has a 4 cm × 3 cm base (4 cm long, 3 cm deep), so it has 3 rows of 4 cubes: $4 \cdot 3 = 12$ cubes.



Step 2. Count the layers

The height is 3 cm, so there are 3 layers stacked on top of each other.

Step 3. Multiply

Total number of cubes: $12 \cdot 3 = 36$ cubes. That is, $V = 4 \cdot 3 \cdot 3 = 36$ (cm^3).

Common mistake. Option (A) 10 is the *sum* $4 + 3 + 3$, and (C) 33 just puts digits together. The volume is the *product* of the three dimensions, not their sum.

Check: Counting another way: each vertical "slice" parallel to the front face has $4 \cdot 3 = 12$ cubes, and there are 3 slices front to back: $12 \cdot 3 = 36$. So the answer is **(E)**.

Answer: (E) 36

Problem 11

A square garden has a perimeter of 36 m. The area of the garden, in m^2, is

(A) 24 (B) 81 (C) 36 (D) 1296 (E) 324

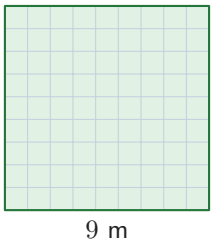
Detailed solution

Understanding the problem. We know the perimeter and want the area. The bridge between the two quantities is the *side length*: from the perimeter find the side, from the side find the area.

Recall. Square with side a: perimeter $P = 4 \cdot a$, area $S = a \cdot a$.

Step 1. Find the side of the square
A square has 4 equal sides, so $a = P \div 4 = 36 \div 4 = 9$ (m).

Step 2. Find the area
$S = 9 \cdot 9 = 81$ (m^2).



9 rows of 9 squares of 1 m^2:
$9 \cdot 9 = 81$ m^2

Common mistake.
- (C) 36: confusing perimeter with area.
- (D) $1296 = 36 \cdot 36$: squaring the perimeter instead of the side.
- (E) $324 = 36 \cdot 9$: multiplying the perimeter by the side.

Check: Side 9 m $\Rightarrow$ perimeter $4 \cdot 9 = 36$ m, as given.
So the answer is **(B)**.

Answer: (B) 81

ThinkAlgo Tip. For a square, *always* find the side first. A small table worth remembering: side 5 $\rightarrow$ perimeter 20, area 25; side 9 $\rightarrow$ perimeter 36, area 81; side 10 $\rightarrow$ perimeter 40, area 100. When the side doubles, the perimeter doubles but the area becomes *four* times as large.

Problem 12

Two isosceles triangles each have at least one angle of 70°. In the first triangle, each of the two remaining angles measures an even number of degrees. In the second triangle, each of the two remaining angles measures an odd number of degrees. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) 280° (B) 250° (C) 220° (D) 200° (E) 300°

Detailed solution

Understanding the problem. This looks long, but it is really the familiar "isosceles triangle with an angle of 70°" problem. We list *all* isosceles triangles with a 70° angle, then use the even/odd conditions to decide which is the first triangle and which is the second.

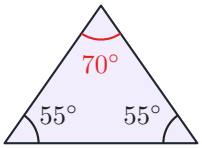
Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°. A given angle may be the apex angle or a base angle – both must be considered.

Step 1. List the isosceles triangles with a 70° angle

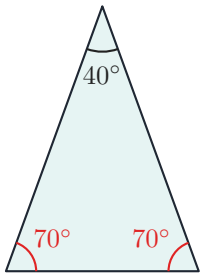
Case 1: 70° *is the apex angle.* Each base angle: $(180° - 70°) \div 2 = 110° \div 2 = 55°$. Triangle: 70°, 55°, 55°.

Case 2: 70° *is a base angle.* The other base angle is also 70°, and the apex angle is $180° - 2 \cdot 70° = 40°$. Triangle: 70°, 70°, 40°.

There are only these two triangles.



70° at the apex



70° at the base

Step 2. Identify the first triangle (both remaining angles even)

Remove one 70° angle:

- triangle $70°, 55°, 55°$ leaves 55° and 55° – both *odd*;
- triangle $70°, 70°, 40°$ leaves 70° and 40° – both *even*.

So the first triangle is 70°, 70°, 40° and the second is 70°, 55°, 55°.

Step 3. Find S and T

The first triangle has equal angles 70° and 70°: $S = 70° + 70° = 140°$. The second triangle has equal angles 55° and 55°: $T = 55° + 55° = 110°$.

Step 4. Add

$S + T = 140° + 110° = 250°$.

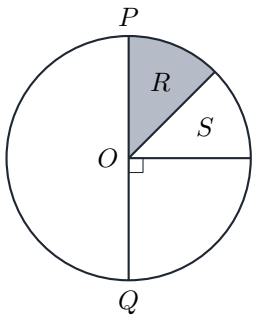
Common mistake. Read carefully: S and T are the sums of the *two equal angles*, not the sums of the two "remaining" angles (so for the first triangle we take $70° + 70°$, not $70° + 40°$).

Check: $70 + 70 + 40 = 180$ and $70 + 55 + 55 = 180$. Both triangles are valid; $140 + 110 = 250$. So the answer is **(B)**.

Answer: (B) 250°

Problem 13

On the circular spinner with centre O shown, PQ is a diameter, the horizontal radius is perpendicular to PQ at O, and the sectors R and S have equal areas. About what percentage of spins will stop in the shaded region?



(A) 50% (B) 22.5% (C) 25% (D) 45% (E) 12.5%

Detailed solution

Understanding the problem. The pointer is equally likely to stop in any direction, so the fraction of spins stopping in the shaded region equals the fraction of the area (also the fraction of the central angle) of that region compared with the whole circle. We need the central angle of sector R.

Recall. One turn around the centre is 360°. A sector with central angle $n°$ is $\frac{n}{360}$ of the area of the circle.

Step 1. Angle of R and S together
Radius OP points up and the horizontal radius points right; these two radii are perpendicular. So R and S together form a quarter of the circle, with central angle 90°.

Step 2. Angle of R alone
R and S have equal areas, so their central angles are equal: each is $90° \div 2 = 45°$.

Step 3. Convert to a percentage

$$\frac{45}{360} = \frac{1}{8} = 0.125 = 12.5\%.$$

Why is it true? Why does "equal areas" imply "equal angles"? The area of a sector is $\frac{n}{360}$ of the area of the circle, proportional to the angle n. Two sectors of the same circle with equal areas satisfy $\frac{n_1}{360} = \frac{n_2}{360}$, so $n_1 = n_2$. □

Common mistake. Option (C) 25% is the share of R and S *together*; (B) 22.5% comes from confusing 45° with 45% and halving. Only R is shaded, and the angle must be divided by 360.

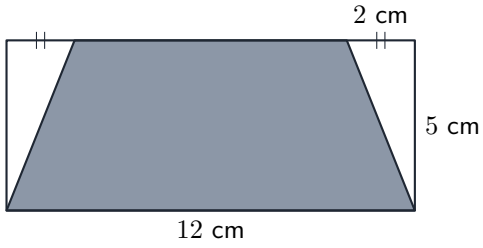
Check: The shaded part is half of a quarter circle, i.e. $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8} = 12.5\%$.
So the answer is **(E)**.

Answer: (E) 12.5%

Note for parents. Fold a paper circle in half, then in quarters, then in eighths: each fold halves the smallest piece. Unfolded, your child will see 8 equal sectors, each with an angle of 45° and a share of 12.5% – exactly the region R in this problem.

Problem 14

In the rectangle below (12 cm long, 5 cm wide), the two segments marked alike on the top side are each 2 cm long. The area of the shaded region is



(A) 60 cm^2 (B) 20 cm^2 (C) 30 cm^2 (D) 40 cm^2 (E) 50 cm^2

Detailed solution

Understanding the problem. The shaded region is a trapezoid: its longer parallel side is the bottom side of the rectangle, and its shorter parallel side lies on the top side. The two segments at the ends of the top side are marked alike, which means they are *equal*, each 2 cm. We solve it in two ways to check ourselves.

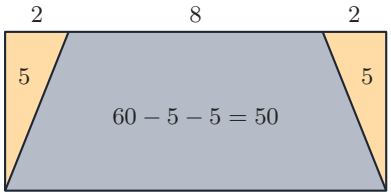
Step 1. Method 1 – rectangle minus two triangles

Recall. Area of a rectangle = length × width. The area of a right triangle is half the product of its two legs.

Area of the rectangle: $12 \cdot 5 = 60$ (cm^2).
The two unshaded parts are right triangles in the corners; each has legs 2 cm (on the top side) and 5 cm (a side of the rectangle):

$$S_{\text{one triangle}} = \frac{2 \cdot 5}{2} = 5 \text{ (cm}^2\text{)}.$$

Shaded area: $60 - 5 - 5 = 50$ (cm^2).



Step 2. Method 2 – the trapezoid formula

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$.

Longer parallel side $a = 12$ cm; shorter parallel side $b = 12 - 2 - 2 = 8$ cm; height $h = 5$ cm (the distance between the top and bottom sides of the rectangle).

$$S = \frac{(12+8)\cdot 5}{2} = \frac{20\cdot 5}{2} = \frac{100}{2} = 50 \text{ (cm}^2\text{)}.$$

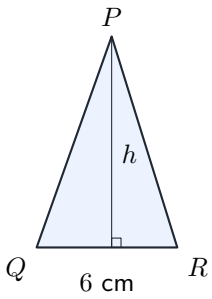
Common mistake. Option (A) 60 cm^2 is the area of the *whole* rectangle. Also remember that the height of the trapezoid is 5 cm (the perpendicular distance), not the length of a slanted side.

Check: Both methods give 50 cm^2.
So the answer is **(E)**.

Answer: (E) 50 cm^2

Problem 15

Triangle PQR has area 27 cm^2 and base $QR = 6$ cm. The height h of triangle PQR (to the base QR) is



(A) 9 cm (B) 18 cm (C) 4.5 cm (D) 2.25 cm (E) 7 cm

Detailed solution

Understanding the problem. We know the area and the base and need the height: this is the triangle area formula "worked backwards".

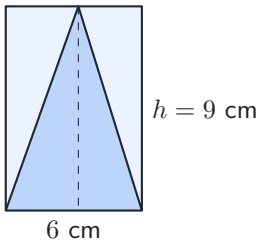
Recall. $S_{\text{triangle}} = \dfrac{a \cdot h}{2}$, where a is the base and h the height to that base. Hence $h = \dfrac{2 \cdot S}{a}$.

Step 1. Substitute into the formula
$27 = \dfrac{6 \cdot h}{2}$.

Step 2. Get rid of the division by 2
Multiply both sides by 2: $6 \cdot h = 54$.

Step 3. Find h
$h = 54 \div 6 = 9$ (cm).

Why is it true? Why double the area? The triangle is half of the rectangle with the same base 6 cm and the same height h. That rectangle has area $2 \cdot 27 = 54$ cm^2 and one side 6 cm, so its other side (which is h) is $54 \div 6 = 9$ cm. □



rectangle: 54 cm^2
triangle: 27 cm^2

Common mistake. Option (C) 4.5 cm is the result of $27 \div 6$ – forgetting that the area of a triangle is only *half* of base times height.

Check: $\dfrac{6 \cdot 9}{2} = \dfrac{54}{2} = 27$ (cm^2). Correct.
So the answer is **(A)**.

Answer: (A) 9 cm

Problem 16

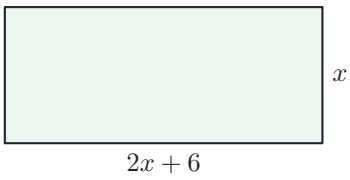
A rectangular garden has a length that is 6 m more than *twice* its width. The perimeter of the garden is 120 m. How many metres wide is the garden?

(A) 8 m (B) 18 m (C) 27 m (D) 38 m (E) 22 m

Detailed solution

Understanding the problem. We know two things: (1) length = twice the width, plus 6 m; (2) perimeter = 120 m. We must find the width. We introduce an unknown and form an equation.

Step 1. Choose the unknown and express the length in terms of it
Let the width of the garden be x (m), with $x > 0$. "Twice the width" is $2x$; the length is 6 m more than that, so the length is $2x + 6$ (m).



Recall. Perimeter of a rectangle = (length + width) · 2.

Step 2. Form the equation
The perimeter is 120 m, so

$$\big((2x + 6) + x\big) \cdot 2 = 120.$$

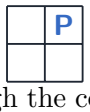
Step 3. Solve the equation
Divide both sides by 2: $(2x + 6) + x = 60$. Collect like terms: $3x + 6 = 60$. Subtract 6 from both sides: $3x = 60 - 6 = 54$. Divide both sides by 3: $x = 54 \div 3 = 18$ (this satisfies $x > 0$).
So the width is 18 m and the length is $2 \cdot 18 + 6 = 42$ (m).

Common mistake. Many students forget that the perimeter contains *two* lengths and *two* widths, write $3x + 6 = 120$ and get $x = 38$ – exactly the "trap" option (D).

Check: Length 42 m, width 18 m: $42 = 2 \cdot 18 + 6$ (correct) and the perimeter is $(42 + 18) \cdot 2 = 60 \cdot 2 = 120$ m (correct).
A width of 18 m is option **(B)**.

Answer: (B) 18 m

Problem 17

The letter P is written in a 2×2 grid of squares as shown: 

. A sequence of rotations about the centre of the grid and reflections in the two lines through the centre of the grid (the horizontal line and the vertical line) is carried out, and the result is: 

.

If *exactly the same sequence of transformations* is applied to the grid 

, which picture do we get?

(A) 

(B) 

(C) 

(D) 

(E) 

Detailed solution

Understanding the problem. We do not know which individual steps the sequence contains, but we know its *final effect* on the letter P. Idea: imagine the grid is a square *card* made of clear plastic with the letter drawn on it. Every rotation or reflection just turns or flips the card and puts it back in the same place. The whole sequence is therefore also just *one way of putting the card back*. We read off that way from the picture of P, and then apply it to A.

Step 1. See where the letter P went

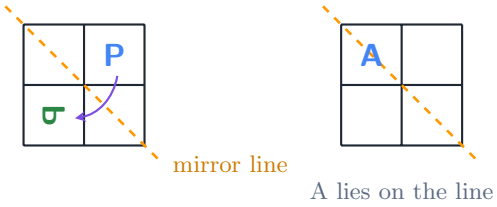
The letter P in the *top-right* cell has moved to the *bottom-left* cell. At the same time, the straight stroke of P (its "stem"), which was vertical, is now horizontal at the top; the round "bowl", which was to the right of the stem, is now underneath it.

Step 2. Find out how the card was put back

There are two ways of putting the card back that move the top-right cell to the bottom-left cell:

- a 180° rotation: then P would be upside down (its stem still *vertical*, looking like a "d");
- a reflection in the diagonal joining the top-left corner to the bottom-right corner (the dashed line in the picture below): then the vertical stem of P becomes *horizontal*.

In the given picture the stem is horizontal, so the sequence acts like the **reflection in the top-left – bottom-right diagonal**.



Recall. A sequence of rotations about the centre and reflections in the horizontal and vertical lines can have the same effect as a reflection in a diagonal. For example: *a 90° clockwise rotation followed by a reflection in the vertical line*. Check with P: the 90° rotation takes P from the top-right cell down to the bottom-right cell, the reflection in the vertical line takes it to the bottom-left cell, and the stem of P is horizontal – exactly as in the given picture.

Step 3. Apply it to the letter A

The top-left cell lies *right on* the mirror line (the diagonal passes through it), so the letter A *stays in the top-left cell*. This diagonal reflection turns the direction "up" into the direction "left". The tip of A, which pointed up, now points left. The letter A is symmetric left-to-right, so we need not worry about it being "mirrored".

Common mistake. Do not look only at the position of the cell! P moved from the top-right to the bottom-left cell, but that does *not* mean A also "moves diagonally down". First find the transformation of *the whole card*, then apply it.

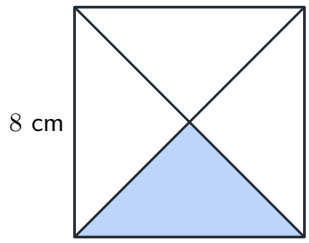
Check: Use the concrete sequence "90° clockwise rotation, then reflection in the vertical line": the rotation moves A from the top-left to the top-right cell with its tip pointing right; the

reflection in the vertical line brings A back to the top-left cell with its tip pointing left. This is picture (B).

Answer: (B) A in the top-left cell, tip pointing left

Problem 18

In a square with side 8 cm, the two diagonals are drawn. What is the area of the shaded region?



(A) 4 cm^2 (B) 8 cm^2 (C) 16 cm^2 (D) 56 cm^2 (E) 64 cm^2

Detailed solution

Understanding the problem. The two diagonals divide the square into 4 triangles. The shaded region is one of these four triangles. We need its area.

Step 1. Compute the area of the square

Recall. The area of a square with side a is $a \times a$.

$A_{\text{square}} = 8 \times 8 = 64$ (cm^2).

Step 2. Show that the four triangles have equal areas

Why is it true? The two diagonals of a square are equal and bisect each other at a point O, so O is the same distance from all four vertices. Rotating the square 90° about O moves each triangle exactly onto the next one. So the four triangles are identical, and each is $\frac{1}{4}$ of the square. □

Step 3. Compute the shaded area

$A = 64 \div 4 = 16$ (cm^2).

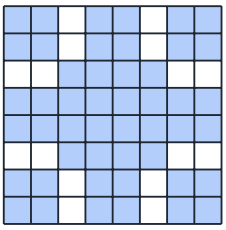
Check: Direct calculation: the shaded triangle has the bottom side as its base, 8 cm, and its height is the distance from the centre O to that side, which is half the side, 4 cm. Area $= \dfrac{8 \times 4}{2} = 16$ cm^2. It matches.

Option **(C)**.

Answer: (C) 16 cm^2

Problem 19

The grid below is made of identical small squares. What percentage of the area of the whole grid is shaded?



(A) 84% (B) 80% (C) 90% (D) 75% (E) 66%

Detailed solution

Understanding the problem. To find a percentage we need to know how many squares there are in total and how many are shaded. Counting the *white* squares is quicker because there are fewer of them.

Step 1. Count all the squares

The grid has 8 rows of 8 squares: $8 \times 8 = 64$ squares.

Step 2. Count the white squares

The white squares form 8 "pairs" of two adjacent squares, placed symmetrically on the four sides:

- top two rows: columns 3 and 6 have a white square in both rows: 4 squares;
- bottom two rows: likewise, 4 squares;
- row 3 (from the top): 2 white squares at the left edge and 2 at the right edge: 4 squares;
- row 6: likewise, 4 squares.

Total number of white squares: $4 + 4 + 4 + 4 = 16$.

Step 3. Count the shaded squares and convert to a percentage

Number of shaded squares: $64 - 16 = 48$.

Recall. The percentage that a is of b equals $\dfrac{a}{b} \times 100\%$.

$$\frac{48}{64} = \frac{3}{4} = \frac{75}{100} = 75\%.$$

ThinkAlgo Tip. You do not even need to count everything: the white squares make up $\dfrac{16}{64} = \dfrac{1}{4} = 25\%$, so the shaded squares make up $100\% - 25\% = 75\%$.

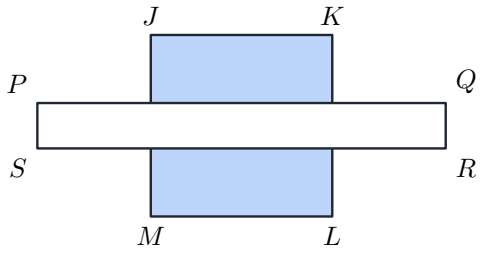
Check: 75% of 64 is $64 \times \dfrac{3}{4} = 48$ squares, and $48 + 16 = 64$. Correct.

Option **(D)**.

Answer: (D) 75%

Problem 20

$JKLM$ is a square and $PQRS$ is a rectangle; they overlap as shown in the diagram. Given that JK is parallel to PQ, $JK = 8$ and $PS = 2$, find the total area of the shaded regions.



(A) 32 (B) 16 (C) 56 (D) 48 (E) 62

Detailed solution

Understanding the problem. The shaded part is the part of the square that is *not* covered by the rectangular strip. So: shaded area = area of the square − area of the overlap of the two figures.

Step 1. Area of the square $JKLM$

Recall. Area of a square = side × side; area of a rectangle = length × width.

$A_{JKLM} = 8 \times 8 = 64$.

Step 2. Identify the overlap

Since $JK \parallel PQ$, the sides of the strip $PQRS$ are parallel to the side JK of the square, i.e. the strip runs "horizontally" across the square. The strip is longer than the square, so it crosses the whole width of the square. The overlap is a rectangle with:

- length equal to the side of the square: 8;
- width equal to the width of the strip: $PS = 2$.

Area of the overlap: $8 \times 2 = 16$.

Step 3. Compute the shaded area

$64 - 16 = 48$.

Common mistake. We do not need to know how high or low the strip is, or how big each shaded piece is: their total is always $64 - 16 = 48$. Many students only compute $8 \times 2 = 16$ (option B) – that is the *covered* part, not the shaded part.

Check: Suppose the strip is 3 below the top side JK; the top piece has area $8 \times 3 = 24$, the bottom piece has height $8 - 3 - 2 = 3$, area 24. Total 48. Correct.
Option **(D)**.

Answer: (D) 48

Problem 21

A triangle can be built with side lengths 4, 5, 8, but *no* triangle can be built with side lengths 4, 5, 10. Using only the lengths 2, 3, 5, 7, 11 (each length may be used more than once), how many different triangles with *exactly two equal sides* can be built?

(A) 8 (B) 5 (C) 20 (D) 10 (E) 14

Detailed solution

Understanding the problem. A triangle with exactly two equal sides is isosceles but not equilateral: two legs of length a and a base of length b with $b \neq a$. We must count the pairs (a, b) taken from $\{2, 3, 5, 7, 11\}$ such that the three segments a, a, b really form a triangle.

> **Recall. Triangle inequality.** Three segments are the sides of a triangle exactly when the sum of any two of them is greater than the third. In practice it is enough to check that *the two shorter segments add up to more than the longest one.* Example: $4 + 5 = 9 > 8$, so it works; $4 + 5 = 9 < 10$, so it does not.

Step 1. Simplify the condition for an isosceles triangle a, a, b

The sums to check are $a + a > b$ and $a + b > a$. The second is always true (since $b > 0$). So we only need

$$2a > b \quad \text{(twice the leg is greater than the base).}$$

Step 2. Make a table by the leg a

Leg a	$2a$	Base $b \neq a$ with $b < 2a$	Number of triangles
2	4	3	1
3	6	2, 5	2
5	10	2, 3, 7	3
7	14	2, 3, 5, 11	4
11	22	2, 3, 5, 7	4

For instance, with $a = 3$: the base $b = 7$ does not work because $3 + 3 = 6 < 7$; $b = 11$ does not work either. With $a = 5$: $b = 11$ does not work because $5 + 5 = 10 < 11$.

Step 3. Add up

$1 + 2 + 3 + 4 + 4 = 14$ triangles.

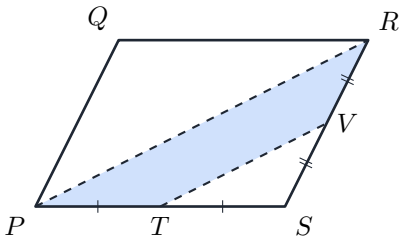
> **Common mistake.** Triangles such as $2, 2, 5$ or $3, 3, 7$ "look" possible, but in fact the two legs are too short to meet. Also, the triangle $5, 5, 5$ has *three* equal sides, so it is not counted.

Check: Count the other way: there are $5 \times 4 = 20$ pairs (a, b) with $b \neq a$. The rejected pairs (because $b \geq 2a$) are $(2, 5), (2, 7), (2, 11), (3, 7), (3, 11), (5, 11)$: 6 pairs. $20 - 6 = 14$. It matches. Option **(E)**.

Answer: (E) 14

Problem 22

Parallelogram $PQRS$ has area 40. Points T and V are the midpoints of sides PS and RS respectively. Find the area of quadrilateral $PRVT$.



(A) 10 (B) 12 (C) 15 (D) 16 (E) 18

Detailed solution

Understanding the problem. Quadrilateral $PRVT$ is what remains when the small triangle TVS at corner S is removed from triangle PRS. So we compute A_{PRS} and A_{TVS}.

Step 1. Area of triangle PRS

Recall. A diagonal divides a parallelogram into two identical triangles, so each triangle has half the area of the parallelogram.

$A_{PRS} = 40 \div 2 = 20$.

Step 2. Area of triangle TRS (join T to R)

Recall. Two triangles with *the same height* have areas in the same ratio as their bases. (Because the area of a triangle $= \frac{1}{2} \times$ base $\times$ height.)

Triangles TRS and PRS share the vertex R, and their bases TS and PS lie on the same line, so they have the same height from R. Since T is the midpoint of PS, $TS = \frac{1}{2}PS$. Therefore $A_{TRS} = \frac{1}{2} \cdot 20 = 10$.

Step 3. Area of triangle TVS

Triangles TVS and TRS share the vertex T, and their bases VS and RS lie on the line RS. Since V is the midpoint of RS, $VS = \frac{1}{2}RS$, so $A_{TVS} = \frac{1}{2} \cdot 10 = 5$.

Step 4. Area of quadrilateral $PRVT$

$A_{PRVT} = A_{PRS} - A_{TVS} = 20 - 5 = 15$.

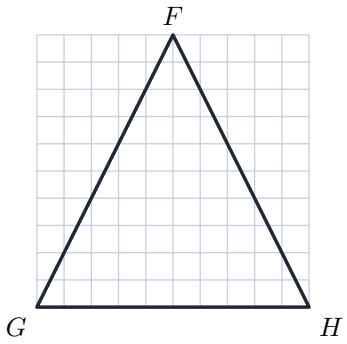
ThinkAlgo Tip. A "half – half" triangle at a corner has area $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ of the big triangle. This is a very useful fact!

Check: Place the figure on a grid: $P(0,0)$, $S(6,0)$, $R(8,4)$, $Q(2,4)$ (area $6 \times 4 = 24$, not 40, but the ratios are the same). Then $T(3,0)$, $V(7,2)$; $A_{PRS} = 12$, $A_{TVS} = \frac{1}{2} \cdot 3 \cdot 2 = 3$, so $A_{PRVT} = 9 = \frac{3}{8} \cdot 24$. For the given figure: $\frac{3}{8} \cdot 40 = 15$. It matches. So the answer is **(C)**.

Answer: (C) 15

Problem 23

On a 10×10 grid of squares (each square has side 1), triangle FGH is drawn as shown. Point P can be any of the 41 points where gridlines cross *inside* (not on the sides of) triangle FGH. For each position of P we get exactly three triangles: FPG, GPH, HPF. Of all these 123 triangles, how many have an area *exactly half* the area of triangle FGH?



(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

Detailed solution

Understanding the problem. Use coordinates: $G(0,0)$, $H(10,0)$, $F(5,10)$. Triangle FGH has base $GH = 10$ and height 10, so

$$A_{FGH} = \frac{10 \times 10}{2} = 50, \qquad \text{and half of it is } 25.$$

We count, for each type of triangle GPH, FPG, HPF in turn, the positions of P that give area 25.

Recall. Two triangles with *the same base* have areas in the same ratio as their heights. Points that are the same distance from a line (on the same side) lie on a line parallel to it.

Step 1. Triangle GPH

Triangle GPH has the same base GH as triangle FGH. For half the area, the height from P must be half of the height 10, i.e. 5. So P lies on the 5th horizontal gridline (counting from GH). On this line, side FG passes through the point with x-coordinate 2.5 and side FH through the point with x-coordinate 7.5 (the midpoints of those sides). The grid points *inside* have $x = 3, 4, 5, 6, 7$:

$$(3,5),\ (4,5),\ (5,5),\ (6,5),\ (7,5) \quad \Rightarrow \quad 5 \text{ triangles.}$$

Step 2. Triangle FPG

This time the common base is FG. Let $M(5,0)$ be the midpoint of GH and $N(7.5,5)$ the midpoint of FH.

Why is it true? Triangles FMG and FHG have the same height from F, and the base $GM = \frac{1}{2}GH$, so $A_{FMG} = \frac{1}{2}A_{FHG} = 25$. So point M is exactly half as far from FG as H is. The same holds for N (the midpoint of FH). Hence $A_{FPG} = 25$ exactly when P is the same distance from FG as M, i.e. when P lies on the line MN (parallel to FG). □

Going from G to F we move 5 right and 10 up, i.e. "1 right, 2 up". Line MN is parallel to FG, so from $M(5,0)$ we also go "1 right, 2 up": we meet $(6,2)$, $(7,4)$, and then $N(7.5,5)$ (not a grid point). The two points $(6,2)$, $(7,4)$ are inside the triangle $\Rightarrow$ 2 triangles.

Step 3. Triangle HPF

The figure is symmetric about the vertical line $x = 5$, so the result is the same as in step 2: $P = (4,2)$ or $(3,4) \Rightarrow$ 2 triangles.

Step 4. Add up

$5 + 2 + 2 = 9$ triangles.

Check: Check $P(6,2)$ with the "enclosing rectangle" method: the rectangle from $x = 0$ to 6 and $y = 0$ to 10 has area 60. Remove the three right triangles outside triangle FPG: $\frac{5\cdot 10}{2} = 25$, $\frac{1\cdot 8}{2} = 4$, $\frac{6\cdot 2}{2} = 6$. What remains is $60 - 35 = 25$, exactly half of 50.

Common mistake. Do not forget the two types FPG and HPF! If you only consider triangles with the horizontal base GH, you will wrongly choose option (A) 5.

Option **(E)**.

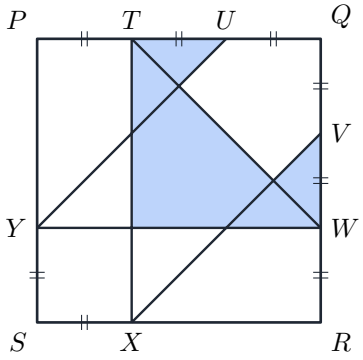
Answer: (E) 9

Problem 24

Points T, U, V, W, X, Y lie on the sides of square $PQRS$ as shown, with

$$PT = TU = UQ = QV = VW = WR = XS = SY.$$

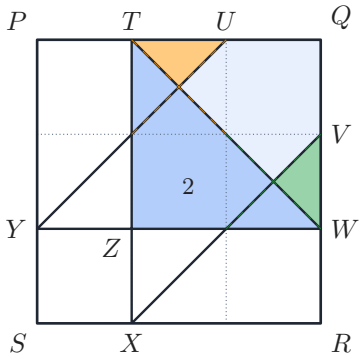
What fraction of the area of square $PQRS$ is shaded?



(A) $\frac{5}{18}$ (B) $\frac{1}{3}$ (C) $\frac{2}{9}$ (D) $\frac{1}{4}$ (E) $\frac{1}{6}$

Detailed solution

Understanding the problem. Each side of the square is divided into 3 equal parts. Let each part have length 1 unit; then the square has side 3 and area $3 \times 3 = 9$. A nice idea: draw extra lines dividing the square into a 3×3 grid of unit squares (dotted lines); then every line in the figure becomes very "tidy".



Step 1. Recognise the three pieces of the shaded region
Let Z be the point where TX meets YW. The shaded region consists of three non-overlapping pieces: the big triangle TZW (dark blue), the small triangle next to side TU (orange) and the small triangle next to side VW (green).

Step 2. Triangle TZW
$TQWZ$ is a 2×2 square (made of 4 unit squares) and TW is its diagonal.

> **Recall.** A diagonal divides a square (rectangle) into two triangles of equal area.

So $A_{TZW} = \frac{1}{2} \cdot 4 = 2$.

Step 3. The two small triangles
Look at the unit square whose top side is TU. Line YU goes "1 right, 1 up" through grid corners, so it contains one diagonal of this unit square; line TW goes "1 right, 1 down" from T, so it contains the other diagonal. The two diagonals of a square divide it into 4 equal triangles, and the orange triangle is one of them: area $\frac{1}{4} \cdot 1 = \frac{1}{4}$. In exactly the same way, the green triangle lies in the unit square whose right side is VW, cut by the two diagonals XV and TW: its area is also $\frac{1}{4}$.

Step 4. Compute the fraction
Shaded area: $2 + \frac{1}{4} + \frac{1}{4} = \frac{5}{2}$. The required fraction:

$$\frac{5}{2} \div 9 = \frac{5}{2} \cdot \frac{1}{9} = \frac{5}{18}.$$

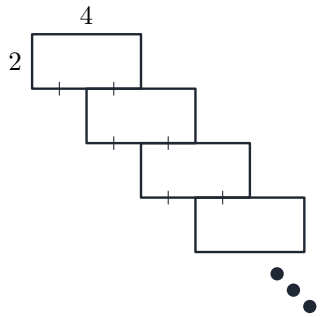
> **Common mistake.** Do not "estimate by eye"! The shaded part looks close to $\frac{1}{3}$, but the exact calculation gives $\frac{5}{18}$ (a little less than $\frac{6}{18} = \frac{1}{3}$).

Check: Put $S(0,0)$ and side 3; computing the area of the shaded polygon by adding the triangles gives the same result $\frac{5}{2}$, and $\frac{5}{2} = \frac{5}{18} \cdot 9$. Correct.
Option **(A)**.

Answer: (A) $\frac{5}{18}$

Problem 25

Rectangles measuring 4×2 are arranged by the following rule: from the second rectangle on, the top-left vertex of each rectangle is placed exactly at the midpoint of the bottom side of the rectangle just above it (as shown). When 10 rectangles in total are arranged in this way, what is the perimeter of the whole figure?



(A) 48 (B) 64 (C) 90 (D) 84 (E) 100

Detailed solution

Understanding the problem. Each new rectangle is shifted 2 right (half its length) and 2 down (its width), so the figure is a "staircase". We find its perimeter in two ways.

Step 1. Method 1: count each kind of side

Vertical sides. The left and right sides (length 2) of every rectangle are on the boundary. Total: $10 \times (2 + 2) = 40$.

Horizontal sides on top. The first rectangle shows its whole top side: 4. For each later rectangle, the left half of its top side is against the bottom side of the previous one, so only the right half shows: 2. Total: $4 + 9 \times 2 = 22$.

Horizontal sides at the bottom. Each of the first 9 rectangles shows only the left half of its bottom side: 2; the last one shows its whole bottom side: 4. Total: $9 \times 2 + 4 = 22$. Perimeter: $40 + 22 + 22 = 84$.

Step 2. Method 2: "straighten" the staircase

Why is it true? Walking along the top-right boundary from the top-left corner of the first rectangle to the bottom-right corner of the last one, we only go right or down: the horizontal pieces add up to the width of the whole figure and the vertical pieces to its height. The same holds for the bottom-left boundary. So the staircase *has the same perimeter as the smallest rectangle enclosing it.* □

Width: $4 + 9 \times 2 = 22$; height: $10 \times 2 = 20$. Perimeter: $(22 + 20) \times 2 = 84$.

Common mistake. Not $10 \times 12 = 120$: sides where two rectangles touch lie *inside* the figure.

Check: With 2 rectangles: $8 + (4 + 2) + (2 + 4) = 20$, the perimeter of the enclosing 6×4 rectangle. For n rectangles: $2\big((4 + 2(n - 1)) + 2n\big) = 8n + 4$; $n = 10$ gives 84, option **(D)**.

Answer: (D) 84

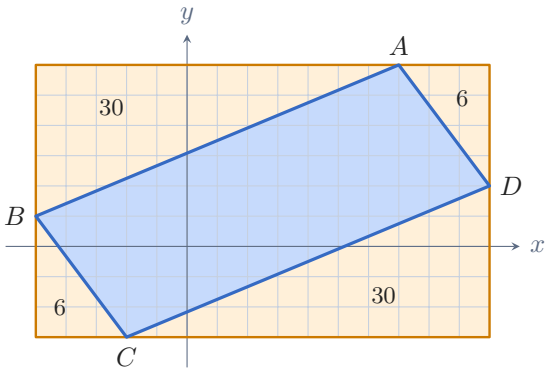
Problem 26

A quadrilateral has vertices $A(7,6)$, $B(-5,1)$, $C(-2,-3)$ and $D(10,2)$ in the coordinate plane. The area of the quadrilateral (in square units) is

(A) 60 (B) 63 (C) 67 (D) 70 (E) 72

Detailed solution

Understanding the problem. The quadrilateral has "slanted" sides, and there is no ready-made formula. A standard technique: *enclose the quadrilateral in a rectangle with sides parallel to the axes*, then subtract the right triangles in the four corners.



Step 1. The enclosing rectangle

The smallest x-coordinate is -5, the largest is 10; the smallest y-coordinate is -3, the largest is 6.

> **Recall.** The distance between two points on a number line is the larger number minus the smaller: from -5 to 10 it is $10-(-5)=15$.

The rectangle has length $10-(-5)=15$, width $6-(-3)=9$, and area $15\times 9=135$.

Step 2. The four corner right triangles

> **Recall.** Area of a right triangle $=\frac{1}{2}\times$ the product of the two legs.

- Top-left corner (between B and A): legs $7-(-5)=12$ and $6-1=5$: area $\frac{12\cdot 5}{2}=30$.
- Bottom-left corner (between B and C): legs $-2-(-5)=3$ and $1-(-3)=4$: area $\frac{3\cdot 4}{2}=6$.
- Bottom-right corner (between C and D): legs $10-(-2)=12$ and $2-(-3)=5$: area 30.
- Top-right corner (between D and A): legs $10-7=3$ and $6-2=4$: area 6.

Total: $30+6+30+6=72$.

Step 3. Area of the quadrilateral

$135-72=63$.

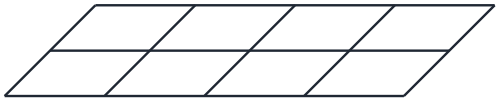
> **Common mistake.** With negative coordinates it is easy to get a length wrong, e.g. writing $7-5=2$ instead of $7-(-5)=12$. Count squares on the picture to check.

Check: In fact $ABCD$ is a parallelogram ($A \to B$ and $D \to C$ are both "12 left, 5 down"). Triangle ABD has half the area of the quadrilateral; enclose it in a 15×5 rectangle (from $y = 1$ to 6): $75 - 30 - \frac{15 \cdot 1}{2} - 6 = 31.5$. Doubling: 63. It matches.
Option **(B)**.

Answer: (B) 63

Problem 27

The figure below is made of 8 identical small parallelograms joined together. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?



(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Detailed solution

Understanding the problem. Every parallelogram in the figure has two sides on two *horizontal* lines and two sides on two *slanted* lines of the figure. We count systematically so as not to miss or repeat any.

Step 1. Count by height and width (in small cells)
Parallelograms 1 *row high* (there are 2 rows to choose from). A row has 4 cells side by side:

- 1 cell wide: 4; 2 cells wide: 3; 3 cells wide: 2; 4 cells wide: 1;
- each row gives $4 + 3 + 2 + 1 = 10$, and the two rows give 20.

Parallelograms 2 *rows high*: likewise $4 + 3 + 2 + 1 = 10$. Total: $20 + 10 = 30$.

Step 2. A faster count: choose the lines

Why is it true? A parallelogram is completely determined by choosing 2 of the 3 horizontal lines (top and bottom sides) and 2 of the 5 slanted lines (left and right sides). Conversely, each such choice gives exactly one parallelogram. □

Recall. The number of ways to choose 2 lines out of k lines is $\dfrac{k \times (k-1)}{2}$ (each pair is counted twice if we pick a "first line" and a "second line").

Choose 2 of the 3 horizontal lines: $\frac{3 \cdot 2}{2} = 3$ ways. Choose 2 of the 5 slanted lines: $\frac{5 \cdot 4}{2} = 10$ ways. Number of parallelograms: $3 \times 10 = 30$.

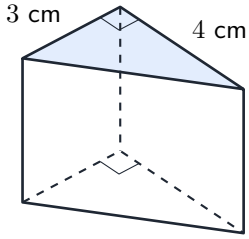
Common mistake. Students often forget the parallelograms 2 rows high, or the biggest one (the whole figure).

Check: The two counting methods give the same result, 30. So the answer is **(B)**.

Answer: (B) 30

Problem 28

A right triangular prism has volume 120 cm^3. Its base is a right triangle whose two legs are 3 cm and 4 cm (as shown). What is the height of the prism, in centimetres?



(A) 12 (B) 20 (C) 10 (D) 16 (E) 8

Detailed solution

Understanding the problem. We know the volume and the base (a right triangle). We must find the height of the prism, i.e. the length of a lateral edge (joining the top base to the bottom base).

Recall. Volume of a right prism = area of the base × height: $V = A_{\text{base}} \cdot h$. Therefore $h = V \div A_{\text{base}}$.

Step 1. Area of the base

The base is a right triangle; taking one leg as the base, the other leg is the height:

$$A_{\text{base}} = \frac{3 \times 4}{2} = 6 \ (\text{cm}^2).$$

Step 2. Height of the prism

$h = 120 \div 6 = 20$ (cm).

Common mistake. Do not confuse "the height of the prism" (the distance between the two bases) with a height of the base triangle. And do not forget to divide by 2 for the area of the triangle: using $3 \times 4 = 12$ gives 10 – the "trap" option (C).

Check: $6 \times 20 = 120$ cm^3. Correct. So the answer is **(B)**.

Answer: (B) 20 cm

Problem 29

Small cubes with edge 1 are used to build a rectangular box of size $8 \times 8 \times n$ (n a positive integer). Let A be the surface area of the box and let B be the total surface area of all the small cubes (computed for each cube separately and then added up). Find the sum of all values of n for which $\dfrac{B}{A}$ is an integer.

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Detailed solution

Understanding the problem. We write A and B in terms of n, form the ratio $\dfrac{B}{A}$, and find the values of n that make it an integer.

Step 1. Compute A

Recall. A rectangular box of size $a \times b \times c$ has 6 faces forming three pairs of equal faces, so its surface area is $2(ab + bc + ca)$.

$A = 2(8 \cdot 8 + 8 \cdot n + 8 \cdot n) = 2(64 + 16n) = 128 + 32n$.

Step 2. Compute B

Number of small cubes: $8 \cdot 8 \cdot n = 64n$. Each cube has 6 faces of area 1, so $B = 64n \cdot 6 = 384n$.

Step 3. Simplify the ratio

$$\frac{B}{A} = \frac{384n}{128 + 32n} = \frac{32 \cdot 12n}{32 \cdot (4 + n)} = \frac{12n}{n+4}.$$

We "split off the whole part": since $12n = 12(n+4) - 48$,

$$\frac{12n}{n+4} = \frac{12(n+4) - 48}{n+4} = 12 - \frac{48}{n+4}.$$

Step 4. When is the ratio an integer?

$12 - \dfrac{48}{n+4}$ is an integer exactly when $\dfrac{48}{n+4}$ is an integer, i.e. when $n+4$ is a *divisor* of 48.

Recall. The positive divisors of 48: $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$.

Since $n \geq 1$, we have $n + 4 \geq 5$, so we take only the divisors $6, 8, 12, 16, 24, 48$, which give

$$n = 2,\ 4,\ 8,\ 12,\ 20,\ 44.$$

Step 5. Add them up

$2 + 4 + 8 + 12 + 20 + 44 = 90$.

n	2	4	8	12	20	44
$B/A = \dfrac{12n}{n+4}$	$\frac{24}{6} = 4$	$\frac{48}{8} = 6$	$\frac{96}{12} = 8$	$\frac{144}{16} = 9$	$\frac{240}{24} = 10$	$\frac{528}{48} = 11$

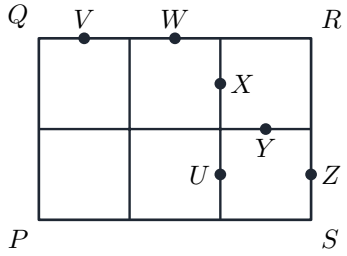
Common mistake. Do not forget that $n+4$ can be 48 (i.e. $n = 44$), a rather large value. And remember to reject the divisors $1, 2, 3, 4$, because then $n \leq 0$.

Check: The table shows that all 6 values give an integer ratio. Try another value, $n = 1$: $\frac{12}{5}$ is not an integer. Correct.
Option **(B)**.

Answer: (B) 90

Problem 30

Rectangle $PQRS$ is made up of six identical squares. Points U, V, W, X, Y, Z are midpoints of sides of the squares, as shown. Which of the following triangles has the greatest area?



(A) PVU (B) PXZ (C) PVX (D) PYS (E) PQW

Detailed solution

Understanding the problem. We need to compute (or compare) the areas of 5 triangles. To avoid fractions, let each small square have side 2; then all the midpoints have whole-number coordinates. Put $P(0,0)$, $S(6,0)$, $R(6,4)$, $Q(0,4)$; then $V(1,4)$, $W(3,4)$, $X(4,3)$, $Y(5,2)$, $U(4,1)$, $Z(6,1)$.

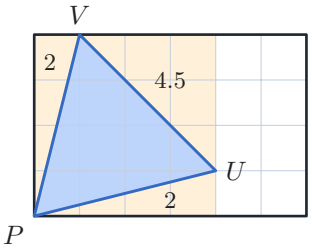
Recall. Area of a triangle $= \frac{1}{2} \times$ base $\times$ height. For a "slanted" triangle, enclose it in a rectangle with sides along the grid and subtract the corner right triangles (as in Problem 26).

Step 1. Two triangles with a side on a side of the rectangle

- PYS: base $PS = 6$, height = distance from Y to PS, which is 2: $A = \frac{6 \cdot 2}{2} = 6$.
- PQW: base $PQ = 4$, height = distance from W to PQ, which is 3: $A = \frac{4 \cdot 3}{2} = 6$.

Step 2. Three "slanted" triangles

- PVU: enclosing rectangle $4 \times 4 = 16$; subtract $\frac{1 \cdot 4}{2} = 2$, $\frac{3 \cdot 3}{2} = 4.5$, $\frac{4 \cdot 1}{2} = 2$. What remains is $16 - 8.5 = 7.5$.
- PXZ: enclosing rectangle $6 \times 3 = 18$; subtract $\frac{4 \cdot 3}{2} = 6$, $\frac{2 \cdot 2}{2} = 2$, $\frac{6 \cdot 1}{2} = 3$. What remains is $18 - 11 = 7$.
- PVX: enclosing rectangle $4 \times 4 = 16$; subtract $\frac{1 \cdot 4}{2} = 2$, $\frac{3 \cdot 1}{2} = 1.5$, $\frac{4 \cdot 3}{2} = 6$. What remains is $16 - 9.5 = 6.5$.



Step 3. Compare
$7.5 > 7 > 6.5 > 6 = 6$. The largest triangle is PVU.

Common mistake. Do not guess by eye: triangle PXZ looks "big" because it stretches all the way to side RS, but its area is only 7, less than that of PVU.

Check: If the side of each square is 1, all areas are divided by 4 (1.875; 1.75; 1.625; 1.5; 1.5), and the order does not change.
Option **(A)**.

Answer: (A) triangle PVU

Extension and challenge

Problem 31

In the coordinate plane Oxy, consider the point $A(-3,\ 2)$.

a) Let B be the reflection of A in the x-axis, C the reflection of A in the y-axis, and D the reflection of A in the origin O. Find the coordinates of B, C, D and state the quadrant of each point.

b) The four points A, C, D, B are the vertices of a rectangle. Find the perimeter and the area of that rectangle.

Detailed solution

Understanding the problem. "Reflection in an axis" is like looking in a mirror placed along that axis: the image lies on the other side of the axis, at the same distance from it as the original point. We check which coordinate changes sign and which stays the same.

> **Recall.** Reflection in the x-axis: $(x,\ y) \to (x,\ -y)$. Reflection in the y-axis: $(x,\ y) \to (-x,\ y)$. Reflection in the origin O (also a 180° rotation about O): $(x,\ y) \to (-x,\ -y)$.

Step 1. Point B – reflection in Ox

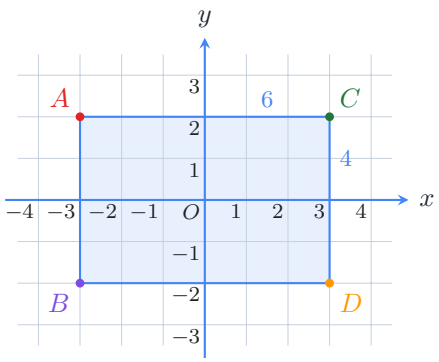
Mirroring in the x-axis: the point keeps its left–right position (the x-coordinate is unchanged) and only moves from above to below (the y-coordinate changes sign). $B(-3,\ -2)$, of the form $(-,-)$: quadrant III.

Step 2. Point C – reflection in Oy

Mirroring in the y-axis: the y-coordinate is unchanged, the x-coordinate changes sign. $C(3,\ 2)$, of the form $(+,+)$: quadrant I.

Step 3. Point D – reflection in the origin O

Both coordinates change sign: $D(3,\ -2)$, of the form $(+,-)$: quadrant IV.



Step 4. Find the dimensions of the rectangle

$A(-3,\ 2)$ and $C(3,\ 2)$ have the same y-coordinate: $AC = 3 - (-3) = 6$. $A(-3,\ 2)$ and $B(-3,\ -2)$ have the same x-coordinate: $AB = 2 - (-2) = 4$.

Step 5. Find the perimeter and the area

Recall. Rectangle with length a, width b: $P = 2 \cdot (a + b)$, $S = a \cdot b$.

$P = 2 \cdot (6 + 4) = 20$ (units); $S = 6 \cdot 4 = 24$ (square units).

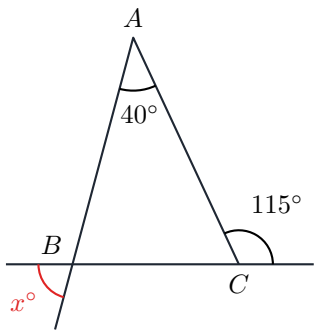
ThinkAlgo Tip. This rectangle has both coordinate axes as lines of symmetry. Each quadrant contains exactly one quarter of it, a 3×2 rectangle of area 6; and $4 \cdot 6 = 24$.

Check: Each quadrant contains one point: A (II), C (I), B (III), D (IV), as in the figure.

Answer: a) $B(-3, -2)$**: III;** $C(3, 2)$**: I;** $D(3, -2)$**: IV. b) Perimeter 20, area 24.**

Problem 32

In the figure, triangle ABC has $\widehat{BAC} = 40°$; side BC is extended in both directions and side AB is extended beyond B. The exterior angle at vertex C is 115°. The value of x is



(A) 65 (B) 75 (C) 105 (D) 115 (E) 140

Detailed solution

Understanding the problem. Angle $x°$ at vertex B is formed by two extensions (of BC and of BA), so it is *vertically opposite* to the interior angle $\widehat{ABC}$. So we only need $\widehat{ABC}$. We solve it in two ways.

Step 1. Method 1 – via the interior angle at C

Recall. Two supplementary angles add up to 180°; the three interior angles of a triangle add up to 180°.

The interior angle at C is supplementary to the exterior angle 115°: $\widehat{ACB} = 180° - 115° = 65°$. Interior angle at B: $\widehat{ABC} = 180° - 40° - 65° = 75°$.

Step 2. Method 2 – the exterior angle property

Recall. An exterior angle of a triangle equals the sum of the two interior angles not adjacent to it.

The exterior angle at C equals $\widehat{A} + \widehat{B}$: $115° = 40° + \widehat{ABC}$, so $\widehat{ABC} = 115° - 40° = 75°$.

Step 3. Deduce x

Recall. Vertically opposite angles are equal.

$x° = \widehat{ABC} = 75°$, so $x = 75$.

Common mistake.

- (A) 65 is the interior angle at C, not at B.
- (C) 105 is the angle supplementary to $\widehat{ABC}$ – confusing "supplementary" with "vertically opposite".

Check: $40° + 75° + 65° = 180°$ and $40° + 75° = 115°$ (exterior angle at C). Both conditions hold.
So the answer is **(B)**.

Answer: (B) 75

Problem 33

a) In an isosceles triangle, the apex angle is 4 times each base angle. Find the angles of the triangle.

b) An isosceles triangle has an angle of 36°. Find all possibilities for the other two angles. For each possibility, find the difference between the largest and the smallest angle of the triangle.

c) Is there an isosceles triangle with a base angle of 92°? Why?

Detailed solution

Understanding the problem. All three parts rest on two facts: an isosceles triangle has two equal base angles, and the three angles add up to 180°.

Recall. Isosceles triangle: apex angle d, each base angle g, then $d + 2g = 180°$.

Step 1. Part a – introduce an unknown and form an equation
Let each base angle measure x degrees, with $0 < x < 90$. The apex angle is 4 times each base angle, so it measures $4x$. The three angles of a triangle add up to 180°, which gives the equation

$$x + x + 4x = 180.$$

Collect the left-hand side: $6x = 180$. Divide both sides by 6: $x = 180 \div 6 = 30$. So each base angle is 30° and the apex angle is $4 \cdot 30° = 120°$.

Step 2. **Part b – two cases**
Case 1: $36°$ *is the apex angle.* Each base angle: $(180° - 36°) \div 2 = 144° \div 2 = 72°$. Triangle $36°$, $72°$, $72°$. Difference between largest and smallest: $72° - 36° = 36°$.
Case 2: $36°$ *is a base angle.* The other base angle is $36°$, the apex angle $180° - 72° = 108°$. Triangle $36°$, $36°$, $108°$. Difference: $108° - 36° = 72°$.

Step 3. **Part c – reasoning**

Why is it true? If one base angle were $92°$, the other base angle would also be $92°$. Then the two base angles alone would add up to $184°$, more than $180°$ – impossible, since *all three* angles add up to only $180°$. So no such isosceles triangle exists. (In general: a base angle of an isosceles triangle is always less than $90°$.) □

Check: $30 + 30 + 120 = 180$; $36 + 72 + 72 = 180$; $36 + 36 + 108 = 180$.

Answer: a) $30°$, $30°$, $120°$. b) $72°$ and $72°$ (difference $36°$), or $36°$ and $108°$ (difference $72°$). c) No.

Problem 34

In the coordinate plane, consider the point $A(4, 1)$.

a) Rotating A about the origin O through $90°$ counterclockwise gives the point B; rotating B a further $90°$ counterclockwise gives C; rotating C a further $90°$ counterclockwise gives E. Find the coordinates of B, C, E.

b) Rotating A about O through $270°$ *clockwise* gives which of these three points?

c) The quadrilateral $ABCE$ is a square. Find its area.

Detailed solution

Understanding the problem. We use the rule for a $90°$ rotation in the coordinate plane, then find the area of the "tilted" square by placing it inside a big square whose sides run along the grid lines.

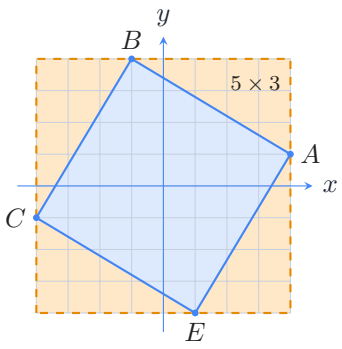
Recall. Rotation of $90°$ counterclockwise about O: $(x, y) \to (-y, x)$. Rotating α clockwise is the same as rotating $360° - \alpha$ counterclockwise.

Why is it true? Why $(x, y) \to (-y, x)$? The point $A(4, 1)$ is reached by going "right 4, up 1". Turn this whole path $90°$ counterclockwise: "right" becomes "up", and "up" becomes "left". So the image is reached by "up 4, left 1", i.e. the point $(-1, 4)$. □

Step 1. **Part a – apply the rule three times**
$A(4, 1) \to B(-1, 4) \to C(-4, -1) \to E(1, -4)$.
(Check: C is the image of A under a $180°$ rotation, which is indeed $(-4, -1)$, both coordinates changing sign.)

Step 2. **Part b**
Rotating $270°$ clockwise is the same as rotating $360° - 270° = 90°$ counterclockwise, so we get exactly the point $B(-1, 4)$.

Step 3. Part c – put the square in a frame



Square $ABCE$ fits inside the big square with vertices $(\pm 4, \pm 4)$, of side $4 - (-4) = 8$ and area $8 \cdot 8 = 64$.
The extra parts are 4 equal right triangles in the four corners. For example, the triangle in the top-right corner has vertices $A(4,\ 1)$, $(4,\ 4)$, $B(-1,\ 4)$: its horizontal leg is $4 - (-1) = 5$, its vertical leg is $4 - 1 = 3$, and its area is $\dfrac{5 \cdot 3}{2} = 7.5$.
Area of square $ABCE$: $64 - 4 \cdot 7.5 = 64 - 30 = 34$ (square units).

ThinkAlgo Tip. "Frame it and subtract" is a very powerful way to find the area of a polygon whose vertices are grid points: draw the smallest surrounding rectangle, then subtract the right triangles (trapezoids, rectangles) outside the polygon.

Check: The four equal right triangles (each 5×3) together with square $ABCE$ fill the 8×8 frame: $4 \cdot 7.5 + 34 = 64$. Also, the frame side $8 = 5 + 3$ is exactly the sum of the two legs.

Answer: a) $B(-1,\ 4)$, $C(-4,\ -1)$, $E(1,\ -4)$. **b) Point** B. **c)** 34 **square units.**

Problem 35

Fill in the blanks:

a) 2.4 km = ... m; b) 75 mm = ... cm; c) 3 m^2 = ... cm^2;

d) 2500 cm^3 = ... litres.

e) A rectangular board is 1.2 m long and 50 cm wide. Find the area of the board in cm^2 and in m^2.

Detailed solution

Understanding the problem. Each part is a unit conversion. Key point: lengths go up by a factor of 10 per step, areas by 100, volumes by 1000.

Recall. 1 km = 1000 m; 1 cm = 10 mm; 1 m^2 = 10 000 cm^2; 1 litre = 1 dm^3 = 1000 cm^3. Larger to smaller: multiply; smaller to larger: divide.

Step 1. Part a
km is larger than m, so multiply: $2.4 \cdot 1000 = 2400$. So 2.4 km = 2400 m.

Step 2. Part b
mm is smaller than cm, so divide: $75 \div 10 = 7.5$. So 75 mm = 7.5 cm.

Step 3. Part c
$3\ \text{m}^2 = 3 \cdot 10\,000\ \text{cm}^2 = 30\,000\ \text{cm}^2$.

Common mistake. Converting an area by multiplying only by 100 (as for lengths in m to cm) gives 300 – wrong. A 1 m × 1 m square is 100 cm × 100 cm and contains 10 000 squares of 1 cm^2.

Step 4. Part d
$2500\ \text{cm}^3 = 2500 \div 1000\ \text{dm}^3 = 2.5\ \text{dm}^3 = 2.5$ litres.

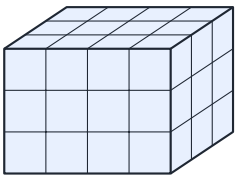
Step 5. Part e – use the same unit first
The two dimensions are in different units, so convert before multiplying: 1.2 m = 120 cm. Area: $120 \cdot 50 = 6000$ (cm^2). In m^2: $6000 \div 10\,000 = 0.6$ (m^2).

Check: Directly in metres: 50 cm = 0.5 m, so $1.2 \cdot 0.5 = 0.6$ (m^2). It matches.

Answer: a) 2400; **b)** 7.5; **c)** 30 000; **d)** 2.5; **e)** 6000 $\text{cm}^2 = 0.6\ \text{m}^2$.

Problem 36

Anna built a 4 cm × 3 cm × 3 cm box from 36 small cubes of edge 1 cm (as in Problem 10). Anna paints all the outside faces of the box and then takes the small cubes apart. How many small cubes have exactly 3 painted faces? Exactly 2? Exactly 1? No painted face?



Detailed solution

Understanding the problem. The number of painted faces of a small cube depends on its *position*: at a corner, along an edge, in the middle of a face, or completely inside. We count each type.

Step 1. Corner cubes – 3 painted faces
Each corner cube of the box shows 3 faces (one front/back, one top/bottom, one left/right). The box has 8 corners, so **8** cubes have 3 painted faces.

Step 2. Cubes along an edge (not at the ends) – 2 painted faces
The box has 12 edges: 4 edges of length 4 cm, 4 edges of length 3 cm (height) and 4 edges of length 3 cm (depth). Along an edge of n cubes, remove the two end cubes (already counted in step 1): $n - 2$ cubes remain.

- 4 edges of length 4: each $4 - 2 = 2$ cubes, total $4 \cdot 2 = 8$;
- 8 edges of length 3: each $3 - 2 = 1$ cube, total $8 \cdot 1 = 8$.

So $8 + 8 = \mathbf{16}$ cubes have 2 painted faces.

Step 3. Cubes in the middle of a face – 1 painted face

On an $m \times n$ face, removing the outer border leaves $(m-2)\cdot(n-2)$ cubes.

- 2 faces 4×3 (front, back): each $2 \cdot 1 = 2$ cubes, total 4;
- 2 faces 4×3 (top, bottom): each $2 \cdot 1 = 2$ cubes, total 4;
- 2 faces 3×3 (left, right): each $1 \cdot 1 = 1$ cube, total 2.

So $4 + 4 + 2 = \mathbf{10}$ cubes have 1 painted face.

Step 4. Cubes completely inside – unpainted

Removing one layer of "skin" on each side, the inner core measures $(4-2)\times(3-2)\times(3-2) = 2\times1\times1$, i.e. **2** cubes. Summary of the results:

Painted faces	3	2	1	0	Total
Small cubes	8	16	10	2	36

Note for parents. If you have a set of toy cubes (or a Rubik's-type cube), build the $4 \times 3 \times 3$ box with your child, stick coloured paper on the outside faces, then take it apart and count. Your child will see for themselves why a corner cube has 3 coloured faces.

Check: The four types must add up to the number of cubes: $8 + 16 + 10 + 2 = 36 = 4 \cdot 3 \cdot 3$. Correct.

Answer: 3 faces: 8 cubes; 2 faces: 16 cubes; 1 face: 10 cubes; unpainted: 2 cubes.

Problem 37

A rectangle has the same perimeter as a square of area 81 cm^2. The length of the rectangle is twice its width. The area of the rectangle is

(A) 81 cm^2 (B) 72 cm^2 (C) 54 cm^2 (D) 36 cm^2 (E) 64 cm^2

Detailed solution

Understanding the problem. The chain of reasoning: area of the square → side of the square → perimeter (shared by both figures) → half-perimeter of the rectangle → length and width → area.

Recall. Square with side a: $S = a \cdot a$, $P = 4a$. Rectangle: $P = 2 \cdot (\text{length} + \text{width})$, $S = \text{length} \cdot \text{width}$.

Step 1. Side of the square

We need a number a with $a \cdot a = 81$: $a = 9$ cm (since $9 \cdot 9 = 81$).

Step 2. Perimeter

Perimeter of the square $= 4 \cdot 9 = 36$ cm, which is also the perimeter of the rectangle.

Step 3. Sum of length and width

Half-perimeter of the rectangle: $36 \div 2 = 18$ cm = length + width.

Step 4. Introduce an unknown and form an equation
Let the width of the rectangle be x cm, with $x > 0$. The length is twice the width, so it is $2x$ cm. By step 3, length + width = 18, which gives the equation

$$x + 2x = 18.$$

Collect terms: $3x = 18$, so $x = 18 \div 3 = 6$. So the width is 6 cm and the length is $2 \cdot 6 = 12$ cm.

Step 5. Area
$S = 12 \cdot 6 = 72$ (cm^2).

ThinkAlgo Tip. Both figures have perimeter 36 cm, but the square's area 81 cm^2 is larger than the rectangle's (72 cm^2). Among all rectangles with the same perimeter, the square has the largest area.

Check: Perimeter of the rectangle: $2 \cdot (12 + 6) = 36$ cm = perimeter of the square; $12 = 2 \cdot 6$. Correct.
So the answer is **(B)**.

Answer: (B) 72 cm^2

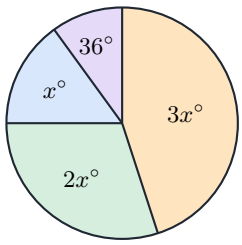
Problem 38 *

A circular spinner is divided into four regions with central angles 36°, x°, $2x$° and $3x$° (see figure).

a) Find x.

b) What percentage of the spinner's area does each region take up?

c) About what percentage of spins will *not* stop in the largest region?



Detailed solution

Understanding the problem. The four central angles together make a full turn. This gives an equation for x; then we convert each angle to a percentage.

Recall. The angles around a point add up to 360°. A region with central angle n° is $\frac{n}{360}$ of the circle.

Step 1. Part a – set up the equation
$36 + x + 2x + 3x = 360$, i.e. $36 + 6x = 360$. Subtract 36 from both sides: $6x = 324$. Divide by 6: $x = 54$.

Step 2. Part b – the angles and percentages
The four angles: 36°, 54°, 108°, 162°.

$$\frac{36}{360} = 10\%, \qquad \frac{54}{360} = 15\%, \qquad \frac{108}{360} = 30\%, \qquad \frac{162}{360} = 45\%.$$

(Shortcut: $\frac{36}{360} = \frac{1}{10}$; the other angles are 1.5, 3 and 4.5 times 36°, so the percentages are 15%, 30%, 45%.)

Step 3. Part c – use the complement
The largest region is the 162° one, with 45%. The pointer does *not* stop there in about $100\% - 45\% = 55\%$ of spins.

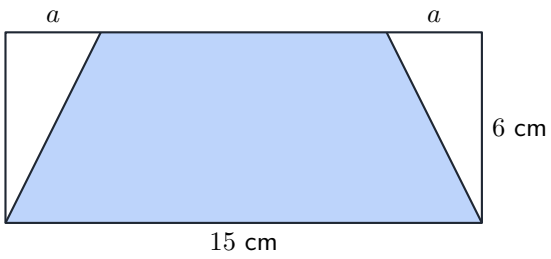
Common mistake. The "55%" is only *approximately* the share of spins after a great many spins. In just a few spins the actual count may differ; the area ratio tells how *often* the pointer stops somewhere in the long run.

Check: $36 + 54 + 108 + 162 = 360$ and $10\% + 15\% + 30\% + 45\% = 100\%$.

Answer: a) $x = 54$. **b)** 10%, 15%, 30%, 45%. **c) About** 55%.

Problem 39

A rectangle is 15 cm long and 6 cm wide. On its top side, two segments of the same length a cm are taken at the two ends, and the two right triangles at the top corners are cut off, leaving the shaded trapezoid shown. The area of the trapezoid is 72 cm^2. Find a and the length of the shorter parallel side of the trapezoid.



Detailed solution

Understanding the problem. Problem 14 gave the dimensions and asked for the area; this problem goes the other way: the area is given and a dimension is asked for. We express the trapezoid's area in terms of a as "rectangle minus two triangles", then find a.

Recall. $S_{\text{rectangle}} = \text{length} \times \text{width}$; $S_{\text{right triangle}} = \dfrac{\text{product of the two legs}}{2}$.

Step 1. Area of the rectangle
$15 \cdot 6 = 90$ (cm^2).

Step 2. **Total area of the two cut-off triangles**
Area of the two triangles $= 90 - 72 = 18$ (cm^2), so each triangle has area $18 \div 2 = 9$ (cm^2).

Step 3. **Find** a
Each right triangle has legs a and 6:

$$\frac{a \cdot 6}{2} = 9 \quad \Rightarrow \quad 3 \cdot a = 9 \quad \Rightarrow \quad a = 3.$$

Step 4. **Shorter parallel side of the trapezoid**
Shorter side $= 15 - a - a = 15 - 3 - 3 = 9$ cm.

Check: By the trapezoid formula: $\dfrac{(15+9)\cdot 6}{2} = \dfrac{24 \cdot 6}{2} = 72$ (cm^2). Correct.

ThinkAlgo Tip. It can also be solved with the trapezoid formula: the shorter side b satisfies $\dfrac{(15+b)\cdot 6}{2} = 72$, i.e. $(15+b)\cdot 3 = 72$, $15+b = 24$, $b = 9$; then $a = (15-9) \div 2 = 3$.

Answer: $a = 3$ cm; shorter parallel side 9 cm.

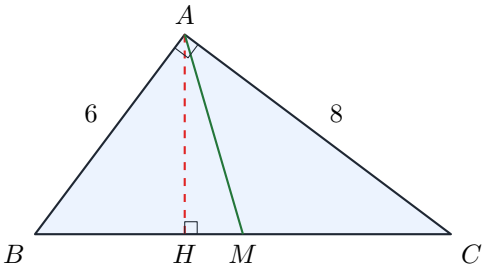
Problem 40 *

Triangle ABC has a right angle at A, with $AB = 6$ cm, $AC = 8$ cm and $BC = 10$ cm.

a) Find the area of triangle ABC.

b) Find the length of the height AH drawn from A to side BC.

c) Let M be the midpoint of BC. Find the area of triangle ABM.



Detailed solution

Understanding the problem. Any side of a triangle may be chosen as the base, as long as we use the height belonging to that base. The area is always the same. We compute the area the easy way (base AB, height AC), then use it to find the height to the base BC.

Recall. $S = \dfrac{\text{base} \cdot \text{height}}{2}$, so height $= \dfrac{2 \cdot S}{\text{base}}$. In a right triangle, each leg is the height to the other leg.

Step 1. Part a – choose base AB
Since angle A is a right angle, AC is perpendicular to AB: with AB as the base, AC is the height.

$$S_{ABC} = \frac{6 \cdot 8}{2} = 24 \text{ (cm}^2\text{)}.$$

Step 2. Part b – choose base BC
With $BC = 10$ cm as the base, the matching height is AH. The area is still 24 cm^2:

$$\frac{10 \cdot AH}{2} = 24 \quad \Rightarrow \quad 10 \cdot AH = 48 \quad \Rightarrow \quad AH = 4.8 \text{ cm}.$$

Step 3. Part c – the midpoint halves the area

Recall. Two triangles with equal bases and a common height have equal areas.

Triangles ABM and ACM have bases $BM = MC = 5$ cm (since M is the midpoint) and the common height AH (both drawn from A to the line BC). So they have equal areas, each half of triangle ABC:

$$S_{ABM} = \frac{5 \cdot 4.8}{2} = 12 \text{ (cm}^2\text{)} = \frac{24}{2}.$$

Common mistake. Do not use "base BC times side AB" – side AB is not perpendicular to BC, so it is not the height to the base BC.

Check: $AH = 4.8$ is smaller than both $AB = 6$ and $AC = 8$ – sensible, because the perpendicular is the shortest path from A to the line BC. And $\frac{10 \cdot 4.8}{2} = 24$.

Answer: a) 24 cm^2. b) $AH = 4.8$ cm. c) 12 cm^2.

Problem 41 ⋆

(Extension of Problem 16.) We only consider rectangles whose two side lengths are *positive integers* (in cm); a square also counts as a rectangle.

a) Find all rectangles whose area (in cm^2) is numerically *equal* to their perimeter (in cm).

b) Find all rectangles whose area is numerically *twice* their perimeter.

Detailed solution

Understanding the problem. Call the sides a and b (positive integers). Without loss of generality, assume $a \le b$ (a is the shorter side). The area is ab and the perimeter is $2(a+b)$. This is a "whole-number solutions" problem: one equation with two unknowns. Trick: use $a \le b$ to *trap* a in a small range, then try each value.

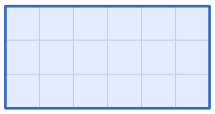
a) Condition: $ab = 2a + 2b$.

Step 1. An upper bound for a
Since $a \le b$, we have $2a + 2b \le 2b + 2b = 4b$. Hence $ab \le 4b$. Dividing both sides by $b > 0$: $a \le 4$.

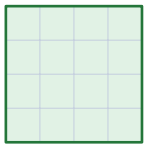
Step 2. A lower bound for a
Since $2a > 0$, we have $ab = 2a + 2b > 2b$. Dividing both sides by $b > 0$: $a > 2$. So $a = 3$ or $a = 4$.

Step 3. Try each value

- $a = 3$: $3b = 6 + 2b \Rightarrow b = 6$. The 3×6 rectangle.
- $a = 4$: $4b = 8 + 2b \Rightarrow 2b = 8 \Rightarrow b = 4$. The 4×4 square.



$A = 18,\ P = 18$



$A = 16,\ P = 16$

b) Condition: $ab = 2 \cdot 2(a + b) = 4a + 4b$.

Step 4. Trap a as before

$ab = 4a + 4b \leq 8b \Rightarrow a \leq 8$; and $ab = 4a + 4b > 4b \Rightarrow a > 4$. So $a \in \{5, 6, 7, 8\}$.

Step 5. Try each value

From $ab = 4a + 4b$ we get $ab - 4b = 4a$, i.e. $b(a - 4) = 4a$, so $b = \dfrac{4a}{a - 4}$.

a	5	6	7	8
$b = \dfrac{4a}{a-4}$	$\dfrac{20}{1} = 20$	$\dfrac{24}{2} = 12$	$\dfrac{28}{3}$ (not an integer)	$\dfrac{32}{4} = 8$

There are three rectangles: 5×20, 6×12, 8×8.

Why is it true? Why have we missed none? Every rectangle that works has its shorter side a in the range we trapped, and for each a the side b is determined uniquely by the equation. We tried all those values of a. □

ThinkAlgo Tip. Another way for part a): $ab - 2a - 2b + 4 = 4$, i.e. $(a - 2)(b - 2) = 4 = 1 \cdot 4 = 2 \cdot 2$. Hence $a - 2 = 1$, $b - 2 = 4$ or $a - 2 = b - 2 = 2$. Again we get 3×6 and 4×4.

Check: 3×6: $A = 18$, $P = 2 \cdot 9 = 18$. 4×4: $16 = 16$. 5×20: $A = 100$, $P = 50$; 6×12: $A = 72$, $P = 36$; 8×8: $A = 64$, $P = 32$ – all twice.

Answer: a) 3×6 and 4×4. b) 5×20, 6×12, 8×8.

Problem 42 *

(Extension of Problem 17.) The letter P is written in the top-right cell of a 2×2 grid: 

. We may use three transformations, as many times as we like and in any order:

- Q: rotate the grid 90° clockwise about its centre;
- D: reflect the grid in the *vertical* line through the centre;
- N: reflect the grid in the *horizontal* line through the centre.

a) Draw the result of doing D and then N. Can that result be obtained using only Q?

b) Show that doing D and then Q has the same effect as a reflection in one of the *diagonals* of the grid. Which diagonal?

c) How many different pictures can be obtained altogether (including the starting picture)? Explain why there cannot be more.

Detailed solution

Understanding the problem. As in Problem 17, think of the grid as a clear square *card* with the letter P drawn on it. Each transformation lifts the card and puts it back exactly in its place. To keep track we record two things: which cell P is in, and how it has been turned or flipped.

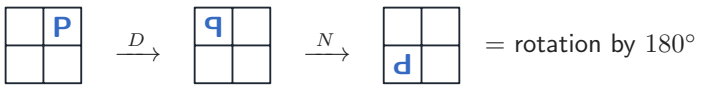
a)

Step 1. Do D

Reflecting in the vertical line: the top-right cell goes to the top-left cell, and P is "mirrored" (its bowl now faces left).

Step 2. Then do N

Reflecting in the horizontal line: the top-left cell goes down to the bottom-left cell, and the letter is flipped top-to-bottom. Result: P is in the bottom-left cell, *upside down* (looking like a "d"), as in the third picture below.



Step 3. Compare with the Q moves

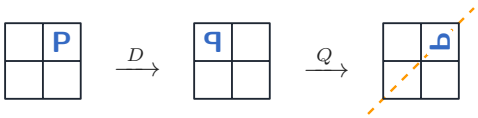
Doing Q twice (a 180° rotation): the top-right cell goes to the opposite cell through the centre, the bottom-left cell, and the letter is upside down. This is the same result. So **D then N has the same effect as Q then Q.**

> **Recall.** Two reflections in two perpendicular lines (both through the centre) have the same effect as a 180° rotation about the centre.

b)

Step 4. Do D and then Q

D: P goes to the top-left cell, mirrored. Q (90° clockwise): the top-left cell goes to the top-right cell. Result: P is *back in the top-right cell*, but its stem is horizontal and its bowl is on top:



Step 5. Recognise the diagonal reflection

The top-right cell has not moved, which means it lies on the mirror line. The diagonal through the top-right cell (and the bottom-left cell) is the diagonal joining the **bottom-left corner to the top-right corner** (dashed). Reflecting in this line turns "up" into "right" and vice versa, so the (vertical) stem of P becomes horizontal and the bowl of P (on the right) moves to the top – exactly as in the picture. A further check: a letter in the top-left cell moves to the bottom-right cell in both cases.

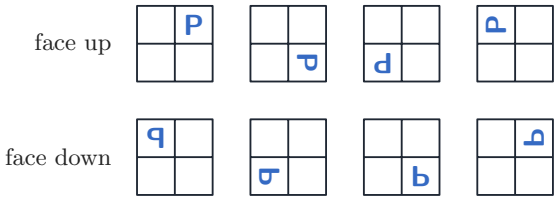
c)

Step 6. There cannot be more than 8 pictures

Why is it true? Mark the top-right corner of the card with a star. Each time the card is put back, the star can only be in 1 of the 4 corners of the frame. In addition, the card can only be *face up* or *face down* (each reflection D or N switches face up and face down; the rotation Q does not). Once we know where the star is and whether the card is face up or face down, the position of the card – and hence the whole picture – is completely determined. So there are at most $4 \times 2 = 8$ pictures. □

Step 7. All 8 pictures occur

Four "face up" pictures: rotations by 0°, 90°, 180°, 270° (using Q from 0 to 3 times). Four "face down" pictures: each of the above followed by D. The letter P has no line of symmetry, so these 8 pictures are all different:

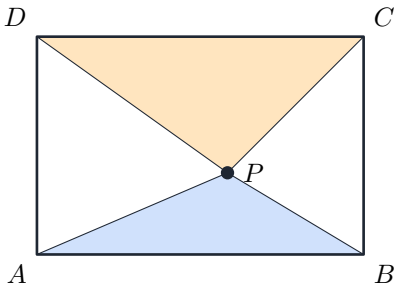


Common mistake. Do not count "4 cells × 8 orientations = 32". The cell and the orientation of the letter are *tied together*, because both move with the same card.

Answer: a) P in the bottom-left cell, upside down; the same as Q then Q (a 180° rotation). b) The diagonal joining the bottom-left corner to the top-right corner. c) 8 pictures.

Problem 43

(Extension of Problem 18.) Rectangle $ABCD$ has $AB = 12$ cm and $BC = 8$ cm. Point P lies inside the rectangle. (We write S_{XYZ} for the area of triangle XYZ.)



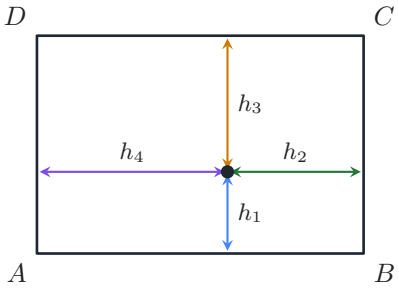
a) Prove that wherever P is, we always have $S_{PAB} + S_{PCD} = 48$ cm^2.

b) Given $S_{PAB} = 18$ cm^2 and $S_{PBC} = 20$ cm^2, find S_{PCD}, S_{PDA} and the distances from P to the four sides of the rectangle.

c) Is there a point P inside the rectangle with $S_{PAB} = 20$, $S_{PBC} = 30$ and $S_{PCD} = 30$ (cm^2)?

Detailed solution

Understanding the problem. Each of the four triangles has a side of the rectangle as its base and P as its apex. The height of each triangle is just *the distance from P to that side.* Call the distances from P to AB, BC, CD, DA respectively h_1, h_2, h_3, h_4.



a)

Step 1. The sum of two heights

Draw through P the line perpendicular to AB; since $AB \parallel CD$ it is also perpendicular to CD. The part between AB and CD has length $BC = 8$, and P divides it into two parts h_1 and h_3. So $h_1 + h_3 = 8$. Similarly $h_2 + h_4 = AB = 12$.

Step 2. Add the two areas

Recall. Area of a triangle $= \frac{1}{2} \times$ base $\times$ height.

$$S_{PAB} + S_{PCD} = \frac{12 \cdot h_1}{2} + \frac{12 \cdot h_3}{2} = 6(h_1 + h_3) = 6 \cdot 8 = 48 \text{ (cm}^2\text{)}.$$

The result does not depend on the position of P. (It is exactly half the area of the rectangle, $12 \cdot 8 = 96$.) Similarly, $S_{PBC} + S_{PDA} = \frac{8}{2}(h_2 + h_4) = 4 \cdot 12 = 48$ cm^2.

b)

Step 3. Use part a

$S_{PCD} = 48 - 18 = 30$ cm^2, $\quad S_{PDA} = 48 - 20 = 28$ cm^2.

Step 4. Compute the four distances

From $A = \frac{1}{2} \cdot$ base $\cdot h$ we get $h = 2A \div$ base.

$$h_1 = \frac{2 \cdot 18}{12} = 3, \quad h_2 = \frac{2 \cdot 20}{8} = 5, \quad h_3 = \frac{2 \cdot 30}{12} = 5, \quad h_4 = \frac{2 \cdot 28}{8} = 7 \text{ (cm)}.$$

c)

Step 5. Check the compulsory condition

By part a), we always have $S_{PAB} + S_{PCD} = 48$. But $20 + 30 = 50 \neq 48$. So **there is no** such point P.

ThinkAlgo Tip. Problem 18 is a special case: P is the centre of a square, so the four triangles are equal. Here P is arbitrary, yet "the sum of two opposite triangles" is still always half of the figure.

Check: $h_1 + h_3 = 3 + 5 = 8 = BC$ and $h_2 + h_4 = 5 + 7 = 12 = AB$. With $A(0,0)$ we get $P(7,3)$ – exactly the point in the picture.

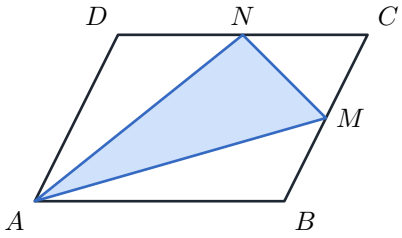
Answer: b) $S_{PCD} = 30$ **cm**^{2}**,** $S_{PDA} = 28$ **cm**^{2}**; the distances to** AB, BC, CD, DA **are** $3, 5, 5, 7$ **cm. c) No.**

Problem 44 *

(Extension of Problem 22.) Parallelogram $ABCD$ has area 48 cm^2.

a) Let M and N be the midpoints of BC and CD respectively. Find the area of triangle AMN.

b) Keep N as the midpoint of CD, but take M on side BC with $BM = 3\,MC$. Find the area of triangle AMN.



Detailed solution

Understanding the problem. Triangle AMN sits "diagonally" inside the parallelogram. We take the whole parallelogram and remove the three corner triangles ABM, MCN and NDA (as usual, S_{XYZ} denotes the area of triangle XYZ):

$$S_{AMN} = S_{ABCD} - S_{ABM} - S_{MCN} - S_{NDA}.$$

Recall. (1) Each diagonal divides a parallelogram into two triangles of equal area (= 24 cm^2). (2) Two triangles with a common apex whose bases lie on the same line have areas in the same ratio as their bases.

a)

Step 1. Triangle ABM
Compared with triangle ABC (= 24): common apex A, base $BM = \frac{1}{2}BC$. So $S_{ABM} = \frac{1}{2} \cdot 24 = 12$.

Step 2. Triangle NDA
Compared with triangle ACD (= 24): common apex A, base $DN = \frac{1}{2}DC$. So $S_{NDA} = 12$.

Step 3. Triangle MCN **(in two steps, as in Problem 22)**
Triangle BCD has area 24. Triangle MCD shares the apex D with it, with base $CM = \frac{1}{2}CB$: $S_{MCD} = 12$. Triangle MCN shares the apex M with triangle MCD, with base $CN = \frac{1}{2}CD$: $S_{MCN} = \frac{1}{2} \cdot 12 = 6$.

Step 4. Result
$S_{AMN} = 48 - 12 - 6 - 12 = 18$ cm^2.

b)

Step 5. Redo it with $MC = \frac{1}{4}BC$
Since $BM = 3\,MC$, we have $BC = 4\,MC$, i.e. $MC = \frac{1}{4}BC$ and $BM = \frac{3}{4}BC$.

- $S_{ABM} = \frac{3}{4} \cdot 24 = 18$;
- $S_{NDA} = 12$ (unchanged);
- $S_{MCD} = \frac{1}{4} \cdot 24 = 6$, then $S_{MCN} = \frac{1}{2} \cdot 6 = 3$.

$S_{AMN} = 48 - 18 - 3 - 12 = 15$ cm^2.

ThinkAlgo Tip. The "corner" rule: if on two sides of a triangle, measured from the same vertex, we take the parts $\frac{p}{q}$ and $\frac{r}{s}$ of those sides, then the small triangle at that corner has area $\frac{p}{q} \cdot \frac{r}{s}$ of the big triangle. In part b): $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$ of 24 is 3.

Check: Put $A(0,0)$, $B(8,0)$, $D(2,6)$, $C(10,6)$ (area $8 \cdot 6 = 48$). Part a): $M(9,3)$, $N(6,6)$; enclose triangle AMN in a $9 \times 6 = 54$ rectangle and subtract $\frac{9 \cdot 3}{2} = 13.5$, $\frac{3 \cdot 3}{2} = 4.5$, $\frac{6 \cdot 6}{2} = 18$: 18 remains. Correct.

Answer: a) 18 cm^2. b) 15 cm^2.

Problem 45 *

(Extension of Problem 21.) Consider triangles whose three side lengths are positive integers and whose perimeter is 15 (two triangles with the same three side lengths count as one).

a) Let c be the longest side. Prove that $5 \le c \le 7$.

b) How many such triangles are there? List them.

c) How many of them are isosceles? How many have *exactly* two equal sides?

Detailed solution

Understanding the problem. Call the sides $a \le b \le c$ (in increasing order, so that nothing is counted twice). We have $a + b + c = 15$, and by the triangle inequality we only need to check $a + b > c$.

Recall. Three segments $a \le b \le c$ are the sides of a triangle exactly when $a + b > c$ (the two smaller sides add up to more than the largest one).

a)

Step 1. Upper bound
$a + b = 15 - c$. The condition $a + b > c$ becomes $15 - c > c$, i.e. $2c < 15$, $c < 7.5$. Since c is an integer, $c \le 7$.

Step 2. Lower bound
c is the largest side, so $a \le c$ and $b \le c$, hence $15 = a + b + c \le 3c$, i.e. $c \ge 5$.

b)

Step 3. List them by c
For each c we have $a + b = 15 - c$ and $a \le b \le c$.

c	$a+b$	Pairs (a,b) with $a \le b \le c$	Number of triangles
7	8	$(1,7),\ (2,6),\ (3,5),\ (4,4)$	4
6	9	$(3,6),\ (4,5)$ (not $(2,7)$ since $7>6$)	2
5	10	$(5,5)$	1

Altogether $4+2+1=7$ triangles:

$$(1,7,7),\ (2,6,7),\ (3,5,7),\ (4,4,7),\ (3,6,6),\ (4,5,6),\ (5,5,5).$$

Common mistake. For $c=7$, the pair $(1,7)$ gives the triangle $1,7,7$ – very "thin" but still valid because $1+7=8>7$. On the other hand, no triangle has a side 8, because the other two sides would add up to $7<8$.

c)

Step 4. Pick out the isosceles triangles

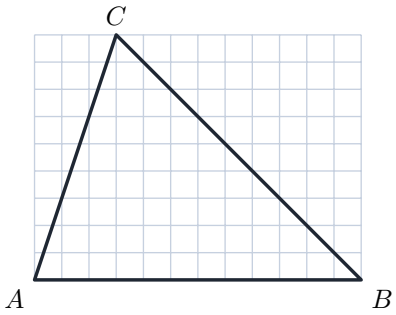
An isosceles triangle has at least two equal sides: $(1,7,7)$, $(4,4,7)$, $(3,6,6)$, $(5,5,5)$ – that is 4 triangles. Among them $(5,5,5)$ is equilateral (three equal sides), so 3 triangles have exactly two equal sides.

Check: Each triple adds up to 15 and satisfies $a+b>c$: $8>7$, $8>7$, $8>7$, $8>7$, $9>6$, $9>6$, $10>5$. Correct.

Answer: a) $5 \le c \le 7$. b) 7 triangles. c) 4 isosceles triangles, 3 of which have exactly two equal sides.

Problem 46 ★★

(Extension of Problem 23.) On a square grid (each square has side 1), triangle ABC has vertices $A(0,0)$, $B(12,0)$, $C(3,9)$. Exactly 43 points where gridlines cross lie *inside* the triangle (not on its sides). For each such point P we get three triangles PAB, PBC, PCA.



a) Find the area of triangle ABC.

b) Find the point P for which the three triangles PAB, PBC, PCA have equal areas.

c) Of all the $3 \cdot 43 = 129$ triangles, how many have an area of exactly $\frac{1}{3}$ of the area of triangle ABC? For how many positions of P is at least one of the three triangles like that?

Detailed solution

Understanding the problem. This is the "one-third version" of Problem 23. The key idea is the same: for a triangle with a fixed base, the area depends only on the height, and points with the same height lie on a line parallel to the base.

a)

Step 1. Area of triangle ABC

The base $AB = 12$ lies on the x-axis and the height is the y-coordinate of C, which is 9. Writing S_{XYZ} for the area of triangle XYZ: $S_{ABC} = \frac{12 \cdot 9}{2} = 54$. One third is 18.

c) (done first; part b) follows immediately)

Step 2. Triangle PAB with area 18

The base is $AB = 12$, so the height must be $2 \cdot 18 \div 12 = 3$, i.e. P lies on the horizontal line $y = 3$. On this line: side AC (going "1 right, 3 up") passes through $(1, 3)$; side BC (going "1 left, 1 up" from B) passes through $(9, 3)$. The points inside: $x = 2, 3, \ldots, 8$, i.e. 7 points.

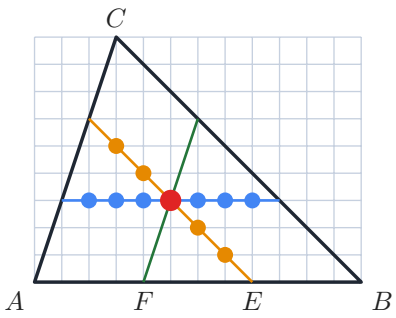
Step 3. Triangle PBC with area 18

Why is it true? Take E on AB with $EB = \frac{1}{3}AB = 4$, i.e. $E(8, 0)$. Triangles EBC and ABC share the apex C, and the base $EB = \frac{1}{3}AB$, so $S_{EBC} = \frac{1}{3} \cdot 54 = 18$. Hence $S_{PBC} = 18$ exactly when P is as far from BC as E is, i.e. when P lies on the line through E parallel to BC (on the inside of the triangle). □

BC goes in the direction "1 left, 1 up". Starting from $E(8, 0)$ in that direction: $(7, 1)$, $(6, 2)$, $(5, 3)$, $(4, 4)$, $(3, 5)$, then $(2, 6)$ – this point is on AC (on AC the y-coordinate is 3 times the x-coordinate), so it does not count. That gives 5 points.

Step 4. Triangle PCA with area 18

Similarly, take F on AB with $AF = 4$, i.e. $F(4, 0)$; P must lie on the line through F parallel to AC (direction "1 right, 3 up"): $(5, 3)$, then $(6, 6)$ – which is on BC (since $6 + 6 = 12$), so it does not count. That gives 1 point.



Step 5. Count

Number of triangles: $7 + 5 + 1 = 13$. The point $(5, 3)$ lies on all three lines (the red point), so it was counted three times. Number of different positions of P: $13 - 2 = 11$.

b)

Step 6. Three equal triangles

The three triangles have total area 54, so they are equal when each has area 18, i.e. P must lie on *all three* lines we found. There is only one such point: $P(5, 3)$.

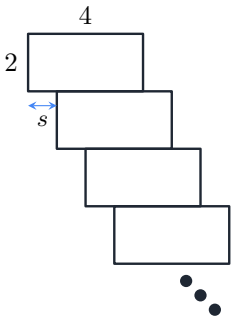
ThinkAlgo Tip. The point $(5, 3)$ is the *centroid* of the triangle: $x = \frac{0+12+3}{3} = 5$, $y = \frac{0+0+9}{3} = 3$. You will study the centroid in detail in the chapter on triangles.

Check: For $P(5, 3)$: $S_{PAB} = \frac{12 \cdot 3}{2} = 18$. S_{PCA}: enclosing rectangle $5 \times 9 = 45$, minus $\frac{3 \cdot 9}{2} = 13.5$ (top-left corner), $\frac{2 \cdot 6}{2} = 6$ (top-right corner), $\frac{5 \cdot 3}{2} = 7.5$ (bottom-right corner): 18 remains. So $S_{PBC} = 54 - 18 - 18 = 18$. Correct.

Answer: a) 54. **b)** $P(5, 3)$. **c) 13 triangles, at 11 positions of P.**

Problem 47 *

(Extension of Problem 25.) Again we arrange 4×2 rectangles in a "staircase" as in Problem 25, but this time the top-left vertex of each rectangle (from the second one on) is placed on the bottom side of the rectangle just above it, at distance s from that rectangle's bottom-left vertex, where s is 1, 2 or 3. (Problem 25 is the case $s = 2$.)



a) For $s = 1$ and 10 rectangles, find the perimeter of the whole figure.

b) Find a formula for the perimeter in terms of the number of rectangles n and the distance s.

c) Can we choose $s \in \{1, 2, 3\}$ and a number of rectangles n so that the perimeter is exactly 100? Find all the ways.

Detailed solution

Understanding the problem. In Problem 25 we saw that the perimeter of a "staircase" equals the perimeter of the smallest rectangle enclosing it. That argument works for every s (the top-right boundary only goes right and down; so does the bottom-left boundary). So we only need the width and the height of the whole figure.

Recall. A "staircase" shape (whose boundary on each side only goes right/down) has the same perimeter as its enclosing rectangle: $2 \times (\text{width} + \text{height})$.

b) (done first, since part a) follows from it)

Step 1. Width and height

Each new rectangle is shifted s to the right. The first has width 4 and each later one adds s: width $= 4 + (n - 1)s$. Each new rectangle is 2 lower: height $= 2n$.

Step 2. The formula

$$P = 2\big(4 + (n-1)s + 2n\big).$$

a)

Step 3. Substitute $s = 1$, $n = 10$

Width $4 + 9 \cdot 1 = 13$, height 20; $P = 2(13 + 20) = 66$.

c)

Step 4. Consider each value of s

- $s = 1$: $P = 2(4 + n - 1 + 2n) = 2(3n + 3) = 6n + 6$. $6n + 6 = 100 \Rightarrow 6n = 94$: not divisible, rejected.
- $s = 2$: $P = 2(4 + 2n - 2 + 2n) = 2(4n + 2) = 8n + 4$. $8n + 4 = 100 \Rightarrow n = 12$. Accepted.
- $s = 3$: $P = 2(4 + 3n - 3 + 2n) = 2(5n + 1) = 10n + 2$. $10n + 2 = 100 \Rightarrow 10n = 98$: rejected.

There is only one way: $s = 2$ and $n = 12$.

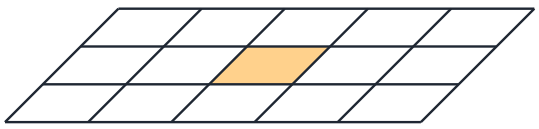
Common mistake. Remember that the height $2n$ does not depend on s; only the width changes. Many students wrongly add in the hidden pieces inside the figure.

Check: Part a) by direct counting: vertical sides $10 \cdot 4 = 40$; horizontal sides on top $4 + 9 \cdot 1 = 13$; at the bottom $9 \cdot 1 + 4 = 13$. Total 66. Correct. Part c): $s = 2$, $n = 12$: $8 \cdot 12 + 4 = 100$. Correct.

Answer: a) 66. **b)** $P = 2\big(4 + (n-1)s + 2n\big)$. **c) Only** $s = 2$, $n = 12$.

Problem 48

(Extension of Problem 27.) The figure below consists of 15 identical small parallelograms arranged in 3 rows of 5. The small parallelogram in the very middle is shaded.



a) How many parallelograms are there in the figure altogether?

b) How many parallelograms contain the shaded small parallelogram?

c) How many parallelograms are made of *exactly* 4 small parallelograms?

Detailed solution

Understanding the problem. The figure has 4 horizontal lines and 6 slanted lines. As in Problem 27, each parallelogram corresponds to a choice of 2 horizontal lines (top and bottom sides) and 2 slanted lines (left and right sides).

Recall. The number of ways to choose 2 lines out of k lines is $\dfrac{k(k-1)}{2}$.

a)

Step 1. Count

Choose 2 of the 4 horizontal lines: $\frac{4\cdot 3}{2} = 6$ ways; choose 2 of the 6 slanted lines: $\frac{6\cdot 5}{2} = 15$ ways. Number of parallelograms: $6 \times 15 = 90$.

b)

Step 2. Condition for containing the shaded cell

Number the horizontal lines from the bottom up as $0, 1, 2, 3$; the shaded cell lies between lines 1 and 2. A parallelogram contains this cell when its bottom side is line 0 or 1 (2 ways) and its top side is line 2 or 3 (2 ways). Number the slanted lines from left to right as $0, 1, \ldots, 5$; the shaded cell lies between lines 2 and 3. The left side is line 0, 1 or 2 (3 ways); the right side is line 3, 4 or 5 (3 ways).

Step 3. Multiply

$2 \times 2 \times 3 \times 3 = 36$ parallelograms.

Why is it true? Why multiply? The four choices (bottom, top, left, right) are independent: every combination gives exactly one parallelogram containing the shaded cell, and different combinations give different parallelograms. □

c)

Step 4. "Sizes" made of exactly 4 cells

A parallelogram h rows high and w columns wide consists of $h \times w$ cells. We need $h \times w = 4$ with $h \le 3$, $w \le 5$:

- $h = 1$, $w = 4$: choose the row (3 ways) and 4 adjacent columns out of 5 (2 ways): $3 \cdot 2 = 6$;
- $h = 2$, $w = 2$: choose 2 adjacent rows (2 ways) and 2 adjacent columns (4 ways): $2 \cdot 4 = 8$;
- $h = 4$, $w = 1$: impossible, since there are only 3 rows.

Total: $6 + 8 = 14$.

Common mistake. Do not forget the 2×2 type! And there are no parallelograms 4 rows high.

Check: Part a) counted by rows: 1 row high: $3 \cdot 15 = 45$; 2 rows high: $2 \cdot 15 = 30$; 3 rows high: 15 (each band has $5 + 4 + 3 + 2 + 1 = 15$ choices of columns). Total 90. It matches.

Answer: a) 90. **b)** 36. **c)** 14.

Problem 49 ★★

(Extension of Problems 28 and 29.) A $4 \times 5 \times 6$ rectangular box is built from 120 small cubes with edge 1. All the outside faces of the box are painted, and then the small cubes are separated.

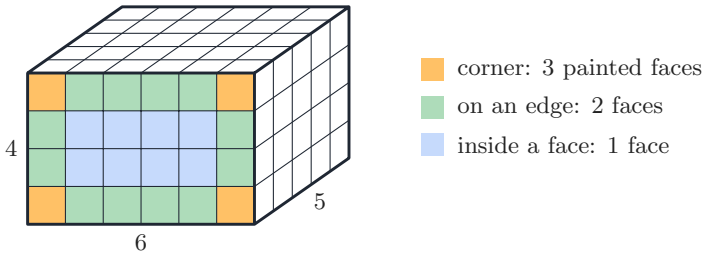
a) How many small cubes have exactly 3 painted faces? exactly 2? exactly 1? no painted face at all?

b) What is the total number of painted small faces (1×1 squares)? Explain why the result equals the surface area of the box.

c) An $n \times n \times n$ cube ($n \geq 3$) is painted and separated in the same way. Find n such that the number of small cubes with exactly 1 painted face equals the number of small cubes with no painted face.

Detailed solution

Understanding the problem. The number of painted faces of a small cube depends on its *position*: at a corner, on an edge, on a face, or completely inside. Look at the front face (size 6×4) to see this clearly:



a)

Step 1. Cubes at the corners: 3 painted faces
The box has 8 corners, each with exactly one cube: 8 cubes.

Step 2. Cubes on an edge (not at its ends): 2 painted faces
The box has 12 edges: 4 of length 4, 4 of length 5, 4 of length 6. An edge of length L contains L cubes; removing the 2 cubes at its ends (already counted as corners) leaves $L - 2$ cubes:

$$4 \cdot (4-2) + 4 \cdot (5-2) + 4 \cdot (6-2) = 8 + 12 + 16 = 36.$$

Step 3. Cubes in the middle of a face: 1 painted face
On a $p \times q$ face, removing the outer border leaves $(p-2)(q-2)$ squares. The box has 2 faces 4×5, 2 faces 4×6 and 2 faces 5×6:

$$2 \cdot (2 \cdot 3) + 2 \cdot (2 \cdot 4) + 2 \cdot (3 \cdot 4) = 12 + 16 + 24 = 52.$$

Step 4. Cubes completely inside: not painted
Peel off a "shell" of thickness 1 on every side; what remains is a box of $(4-2) \times (5-2) \times (6-2) = 2 \times 3 \times 4 = 24$ cubes.

Check: $8 + 36 + 52 + 24 = 120$ – exactly the total number of cubes.

b)

Step 5. Count the painted small faces
$8 \cdot 3 + 36 \cdot 2 + 52 \cdot 1 + 24 \cdot 0 = 24 + 72 + 52 = 148$.

Recall. The surface area of an $a \times b \times c$ box is $2(ab + bc + ca)$.

$2(4 \cdot 5 + 5 \cdot 6 + 6 \cdot 4) = 2(20 + 30 + 24) = 148$. The two results agree because each 1×1 square on the outside of the box is exactly one painted small face of exactly one cube – we are simply counting the same collection in two ways.

c)

Step 6. Write the equation

For an $n \times n \times n$ cube: each face has $(n-2)^2$ middle squares, so the 6 faces give $6(n-2)^2$ cubes with exactly 1 painted face; the inside is a cube of $(n-2)^3$ small cubes. We need:

$$(n-2)^3 = 6(n-2)^2.$$

Step 7. Solve

Since $n \geq 3$, $(n-2)^2 > 0$; dividing both sides by $(n-2)^2$: $n-2=6$, i.e. $n=8$.

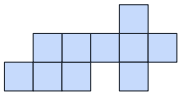
Common mistake. The condition $n \geq 3$ matters: for $n=2$ both numbers are 0 ("equal" in a trivial way), and we must not divide by 0.

Check: $n=8$: $6 \cdot 6^2 = 216$ and $6^3 = 216$. Correct.

Answer: a) 8; 36; 52; 24 cubes. b) 148. c) $n=8$.

Problem 50 ★★

(Extension of Problems 19 and 25.) Ten unit squares (side 1) are joined to form a flat shape such that: adjacent squares share a whole side, and the shape is "in one piece" (from any square one can reach any other square through shared sides). The perimeter of the shape is the total length of the square sides that are not covered by another square.



an example

a) Show that the perimeter is always an even number.

b) What is the greatest possible perimeter?

c) What is the smallest possible perimeter?

d) Which numbers can be the perimeter of such a shape?

Detailed solution

Understanding the problem. Let k be the number of *shared sides* (sides where two squares are joined). Key idea: each shared side removes two square sides from the perimeter.

a)

Step 1. A formula for the perimeter

10 squares have $10 \cdot 4 = 40$ sides in total. Each shared side is a side of two squares and is not on the perimeter; every other square side is on the perimeter. Therefore

$$P = 40 - 2k.$$

40 and $2k$ are both even, so P is even.

b)

Step 2. A shape in one piece has at least 9 shared sides

Why is it true? Rebuild the shape by placing one square at a time: start with any square, and each time add a square next to one already placed (this is always possible because the shape is in one piece). Each added square creates at least 1 new shared side. There are 9 additions, so $k \geq 9$. □

Step 3. Conclusion

$P = 40 - 2k \leq 40 - 18 = 22$. Equality holds for a straight row 1×10: perimeter $2(10 + 1) = 22$. So the greatest perimeter is 22.

c)

Step 4. Compare with the enclosing rectangle

Let the smallest rectangle (with sides along the grid) containing the whole shape be $a \times b$ (a columns, b rows).

Why is it true? Every column of the enclosing rectangle contains at least one square. The highest square in the column has its top side on the perimeter, and the lowest square has its bottom side on the perimeter: each column contributes at least 2 units of horizontal sides. Likewise each row contributes at least 2 units of vertical sides. So $P \geq 2a + 2b = 2(a + b)$. □

Step 5. Bound $a + b$

The enclosing rectangle contains 10 squares, so $ab \geq 10$. If $a + b \leq 6$ then $ab \leq 9$ (pairs with sum 6: $1 \cdot 5 = 5$, $2 \cdot 4 = 8$, $3 \cdot 3 = 9$; smaller sums give even smaller products) – impossible. So $a + b \geq 7$ and $P \geq 14$.

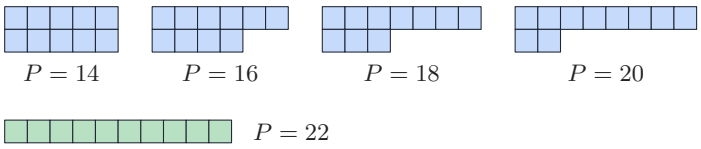
Step 6. A shape with perimeter 14 exists

The 2×5 rectangle: perimeter $2(2 + 5) = 14$. So the smallest perimeter is 14.

d)

Step 7. The possible values

From a), b), c): P is even and $14 \leq P \leq 22$, so $P \in \{14, 16, 18, 20, 22\}$. Each value is achieved by the following "two-row" shapes (using the idea of Problem 25: the boundary is a staircase, so the perimeter equals that of the enclosing rectangle):



Top row $5, 6, 7, 8$ squares, bottom row $5, 4, 3, 2$ squares: the enclosing rectangles are 5×2, 6×2, 7×2, 8×2, with perimeters $14, 16, 18, 20$; and the straight row gives 22.

Common mistake. "Compact" does not automatically mean a smaller perimeter: a 3×3 square with one extra square sticking out has perimeter $12 + 2 = 14$, the same as the 2×5 rectangle, not less.

Check: The $6 + 4$ shape: $k = 5 + 3 + 4 = 12$, $P = 40 - 24 = 16$. The $8 + 2$ shape: $k = 7 + 1 + 2 = 10$, $P = 40 - 20 = 20$. Correct.

Answer: a) $P = 40 - 2k$ is even. b) 22. c) 14. d) 14, 16, 18, 20, 22.

3 Counting and Probability

The big idea. In this topic every problem comes back to *one core skill*: **counting correctly** – missing nothing and counting nothing twice.

- **Counting** answers the question "*how many?*": how many numbers, how many seating arrangements, how many rectangles in a figure, how many toothpicks...
- **Probability** answers the question "*how likely is it?*". When all outcomes are equally likely, we only need to *count twice*:

$$\text{Probability} = \frac{\text{number of favourable outcomes}}{\text{number of all possible outcomes}}.$$

- Our counting tools: **systematic listing**, **tables**, **tree diagrams**, the **multiplication rule**, **splitting into cases** (the addition rule), **finding a pattern** and **reasoning by elimination**.

Always ask yourself three questions before starting: *(1) How many possible outcomes are there altogether? (2) Are these outcomes "fair" (equally likely)? (3) How many of them satisfy the requirement?*



① Experiments, outcomes, events and probability

Key words.

- A **random experiment** is an action for which we know in advance *all the possible outcomes*, but not *which outcome will happen*. Examples: rolling a die, tossing a coin, spinning a spinner, drawing a marble from a bag without looking.
- The **set of possible outcomes** is the list of *all* outcomes. Rolling a die: $\{1, 2, 3, 4, 5, 6\}$ – 6 outcomes. Tossing a coin: {heads, tails} – 2 outcomes.
- An **event** is something we are interested in, for example "rolling an even number". The outcomes that make the event happen are called **favourable outcomes** for the event. The event "rolling an even number" has 3 favourable outcomes: $2, 4, 6$.
- Outcomes are **equally likely** if none of them is "easier to get" than another (a fair die, a fair coin, equal sectors on a spinner, marbles of identical shape and size).

Probability formula (when the outcomes are equally likely):

$$P(\text{event}) = \frac{\text{number of outcomes favourable to the event}}{\text{number of all possible outcomes}}.$$

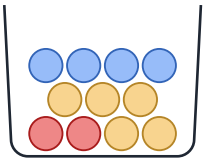
Properties. The number of favourable outcomes is between 0 and the total number of outcomes, so always $0 \le P \le 1$.

- $P = 0$: an *impossible* event (for example rolling a 7 on a die).
- $P = 1$: a *certain* event (for example rolling a number less than 7).

Why is it true? Why a division? Suppose there are n equally likely outcomes. The total "chance" 1 (one of the outcomes is sure to happen) is shared *equally* among n outcomes, so each outcome has probability $\frac{1}{n}$. If the event has k favourable outcomes, we add k equal parts: $\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n} = \frac{k}{n}$. □



Spinner with 6 equal sectors: each sector has probability $\frac{1}{6}$



Bag with 2 red, 5 yellow, 4 blue: $P(\text{red}) = \frac{2}{11}$

Common mistake. Count *objects*, not *kinds*. The bag above has 3 *colours* but 11 *marbles*. The probability of drawing a red marble is $\frac{2}{11}$, **not** $\frac{1}{3}$. Each marble is one equally likely outcome; the three colours are not equally likely (yellow has more marbles, so it is easier to draw).

Common mistake. Use the formula only when the outcomes are equally likely. On a spinner with big and small sectors, the sectors are *not* equally likely. Then the probability of landing in a sector equals *the fraction of the whole circle* that the sector takes up, that is $\frac{\text{central angle of the sector}}{360°}$. For example a sector with angle 90° has probability $\frac{90}{360} = \frac{1}{4}$.

ThinkAlgo Tip. In multiple-choice questions the answer may be written in *unsimplified* form, such as $\frac{2}{6}$ instead of $\frac{1}{3}$. Compare the values of the fractions, not just how they look. On the other hand, when you write your own solution, simplify your final answer.

② Complementary events. Finding a quantity from a probability

Complementary event. The event "A does *not* happen" is called the complement of A. We have

$$P(\text{not } A) = 1 - P(A).$$

Example: rolling a die, $P(\text{a } 6) = \frac{1}{6}$, so $P(\text{not a } 6) = 1 - \frac{1}{6} = \frac{5}{6}$.

Finding a quantity from a probability. From the probability formula we get

$$\text{number of favourable outcomes} = P \times \text{number of all outcomes.}$$

Example: a bag has 20 sweets and the probability of drawing a lemon sweet is $\frac{3}{4}$; then there are $20 \times \frac{3}{4} = 15$ lemon sweets.

When the quantities change (drawn and *not replaced*, eaten, added…): **recount both the numerator and the denominator** before computing the probability for the next draw.

Why is it true? Suppose there are n outcomes, of which k are favourable to A. Each of the remaining $n - k$ outcomes makes A *not* happen. So $P(\text{not } A) = \frac{n-k}{n} = \frac{n}{n} - \frac{k}{n} = 1 - P(A)$. □

Common mistake. A box has 10 red and 10 blue marbles. A student takes out 1 blue marble and does *not put it back*. What is the probability that the next marble is red?

- **Wrong:** "still $\frac{10}{20}$" (forgetting that the box now has one marble fewer).
- **Right:** 19 marbles are left, and all 10 red ones are still there, so the probability is $\frac{10}{19}$.

Note for parents. Do a real experiment with sweets or bottle caps of different colours: put them in a cloth bag, count together "how many are there altogether, how many are red", then take one out and ask the same two questions again. Your child will see for themselves that both the numerator and the denominator change.

③ Systematic listing: lists, tables and tree diagrams

When the number of possibilities is not too large, the safest method is to **list them all**. To list without missing or repeating anything, we need **an order**:

- An **ordered list** (from smallest to largest, alphabetically…). Example: the two-digit numbers with different digits made from 1, 2, 3: 12, 13, 21, 23, 31, 32 – ordered by the tens digit.
- A **two-way table** when the experiment has *two stages* (rolling two dice, choosing a main dish and a dessert…): each row is an outcome of stage 1, each column an outcome of stage 2, each cell a combined outcome.
- A **tree diagram** when there are *two, three, four stages in a row*: each stage branches out; each "path" from the root to the end of a branch is one outcome.

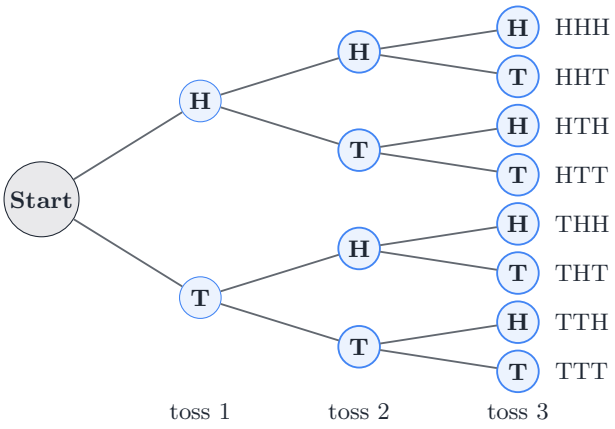
Table for rolling two dice (one red, one blue). Each cell shows the *sum* of the two numbers. There are 6 rows and 6 columns, so $6 \times 6 = 36$ equally likely outcomes.

red ↓ \ blue →	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

Orange cells: sum equals 7. There are 6 such cells, so $P(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}$.

Common mistake. (1, 2) and (2, 1) are two different outcomes! Red 1, blue 2 is different from red 2, blue 1. If you only count "pairs of numbers" without order, the pairs are *no longer* equally likely (the pair $\{1, 2\}$ can happen in 2 ways, the pair $\{1, 1\}$ in only 1 way). That is why the sums $2, 3, \ldots, 12$ are *not* equally likely: a sum of 7 can happen in 6 ways, a sum of 2 in only 1 way.

Tree diagram for tossing a coin three times (H: heads, T: tails). Each toss has 2 branches, so there are $2 \times 2 \times 2 = 8$ paths – that is, 8 equally likely outcomes.



Example: the event "exactly two heads" consists of HHT, HTH, THH – 3 outcomes, so its probability is $\frac{3}{8}$.

ThinkAlgo Tip. A problem like "walk on a grid by tossing a coin" (heads: move up, tails: move right) is really a problem about counting sequences of heads and tails: after 4 tosses there are $2^4 = 16$ equally likely sequences, and the *end point* depends only on the *number of heads*, not on their order.

④ The multiplication rule and the addition rule

Multiplication rule. If a task consists of two successive stages, stage 1 can be done in m ways, and *for each* way of doing stage 1, stage 2 can be done in n ways, then the whole task can be done in $m \times n$ ways. (With three or four stages: keep multiplying.)

Addition rule (splitting into cases). If the ways are split into cases that *do not overlap*: case 1 has a ways, case 2 has b ways, ..., then the total number of ways is $a + b + \ldots$

Common situations with digits.

- Three-digit numbers made from 3 given non-zero digits, *no repetition*: $3 \times 2 \times 1 = 6$ numbers.
- Three-digit numbers made from 3 given non-zero digits, *repetition allowed*: $3 \times 3 \times 3 = 27$ numbers.
- The first (leading) digit **cannot be** 0: 9 choices (1 to 9); the other places have 10 choices (0 to 9).
- A three-digit **palindrome** (reads the same forwards and backwards) has the form $\overline{aba}$: 9 ways to choose a, 10 ways to choose b, and the last digit is "forced" to equal a (only 1 way). There are $9 \times 10 = 90$ such numbers.



Number of three-digit palindromes: $9 \times 10 \times 1 = 90$.

Why is it true? Why is the multiplication rule true? Draw a table with m rows (the ways of doing stage 1) and n columns (the ways of doing stage 2). Each cell of the table is *exactly one* way of doing the whole task, and each way of doing the whole task is in *exactly one* cell. The table has m rows of n cells, so $m \times n$ cells. A tree diagram gives the same result: m big branches, each splitting into n small branches. □

Choosing two objects (order does not matter). Choosing 2 of n students for classroom duty: if we care about order (first person, second person) there are $n \times (n-1)$ ways; but each pair is counted *twice* (Anna–Ben and Ben–Anna), so the number of pairs is

$$\frac{n \times (n-1)}{2}.$$

Example: there are $\dfrac{10 \times 9}{2} = 45$ pairs of different numbers taken from 10 numbers.

Common mistake. Double counting. When the question asks "how many *pairs*", "how many *groups*" (who comes first does not matter) and you use the multiplication rule with order, each group is counted several times – you must divide. On the other hand, when the question is about *numbers* (the order of the digits matters) or *seating in a row*, order matters and you do *not* divide.

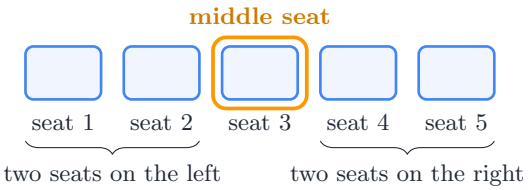
ThinkAlgo Tip. When splitting into cases, ask: "Do the cases *cover* every possibility? Do they *overlap*?" For example, for the sum $m + n$ to be even, m and n must be *both odd* or *both*

even – these two cases are separate and together they cover every possibility.

⑤ Reasoning by elimination: seating and colouring

Some problems do not ask "how many" but "*who can*, *who cannot*", "*what is the least number needed*". Then we use **reasoning**:

- **Draw a diagram** of the positions (seats, cells, faces…) and try filling it in.
- Two people who **must sit together** can be treated as **one block** taking two neighbouring seats.
- To show "*X can* sit in the middle seat": just **give one arrangement** that satisfies all the conditions.
- To show "*X cannot* sit in the middle seat": **suppose** X sits in the middle seat and show that *every* remaining arrangement breaks a condition (leading to a contradiction). This is called **proof by contradiction**.
- A "what is the *least* number" problem has **two parts**: (1) show a way that works with that number; (2) explain why fewer is impossible.



Key observation: once someone sits in the middle seat, the other four seats split into *two pairs of neighbouring seats* (seats 1–2 and seats 4–5). The two seats on either side of the middle seat (seat 2 and seat 4) are *not* next to each other.

Colouring so that neighbouring parts have different colours. Two faces of a cube are "neighbours" if they share an edge. Each face is a neighbour of 4 faces and is *not* a neighbour of exactly 1 face – the **opposite** face. A cube has 3 pairs of opposite faces (top–bottom, front–back, left–right).

- 3 *colours are enough:* paint each pair of opposite faces one colour; two faces of the same colour are opposite, so they are not neighbours.
- *Fewer than* 3 *colours is impossible:* the three faces meeting at a corner are neighbours of each other in pairs, so they need three different colours.

ThinkAlgo Tip. For a multiple-choice question "Who *cannot*…?", you can work backwards: for each option, try to find *one* valid arrangement. Every option for which you find an arrangement is eliminated; the remaining option is the answer – but remember to explain why it really cannot be arranged.

⑥ Counting shapes in a figure

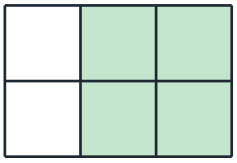
Principle. Counting shapes is a listing problem. To miss nothing and count nothing twice, **split into types** (by size, by position) and then **add up**.

The "choose two lines" trick. A rectangle in a grid is completely determined by *two horizontal lines* (top and bottom sides) and *two vertical lines* (left and right sides). Therefore:

- A strip of 1 row and n squares has $n+1$ vertical lines; the number of rectangles is the number of ways to choose 2 vertical lines: $\frac{(n+1)\times n}{2}$. Example: a strip of 4 squares has $\frac{5\times 4}{2} = 10$ rectangles.
- A grid with a horizontal lines and b vertical lines (all running the full length) has $\frac{a(a-1)}{2} \times \frac{b(b-1)}{2}$ rectangles (multiplication rule!).
- The same counting works for *parallelograms* in a grid whose slanted lines are parallel to each other.



Choose vertical lines 2 and 4
⇒ the orange rectangle

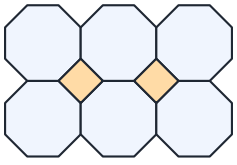


3 horizontal, 4 vertical lines:
$3 \times 6 = 18$ rectangles

Why is it true? Why does a strip of n squares have $\frac{(n+1)\times n}{2}$ rectangles? Choose the left side: $n+1$ ways; choose the right side (different from the left side): n ways – that gives $(n+1)\times n$ *ordered* choices. But choosing "line 2 then line 4" or "line 4 then line 2" gives the same rectangle, so each rectangle is counted 2 times; divide by 2. Counting by size gives the same result: a strip of 4 squares has 4 rectangles of width 1, 3 of width 2, 2 of width 3 and 1 of width 4, in total $4+3+2+1 = 10$. □

Common mistake. When the shapes **partly overlap** (not a grid with full-length lines), the "choose two lines" trick *does not work*, because many segments do not run all the way across. Check each shape: **are all four sides really drawn?** Do not count "imaginary rectangles" that are missing a side.

Counting in a tiling. A tiling made of regular octagons and small squares: if there are m rows with n octagons in each row, there are $m \times n$ octagons. Each small square sits where *four octagons meet*; between n columns there are $n-1$ gaps and between m rows there are $m-1$ gaps, so there are $(m-1)\times(n-1)$ squares.



2 rows, 3 columns: $2 \times 3 = 6$ octagons,
$(2-1)\times(3-1) = 2$ squares.

⑦ Counting with patterns: evenly spaced sequences, constant rates

Counting the whole numbers from a to b (including a and b): there are $b - a + 1$ numbers. For example from 1 to 11 there are 11 numbers, of which 6 are odd $(1, 3, 5, 7, 9, 11)$ and 5 are even.

Evenly spaced sequence (each number is a fixed amount d more than the one before):

$$\text{number of terms} = \frac{\text{last} - \text{first}}{d} + 1, \qquad \text{sum} = \frac{(\text{first} + \text{last}) \times \text{number of terms}}{2}.$$

Quantities that change at a constant rate (moving steadily, flowing steadily, counting steadily…): amount = rate × time. For example, if 3 sheep pass each minute, then in 14 minutes $3 \times 14 = 42$ sheep pass.

Figures made of sticks. Count the *horizontal sticks* and the *vertical sticks* separately and then add – usually easier than counting square by square.

Events that repeat in cycles. If one thing repeats every a minutes and another every b minutes, and they start together, they coincide again every $\text{LCM}(a, b)$ minutes.

Why is it true? Gauss's way of adding. Compute $S = 1 + 2 + 3 + \cdots + 10$. Write the sum twice, once forwards and once backwards:

$$\begin{array}{rcccccc} S & = & 1 & +2 & +3 & +\cdots+ & 10 \\ S & = & 10 & +9 & +8 & +\cdots+ & 1 \\ \hline 2S & = & 11 & +11 & +11 & +\cdots+ & 11 \end{array}$$

There are 10 columns, each with sum 11, so $2S = 10 \times 11 = 110$ and $S = 55$. In general: sum $= \dfrac{(\text{first} + \text{last}) \times \text{number of terms}}{2}$. □

Common mistake. The "off by one" error. From 3 to 10 there are $10 - 3 + 1 = 8$ numbers, *not* 7. A fence 10 m long with a post every 1 m (including both ends) needs 11 posts, not 10. Test a formula on a small example to check it.

Note for parents. For "toothpick" problems, let your child actually build the small figures (1 square, 2 squares, 3 squares…) with toothpicks, write the number of toothpicks in a table, and then ask: "each time we add one square, how many toothpicks do we add?". A child who finds the pattern alone remembers it much longer than a memorised formula.

Summary: read the clues, choose the tool.

Clue in the problem	Tool to use
"choose one at random", "what is the probability"	count all outcomes and favourable outcomes, then divide
probability given, quantity asked	favourable $= P \times$ total
something was taken out/eaten before the next draw	recount numerator and denominator
"at least one", "not"	complement: $1 - P$
two dice, two independent choices	two-way table
tossing a coin several times, several stages in a row	tree diagram
"how many numbers can be made", "how many ways"	multiplication rule (choose restricted positions first)
"how many pairs" (order does not matter)	multiplication rule, then divide by 2
conditions of the form "either... or..."	split into cases, add
"who can / cannot"	try one arrangement / proof by contradiction
"how many shapes..."	split by size, or choose two lines
rows of seats, toothpicks, constant rates	find the pattern, evenly spaced sequences

Example 1

A box contains 3 red marbles, 5 blue marbles and 4 yellow marbles, all of the same size. One marble is drawn at random.

a) Find the probability of drawing a blue marble.

b) Find the probability of drawing a marble that is *not* red.

c) Find the probability of drawing a purple marble.

Guided solution.

Step 1. Count all possible outcomes

Each marble is one outcome; the marbles have the same size, so the outcomes are equally likely. There are $3 + 5 + 4 = 12$ marbles, i.e. 12 possible outcomes.

Step 2. Part a

There are 5 blue marbles, i.e. 5 favourable outcomes. $P(\text{blue}) = \dfrac{5}{12}$.

Step 3. Part b – use the complement

$P(\text{red}) = \dfrac{3}{12} = \dfrac{1}{4}$, so $P(\text{not red}) = 1 - \dfrac{1}{4} = \dfrac{3}{4}$.

Another way (direct counting): 5 blue + 4 yellow = 9 marbles are not red, so $P = \frac{9}{12} = \frac{3}{4}$ – the same result.

Step 4. Part c

There are no purple marbles in the box: 0 favourable outcomes. $P(\text{purple}) = \dfrac{0}{12} = 0$ – an impossible event.

Answer: a) $\frac{5}{12}$; b) $\frac{3}{4}$; c) 0.

Example 2

A bag contains 30 sweets, some strawberry and some orange. If one sweet is drawn at random, the probability that it is strawberry is $\frac{2}{5}$.

a) How many strawberry sweets and how many orange sweets are in the bag?

b) Anna takes out 2 strawberry sweets and eats them. Then Ben draws one sweet at random from the bag. Find the probability that Ben gets a strawberry sweet.

Guided solution.

Step 1. Find the number of strawberry sweets

Recall. Number of favourable outcomes = probability × number of all outcomes.

Strawberry sweets: $30 \times \frac{2}{5} = \frac{30 \times 2}{5} = \frac{60}{5} = 12$. Orange sweets: $30 - 12 = 18$.

Check: $\frac{12}{30} = \frac{2}{5}$ ✓.

Step 2. Recount after Anna eats 2 sweets

	strawberry	orange	total
At the start	12	18	30
After Anna eats 2 strawberry	$12 - 2 = 10$	18	$30 - 2 = 28$

Step 3. Compute the probability for Ben

$P(\text{strawberry}) = \frac{10}{28} = \frac{5}{14}$.

Common mistake. Do not answer $\frac{10}{30}$: Anna has eaten 2 sweets, so the total number of sweets (the denominator) has gone down too.

Answer: a) 12 strawberry, 18 orange; b) $\frac{5}{14}$.

Example 3

A red die and a blue die (both fair) are rolled together. Find the probability of each event:

a) The sum of the two numbers is 8.

b) Both dice show the same number.

c) The sum of the two numbers is greater than or equal to 10.

Guided solution.

Step 1. Make a table of all outcomes

Rows are the numbers on the red die, columns the numbers on the blue die, and each cell shows the sum. The table has $6 \times 6 = 36$ cells, corresponding to 36 equally likely outcomes.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

sum equals 8 (5 cells)
both numbers equal (6 cells)
sum ≥ 10 (6 cells)

Step 2. **Part a**

The cells with sum 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$ – 5 outcomes (the first number is the red die). $P = \dfrac{5}{36}$.

Step 3. **Part b**

Equal numbers lie on the diagonal: $(1,1), (2,2), \ldots, (6,6)$ – 6 outcomes. $P = \dfrac{6}{36} = \dfrac{1}{6}$.

Step 4. **Part c**

Sum 10: $(4,6), (5,5), (6,4)$; sum 11: $(5,6), (6,5)$; sum 12: $(6,6)$. Altogether $3+2+1 = 6$ outcomes. $P = \dfrac{6}{36} = \dfrac{1}{6}$.

ThinkAlgo Tip. The 6×6 table also shows *symmetry*: the number of cells above the diagonal (red smaller than blue) equals the number of cells below the diagonal (red greater than blue).

Answer: a) $\frac{5}{36}$; b) $\frac{1}{6}$; c) $\frac{1}{6}$.

Example 4

A fair coin is tossed three times in a row. Find the probability of getting: a) exactly two heads; b) at least one head.

Guided solution.

Step 1. **Draw a tree diagram and list**

Each toss has 2 outcomes (H or T). By the tree diagram in section 3, there are $2 \times 2 \times 2 = 8$ equally likely outcomes:

$$\text{HHH},\ \text{HHT},\ \text{HTH},\ \text{HTT},\ \text{THH},\ \text{THT},\ \text{TTH},\ \text{TTT}.$$

Step 2. **Part a**

The outcomes with exactly two H's: HHT, HTH, THH – 3 outcomes. $P = \frac{3}{8}$.

Step 3. **Part b – complement**

The complement of "at least one head" is "no heads at all", which is only TTT (1 outcome). So $P(\text{at least one H}) = 1 - \frac{1}{8} = \frac{7}{8}$.

> **ThinkAlgo Tip.** When you see the words "*at least one*", think immediately of the complement "*none at all*" – it usually has very few outcomes and is easy to count.

Answer: a) $\frac{3}{8}$; b) $\frac{7}{8}$.

Example 5

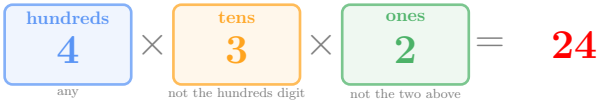
Using the digits $1, 2, 3, 4$, how many three-digit numbers can be made if: a) digits may be repeated; b) the digits must be different; c) the digits are different and the number is even?
d) One of the numbers from part b) is chosen at random. Find the probability that it is even.

Guided solution.

Step 1. **Part a – repetition allowed**

Hundreds: 4 ways; tens: still 4 ways; ones: 4 ways. By the multiplication rule: $4 \times 4 \times 4 = 64$ numbers.

Step 2. **Part b – no repetition**



Hundreds: 4 ways. The tens digit must differ from the digit already used: 3 ways. The ones digit must differ from both digits used: 2 ways. There are $4 \times 3 \times 2 = 24$ numbers.

Step 3. **Part c – even, no repetition: choose the ones digit FIRST**

An even number ends in 2 or 4: 2 ways. Then the hundreds digit is one of the 3 remaining digits: 3 ways; the tens digit: 2 ways. There are $2 \times 3 \times 2 = 12$ numbers.

ThinkAlgo Tip. A position with a *condition* (must be even, must not be 0...) should be **chosen first**. If you choose the hundreds digit first, the number of choices for the ones digit is sometimes 1 and sometimes 2 (depending on whether the hundreds digit used an even or an odd digit) – very easy to get wrong.

Step 4. Part d

$P(\text{even}) = \frac{12}{24} = \frac{1}{2}$.

Check: List the numbers starting with 1: $123, 124, 132, 134, 142, 143$ – exactly $6 = 3 \times 2$ numbers, of which $124, 132, 134, 142$ are even (4 numbers). The four possible first digits behave alike, so there are $4 \times 6 = 24$ numbers, ... matching part b.

Answer: a) 64; b) 24; c) 12; d) $\frac{1}{2}$.

Example 6

Four friends Anna, Ben, Chloe and David sit on four chairs in a row, numbered $1, 2, 3, 4$ from left to right. We know that: David sits on chair 1; Ben and Chloe sit next to each other; Anna does not sit next to Ben; Anna does not sit on chair 4. Who sits on chair 4?

Guided solution.

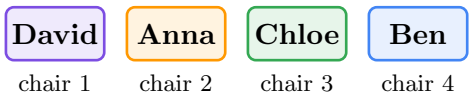
Step 1. Place David

David is on chair 1, leaving chairs $2, 3, 4$ for Anna, Ben and Chloe.

Step 2. Consider Anna's position

Anna does not sit on chair 4, so Anna sits on chair 2 or chair 3.

- *If Anna sits on chair* 3*:* Ben and Chloe take chairs 2 and 4 – these chairs are not next to each other, which breaks the condition "Ben and Chloe sit next to each other". Rejected.
- *If Anna sits on chair* 2*:* Ben and Chloe take chairs 3 and 4 (next to each other, good). Anna must not be next to Ben, and chair 3 is next to chair 2, so Ben is not on chair 3. So Ben is on chair 4 and Chloe on chair 3.



Check: David on chair 1 ✓; Ben (chair 4) next to Chloe (chair 3) ✓; Anna (chair 2) is next to David and Chloe, not next to Ben ✓; Anna is not on chair 4 ✓.

Answer: Ben sits on chair 4.

Example 7

a) The figure on the left is a strip of 4 squares. How many rectangles (including squares) are there altogether?
b) The figure on the right is a 2×2 grid of squares. How many rectangles are there? How many of them are squares?



Guided solution.

Step 1. **Part a – count by width**
Measure the width in squares:

Width	1 square	2 squares	3 squares	4 squares
Number of shapes	4	3	2	1

Total: $4+3+2+1=10$ rectangles. *Check with the "choose two vertical lines" trick:* there are 5 vertical lines, and the number of ways to choose 2 of them is $\frac{5\times 4}{2}=10$ ✓.

Step 2. **Part b – choose two horizontal and two vertical lines**
The grid has 3 horizontal and 3 vertical lines. Choose 2 horizontal lines: $\frac{3\times 2}{2}=3$ ways; choose 2 vertical lines: 3 ways. By the multiplication rule there are $3\times 3=9$ rectangles.

Step 3. **Count the squares separately**
1×1 squares: 4; 2×2 squares: 1 (the whole grid). There are $4+1=5$ squares.

Check: Counting by size: 1×1: 4; 1×2 (lying flat): 2; 2×1 (standing): 2; 2×2: 1. Total $4+2+2+1=9$ ✓.

Answer: a) 10 rectangles; b) 9 rectangles, of which 5 are squares.

Example 8

Toothpicks are used to make a row of joined squares as shown below.



a) How many toothpicks are needed for a row of 20 squares?
b) Lucy builds, one after another and separately, a row of 1 square, a row of 2 squares, ..., a row of 10 squares. How many toothpicks does Lucy use altogether?

Guided solution.

Step 1. **Find the pattern**

Squares	1	2	3	n
Toothpicks	4	7	10	$3n+1$

Explanation: count each kind of toothpick separately. Horizontal toothpicks: each square has 1 on top and 1 at the bottom, that is $2n$ toothpicks. Vertical toothpicks: n squares have $n+1$ vertical toothpicks (just as n gaps need $n+1$ posts). Total $2n+(n+1)=3n+1$.

Step 2. **Part a**

$n=20$: $3\times 20+1=61$ toothpicks.

Step 3. **Part b – sum of an evenly spaced sequence**

The numbers of toothpicks are $4, 7, 10, \ldots, 31$ (a row of 10 squares needs $3\times 10+1=31$ toothpicks). This is an evenly spaced sequence with step 3 and 10 terms.

> **Recall.** Sum of an evenly spaced sequence $=\dfrac{(\text{first}+\text{last})\times \text{number of terms}}{2}$.

Sum: $\dfrac{(4+31)\times 10}{2}=\dfrac{350}{2}=175$ toothpicks.

Check: $4+7+10+13+16+19+22+25+28+31$: pair first with last, $4+31=7+28=10+25=13+22=16+19=35$, giving 5 pairs, $5\times 35=175$ ✓.

Answer: a) 61 toothpicks; b) 175 toothpicks.

Core practice

Problem 1

The spinner shown is divided into 6 sectors of equal size, each marked with a letter. The spinner is spun once. What is the probability that the arrow lands on a sector marked with the letter P?

(A) $\frac{3}{6}$ (B) $\frac{4}{6}$ (C) $\frac{5}{6}$ (D) $\frac{2}{6}$ (E) $\frac{1}{6}$



Detailed solution

Understanding the problem. The spinner has 6 *equal* sectors, so when it is spun the arrow is equally likely to stop in each sector. We need the probability that the arrow points to a sector marked P.

Recall. When the outcomes are equally likely,

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{number of all outcomes}}.$$

Step 1. Count all possible outcomes

The arrow can stop in any one of the 6 sectors, so there are 6 possible outcomes.

Step 2. Count the favourable outcomes

Reading the letters clockwise, starting from the top-left sector: P, R, P, Q, R, Q. Exactly 2 sectors are marked P (the top-left one and the right one).

Step 3. Compute the probability

$P(\text{letter } P) = \frac{2}{6} \left(= \frac{1}{3}\right)$.

Common mistake. There are 3 different letters (P, Q, R), but the probability is *not* computed as "1 letter out of 3 letters". Here $\frac{1}{3} = \frac{2}{6}$ only by coincidence, because each letter appears exactly 2 times; if the letters took up different numbers of sectors, that way of thinking would be wrong. Always count *sectors*, not *kinds of letters*.

Check: Each of the letters P, Q, R takes 2 sectors: $\frac{2}{6} + \frac{2}{6} + \frac{2}{6} = 1$ – the probabilities of all possibilities add up to 1, as they should.

The value $\frac{2}{6}$ matches option **(D)**. (Option (A) $\frac{3}{6}$ comes from miscounting 3 sectors; (E) $\frac{1}{6}$ is the probability of *one* particular sector.)

Answer: (D) $\frac{2}{6}$

Problem 2

Using the three digits 4, 7 and 9, each digit used *at most once*, how many three-digit whole numbers with different digits can be made?

(A) 6 (B) 3 (C) 5 (D) 12 (E) 9

Detailed solution

Understanding the problem. A three-digit number using exactly the digits $4, 7, 9$, with no digit repeated. So each number is an *arrangement* of these three digits in the three places: hundreds, tens, ones.

Step 1. Method 1 – systematic listing

Order by the hundreds digit (from smallest to largest):

- Hundreds digit 4: the other two digits $7, 9$ give 479 and 497.
- Hundreds digit 7: 749 and 794.
- Hundreds digit 9: 947 and 974.

There are $2 + 2 + 2 = 6$ numbers.

Step 2. Method 2 – multiplication rule

Recall. Multiplication rule: if a task has several successive stages, each with a number of ways (not depending on the choices made before), then the number of ways to do the whole task is the *product* of those numbers.

$$\underbrace{3}_{\text{hundreds}\ (4 \text{ or } 7 \text{ or } 9)} \times \underbrace{2}_{\text{tens}\ (2 \text{ digits left})} \times \underbrace{1}_{\text{ones}\ (\text{the last digit})} = 6$$

Hundreds: 3 choices. Tens: only 2 unused digits remain, 2 choices. Ones: exactly 1 digit remains, 1 choice. Number of numbers: $3 \times 2 \times 1 = 6$.

Common mistake. If repeated digits were allowed there would be $3 \times 3 \times 3 = 27$ numbers (such as 444, 477...). The problem forbids repetition, so after each stage the number of choices goes down by 1.

Check: The six numbers $479, 497, 749, 794, 947, 974$ are all different and all use the three digits. Both methods give the same result.

The answer 6 is option **(A)**. (Option (B) 3 is just the number of digits given.)

Answer: (A) 6

Problem 3

An envelope contains 2 sunflower seeds, 3 mung bean seeds and 4 pumpkin seeds. Carrie takes one seed from the envelope at random. What is the probability that Carrie takes a sunflower seed?

(A) $\frac{2}{9}$ (B) $\frac{5}{9}$ (C) $\frac{9}{7}$ (D) $\frac{7}{9}$ (E) $\frac{1}{9}$

Detailed solution

Understanding the problem. The envelope has three kinds of seeds in different numbers. Carrie takes *one seed* at random: every seed has the same chance of being taken, so each *seed* is one equally likely outcome.

Step 1. Count all possible outcomes
Total number of seeds: $2 + 3 + 4 = 9$, i.e. 9 outcomes.

Step 2. Count the favourable outcomes
There are 2 sunflower seeds, so 2 favourable outcomes.

Step 3. Compute the probability

Recall. $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$.

$P(\text{sunflower}) = \dfrac{2}{9}$.

Common mistake. There are 3 *kinds* of seeds but the answer is not $\dfrac{1}{3}$, because the pumpkin seeds are more numerous and so more likely to be taken. Remember too that a probability is never greater than 1, so option (C) $\dfrac{9}{7}$ is certainly wrong.

Check: $P(\text{sunflower}) + P(\text{mung bean}) + P(\text{pumpkin}) = \dfrac{2}{9} + \dfrac{3}{9} + \dfrac{4}{9} = \dfrac{9}{9} = 1$.

The result $\dfrac{2}{9}$ is option **(A)**. (Option (D) $\dfrac{7}{9}$ is the probability of *not* taking a sunflower seed.)

Answer: (A) $\dfrac{2}{9}$

Problem 4

A cinema has 11 rows of seats, numbered from 1 to 11. Each odd-numbered row has 15 seats and each even-numbered row has 16 seats. How many seats are there in the cinema altogether?

(A) 176 (B) 186 (C) 165 (D) 170 (E) 171

Detailed solution

Understanding the problem. The number of seats in a row depends on whether the row number is odd or even. So we need to know how many of the numbers from 1 to 11 are odd and how many are even.

Step 1. Count the odd rows and the even rows
Odd rows: $1, 3, 5, 7, 9, 11$ – 6 rows.
Even rows: $2, 4, 6, 8, 10$ – 5 rows.
Check: $6 + 5 = 11$ rows ✓.

Recall. Number of terms of an evenly spaced sequence $= \dfrac{\text{last} - \text{first}}{\text{step}} + 1$. The odd numbers $1, 3, \ldots, 11$ have step 2: $\dfrac{11-1}{2} + 1 = 6$ numbers. The even numbers $2, 4, \ldots, 10$: $\dfrac{10-2}{2} + 1 = 5$ numbers.

Common mistake. Do not split the 11 rows into "5.5 odd rows and 5.5 even rows". The list starts *and* ends with an odd number (1 and 11), so there is exactly 1 more odd number than even numbers.

Step 2. **Seats in the odd rows**
$6 \times 15 = 90$ seats.

Step 3. **Seats in the even rows**
$5 \times 16 = 80$ seats.

Step 4. **Add up**
Total number of seats: $90 + 80 = 170$.

Check: *Another way – pairing:* pair row 1 with row 2, row 3 with row 4, ..., row 9 with row 10: that gives 5 pairs, each with $15 + 16 = 31$ seats, i.e. $5 \times 31 = 155$ seats; row 11 with 15 seats is left over. Total $155 + 15 = 170$ ✓.
The answer 170 is option **(D)**. (Mixing up and taking 5 odd rows and 6 even rows gives 171 – option (E), a very common "trap".)

Answer: (D) 170

Problem 5

At a class party, each student in turn draws a wrapped present at random from a bag. The presents are only books and calculators; there are 27 presents in the bag altogether. Meghan draws first. The probability that Meghan draws a book is $\dfrac{2}{3}$. How many books are in the bag?

(A) 15 (B) 9 (C) 21 (D) 7 (E) 18

Detailed solution

Understanding the problem. A "backwards" problem: we know the *probability* and the *total* number of presents, and need the *number of favourable outcomes* (the number of books). Since Meghan draws first, all 27 presents are still in the bag.

Recall. From $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$ we get: number of favourable outcomes $= P \times$ number of all outcomes.

Step 1. **Call the number of books an unknown**
Let the number of books be s. The probability of drawing a book is $\dfrac{s}{27}$. By the problem: $\dfrac{s}{27} = \dfrac{2}{3}$.

Step 2. Solve the equation

Multiply both sides of $\frac{s}{27} = \frac{2}{3}$ by 27:

$$s = 27 \times \frac{2}{3} = \frac{27 \times 2}{3} = \frac{54}{3} = 18.$$

Another way – find the number of calculators first: let the number of calculators be c; then $s + c = 27$. The event "drawing a calculator" is the complement of "drawing a book", so $\frac{c}{27} = 1 - \frac{2}{3} = \frac{1}{3}$, which gives $c = 27 \times \frac{1}{3} = 9$ and $s = 27 - c = 27 - 9 = 18$. Both ways give the same result.

Check: There are 18 books and $27 - 18 = 9$ calculators. Probability of drawing a book: $\frac{18}{27} = \frac{18 \div 9}{27 \div 9} = \frac{2}{3}$ ✓.

Common mistake. The 9 in option (B) is the number of *calculators*, not books – read the question carefully before choosing.

The answer 18 is option **(E)**.

Answer: (E) 18

Problem 6

A crate of fruit contains 20 apples, 10 oranges and no other fruit. One fruit is taken from the crate at random. What is the probability that it is an orange?

(A) $\frac{1}{10}$ (B) $\frac{1}{20}$ (C) $\frac{1}{30}$ (D) $\frac{1}{3}$ (E) $\frac{2}{3}$

Detailed solution

Understanding the problem. Every fruit in the crate has the same chance of being taken. A favourable outcome is taking one of the oranges.

Step 1. Count all outcomes

Total number of fruits: $20 + 10 = 30$.

Step 2. Count the favourable outcomes

There are 10 oranges.

Step 3. Compute and simplify

Recall. $P = \frac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$; simplify a fraction by dividing the numerator and denominator by their GCD.

$$P(\text{orange}) = \frac{10}{30} = \frac{10 \div 10}{30 \div 10} = \frac{1}{3}.$$

Common mistake. Two common mistakes: (1) using the number of apples (20) as the denominator instead of the *total number of fruits* (30), giving $\frac{10}{20}$; (2) writing $\frac{1}{10}$ (option A) thinking "there are 10 oranges so each is $\frac{1}{10}$". The denominator is always *all* the outcomes.

Check: $P(\text{apple}) = \frac{20}{30} = \frac{2}{3}$ and $\frac{1}{3} + \frac{2}{3} = 1$ ✓. It also makes sense: there are twice as many apples as oranges, so an apple is twice as likely as an orange.
The result $\frac{1}{3}$ is option **(D)**. (Option (E) $\frac{2}{3}$ is the probability of taking an apple.)

Answer: (D) $\frac{1}{3}$

Problem 7

A bag contains 2 red balls, 5 yellow balls and 4 blue balls, all the same size. Without looking, one ball is taken from the bag. The probability of taking a yellow ball is

(A) $\frac{2}{11}$ (B) $\frac{5}{11}$ (C) $\frac{4}{11}$ (D) $\frac{6}{11}$ (E) $\frac{7}{11}$

Detailed solution

Understanding the problem. "Without looking" and "the same size" guarantee that every ball has the same chance of being taken.

Step 1. Count all outcomes
Total number of balls: $2 + 5 + 4 = 11$.

Step 2. Count the favourable outcomes
There are 5 yellow balls.

Step 3. Compute the probability

Recall. $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$.

$P(\text{yellow}) = \frac{5}{11}$ (this fraction is already in lowest terms because 5 and 11 have no common factor other than 1).



5 of the 11 balls are yellow

Check: $\frac{2}{11} + \frac{5}{11} + \frac{4}{11} = \frac{11}{11} = 1$ ✓.
The distractors: (A) $\frac{2}{11}$ is the probability of red, (C) $\frac{4}{11}$ the probability of blue, (D) $\frac{6}{11}$ the probability of *not* taking a yellow ball. The correct answer is **(B)**.

Answer: (B) $\frac{5}{11}$

Problem 8

You are doing a multiple-choice test. On one question you do not know the answer, so you choose an option at random. If the question has five options (A, B, C, D, E) and only one of them is correct, what is the probability that you guess correctly?

(A) $\frac{1}{5}$ (B) $\frac{5}{5}$ (C) $\frac{4}{5}$ (D) $\frac{2}{5}$ (E) $\frac{3}{5}$

Detailed solution

Understanding the problem. "Choose at random" means the five options are equally likely to be chosen. The question has exactly *one* correct option.

Step 1. Count all outcomes

There are 5 options to choose from: A, B, C, D, E.

Step 2. Count the favourable outcomes

Only 1 option is correct, so there is 1 favourable outcome.

Step 3. Compute the probability

> **Recall.** $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$.

$P(\text{correct}) = \frac{1}{5}$.

> **ThinkAlgo Tip.** The probability of guessing *wrong* is $1 - \frac{1}{5} = \frac{4}{5}$ – option (C). So when guessing you are 4 times more likely to be wrong than right. And (B) $\frac{5}{5} = 1$ means "certainly correct" – absurd for a guess.

Check: If you did 100 such questions and guessed every one, you would get about $100 \times \frac{1}{5} = 20$ right on average – matching the feeling "1 right out of every 5".

The answer is **(A)**.

Answer: (A) $\frac{1}{5}$

Problem 9

A fair six-sided die has the numbers 1 to 6 on its faces. The die is rolled once. What is the probability that it shows a 5?

(A) $\frac{2}{6}$ (B) $\frac{1}{6}$ (C) $\frac{5}{6}$ (D) $\frac{3}{6}$ (E) $\frac{4}{6}$

Detailed solution

Understanding the problem. The die is *fair*, so the six faces are equally likely. The event we care about: "the face showing is 5".

Step 1. List the possible outcomes
$\{1, 2, 3, 4, 5, 6\}$ – 6 equally likely outcomes.

Step 2. Count the favourable outcomes
Only one face shows 5: 1 favourable outcome.

Step 3. Compute the probability

Recall. $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$.

$P(5) = \dfrac{1}{6}$.

Common mistake. Do not confuse "the number 5" with "the number of favourable outcomes". Many students write $\frac{5}{6}$ (option C) because they see a 5 in the question. The 5 is only the *name* of the face; the number of faces with that name is only 1. ($\frac{5}{6}$ is the probability of rolling a number *from* 1 *to* 5, i.e. anything but 6.)

Check: Each face has probability $\frac{1}{6}$, and the six faces together give $\frac{6}{6} = 1$ ✓.
The answer is **(B)**.

Answer: (B) $\frac{1}{6}$

Problem 10

Five friends Andy, Jen, Sally, Mike and Tom sit in a row of five seats. Andy does *not* sit next to Jen, and Sally sits next to Mike. Who *cannot* sit in the middle seat?

(A) Andy (B) Jen (C) Sally (D) Mike (E) Tom

Detailed solution

Understanding the problem. The five seats are in a row, numbered $1, 2, 3, 4, 5$ from left to right; the middle seat is seat 3. Two conditions: (i) Andy and Jen do *not* sit next to each other; (ii) Sally and Mike sit next to each other. We must find the friend who can *never* sit in seat 3.

Recall. To show that a friend *can* sit in the middle: just give *one* arrangement satisfying every condition. To show that a friend *cannot*: suppose they sit in the middle and show that every arrangement then breaks a condition (proof by contradiction).

Step 1. Andy, Jen, Sally and Mike can each sit in the middle
Here is one arrangement for each of them (seats $1 \to 5$):

In the middle	seat 1	seat 2	**seat 3**	seat 4	seat 5	check
Andy	Sally	Mike	**Andy**	Tom	Jen	Andy (3), Jen (5) not neighbours
Jen	Sally	Mike	**Jen**	Tom	Andy	Jen (3), Andy (5) not neighbours
Sally	Andy	Mike	**Sally**	Tom	Jen	Sally (3) next to Mike (2)
Mike	Andy	Sally	**Mike**	Tom	Jen	Mike (3) next to Sally (2)

In all four arrangements Sally and Mike are always neighbours, and Andy and Jen are always apart (Andy in seat 1 or 3, Jen in seat 5, or the other way round). So options (A), (B), (C), (D) are eliminated.

Step 2. Show that Tom cannot sit in the middle

Suppose Tom sits in seat 3. The other four seats split into two pairs of neighbouring seats: the left pair (seats 1–2) and the right pair (seats 4–5). Note that seats 2 and 4 are *not* neighbours (Tom is between them).



- Sally and Mike must be neighbours, and the only neighbouring seats (not counting Tom's) are seats 1–2 or seats 4–5. So Sally and Mike take up one whole pair.
- Then Andy and Jen are forced into the other pair – that is, *next to each other*. This contradicts the condition "Andy does not sit next to Jen".

So the assumption is false: Tom cannot sit in the middle seat.

Common mistake. Do not choose an answer just because "I tried a few ways and none worked". For a "cannot" question you must argue that *every* way fails, as in Step 2.

Check: You could check by listing all $5 \times 4 \times 3 \times 2 \times 1 = 120$ arrangements: no valid one puts Tom in the middle, while each of the other friends has a valid one.
The answer is **(E)**.

Answer: (E) Tom

Problem 11

A box of jelly beans contains 15 red, 20 blue and 16 green jelly beans. Jack takes a green jelly bean and eats it. Then he takes a blue jelly bean and eats it too. If each jelly bean left in the box is equally likely to be chosen, what is the probability that the next jelly bean Jack takes is red?

(A) $\frac{15}{31}$ (B) $\frac{34}{49}$ (C) $\frac{15}{49}$ (D) $\frac{2}{7}$ (E) $\frac{1}{15}$

Detailed solution

Understanding the problem. The box has lost 2 jelly beans before the draw we care about. So we must *recount* the jelly beans in the box just before the third draw.

Recall. When objects are taken out and not replaced, update both the *number of favourable outcomes* (numerator) and the *total number of outcomes* (denominator) before computing the probability of the next draw.

Step 1. Jelly beans at the start

$15 + 20 + 16 = 51$.

Step 2. Jelly beans after Jack eats 2

	red	blue	green	total
At the start	15	20	16	51
After eating 1 green	15	20	15	50
After eating 1 blue	15	19	15	49

The number of red jelly beans does not change (15) because Jack has not eaten a red one; the total goes down to 49.

Step 3. Compute the probability

$P(\text{red}) = \dfrac{15}{49}$.

Common mistake. Two common wrong ways: (1) forgetting to subtract the eaten jelly beans, giving $\dfrac{15}{51}$ (not among the options – a signal to check your work!); (2) comparing the red ones only with the remaining green ones, giving $\dfrac{15}{31}$ (option A). The denominator must be *all* the jelly beans left in the box.

Check: $P(\text{red}) + P(\text{not red}) = \dfrac{15}{49} + \dfrac{19+15}{49} = \dfrac{15}{49} + \dfrac{34}{49} = 1$ ✓. (By the way: $\dfrac{34}{49}$ in option (B) is the probability of *not* getting a red one.)

The answer is **(C)**.

Answer: (C) $\dfrac{15}{49}$

Problem 12

Nicholas is counting the sheep in a flock as they cross a road. The sheep start crossing at 2:00 p.m., at a steady rate of 3 sheep per minute. After counting 42 sheep, Nicholas falls asleep. He sleeps for exactly an hour and a half and then wakes up, and at that moment the number of sheep that have crossed since 2:00 p.m. is exactly half of the whole flock. How many sheep are in the whole flock?

(A) 630 (B) 621 (C) 582 (D) 624 (E) 618

Detailed solution

Understanding the problem. The sheep cross *steadily*, 3 per minute. The time from 2:00 p.m. until Nicholas wakes up has two parts: the time Nicholas spends counting 42 sheep, and the time Nicholas sleeps (1 hour 30 minutes). The number of sheep that cross in this whole period equals *half the flock*. We let the size of the flock be an unknown and form an equation.

> **Recall.** Steady change: amount = rate × time, so time = amount ÷ rate.

Step 1. Choose the unknown
Let the whole flock have x sheep (x a whole number). When Nicholas wakes up, the number of sheep that have crossed since 2:00 p.m. is $\frac{x}{2}$.

Step 2. Time Nicholas spends counting 42 sheep
$42 \div 3 = 14$ minutes. So Nicholas falls asleep at 2:14 p.m.

Step 3. Total time from 2:00 p.m. to waking up
An hour and a half $= 60 + 30 = 90$ minutes. Total time: $14 + 90 = 104$ minutes, so Nicholas wakes up at 2:00 p.m. + 104 minutes = 3:44 p.m.



Step 4. Form the equation
In 104 minutes, $3 \times 104 = 312$ sheep cross. (Or: 42 sheep while counting + $3 \times 90 = 270$ sheep while asleep = 312 sheep.) This is half the flock, so we have the equation

$$\frac{x}{2} = 312.$$

Step 5. Solve the equation
Multiply both sides by 2: $x = 312 \times 2 = 624$.

> **Common mistake.** Do not forget the 42 sheep that crossed *before* Nicholas fell asleep. Counting only the 270 sheep while asleep and doubling gives 540 – which does not fit the problem, because it is the number of sheep crossed "since 2:00 p.m." that is half the flock.

Check: Half the flock $= 624 \div 2 = 312$ sheep, which takes $312 \div 3 = 104$ minutes, i.e. from 2:00 p.m. to 3:44 p.m.; subtracting the 14 minutes of counting leaves exactly 90 minutes of sleep ✓. The answer is **(D)**.

Answer: (D) 624

Problem 13

A positive whole number is called a *palindrome* if its digits read the same from left to right as from right to left. For example, 474 and 222 are palindromes. How many palindromes are there between 100 and 1000?

(A) 10 (B) 90 (C) 100 (D) 900 (E) 1000

Detailed solution

Understanding the problem. The numbers between 100 and 1000 are the *three-digit* numbers (101 to 999; 100 and 1000 are not palindromes, so including them or not changes nothing). A three-digit number $\overline{abc}$ is a palindrome when reading it backwards, $\overline{cba}$, gives the same number, that is $c = a$. So a palindrome has the form $\overline{aba}$.

Step 1. Choose the hundreds digit a
a is the first digit, so $a \neq 0$: $a \in \{1, 2, \ldots, 9\}$ – 9 ways.

Step 2. Choose the tens digit b
b can be any digit from 0 to 9 – 10 ways.

Step 3. The ones digit
The ones digit must equal a – only 1 way (it cannot be chosen freely).

Step 4. Apply the multiplication rule

Recall. Multiplication rule: number of ways = product of the numbers of choices at each stage.

$$\underbrace{9}_{\text{hundreds a, from 1 to 9}} \times \underbrace{10}_{\text{tens b, from 0 to 9}} \times \underbrace{1}_{\text{ones, must equal a}} = 90$$

There are $9 \times 10 \times 1 = 90$ palindromes.

Check: List by a: for $a = 1$ we have $101, 111, 121, \ldots, 191$ – exactly 10 numbers. Each value of a from 1 to 9 gives 10 such numbers, so the total is $9 \times 10 = 90$ ✓.

Common mistake. Option (D) 900 is the number of *all* three-digit numbers (from 100 to 999), not the palindromes. Option (C) 100 comes from forgetting the condition $a \neq 0$ (counting "010", "020"..., which are not three-digit numbers).

The answer is **(B)**.

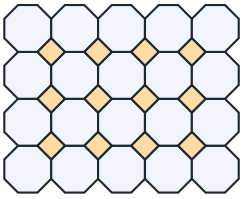
Answer: (B) 90

Problem 14

The figure shown is made only of octagons and small squares. The ratio of the number of octagons to the number of squares is

(A) 1 : 1 (B) 2 : 1 (C) 25 : 12

(D) 5 : 0 (E) 5 : 3



Detailed solution

Understanding the problem. We count the octagons and the squares separately, then form the ratio (octagons : squares) and simplify.

Step 1. Count the octagons

The octagons are arranged in 4 rows of 5. Number of octagons: $4 \times 5 = 20$.

Step 2. Count the squares

Each small square sits where *four* octagons meet (between two neighbouring rows and between two neighbouring columns).

- Between 5 columns of octagons there are $5 - 1 = 4$ vertical gaps.
- Between 4 rows of octagons there are $4 - 1 = 3$ horizontal gaps.

Number of squares: $3 \times 4 = 12$ (exactly what you see when counting directly in the figure: 3 rows of 4).

Common mistake. There are *no* squares on the outer edge of the figure (the notch between two octagons on the edge is not closed). Do not count extra "half squares" around the border.

Step 3. Form and simplify the ratio

Recall. Simplify a ratio $a : b$ by dividing both numbers by their GCD.

Ratio: $20 : 12$. $\text{GCD}(20, 12) = 4$, so $20 : 12 = (20 \div 4) : (12 \div 4) = 5 : 3$.

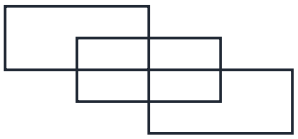
Check: $5 : 3$ means 3 squares for every 5 octagons; $20 = 5 \times 4$ and $12 = 3 \times 4$ ✓.

The distractors: (D) $5 : 0$ forgets to count the squares; (C) $25 : 12$ miscounts the octagons as 5×5. The answer is **(E)**.

Answer: (E) $5 : 3$

Problem 15

The figure shown consists of three identical rectangles overlapping each other. How many rectangles of all sizes are there in the figure?



(A) 11 (B) 15 (C) 7

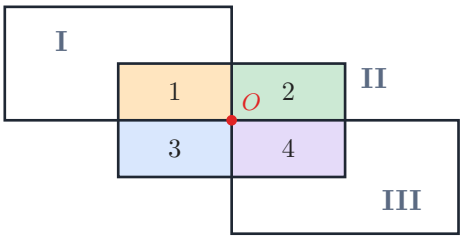
(D) 13 (E) 9

Detailed solution

Understanding the problem. Call the three identical rectangles **I** (top left), **II** (middle) and **III** (bottom right). Where their sides cross, more small rectangles appear. We must count *all* rectangles whose four sides are all drawn in the figure.

Step 1. Look carefully at the figure

Label the points as below. Notice: the bottom side of rectangle I and the top side of rectangle III lie on *the same straight line* and join at the point O – the centre of rectangle II. Likewise the right side of I and the left side of III join at O. So inside rectangle II there is a "plus sign" that divides II into 4 equal small cells.



Step 2. Count type by type

- **The three original large rectangles:** I, II, III – 3 rectangles.
- **The four small cells** inside rectangle II (cells 1, 2, 3, 4) – 4 rectangles. Each cell has all four sides: for example cell 1 has its top and left sides on rectangle II, and its bottom and right sides on rectangle I.
- **Halves of rectangle II** (two neighbouring cells joined): cells $1+2$ (top half), $3+4$ (bottom half), $1+3$ (left half), $2+4$ (right half) – 4 rectangles.

Step 3. Check that there are no others

- The part of rectangle I *outside* rectangle II is L-shaped, and no segment passes through it. A rectangle containing part of this L-shaped region must use the left side and the top side of rectangle I as its sides; these two sides are exactly as long as the sides of rectangle I, so that rectangle is rectangle I itself. Likewise, a rectangle containing part of III outside II can only be rectangle III.
- Every other rectangle must lie entirely inside rectangle II. Rectangle II with its inner "plus sign" is a 2×2 grid of cells, and in Step 2 we have listed them all: 4 cells, 4 halves and rectangle II itself.

Common mistake. Do not count the large rectangle surrounding all three shapes (from the top-left corner of I to the bottom-right corner of III): *none* of its sides is fully drawn. And do not miss the small cells created by the "plus sign" in the centre.

Step 4. Add up

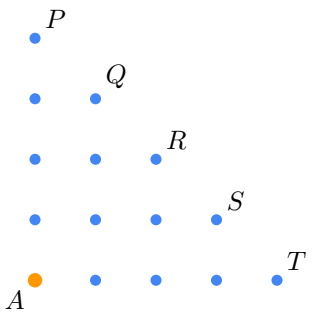
$3+4+4=11$ rectangles.

Check: Another count – inside rectangle II alone (a complete 2×2 grid): by Example 7 there are 9 rectangles (4 cells, 4 halves and II itself). Adding rectangles I and III: $9+2=11$ ✓.
The answer is **(A)**.

Answer: (A) 11

Problem 16

On the dot grid shown, Jane stands at dot A. She tosses a fair coin to decide how to move: if the coin shows *heads* she moves *up* one dot; if it shows *tails* she moves *right* one dot. After exactly four tosses, Jane will stop at one of the five dots P, Q, R, S, T. What is the probability that Jane stops at dot R?



(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{9}{16}$ (D) $\frac{7}{16}$ (E) $\frac{5}{16}$

Detailed solution

Understanding the problem. At each toss Jane moves *up* (↑, for heads) or *right* (→, for tails). After 4 tosses she has made exactly 4 moves. We need the probability that after these 4 moves she is at R.

Step 1. Count all possible outcomes

Each toss has 2 outcomes (heads or tails), so four tosses in a row give $2 \times 2 \times 2 \times 2 = 16$ sequences, for example ↑→→↑. Since the coin is fair, these 16 sequences are *equally likely*.

> **Recall.** *Multiplication principle:* if the first task can be done in a ways, the second in b ways, … then doing them one after another can be done in $a \times b \times \cdots$ ways. The *probability* of an event (when the outcomes are equally likely) $= \dfrac{\text{number of favourable outcomes}}{\text{number of possible outcomes}}$.

Step 2. The final dot depends only on the number of up-moves

If k of the 4 moves are up, then $4 - k$ moves are to the right. The order of the moves does not change the final dot. Starting from A: 4 ups → P; 3 ups → Q; 2 ups → R; 1 up → S; 0 ups → T. So Jane stops at R exactly when she makes **exactly 2 moves up and 2 moves right**.

Step 3. List the favourable sequences

We choose which two of the four turns (turn 1, 2, 3, 4) are ↑:

$$\uparrow\uparrow\rightarrow\rightarrow, \quad \uparrow\rightarrow\uparrow\rightarrow, \quad \uparrow\rightarrow\rightarrow\uparrow, \quad \rightarrow\uparrow\uparrow\rightarrow, \quad \rightarrow\uparrow\rightarrow\uparrow, \quad \rightarrow\rightarrow\uparrow\uparrow .$$

There are 6 sequences (choosing 2 turns out of 4: $\dfrac{4 \times 3}{2} = 6$ ways).

Step 4. Compute the probability

$$P(\text{stop at } R) = \frac{6}{16} = \frac{3}{8}.$$

Step 5. A picture: write the number of paths at each dot
At each dot write the number of move sequences leading from A to that dot. A dot can only be reached from the dot just to its left or the dot just below it, so the number at each dot = the sum of the numbers to its left and below (just like Pascal's triangle):

1 P
1 4 Q
1 3 6 R
1 2 3 4 S
1 1 1 1 1 T

Along the last diagonal: $1+4+6+4+1=16$ (exactly the total number of sequences), and the number at R is 6.

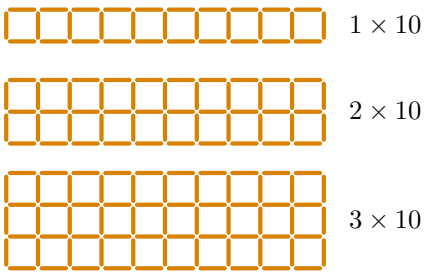
Common mistake. The five dots P, Q, R, S, T are *not* equally likely! Do not answer "one of 5 dots, so the probability is $\frac{1}{5}$". The middle dot (R) can be reached in the most ways; each of the two end dots (P and T) can be reached in only one way.

Check: The probabilities of stopping at P, Q, R, S, T are $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$; they add up to $\frac{16}{16}=1$, as they should.
The value $\frac{3}{8}=\frac{6}{16}$ is option **(B)**.

Answer: (B) $\frac{3}{8}$

Problem 17

Identical toothpicks are used to build rectangular grids of squares, as shown below. For example, the 1×10 grid (1 row of 10 squares) uses 31 toothpicks in total. How many toothpicks are needed for a 43×10 grid (43 rows, each with 10 squares)?



(A) 913 (B) 860 (C) 871 (D) 903 (E) 946

Detailed solution

Understanding the problem. The 43×10 grid has 43 rows of squares, 10 squares in each row. The toothpicks lie in two directions: *horizontal* and *vertical*. We count each kind separately and add (addition principle).

Step 1. Check the method on the 1×10 grid

Horizontal toothpicks: the grid has 2 horizontal lines (top edge and bottom edge), each made of 10 toothpicks: $2 \times 10 = 20$ toothpicks. *Vertical toothpicks:* there are 11 vertical lines (10 squares need 11 walls), each made of 1 toothpick: $11 \times 1 = 11$ toothpicks. Total: $20 + 11 = 31$, exactly as stated.

Recall. A row of 10 squares needs $10 + 1 = 11$ vertical walls (like the fence-post problem: 10 gaps need 11 posts). In the same way, 43 rows of squares have $43 + 1 = 44$ horizontal lines.

Step 2. Count the horizontal toothpicks of the 43×10 grid

There are $43 + 1 = 44$ horizontal lines, each 10 toothpicks long: $44 \times 10 = 440$ toothpicks.

Step 3. Count the vertical toothpicks of the 43×10 grid

There are $10+1 = 11$ vertical lines, each 43 toothpicks long (one toothpick per row): $11 \times 43 = 473$ toothpicks.

Step 4. Add

$440 + 473 = 913$ toothpicks.

Step 5. Second method: add one row at a time

The 1×10 grid has 31 toothpicks. Each time a new row is placed on top, we need 10 more horizontal toothpicks (the top edge of the new row) and 11 vertical ones, that is 21 toothpicks (the bottom edge is already there). Going from 1 row to 43 rows means adding 42 rows:

$$31 + 42 \times 21 = 31 + 882 = 913.$$

Common mistake. Do not compute $43 \times 31 = 1333$: when rows are stacked, two neighbouring rows *share* a line of 10 horizontal toothpicks, so multiplying like that counts those toothpicks twice.

Check: By the formula $(\text{rows} + 1) \times 10 + 11 \times \text{rows}$: the 2×10 grid needs $3 \times 10 + 11 \times 2 = 52$ toothpicks and the 3×10 grid needs $4 \times 10 + 11 \times 3 = 73$; counting directly in the picture also gives 52 and 73. Both methods give the same result, 913.
So the answer is **(A)**.

Answer: (A) 913 toothpicks

Problem 18

In the cafeteria at Braydon's school, the meats are beef and chicken, and the fruits are apple, pear and banana. Braydon is given a random lunch consisting of one meat and one fruit. The probability that Braydon's lunch includes a banana is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{5}$ (E) $\frac{3}{5}$

Detailed solution

Understanding the problem. A lunch = (one meat, one fruit). The lunch is chosen at random, so all possible lunches are equally likely. We need the probability that the lunch includes a banana.

Step 1. List all possible lunches

By the multiplication principle there are $2 \times 3 = 6$ lunches. We make a table (rows: meat, columns: fruit):

	Apple	Pear	Banana
Beef	Beef–Apple	Beef–Pear	Beef–Banana
Chicken	Chicken–Apple	Chicken–Pear	Chicken–Banana

Step 2. Count the lunches with a banana

The "Banana" column has 2 lunches: Beef–Banana and Chicken–Banana.

Step 3. Compute the probability

$$P(\text{banana}) = \frac{2}{6} = \frac{1}{3}.$$

Recall. To simplify a fraction, divide the numerator and the denominator by their greatest common divisor: $\frac{2}{6} = \frac{2 \div 2}{6 \div 2} = \frac{1}{3}$.

ThinkAlgo Tip. The meat has no effect on the fruit. A quicker way to think: the fruit is one of 3 equally likely kinds, and banana is 1 of the 3 kinds, so the probability is $\frac{1}{3}$.

Common mistake. Do not add "2 meats + 3 fruits = 5" and answer $\frac{1}{5}$: the number of lunches is $2 \times 3 = 6$, not $2 + 3 = 5$.

Check: The probabilities of getting an apple, a pear and a banana are all $\frac{1}{3}$; they add up to $\frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$.

So the answer is **(A)**.

Answer: (A) $\frac{1}{3}$

Problem 19

A special six-sided die is made, with one number written on each face (numbers may repeat). When this die is rolled, the probability of getting a multiple of 3 is $\frac{1}{2}$, and the probability of getting an even number is $\frac{1}{3}$. The six numbers on the faces could be

(A) $1, 2, 3, 5, 5, 6$ (B) $1, 2, 3, 3, 5, 6$ (C) $1, 2, 3, 4, 6, 6$

(D) $1, 2, 3, 3, 4, 6$ (E) $2, 3, 3, 3, 5, 6$

Detailed solution

Understanding the problem. The die has 6 equally likely faces. From the two given probabilities we work out the *number of faces* showing a multiple of 3 and the number of faces showing an even number, then check each option.

Step 1. Turn probabilities into numbers of faces

Recall. With 6 equally likely faces: $P = \dfrac{\text{number of favourable faces}}{6}$, so the number of favourable faces $= P \times 6$.

Faces with a multiple of 3: $\frac{1}{2} \times 6 = 3$ faces. Even faces: $\frac{1}{3} \times 6 = 2$ faces.
So we need: **exactly 3 faces that are multiples of 3** and **exactly 2 even faces**.

Step 2. Check each option

Among the numbers 1 to 6: the multiples of 3 are 3 and 6; the even numbers are 2, 4, 6. Note that 6 is both even and a multiple of 3, so it must be counted in *both* columns.

Option	Faces	Multiples of 3	Even faces	OK?
(A)	1, 2, 3, 5, 5, 6	3, 6 (2 faces)	2, 6 (2 faces)	No
(B)	1, 2, 3, 3, 5, 6	3, 3, 6 (3 faces)	2, 6 (2 faces)	Yes
(C)	1, 2, 3, 4, 6, 6	3, 6, 6 (3 faces)	2, 4, 6, 6 (4 faces)	No
(D)	1, 2, 3, 3, 4, 6	3, 3, 6 (3 faces)	2, 4, 6 (3 faces)	No
(E)	2, 3, 3, 3, 5, 6	3, 3, 3, 6 (4 faces)	2, 6 (2 faces)	No

Common mistake. Forgetting that 6 is even, you would count only 2 even faces for (D) and wrongly choose (D).

Check: For (B): $P(\text{multiple of 3}) = \frac{3}{6} = \frac{1}{2}$, $P(\text{even}) = \frac{2}{6} = \frac{1}{3}$. Only option **(B)** works.

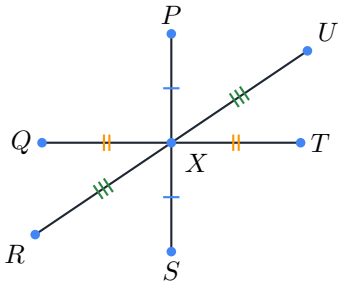
Answer: (B) 1, 2, 3, 3, 5, 6

Problem 20

Three bus routes run along the roads shown in the figure, all passing through stop X (segments with the same number of small marks are equal: $PX = XS$, $QX = XT$, $RX = XU$).

- Bus A runs the route $P \to X \to S \to X \to P$; each round trip takes 12 minutes.
- Bus B runs the route $Q \to X \to T \to X \to Q$; each round trip takes 20 minutes.
- Bus C runs the route $R \to X \to U \to X \to R$; each round trip takes 28 minutes.

At 1:00 p.m., buses A, B, C leave P, Q, R respectively. Each bus travels at a constant speed, turns around instantly at both ends of its route and runs without stopping until 11:00 p.m. Between 5:00 p.m. and 10:00 p.m., at how many moments are two or more buses at stop X at the same time?



(A) 18 (B) 19 (C) 20 (D) 21 (E) 22

Detailed solution

Understanding the problem. We measure time as *the number of minutes after 1:00 p.m.* and call this number t. We need the values of t at which at least two buses are at X, with t between 5:00 p.m. and 10:00 p.m.

Step 1. Convert the time window into minutes

5:00 p.m. is 4 hours after 1:00 p.m.: $t = 4 \times 60 = 240$. 10:00 p.m. is 9 hours after 1:00 p.m.: $t = 9 \times 60 = 540$. So we look at $240 \le t \le 540$.

Step 2. When is bus A at X?

One round trip of bus A consists of 4 equal legs: $P \to X$, $X \to S$, $S \to X$, $X \to P$ (because $PX = XS$). The bus moves at constant speed, so each leg takes $12 \div 4 = 3$ minutes. Bus A reaches X at the end of the first leg and at the end of the third leg of every round trip, that is at

$$t = 3,\ 9,\ 15,\ 21,\ 27,\ \ldots$$

These are exactly the **odd multiples of** 3 (3×1, 3×3, 3×5, ...). At the end of the second and fourth legs the bus is at S or P, not at X.

Step 3. The same for buses B and C

Bus B: each leg takes $20 \div 4 = 5$ minutes, so it is at X when $t = 5, 15, 25, 35, \ldots$ – the **odd multiples of** 5.

Bus C: each leg takes $28 \div 4 = 7$ minutes, so it is at X when $t = 7, 21, 35, 49, \ldots$ – the **odd multiples of** 7.

Step 4. When do two buses meet at X?

Recall. A number divisible by both 3 and 5 is divisible by $\text{lcm}(3,5) = 15$. Similarly: divisible by both 3 and 7 $\Leftrightarrow$ divisible by 21; divisible by both 5 and 7 $\Leftrightarrow$ divisible by 35.

Buses A and B are both at X when t is an odd multiple of 3 and also an odd multiple of 5, that is when t is an **odd number divisible by** 15 (an odd multiple of 15). Similarly: A and C meet when t is an odd multiple of 21; B and C meet when t is an odd multiple of 35.

Why is it true? An odd multiple of 15 is an odd number divisible by 15, so it is an odd number divisible by 3 (an odd multiple of 3) and an odd number divisible by 5 (an odd multiple of 5). Conversely, a number that is an odd multiple of 3 and an odd multiple of 5 is odd and divisible by 15. □

Step 5. List them for $240 \le t \le 540$

Two buses	Values of t (odd multiples)	Count
A and B (15)	$255, 285, 315, 345, 375, 405, 435, 465, 495, 525$	10
A and C (21)	$273, 315, 357, 399, 441, 483, 525$	7
B and C (35)	$245, 315, 385, 455, 525$	5

(For example, in the first row: $15 \times 17 = 255$ is the first odd multiple greater than 240; after that we keep adding 30, because two consecutive odd multiples of 15 are $2 \times 15 = 30$ apart.)

Step 6. Remove the moments counted more than once

Adding the three rows gives $10 + 7 + 5 = 22$, but the moments 315 and 525 appear in *all three rows* (they are odd multiples of $3 \times 5 \times 7 = 105$: at those moments all three buses are at X). Each of these moments is counted 3 times but should be counted only once, so we subtract $2 + 2 = 4$:

$$22 - 4 = 18 \text{ moments.}$$

Common mistake. The question asks for the number of *moments*, not the number of "pairs of buses meeting". When all three buses are at X together, that is just *one* moment. Adding $10 + 7 + 5 = 22$ and choosing (E) is wrong.

Check: In clock times: 5:05, 5:15, 5:33, 5:45, 6:15, 6:45, 6:57, 7:15, 7:25, 7:39, 7:45, 8:15, 8:21, 8:35, 8:45, 9:03, 9:15, 9:45 p.m. – exactly 18 moments, all between 5:00 p.m. and 10:00 p.m. So the answer is **(A)**.

Answer: (A) 18

Problem 21

Each face of a cube is painted with exactly one colour. What is the smallest number of colours needed to paint the cube so that any two faces sharing an edge have different colours?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Detailed solution

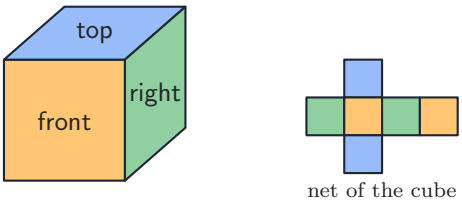
Understanding the problem. Two faces "sharing an edge" are adjacent faces. Two opposite faces (such as the top and the bottom) have no common edge, so they may have the same colour. We must (1) show a colouring that uses few colours, and (2) prove that fewer colours are impossible.

Step 1. Two colours are not enough

Why is it true? Look at any corner of the cube, for example the top–front–right corner. Three faces meet there: the top, the front and the right face. Any two of these three faces share an edge, so the three faces must have three different colours. Hence at least 3 colours are needed. □

Step 2. Three colours are enough

Paint *each pair of opposite faces the same colour*: top and bottom blue, front and back orange, left and right green. Each face is adjacent to 4 faces, which are exactly the four faces *not* opposite to it, and these four faces carry the other two colours. So no two adjacent faces have the same colour.



Step 3. Conclusion

At least 3 colours are needed (step 1) and 3 colours are enough (step 2), so the smallest number of colours is 3.

Common mistake. Finding one colouring with 3 colours is *not enough* to claim that 3 is the smallest; you must explain why 2 colours fail (the three faces at a corner). On the other hand, thinking "6 faces need 6 colours" forgets that opposite faces may share a colour.

Check: On the net, each square shares edges only with squares of other colours (when folded, two squares of the same colour become opposite faces).
So the answer is **(B)**.

Answer: (B) 3 colours

Problem 22

How many different pairs (m, n) can be formed from the integers $1, 2, 3, \ldots, 20$ such that $m < n$ and $m + n$ is even?

(A) 55 (B) 90 (C) 140 (D) 110 (E) 50

Detailed solution

Understanding the problem. We choose two different numbers out of 20; because $m < n$, each pair of two numbers is counted *only once* (the smaller one is m, the larger one is n). Condition: the sum $m + n$ is even.

Step 1. When is the sum of two numbers even?

> **Recall.** even + even = even; odd + odd = even; even + odd = odd.

So $m + n$ is even exactly when m and n have **the same parity** (both even or both odd).

Step 2. Split the 20 numbers into two groups

Even group: $2, 4, 6, \ldots, 20$ – 10 numbers. Odd group: $1, 3, 5, \ldots, 19$ – 10 numbers.

Step 3. Count the pairs in the even group

The first number can be chosen in 10 ways and the second (different from the first) in 9 ways: $10 \times 9 = 90$ *ordered* choices. But each pair, for example $\{2, 4\}$, is counted twice (2 then 4, or 4 then 2), while the condition $m < n$ allows only the order $(2, 4)$. So the number of pairs is $\dfrac{10 \times 9}{2} = 45$.

> **Recall.** The number of ways to choose 2 objects (order does not matter) from k objects is $\dfrac{k \times (k-1)}{2}$.

Step 4. Count the pairs in the odd group and add

In the same way the odd group gives 45 pairs. A pair cannot be in both groups, so by the addition principle: $45 + 45 = 90$ pairs.

> **Common mistake.** Forgetting to divide by 2 gives $90 + 90 = 180$ (not an option); forgetting the odd group gives 45. Always ask yourself: "Did I count a pair twice? Did I miss a case?"

Check: Count using the complement: the total number of pairs with $m < n$ is $\dfrac{20 \times 19}{2} = 190$. The number of pairs with different parity (one even, one odd) is $10 \times 10 = 100$. So the number of pairs with the same parity is $190 - 100 = 90$. It matches!
So the answer is **(B)**.

Answer: (B) 90 pairs

Problem 23 *

Two different two-digit positive integers are called a *reversal pair* if one is obtained from the other by swapping its two digits. For example, 52 and 25 form a reversal pair. The number 2015 has a special property: it is the product of three different prime numbers, two of which form a reversal pair ($2015 = 5 \times 13 \times 31$). Including 2015, how many positive integers less than 10 000 have this same property?

(A) 18 (B) 14 (C) 20 (D) 17 (E) 19

Detailed solution

Understanding the problem. We must count the numbers $N < 10\,000$ of the form $N = p \times q \times r$, where p, q, r are three different primes and two of them (call them p and q) form a two-digit reversal pair. Plan: find all reversal pairs made of two primes, then for each pair find the suitable third primes.

Step 1. Find the reversal pairs in which both numbers are prime

Recall. A prime number is a natural number greater than 1 whose only divisors are 1 and itself. The two-digit primes are: 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

ThinkAlgo Tip. The units digit of the reversal of p is the tens digit of p. A two-digit prime must end in $1, 3, 7$ or 9, so we only need to look at primes starting with $1, 3, 7, 9$: $11, 13, 17, 19, 31, 37, 71, 73, 79, 97$.

Check the reversal of each one:

p	11	13	17	19	31	37	71	73	79	97
reversal	11	31	71	91	13	73	17	37	97	79
prime?	same	yes	yes	no	yes	yes	yes	yes	yes	yes

($91 = 7 \times 13$ is not prime; 11 reversed is still 11, so it does not give two *different* numbers.) So there are exactly 4 pairs: $\{13, 31\}$, $\{17, 71\}$, $\{37, 73\}$, $\{79, 97\}$.

Step 2. Compute the product of each pair

$13 \times 31 = 403$; $17 \times 71 = 1207$; $37 \times 73 = 2701$; $79 \times 97 = 7663$.

Step 3. Find the third prime r for each pair

We need r prime, different from the two numbers of the pair, and the product $< 10\,000$.

- Pair $\{13, 31\}$: we need $403 \times r < 10\,000$. Since $403 \times 24 = 9672$ and $403 \times 25 = 10\,075$, we need $r \leq 23$. The primes ≤ 23 are $2, 3, 5, 7, 11, 13, 17, 19, 23$; remove 13 (already in the pair): 8 primes remain, giving 806; 1209; 2015; 2821; 4433; 6851; 7657; 9269.
- Pair $\{17, 71\}$: we need $1207 \times r < 10\,000$; since $1207 \times 8 = 9656$ and $1207 \times 9 = 10\,863$, $r \leq 8$: $r \in \{2, 3, 5, 7\}$, giving 2414; 3621; 6035; 8449 (4 numbers).
- Pair $\{37, 73\}$: $2701 \times 3 = 8103 < 10\,000$ and $2701 \times 4 > 10\,000$, so $r \in \{2, 3\}$: 5402; 8103 (2 numbers).
- Pair $\{79, 97\}$: even $7663 \times 2 = 15\,326 > 10\,000$, so there are none.

Step 4. Check that no number is counted twice, and add

Why is it true? Every number has only one prime factorisation. If a number appeared in two lists, then among its three prime factors there would be two different reversal pairs; these two pairs would have to share a prime. But the table in step 1 shows that each prime belongs to exactly one pair. So no number is counted twice. □

In total: $8 + 4 + 2 + 0 = 14$ numbers, which is option **(B)**.

Common mistake. Two common mistakes: (1) forgetting that the third prime must be *different* from the two primes of the pair, and so also counting $403 \times 13 = 5239 = 13 \times 13 \times 31$ (wrong, since 13 is repeated); (2) forgetting the prime 2.

Check: $2015 = 403 \times 5$ is in the list; the largest number, $9269 = 13 \times 31 \times 23$, is still less than 10 000.

Answer: (B) 14

Problem 24

The figure is made of 8 identical small parallelograms. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?



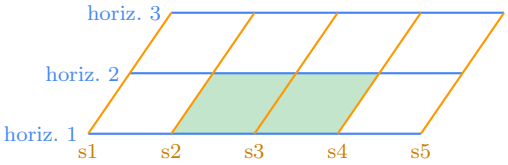
(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Detailed solution

Understanding the problem. The figure has 2 rows, each with 4 small parallelograms. Every parallelogram in the figure is formed from the segments already drawn: two of its sides lie on *horizontal lines* and two lie on *slanted lines*.

Step 1. Count the lines in the figure

There are 3 horizontal lines (bottom, middle, top) and 5 slanted lines (parallel to each other).



For example, the green parallelogram is determined by horizontal lines 1 and 2 and slanted lines s2 and s4.

Step 2. Each parallelogram corresponds to a choice of 2 horizontal lines and 2 slanted lines

Choosing 2 of the 3 horizontal lines as bottom and top sides, and 2 of the 5 slanted lines as left and right sides, gives exactly one parallelogram; conversely, every parallelogram in the figure comes from exactly one such choice.

Recall. The number of ways to choose 2 lines out of k lines is $\dfrac{k \times (k-1)}{2}$.

Choosing 2 horizontal lines: $\dfrac{3 \times 2}{2} = 3$ ways. Choosing 2 slanted lines: $\dfrac{5 \times 4}{2} = 10$ ways.

Step 3. Apply the multiplication principle
Number of parallelograms: $3 \times 10 = 30$.

Step 4. Check by counting by size
Call a parallelogram "height a, width b" if it is made of a rows and b columns of small ones:

	width 1	width 2	width 3	width 4	Total
height 1 (2 positions in rows)	$2 \times 4 = 8$	$2 \times 3 = 6$	$2 \times 2 = 4$	$2 \times 1 = 2$	20
height 2 (1 position in rows)	4	3	2	1	10

Common mistake. Do not just count the 8 small ones plus a few "obvious" big ones. Count systematically: by size (table above) or by choosing lines.

Check: The table gives $20 + 10 = 30$, the same as 3×10.
So the answer is **(B)**.

Answer: (B) 30 parallelograms

Problem 25

A jar contains 2 red marbles and 2 blue marbles and no other marbles. Angie randomly takes 2 marbles out of the jar at the same time. If the two marbles have the same colour, she throws one away and puts the other back into the jar. If they have different colours, she throws away the red marble and puts the blue marble back into the jar. Angie does this three times in total. The probability that the last marble left in the jar is red is

(A) $\frac{1}{2}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$ (E) 0

Detailed solution

Understanding the problem. Each draw removes exactly 1 marble from the jar. After 3 draws, 1 of the 4 marbles is left. We follow the *state* of the jar, written (number of red, number of blue), starting from $(2\text{R}, 2\text{B})$.

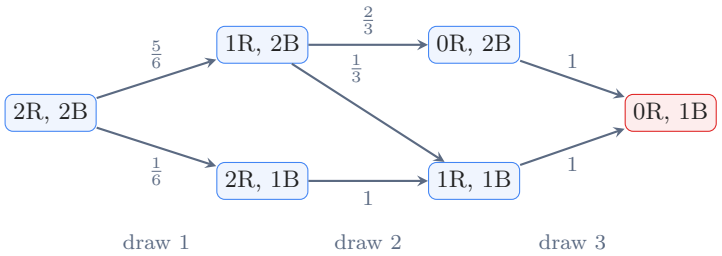
Step 1. First draw, from $(2\mathbf{R}, 2\mathbf{B})$
Call the marbles R_1, R_2, B_1, B_2. There are 6 equally likely ways to take 2 marbles: R_1R_2; B_1B_2; R_1B_1; R_1B_2; R_2B_1; R_2B_2.

- Two reds (1 way): throw away 1 red $\to (1\text{R}, 2\text{B})$.
- Two blues (1 way): throw away 1 blue $\to (2\text{R}, 1\text{B})$.
- Different colours (4 ways): throw away the red $\to (1\text{R}, 2\text{B})$.

So after draw 1: state $(1\text{R}, 2\text{B})$ with probability $\frac{5}{6}$, state $(2\text{R}, 1\text{B})$ with probability $\frac{1}{6}$.

Step 2. Second draw
From $(1\text{R}, 2\text{B})$: 3 ways to draw; 2 ways give different colours $\to (0\text{R}, 2\text{B})$; 1 way gives two blues $\to (1\text{R}, 1\text{B})$.
From $(2\text{R}, 1\text{B})$: 3 ways to draw; 1 way gives two reds $\to (1\text{R}, 1\text{B})$; 2 ways give different colours $\to (1\text{R}, 1\text{B})$. So from $(2\text{R}, 1\text{B})$ we certainly go to $(1\text{R}, 1\text{B})$.

Step 3. Third draw
From $(0\text{R}, 2\text{B})$: two blues are drawn, one is thrown away $\to (0\text{R}, 1\text{B})$.
From $(1\text{R}, 1\text{B})$: different colours are drawn, the red is thrown away $\to (0\text{R}, 1\text{B})$.



Every branch ends at $(0\text{R}, 1\text{B})$: the last marble is **always** blue.

Step 4. Why – without drawing the tree

Why is it true? Look at the number of blue marbles. Whenever at least one of the two marbles drawn is blue, Angie always puts a blue marble back into the jar (two blues: she puts one back; different colours: she puts the blue back). The number of blue marbles decreases only when two blues are drawn, and then the one that was put back is still there. So the jar *always contains at least one blue marble*. When only one marble is left, it must be blue. □

Therefore the event "the last marble is red" is an *impossible event*, with probability 0.

Recall. An impossible event (one that never happens) has probability 0; a certain event has probability 1.

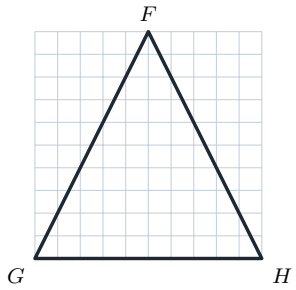
Common mistake. Seeing "2 red, 2 blue" and immediately guessing $\frac{1}{2}$ is a trap: the throwing-away rule is *not symmetric* in the two colours (with different colours, the red one is always thrown away).

Check: Total probability of reaching the final state $(0\text{R}, 1\text{B})$: $\frac{5}{6}\cdot\frac{2}{3}+\frac{5}{6}\cdot\frac{1}{3}+\frac{1}{6}\cdot 1 = \frac{10}{18}+\frac{5}{18}+\frac{3}{18} = 1$.
So the answer is **(E)**.

Answer: (E) 0

Problem 26

On the 10×10 grid of squares shown, triangle FGH is drawn. Point P can be any of the 41 points where grid lines cross that lie *inside* (not on the sides of) triangle FGH. For each position of P, exactly three triangles are formed: FPG, GPH, HPF. Among all $41 \times 3 = 123$ of these triangles, how many have an area equal to exactly half the area of triangle FGH?



(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

Detailed solution

Understanding the problem. Take the side of one small square as the unit and put G at the bottom-left corner. Then G and H are the two bottom corners ($GH = 10$) and F is the midpoint of the top side. Each grid point is named by two numbers (squares to the right, squares up); for example $G = (0, 0)$, $H = (10, 0)$, $F = (5, 10)$.

Step 1. Area of triangle FGH and half of it

> **Recall.** Area of a triangle $= \dfrac{1}{2} \times$ base $\times$ corresponding height.

Area$(FGH) = \frac{1}{2} \times 10 \times 10 = 50$. Half of it is 25.

Step 2. Triangle GPH: the easiest

Triangle GPH has base $GH = 10$, and its height is the distance from P down to GH, that is the "height" h of P (the number of squares above the bottom). Area$(GPH) = \frac{1}{2} \times 10 \times h = 5h$. We need $5h = 25$, so $h = 5$: P lies on the 5th horizontal grid line. At height 5 (half the height) the triangle is half as wide as its base, from 2.5 to 7.5. The grid points strictly *inside* on that segment are $(3, 5), (4, 5), (5, 5), (6, 5), (7, 5)$: **5** points.

Step 3. Triangle FPG: using a parallel line

> **Recall.** Triangles with a common base whose third vertices lie on a line parallel to the base have the same height, so they have the same area.

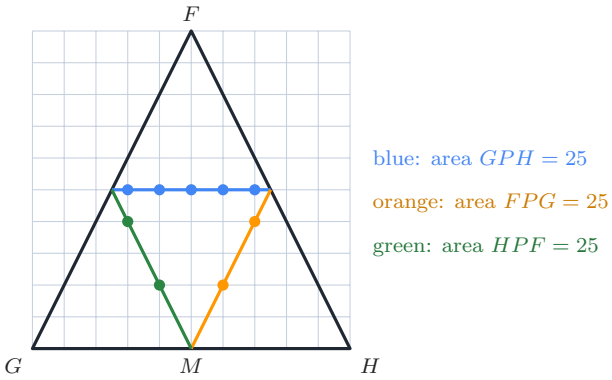
Let $M = (5, 0)$ be the midpoint of GH. Triangle FMG has base $GM = 5$ and height 10, so Area$(FMG) = \frac{1}{2} \times 5 \times 10 = 25$. So every point P on the line through M parallel to FG gives Area$(FPG) = 25$.

Why is it true? If P is farther from FG than that line, the height of triangle FPG (on base FG) is larger, so the area is more than 25; if P is closer to FG, the area is less than 25. So Area$(FPG) = 25$ *only when* P lies on that line. □

Side FG goes from $(0, 0)$ to $(5, 10)$: for every 1 square to the right it goes up 2 squares. The parallel line through $M = (5, 0)$ does the same: it passes through $(5, 0)$, $(6, 2)$, $(7, 4)$ and then meets side FH at (7.5, 5) (the midpoint of FH). The grid points *inside* the triangle are $(6, 2)$ and $(7, 4)$: **2** points. (M lies on side GH, so it is excluded.)

Step 4. Triangle HPF: by symmetry

The figure is symmetric about the vertical line through F, so similarly there are **2** points: $(4, 2)$ and $(3, 4)$.



Step 5. Add

Number of triangles with area 25: $5 + 2 + 2 = 9$. (One point P cannot give two triangles of area 25, because then the third triangle would have area $50 - 25 - 25 = 0$, while P lies strictly inside. So no triangle is missed or counted twice.)

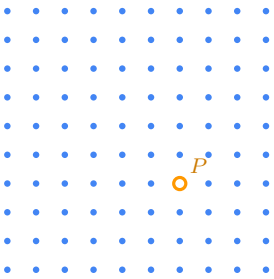
Check: For $P = (6, 2)$: triangle FPG fits in the rectangle from $(0, 0)$ to $(6, 10)$, of area 60. Cut off three right triangles: $\frac{1}{2} \times 6 \times 2 = 6$, $\frac{1}{2} \times 1 \times 8 = 4$ and $\frac{1}{2} \times 5 \times 10 = 25$. What remains is $60 - 6 - 4 - 25 = 25$. Correct!

So the answer is **(E)**.

Answer: (E) 9 triangles

Problem 27

A 10×10 square grid is made of 100 points, as shown. Point P is marked. One of the other 99 points is chosen at random to be Q. The probability that segment PQ is horizontal or vertical is



(A) $\frac{2}{11}$ (B) $\frac{1}{5}$ (C) $\frac{1}{10}$ (D) $\frac{4}{25}$ (E) $\frac{5}{33}$

Detailed solution

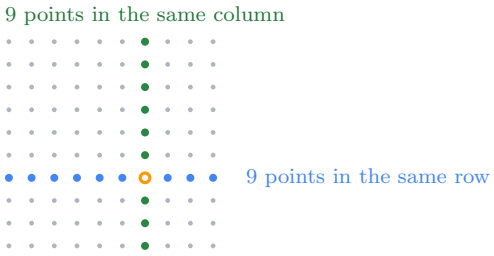
Understanding the problem. Q is chosen at random among 99 points (not counting P), all equally likely. Segment PQ is horizontal when Q is in the same row as P, and vertical when Q is in the same column as P.

Step 1. Number of possible outcomes

There are $100 - 1 = 99$ ways to choose Q.

Step 2. Number of favourable outcomes

The row containing P has 10 points; without P there are 9. The column containing P has 10 points; without P there are 9. A point other than P cannot be in both the same row and the same column as P (the only such point is P itself), so no point is counted twice: $9 + 9 = 18$ points in total.



Step 3. Compute the probability

$$P = \frac{18}{99} = \frac{18 \div 9}{99 \div 9} = \frac{2}{11}.$$

Common mistake. The denominator is 99, not 100, because Q must be different from P. Using 100 gives $\frac{18}{100}$, which is not an option. Also do not count $10 + 10 = 20$ points (that counts P twice).

ThinkAlgo Tip. The position of P in the grid does not matter: every point has exactly 9 points in its row and 9 points in its column.

Check: $\frac{2}{11} \times 99 = 18$, exactly the number of favourable points.
So the answer is **(A)**.

Answer: (A) $\frac{2}{11}$

Problem 28

The sum of all the digits of the integers from 98 to 101 is

$$9 + 8 + 9 + 9 + 1 + 0 + 0 + 1 + 0 + 1 = 38.$$

What is the sum of all the digits of the integers from 1 to 2008?

(A) 30 054 (B) 27 018 (C) 28 036 (D) 30 036 (E) 28 054

Detailed solution

Understanding the problem. Writing out 2008 numbers and adding is far too long. We split them into three blocks: 1 to 999; 1000 to 1999; 2000 to 2008, and find the digit sum of each block by counting *how many times each digit appears*.

Step 1. Block 1 to 999: write leading zeros to get three digits
Write the numbers from 0 to 999 with three digits: $000, 001, 002, \ldots, 998, 999$. Adding leading zeros and the number 000 *does not change* the digit sum. There are 1000 numbers, that is 3000 digits.

Why is it true? In the units place the digits repeat in the cycle $0, 1, 2, \ldots, 9$; 1000 numbers contain 100 cycles, so each digit 0–9 appears exactly 100 times. In the tens place each digit appears 10 times in a row (00–09 have tens digit 0, ...) and this repeats 10 times: again 100 times. In the hundreds place: 000–099 have hundreds digit 0 (100 times), ..., 900–999 have hundreds digit 9 (100 times). So in each place, each digit appears 100 times. □

Hence each digit $0, 1, \ldots, 9$ appears $3 \times 100 = 300$ times. The digit sum is

$$300 \times (0 + 1 + 2 + \cdots + 9) = 300 \times 45 = 13\,500.$$

Recall. $0 + 1 + 2 + \cdots + 9 = \dfrac{9 \times 10}{2} = 45$ (pair them up: $0 + 9 = 1 + 8 = \cdots = 4 + 5 = 9$, and there are 5 pairs).

Step 2. Block 1000 to 1999
Each number has the form $1\,abc$ where abc runs from 000 to 999. The last three digits give 13 500 (as in step 1). The thousands digit is 1, appearing 1000 times, which adds 1000. Total: $13\,500 + 1000 = 14\,500$.

Step 3. Block 2000 to 2008
There are 9 numbers. The thousands digit 2 contributes $9 \times 2 = 18$; the two middle digits are 0; the units digits are $0, 1, \ldots, 8$, contributing $0 + 1 + \cdots + 8 = 36$. Total: $18 + 36 = 54$.

Step 4. Add the three blocks

$$13\,500 + 14\,500 + 54 = 28\,054.$$

Common mistake. Do not forget the last small block 2000–2008 (forgetting it gives 28 000), and do not forget the thousands digit 1 in the second block (forgetting it gives 27 054).

Check: Test the method on a small case, 1 to 99: each digit appears $2 \times 10 = 20$ times, sum $20 \times 45 = 900$. Directly: the units place contributes $10 \times 45 = 450$ and the tens place contributes $10 \times (1 + 2 + \cdots + 9) = 450$, total 900. It matches.
So the answer is **(E)**.

Answer: (E) 28 054

Problem 29

Two fair dice are rolled, one red and one blue (each has six faces numbered 1 to 6). The probability that the number on the red die is greater than the number on the blue die is

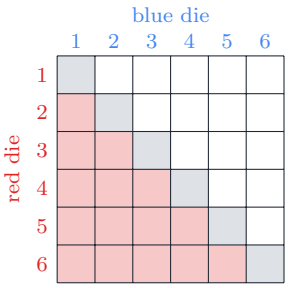
(A) $\frac{18}{36}$ (B) $\frac{25}{36}$ (C) $\frac{15}{36}$ (D) $\frac{12}{36}$ (E) $\frac{17}{36}$

Detailed solution

Understanding the problem. An outcome is a pair (number on the red die, number on the blue die). The dice have different colours, so $(5, 2)$ and $(2, 5)$ are different outcomes.

Step 1. Number of possible outcomes
By the multiplication principle: $6 \times 6 = 36$ equally likely outcomes.

Step 2. Make a table and mark the cells "red > blue"



Red cells: the red number is greater than the blue number; grey cells: the two numbers are equal. Count row by row (red number): red = 1: 0 cells; red = 2: 1 cell; red = 3: 2 cells; red = 4: 3 cells; red = 5: 4 cells; red = 6: 5 cells. Total: $0 + 1 + 2 + 3 + 4 + 5 = 15$ favourable outcomes.

Step 3. Compute the probability
$P(\text{red} > \text{blue}) = \frac{15}{36}$.

Step 4. A quick way: use symmetry
There are 6 outcomes where the two numbers are equal (the diagonal). The other $36 - 6 = 30$ outcomes split equally between "red is greater" and "blue is greater" (swap the roles of the dice), so each kind has $30 \div 2 = 15$ outcomes.

Common mistake. The answer $\frac{18}{36} = \frac{1}{2}$ is a trap: it forgets the cases where the two numbers are equal. "Red is greater" and "blue is greater" are not the only possibilities.

Check: $P(\text{red} > \text{blue}) + P(\text{equal}) + P(\text{blue} > \text{red}) = \frac{15}{36} + \frac{6}{36} + \frac{15}{36} = 1$.
So the answer is **(C)**.

Answer: (C) $\frac{15}{36}$

Problem 30

Two fair dice are rolled, one red and one blue. The probability that the *product* of the two numbers shown is even is

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) $\frac{2}{3}$ (E) $\frac{11}{36}$

Detailed solution

Understanding the problem. As in problem 29, there are $6 \times 6 = 36$ equally likely outcomes. Counting the even products directly is rather long, so we count the *complement*: the outcomes with an odd product.

Step 1. When is the product of two numbers odd?

Recall. odd × odd = odd; even × (any number) = even. So the product of two numbers is odd exactly when *both* numbers are odd.

Step 2. Count the outcomes with an odd product

Odd numbers on each die: $1, 3, 5$ (3 numbers). Both dice odd: $3 \times 3 = 9$ outcomes.

Step 3. Count the outcomes with an even product using the complement

Recall. *Complement*: number of favourable outcomes = number of possible outcomes − number of unfavourable outcomes. Likewise, $P(\text{event}) = 1 - P(\text{event does not happen})$.

Number of outcomes with an even product: $36 - 9 = 27$.

Step 4. Compute the probability

$P(\text{even product}) = \frac{27}{36} = \frac{27 \div 9}{36 \div 9} = \frac{3}{4}$ – this is option **(B)**.

Common mistake. Do not think "even and odd are half and half" and choose $\frac{1}{2}$: just *one* even number makes the product even.

Check: Direct count (red even; or red odd and blue even): $3 \times 6 + 3 \times 3 = 18 + 9 = 27$, it matches.

Answer: (B) $\frac{3}{4}$

Extension and challenge

Problem 31

A pencil case holds 6 red pens, 9 blue pens and 5 black pens, identical except for the colour of the ink. One pen is taken at random.

a) Find the probability of taking a blue pen.

b) Find the probability of taking a pen that is *not* black.

c) How many more black pens must be put into the case so that the probability of taking a black pen is $\frac{1}{2}$?

Detailed solution

Understanding the problem. Each pen is one equally likely outcome. Parts a and b are direct probability calculations; part c is a "backwards" problem: when black pens are added, *both* the number of black pens *and* the total number of pens increase.

Step 1. Total number of pens

$6 + 9 + 5 = 20$ pens.

Step 2. Part a

> **Recall.** $P = \dfrac{\text{number of favourable outcomes}}{\text{number of all outcomes}}$.

$P(\text{blue}) = \dfrac{9}{20}$.

Step 3. Part b – complement

> **Recall.** $P(\text{not } A) = 1 - P(A)$.

$P(\text{black}) = \dfrac{5}{20} = \dfrac{1}{4}$, so $P(\text{not black}) = 1 - \dfrac{1}{4} = \dfrac{3}{4}$.

Check by counting directly: the pens that are not black are $6 + 9 = 15$ pens, $\dfrac{15}{20} = \dfrac{3}{4}$ ✓.

Step 4. Part c – choose an unknown and form an equation

Let x be the number of black pens to be added (x is a natural number). Only black pens are added, so afterwards:

- the number of black pens is $5 + x$;
- the total number of pens is $20 + x$.

The condition in the problem: $P(\text{black}) = \dfrac{5 + x}{20 + x} = \dfrac{1}{2}$.

Step 5. Solve the equation

> **Recall.** For $b \neq 0$, $d \neq 0$: $\dfrac{a}{b} = \dfrac{c}{d}$ exactly when $a \times d = b \times c$ (property of proportions).

From $\dfrac{5+x}{20+x} = \dfrac{1}{2}$ we get $2 \times (5+x) = 1 \times (20+x)$, that is

$$10 + 2x = 20 + x \;\Rightarrow\; 2x - x = 20 - 10 \;\Rightarrow\; x = 10.$$

So 10 black pens must be added.

Remark: $P(\text{black}) = \dfrac{1}{2}$ also means that the number of black pens *equals* the number of non-black pens. There are always 15 non-black pens, so there must be 15 black pens, i.e. we add $15 - 5 = 10$ – in agreement with the solution above.

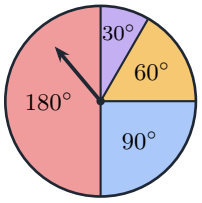
Common mistake. Many students think "to go from $\dfrac{1}{4}$ to $\dfrac{1}{2}$, double the black pens, so add 5". But when 5 black pens are added the total also rises to 25, and the probability is $\dfrac{10}{25} = \dfrac{2}{5}$, not yet $\dfrac{1}{2}$.

Check: Adding 10 black pens: 15 black pens out of $20 + 10 = 30$ pens; $P(\text{black}) = \dfrac{15}{30} = \dfrac{1}{2}$ ✓.

Answer: a) $\dfrac{9}{20}$; **b)** $\dfrac{3}{4}$; **c) add 10 black pens.**

Problem 32

A spinner is divided into four sectors with central angles 180° (red), 90° (blue), 60° (yellow) and 30° (purple), as shown. The spinner is spun once.



a) Find the probability that the arrow lands on blue.

b) Find the probability that the arrow lands on yellow or purple.

c) Leo says: "There are 4 colours, so each colour has probability $\dfrac{1}{4}$." Is Leo right or wrong? Why?

Detailed solution

Understanding the problem. The sectors have *different* sizes, so the four sectors are *not* equally likely. We cannot use "1 sector out of 4". Instead, the chance of the arrow stopping in a sector is proportional to the size of the sector, that is, to its central angle.

Recall. For a spinner with unequal sectors: $P(\text{sector}) = \dfrac{\text{central angle of the sector}}{360°}$. The whole circle has 360°.

Step 1. Check the data

$180° + 90° + 60° + 30° = 360°$ – exactly one full circle ✓.

Step 2. Part a

$P(\text{blue}) = \dfrac{90}{360} = \dfrac{1}{4}$.

Another way to think: we can divide the spinner into 12 equal small sectors of 30° each. Then

red takes 6 small sectors, blue 3, yellow 2 and purple 1; these 12 small sectors are equally likely. $P(\text{blue}) = \frac{3}{12} = \frac{1}{4}$ ✓.

Step 3. Part b

The yellow and purple sectors do not overlap, so their total angle is $60° + 30° = 90°$.

$$P(\text{yellow or purple}) = \frac{90}{360} = \frac{1}{4} \quad \left(\text{or } \frac{2+1}{12} = \frac{3}{12} = \frac{1}{4}\right).$$

Step 4. Part c

Leo is **wrong**. The formula "favourable divided by all" only works when the outcomes are *equally likely*. Here the four colours are not equally likely: the red sector is half of the spinner, so $P(\text{red}) = \frac{180}{360} = \frac{1}{2}$, while $P(\text{purple}) = \frac{30}{360} = \frac{1}{12}$ – very different.

Check: $P(\text{red}) + P(\text{blue}) + P(\text{yellow}) + P(\text{purple}) = \frac{6}{12} + \frac{3}{12} + \frac{2}{12} + \frac{1}{12} = 1$ ✓.

Note for parents. Cut a paper plate into 12 equal "slices of cake" and colour them as in the problem. Counting the slices of each colour, your child will see at once why a big sector has a big probability.

Answer: a) $\frac{1}{4}$; b) $\frac{1}{4}$; c) Wrong, because the sectors are not equal (e.g. $P(\text{red}) = \frac{1}{2}$, $P(\text{purple}) = \frac{1}{12}$).

Problem 33

The school canteen has this lunch menu: one main dish out of 3 (chicken rice, spaghetti, pizza); one drink out of 2 (apple juice, orange juice); one dessert out of 2 (ice cream, yoghurt). Each lunch consists of one main dish, one drink and one dessert.

a) How many different lunches are there?

b) Mia is given a lunch at random. Find the probability that it includes spaghetti.

c) Find the probability that Mia's lunch includes both spaghetti and orange juice.

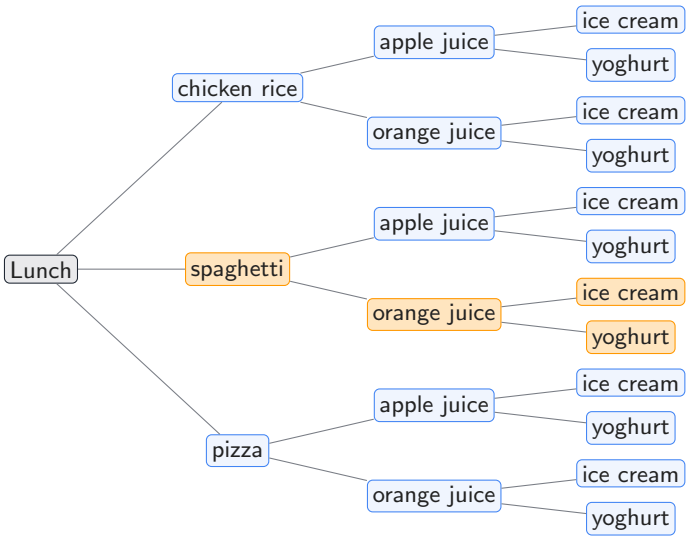
Detailed solution

Understanding the problem. Each lunch is made by three successive choices: main dish, drink, dessert. "Given at random" means all the different lunches are equally likely.

Step 1. Part a – multiplication rule

Recall. Multiplication rule: if a task has stages with m, n and p ways, there are $m \times n \times p$ ways altogether.

Number of lunches: $3 \times 2 \times 2 = 12$.



The tree has 3 big branches, each splitting into 2 branches, each of which splits into 2 again: 12 end points, exactly 12 lunches.

Step 2. **Part b**

Lunches with spaghetti: the main dish is fixed as spaghetti, drink 2 ways, dessert 2 ways: $1 \times 2 \times 2 = 4$ lunches (the middle branch of the tree). $P(\text{spaghetti}) = \frac{4}{12} = \frac{1}{3}$.

ThinkAlgo Tip. A quick way to think: the main dish is chosen at random from 3 dishes, and each dish appears in the same number of lunches, so $P(\text{spaghetti}) = \frac{1}{3}$. The drink and dessert do not matter.

Step 3. **Part c**

Main dish spaghetti, drink orange juice, only the dessert is left to choose: $1 \times 1 \times 2 = 2$ lunches (orange in the tree). $P(\text{spaghetti and orange juice}) = \frac{2}{12} = \frac{1}{6}$.

Check: List them: (spaghetti, orange juice, ice cream) and (spaghetti, orange juice, yoghurt) – exactly 2 lunches ✓.

Answer: a) 12 lunches; b) $\frac{1}{3}$; c) $\frac{1}{6}$.

Problem 34

A hall has 20 rows of seats. The first row has 12 seats, and each row behind has 2 seats more than the row just in front of it.

a) How many seats are in row 20?

b) Which row has exactly 30 seats?

c) How many seats does the hall have altogether?

Detailed solution

Understanding the problem. The numbers of seats in the rows are $12, 14, 16, \ldots$ – an *evenly spaced sequence* with step $d = 2$. We need the 20th term, the position of the term 30, and the sum of 20 terms.

Step 1. Part a – find the pattern

Row	1	2	3	4	k
Seats	12	$12+2$	$12+2\times 2$	$12+3\times 2$	$12+(k-1)\times 2$

From row 1 to row k there are $k-1$ "jumps", each adding 2 seats. Row 20: $12+19\times 2 = 12+38 = 50$ seats.

Common mistake. Not $12+20\times 2 = 52$: from row 1 to row 20 there are only 19 increases (the "off by one" error).

Step 2. Part b

We need $12+(k-1)\times 2 = 30$, so $(k-1)\times 2 = 18$, $k-1 = 9$, $k = 10$. Row 10 has 30 seats.

Step 3. Part c – sum of an evenly spaced sequence

Recall. Sum of an evenly spaced sequence $= \dfrac{(\text{first}+\text{last})\times \text{number of terms}}{2}$ (Gauss's pairing method).

Total number of seats: $\dfrac{(12+50)\times 20}{2} = \dfrac{62\times 20}{2} = 620$ seats.

Why is it true? Pair row 1 with row 20: $12+50 = 62$; row 2 with row 19: $14+48 = 62$; $\ldots$; row 10 with row 11: $30+32 = 62$. There are 10 pairs of 62 seats each, so the total is $10\times 62 = 620$. □

Check: Another way: if every row had 12 seats there would be $20\times 12 = 240$ seats; the "extra" is $0+2+4+\cdots+38 = 2\times(0+1+\cdots+19) = 2\times 190 = 380$; total $240+380 = 620$ ✓.

Answer: a) 50 seats; b) row 10; c) 620 seats.

Problem 35 *

A box contains only red and blue marbles. If one marble is drawn at random, the probability that it is red is $\frac{3}{5}$. If 5 more blue marbles are put into the box, the probability of drawing a red marble drops to $\frac{1}{2}$. How many red marbles and how many blue marbles were in the box at the start?

Detailed solution

Understanding the problem. We do not know the numbers of marbles, only two probabilities (two ratios). We choose unknowns for the numbers of red and blue marbles at the start, turn each condition into an equation, and solve the system of two equations. Note: adding blue marbles

does *not* change the number of red marbles, while the total increases by 5.

Step 1. Choose the unknowns

Let x be the number of red marbles and y the number of blue marbles at the start (x, y non-zero natural numbers). At the start the box holds $x + y$ marbles.

Step 2. Turn the first condition into an equation

$P(\text{red}) = \dfrac{x}{x+y} = \dfrac{3}{5}$. Cross-multiplying: $5x = 3(x+y)$, i.e. $5x = 3x + 3y$, or

$$2x = 3y. \qquad (1)$$

Recall. $\dfrac{a}{b} = \dfrac{c}{d}$ (with b, d non-zero) exactly when $a \times d = b \times c$.

Step 3. Turn the second condition into an equation

After adding 5 blue marbles there are still x red marbles, and the total is $x + y + 5$. So $\dfrac{x}{x+y+5} = \dfrac{1}{2}$. Cross-multiplying: $2x = x + y + 5$, or

$$x = y + 5. \qquad (2)$$

(That is: after adding, the number of red marbles *equals* the number of blue marbles.)

Step 4. Solve the system by substitution

Substitute $x = y + 5$ from (2) into (1):

$$2(y+5) = 3y \;\Rightarrow\; 2y + 10 = 3y \;\Rightarrow\; 3y - 2y = 10 \;\Rightarrow\; y = 10.$$

Then $x = y + 5 = 10 + 5 = 15$.

ThinkAlgo Tip. *Using proportions:* from (1), $\dfrac{x}{3} = \dfrac{y}{2}$; put $\dfrac{x}{3} = \dfrac{y}{2} = k$, so $x = 3k$, $y = 2k$. Substituting into (2): $3k = 2k + 5$, so $k = 5$, $x = 15$, $y = 10$.

Check: At the start: 15 red, 10 blue, $P(\text{red}) = \dfrac{15}{25} = \dfrac{3}{5}$ ✓. Adding 5 blue: 15 red, 15 blue, $P(\text{red}) = \dfrac{15}{30} = \dfrac{1}{2}$ ✓.

Common mistake. Do not try to "subtract probabilities": $\dfrac{3}{5} - \dfrac{1}{2} = \dfrac{1}{10}$ tells us nothing directly about the number of marbles. A probability is a *ratio*; go back to actual quantities.

Answer: At the start there were 15 red and 10 blue marbles.

Problem 36

Recall: a palindrome is a number that reads the same forwards and backwards (for example 3443, 7007).

a) How many four-digit palindromes are there?

b) How many of them are even?

c) A four-digit palindrome is chosen at random. Find the probability that it is even.

d) Show that every four-digit palindrome is divisible by 11.

Detailed solution

Understanding the problem. A four-digit number $\overline{abcd}$ is a palindrome when reading it backwards, $\overline{dcba}$, gives the same number: $d = a$ and $c = b$. So a four-digit palindrome has the form $\overline{abba}$, and only *two* digits are chosen freely: a and b.

Step 1. Part a

Recall. Multiplication rule; the first digit of a number cannot be 0.

Digit	first (a)	second (b)	third	fourth
Condition	from 1 to 9	from 0 to 9	equals b	equals a
Number of choices	9	10	1	1

a: 9 ways (1 to 9); b: 10 ways; the last two digits are forced to equal b and a. There are $9 \times 10 = 90$ numbers.

Step 2. Part b

A number is even when its ones digit is even. The ones digit is a, so a must be even *and* non-zero: $a \in \{2, 4, 6, 8\}$ – 4 ways. b still has 10 ways. There are $4 \times 10 = 40$ even palindromes.

Common mistake. The digit 0 is even, but $a = 0$ is not allowed because a is also the first digit. So there are 4, not 5, choices for a.

Step 3. Part c

$P(\text{even}) = \dfrac{40}{90} = \dfrac{4}{9}$.

Step 4. Part d – proof

Why is it true? Write $\overline{abba}$ using place values: $\overline{abba} = 1000a + 100b + 10b + a = 1001a + 110b$. Since $1001 = 11 \times 91$ and $110 = 11 \times 10$, $\overline{abba} = 11 \times 91 \times a + 11 \times 10 \times b = 11 \times (91a + 10b)$, which is divisible by 11. □

Check: Examples: $3443 = 11 \times 313$; $7007 = 11 \times 637$; $1221 = 11 \times 111$ ✓. Counting check: the palindromes starting with 1 are $1001, 1111, \ldots, 1991$ – exactly 10 numbers; nine first digits give 90 numbers ✓.

Answer: **a)** 90; **b)** 40; **c)** $\dfrac{4}{9}$; **d)** $\overline{abba} = 11 \times (91a + 10b)$.

Problem 37

Using the four digits $0, 1, 2, 3$, four-digit numbers with all digits different are made (each digit used exactly once).

a) How many such numbers can be made?

b) How many of them are even?

c) One of the numbers from part a) is chosen at random. Find the probability that it is even.

Detailed solution

Understanding the problem. Each number is an arrangement of the four digits $0, 1, 2, 3$ in the four places (thousands, hundreds, tens, ones), with the condition that the thousands digit is not 0.

Step 1. Part a – choose the restricted position first

Recall. Multiplication rule. A position with a condition is chosen first.

Place	thousands	hundreds	tens	ones
Condition	not 0	3 digits left	2 digits left	last digit
Number of choices	3	3	2	1

Thousands: 3 ways (1, 2 or 3). Hundreds: 3 digits left (now 0 may be used): 3 ways. Tens: 2 ways. Ones: 1 way. There are $3 \times 3 \times 2 \times 1 = 18$ numbers.

Common mistake. Forgetting the condition "thousands digit not 0" gives $4 \times 3 \times 2 \times 1 = 24$ – which includes 6 "numbers" starting with 0, such as 0123, that are really three-digit numbers.

Step 2. Part b – split into cases by the ones digit

An even number ends in 0 or 2. These two cases differ because the digit 0 affects the thousands place.

- **Case 1: ones digit** 0. The three digits $1, 2, 3$ go freely into the other three places (no more worry about 0 in front): $3 \times 2 \times 1 = 6$ numbers.
- **Case 2: ones digit** 2. $0, 1, 3$ remain. Thousands not 0: 2 ways (1 or 3); hundreds: 2 ways; tens: 1 way. There are $2 \times 2 \times 1 = 4$ numbers.

Recall. Addition rule: for separate cases, add the numbers of ways in each case.

Number of even numbers: $6 + 4 = 10$.

Step 3. Part c

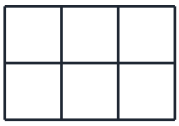
$P(\text{even}) = \dfrac{10}{18} = \dfrac{5}{9}$.

Check: List the even numbers. Case 1: $1230, 1320, 2130, 2310, 3120, 3210$ (6 numbers). Case 2: $1032, 1302, 3012, 3102$ (4 numbers). Total 10 ✓. Odd numbers (ending in 1 or 3): each case is like Case 2, with 4 numbers, total 8; and $10 + 8 = 18$ ✓.

Answer: a) 18 **numbers; b)** 10 **even numbers; c)** $\dfrac{5}{9}$.

Problem 38

The figure shows a rectangular grid of 2 rows and 3 columns of squares.



a) How many rectangles (including squares) are there in the figure altogether?

b) How many squares are there?

c) One of these rectangles is chosen at random. Find the probability that it is a square.

Detailed solution

Understanding the problem. This is a grid with full-length lines, so the "choose two lines" trick works. We will also count by size to check.

Step 1. Part a – choose two horizontal and two vertical lines

Recall. A rectangle in a grid is determined by 2 horizontal and 2 vertical lines. The number of ways to choose 2 of n lines is $\dfrac{n \times (n-1)}{2}$.

The grid has 3 horizontal and 4 vertical lines.

- Choose 2 horizontal lines: $\dfrac{3 \times 2}{2} = 3$ ways.
- Choose 2 vertical lines: $\dfrac{4 \times 3}{2} = 6$ ways.

By the multiplication rule: $3 \times 6 = 18$ rectangles.

Step 2. Check with a table of sizes

Write "height × width" (in squares). A shape w squares wide has $4 - w$ horizontal positions; h squares high, $3 - h$ vertical positions.

Size	1×1	1×2	1×3	2×1	2×2	2×3	Total
Number	$2 \times 3 = 6$	$2 \times 2 = 4$	$2 \times 1 = 2$	$1 \times 3 = 3$	$1 \times 2 = 2$	$1 \times 1 = 1$	18

Both methods give 18 ✓.

Step 3. Part b

The squares are the 1×1 and 2×2 shapes: $6 + 2 = 8$ squares.

Common mistake. There is no 3×3 square because the grid is only 2 squares high. And do not forget the two 2×2 squares (left and right, overlapping in the middle column).

Step 4. Part c

$P(\text{square}) = \dfrac{8}{18} = \dfrac{4}{9}$.

Answer: a) 18; **b)** 8; **c)** $\dfrac{4}{9}$.

Problem 39

Four friends Anna, Ben, Chloe and David are seated at random on four chairs in a row.

a) How many seating arrangements are there?

b) Find the probability that Anna and Ben sit next to each other.

c) Find the probability that Anna and Ben sit at the two ends of the row.

Detailed solution

Understanding the problem. "Seated at random" means all arrangements are equally likely. We count all arrangements, then the favourable ones for each event. Number the chairs $1, 2, 3, 4$ from left to right.

Step 1. Part a

Recall. Seating n people on n chairs: $n \times (n-1) \times \cdots \times 2 \times 1$ ways (multiplication rule; each next chair has one person fewer to choose from).

Chair 1: 4 choices of person; chair 2: 3; chair 3: 2; chair 4: 1. There are $4 \times 3 \times 2 \times 1 = 24$ arrangements.

Step 2. Part b – treat Anna and Ben as one "block"

Glue Anna and Ben together into one "block" taking 2 neighbouring chairs. The table shows that the block has only 3 positions:

	Chair 1	Chair 2	Chair 3	Chair 4
Position 1	Anna and Ben			
Position 2		Anna and Ben		
Position 3			Anna and Ben	

- Position of the block (Anna, Ben): chairs 1–2, 2–3 or 3–4: 3 ways.
- Inside the block, Anna on the left or Ben on the left: 2 ways.
- Chloe and David on the 2 remaining chairs: $2 \times 1 = 2$ ways.

Favourable arrangements: $3 \times 2 \times 2 = 12$. So $P(\text{next to each other}) = \dfrac{12}{24} = \dfrac{1}{2}$.

Common mistake. Do not forget to multiply by 2 for swapping Anna and Ben inside the block: "Anna–Ben" and "Ben–Anna" are two different arrangements.

Step 3. Part c

Anna and Ben on chairs 1 and 4: Anna on chair 1 and Ben on chair 4, or the other way round – 2 ways. Chloe and David on chairs 2, 3: 2 ways. Favourable arrangements: $2 \times 2 = 4$. $P(\text{both ends}) = \dfrac{4}{24} = \dfrac{1}{6}$.

Check: Part b, counted another way: if Anna sits at an end (chair 1 or 4, 2 possibilities) there is only 1 chair next to Anna; if Anna sits in the middle (chair 2 or 3) there are 2 chairs next to Anna. Number of (Anna, Ben) neighbouring chair pairs: $2 \times 1 + 2 \times 2 = 6$; times 2 ways to seat Chloe and David: 12 ✓.

Note for parents. With 4 people, let your child use 4 name cards and actually lay out all 24 arrangements in order (starting with Anna, then starting with Ben...). Listing once helps the child trust the multiplication rule.

Answer: a) 24 arrangements; b) $\frac{1}{2}$; c) $\frac{1}{6}$.

Problem 40 ★★

A box contains 4 red marbles and 6 blue marbles of the same size. Grace draws one marble at random, does *not put it back*, and then draws a second marble at random.

a) Find the probability that both marbles are red.

b) Find the probability that the two marbles have the same colour.

c) Find the probability that the two marbles have different colours.

Detailed solution

Understanding the problem. Grace draws two marbles *one after the other* without replacement. Imagine the marbles numbered 1 to 10 (marbles 1–4 red, marbles 5–10 blue). Each *ordered pair* (first marble, second marble) is one outcome, and these outcomes are equally likely.

Step 1. Count all outcomes

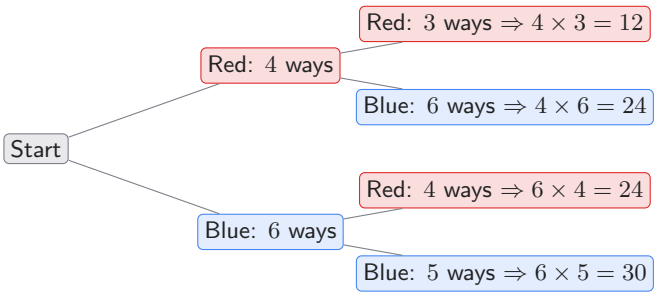
Recall. Multiplication rule: the first draw has 10 ways; for each of them, the second draw has 9 marbles left, so 9 ways.

Number of outcomes: $10 \times 9 = 90$ ordered pairs.

Step 2. Part a – both red

First draw red: 4 ways. Then 3 red marbles are left, second draw red: 3 ways. Favourable outcomes: $4 \times 3 = 12$. $P(\text{red, red}) = \frac{12}{90} = \frac{2}{15}$.

Step 3. Tree diagram with the numbers of ways



Check: $12 + 24 + 24 + 30 = 90$ ✓ – the four branches cover every outcome.

Step 4. Part b – same colour

Same colour has two separate cases: (red, red) with 12 outcomes and (blue, blue) with 30 outcomes.

Recall. Addition rule for cases that do not overlap.

Favourable outcomes: $12 + 30 = 42$. $P(\text{same colour}) = \frac{42}{90} = \frac{7}{15}$.

Step 5. Part c – complement

"Different colours" is the complement of "same colour": $P(\text{different colours}) = 1 - \frac{7}{15} = \frac{8}{15}$.

Check: Direct count: (red, blue) 24 and (blue, red) 24, total 48; $\frac{48}{90} = \frac{8}{15}$ ✓.

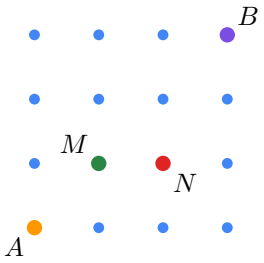
Common mistake. The most common mistake: acting as if all 10 marbles are still there for the second draw, computing $\frac{4}{10} \times \frac{4}{10}$ or counting 4×4 red pairs. Because the marble is *not put back*, the second draw has only 9 marbles and only 3 red ones.

ThinkAlgo Tip. You can also count without order: the number of ways to choose 2 of 10 marbles is $\frac{10 \times 9}{2} = 45$, the number of ways to choose 2 of the 4 red marbles is $\frac{4 \times 3}{2} = 6$, so $P(\text{red, red}) = \frac{6}{45} = \frac{2}{15}$ – the same result. What matters is that the numerator and denominator are counted *the same way* (both ordered or both unordered).

Answer: a) $\frac{2}{15}$; b) $\frac{7}{15}$; c) $\frac{8}{15}$.

Problem 41

On a square dot grid, Ruby stands at dot A. She tosses a fair coin 6 times; each time it shows heads she moves *up* one dot, and each time it shows tails she moves *right* one dot (as in problem 16). The dots M, N, B are marked in the figure.



a) Find the probability that after 6 tosses Ruby stops at dot B.

b) Find the probability that Ruby passes through dot M *and* stops at B.

c) Find the probability that Ruby stops at B *without* passing through dot N.

Detailed solution

Understanding the problem. Each sequence of 6 tosses is a sequence of 6 symbols ↑ (up) or → (right). There are $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$ sequences, all equally likely. Each sequence is a *path* on the grid. We label positions by the pair (number of moves right, number of moves up): $A = (0, 0)$, $M = (1, 1)$, $N = (2, 1)$, $B = (3, 3)$.

Step 1. Count the paths from A to every dot (part a)

Recall. A dot can only be reached from the dot just to its left or the dot just below it. Therefore: number of paths to a dot = number of paths to the dot on its left + number of paths to the dot below (addition principle).

Write the number of paths at each dot, starting from A (write 1); the bottom row and the left column are all 1:

1	4	10	20	B
1	3	6	10	
1	2	3	4	
1	1	1	1	

There are 20 paths from A to B. (Another way: reaching B needs exactly 3 up-moves among the 6 moves; the number of ways to choose 3 turns out of 6 is $\dfrac{6 \times 5 \times 4}{3 \times 2 \times 1} = 20$.)

$$P(a) = \frac{20}{64} = \frac{5}{16}.$$

Step 2. Part b: through M and then to B – multiplication principle

A path through M has two stages: stage $A \to M$ (1 up, 1 right) and stage $M \to B$ (2 up, 2 right). Stage $A \to M$ can be done in 2 ways (↑→ or →↑); stage $M \to B$ in 6 ways (as in problem 16: choose 2 up-turns out of 4). Each way of doing the first stage combined with each way of doing the second gives a different path: $2 \times 6 = 12$ paths.

$$P(b) = \frac{12}{64} = \frac{3}{16}.$$

Step 3. Part c: avoiding N – use the complement

Paths from A through N to B: stage $A \to N$ (2 right, 1 up) has 3 ways; stage $N \to B$ (1 right, 2 up) has 3 ways; in total $3 \times 3 = 9$ paths. So the number of paths from A to B *not* through N is $20 - 9 = 11$.

$$P(c) = \frac{11}{64}.$$

Why is it true? *Check by "blocking" dot N.* Recompute the path numbers but put 0 at N (as if it cannot be used). Row 1: 1, 2, **0**, $0 + 1 = 1$. Row 2: 1, 3, $3 + 0 = 3$, $3 + 1 = 4$. Row 3: 1, 4, $4 + 3 = 7$, $7 + 4 = 11$. At B we get 11, matching the complement. □

Common mistake. Parts b and c ask for probabilities among *all* 64 toss sequences, not among the 20 paths to B. Dividing by 20 answers a different question (a conditional probability, not yet studied).

Check: In part a the possible final dots are $(6,0),(5,1),\ldots,(0,6)$ with $1,6,15,20,15,6,1$ paths; the total is 64. Dot $B=(3,3)$ corresponds to 20.

Answer: a) $\frac{5}{16}$; b) $\frac{3}{16}$; c) $\frac{11}{64}$

Problem 42 **

We build grids of squares from toothpicks as in problem 17. An "$m \times n$" grid has m rows, each with n squares.

a) Find a formula for the number of toothpicks needed for an $m \times n$ grid.

b) A square $k \times k$ grid uses exactly 760 toothpicks. Find k.

c) Sam has exactly 2026 toothpicks and wants to build an $m \times n$ grid (with $1 \le m \le n$) that uses *all* of them. Find all the grids Sam can build.

Detailed solution

Understanding the problem. As in problem 17, we count horizontal and vertical toothpicks separately. Part c leads to an equation in two natural-number unknowns; the trick "multiply by 2 and add 1" turns it into a problem about divisors.

Step 1. Part a: the formula

An $m \times n$ grid has $m+1$ horizontal lines of n toothpicks each: $(m+1) \times n$ horizontal toothpicks. It has $n+1$ vertical lines of m toothpicks each: $(n+1) \times m$ vertical toothpicks. Total:

$$T = (m+1)n + (n+1)m = mn + n + mn + m = 2mn + m + n.$$

Check with the 1×10 grid: $2 \times 1 \times 10 + 1 + 10 = 31$, as in problem 17.

Step 2. Part b: a square $k \times k$ grid

Put $m = n = k$: $T = 2k^2 + 2k = 2k(k+1)$. We need $2k(k+1) = 760$, that is $k(k+1) = 380$. $k(k+1)$ is the product of two consecutive natural numbers. Since $19 \times 20 = 380$, $k = 19$. (As k grows, $k(k+1)$ grows, so there is only one value: $18 \times 19 = 342 < 380 < 420 = 20 \times 21$.)

Step 3. Part c: set up the equation

We need natural numbers $1 \le m \le n$ with $2mn + m + n = 2026$.

ThinkAlgo Tip. Multiply both sides by 2 and add 1: $4mn + 2m + 2n + 1 = 4053$. The left side factorises: $(2m+1)(2n+1) = 4mn + 2m + 2n + 1$ (expand to check). So

$$(2m+1)(2n+1) = 4053.$$

Step 4. Prime factorisation of 4053

The digit sum of 4053 is $4+0+5+3 = 12$, divisible by 3: $4053 = 3 \times 1351$. Next, $1351 = 7 \times 193$. The number 193 is prime: it is not divisible by $2, 3, 5, 7, 11, 13$, and $13 \times 13 = 169 < 193 < 196 = 14 \times 14$, so no more trials are needed.

Recall. To check whether N is prime, it is enough to try dividing by the primes p with $p \times p \le N$.

So $4053 = 3 \times 7 \times 193$. The divisors of 4053 are: 1, 3, 7, 21, 193, 579, 1351, 4053.

Step 5. Find the pairs (m, n)

Since $m \ge 1$, $2m+1 \ge 3$; since $m \le n$, $2m+1 \le 2n+1$. The ways of writing 4053 as a product of two such divisors:

$2m+1$	$2n+1$	m	n	Check $2mn+m+n$
3	1351	1	675	$1350+1+675=2026$
7	579	3	289	$1734+3+289=2026$
21	193	10	96	$1920+10+96=2026$

(The pair 1×4053 is rejected because it gives $m = 0$.)

Common mistake. Trying $m = 1, 2, 3, \ldots$ one by one also works but is very long and it is easy to miss a solution. The "multiply by 2, add 1" trick turns the problem into "find the divisors" and guarantees that *no* solution is missed.

Check: For any other value of m, $2m+1$ is not a divisor of 4053, so n is not a natural number. For example $m = 2$: $5(2n+1) = 4053$ is impossible because 4053 is not divisible by 5.

Answer: a) $2mn+m+n$; **b)** $k = 19$; **c) three grids:** 1×675, 3×289, 10×96

Problem 43

Three fair dice coloured red, blue and yellow are rolled together (each is numbered 1 to 6). Find the probability of each event:

a) the three numbers are all different;

b) the number on the red die is less than the number on the blue die, and the number on the blue die is less than the number on the yellow die;

c) at least one die shows a 6;

d) the largest of the three numbers is exactly 4.

Detailed solution

Understanding the problem. Each outcome is a triple (red, blue, yellow). By the multiplication principle there are $6 \times 6 \times 6 = 216$ equally likely outcomes.

Step 1. Part a: three different numbers

Red die: 6 ways; blue die different from red: 5 ways; yellow die different from both: 4 ways. That gives $6 \times 5 \times 4 = 120$ outcomes.

$$P(a) = \frac{120}{216} = \frac{5}{9}.$$

Step 2. Part b: red < blue < yellow – use symmetry

Take a set of three different numbers, for example $\{2, 3, 5\}$. There are $3 \times 2 \times 1 = 6$ ways to

assign these three numbers to the three dice (235, 253, 325, 352, 523, 532), and *exactly one* of them is increasing ($2 < 3 < 5$). So the 120 outcomes of part a split into groups of 6, each with 1 favourable outcome: there are $120 \div 6 = 20$ favourable outcomes.

$$P(b) = \frac{20}{216} = \frac{5}{54}.$$

Recall. The number of ways to arrange 3 different objects in 3 places is $3 \times 2 \times 1 = 6$.

Step 3. Part c: at least one 6 – complement
The opposite event is "no die shows 6": each die has only 5 outcomes (1 to 5), in total $5 \times 5 \times 5 = 125$. So the number of outcomes with at least one 6 is $216 - 125 = 91$.

$$P(c) = \frac{91}{216}.$$

Common mistake. Do not argue "each die shows 6 with probability $\frac{1}{6}$, so for three dice it is $\frac{3}{6} = \frac{1}{2}$". That addition counts twice the outcomes with two or three sixes (for example $(6, 6, 1)$ is counted twice).

Step 4. Part d: the largest number is exactly 4
"Largest number ≤ 4" means all three numbers are in $\{1, 2, 3, 4\}$: $4 \times 4 \times 4 = 64$ outcomes. "Largest number ≤ 3": $3 \times 3 \times 3 = 27$ outcomes. The largest number is exactly 4 when it is ≤ 4 but not ≤ 3: $64 - 27 = 37$ outcomes.

$$P(d) = \frac{37}{216}.$$

Why is it true? Every outcome whose largest number is ≤ 3 also belongs to the group whose largest number is ≤ 4. Removing the smaller group leaves exactly the outcomes whose largest number is at most 4 but more than 3, that is equal to 4. □

Check: The idea of part d also gives part c: largest number equal to 6 in $6^3 - 5^3 = 216 - 125 = 91$ outcomes, a match. The numbers of outcomes by largest number $1, 2, \ldots, 6$: $1 + 7 + 19 + 37 + 61 + 91 = 216$.

Answer: a) $\frac{5}{9}$; b) $\frac{5}{54}$; c) $\frac{91}{216}$; d) $\frac{37}{216}$

Problem 44 ★★

Special six-sided dice are made with one of the numbers $1, 2, 3, 4, 5, 6$ written on each face (repeats allowed), so that, as in problem 19, the probability of rolling a multiple of 3 is $\frac{1}{2}$ and the probability of rolling an even number is $\frac{1}{3}$. Two dice are considered the same if they carry the same six numbers (regardless of the order of the faces).

a) What is the largest possible sum of the six numbers on such a die? What is the smallest?

b) How many different such dice are there?

Detailed solution

Understanding the problem. As in problem 19, the conditions mean: **exactly 3 faces are multiples of 3** and **exactly 2 faces are even**. The key point is that 6 belongs to both groups, so we sort the numbers 1–6 into four types:

Type	Numbers	Property
I	6	even and a multiple of 3
II	3	odd, multiple of 3
III	2, 4	even, not a multiple of 3
IV	1, 5	odd, not a multiple of 3

Step 1. Express the number of faces of each type in terms of the number of 6s
Let a be the number of faces showing 6. Then:

- the number of multiples of 3 is (type I) + (type II) $= 3$, so type II has $3 - a$ faces;
- the number of even faces is (type I) + (type III) $= 2$, so type III has $2 - a$ faces;
- type IV has $6 - a - (3 - a) - (2 - a) = 1 + a$ faces.

A number of faces cannot be negative, so $2 - a \geq 0$, that is $a \in \{0, 1, 2\}$.

a	type I (the 6s)	type II (the 3s)	type III (2 or 4)	type IV (1 or 5)
0	0	3	2	1
1	1	2	1	2
2	2	1	0	3

Step 2. Part a: largest and smallest sums
For the largest sum, choose all 4s in type III and all 5s in type IV. The sum is

$$6a + 3(3 - a) + 4(2 - a) + 5(1 + a) = 22 + 4a,$$

largest when $a = 2$: sum 30 (the die $3, 5, 5, 5, 6, 6$). For the smallest sum, choose all 2s in type III and all 1s in type IV. The sum is

$$6a + 3(3 - a) + 2(2 - a) + 1 \cdot (1 + a) = 14 + 2a,$$

smallest when $a = 0$: sum 14 (the die $1, 2, 2, 3, 3, 3$).

Recall. Collecting terms: $6a - 3a - 4a + 5a = 4a$ and $9 + 8 + 5 = 22$; similarly $6a - 3a - 2a + a = 2a$ and $9 + 4 + 1 = 14$.

Step 3. Part b: count the dice
Once a is known, we only have to decide how many of the type III faces are 4 (the rest are 2), and how many of the type IV faces are 5 (the rest are 1).

- $a = 0$: type III has 2 faces, the number of 4s can be $0, 1, 2$: 3 ways; type IV has 1 face: 2 ways. That gives $3 \times 2 = 6$ dice.
- $a = 1$: type III has 1 face: 2 ways; type IV has 2 faces: 3 ways. That gives $2 \times 3 = 6$ dice.

- $a = 2$: type III has 0 faces: 1 way; type IV has 3 faces (the number of 5s is $0, 1, 2, 3$): 4 ways. That gives $1 \times 4 = 4$ dice.

Total: $6 + 6 + 4 = 16$ dice.

Common mistake. Since the order of the faces does not matter, two type III faces "2, 4" or "4, 2" are *one* case. Counting the *number* of 4s (0, 1 or 2) instead of arranging them avoids double counting.

Check: Two dice from the list: with $a = 1$, one 4 and two 1s: the die $1, 1, 3, 3, 4, 6$ has 3 multiples of 3 $(3, 3, 6)$ and 2 even faces $(4, 6)$. Correct. The die $1, 2, 2, 3, 3, 3$ (sum 14) also has 3 multiples of 3 and 2 even faces.

Answer: a) Largest 30, smallest 14; b) 16 dice

Problem 45 *

Three lights flash: the first flashes every 4 seconds, the second every 6 seconds, the third every 10 seconds. At the start (0 seconds) all three flash together. Consider the next 10 minutes (from second 1 to the end of second 600).

a) How many times do all three lights flash together?

b) At how many moments do *exactly two* lights flash together?

c) At how many moments does at least one light flash?

Detailed solution

Understanding the problem. The first light flashes at the seconds that are multiples of 4; the second at multiples of 6; the third at multiples of 10 (the same idea as in problem 20: moments that "coincide are found with common multiples). We count multiples from 1 to 600.

Step 1. Count how many times each light flashes

Recall. The number of multiples of d among 1 to N is the whole-number part of $N \div d$ (the quotient in division with remainder).

Light 1: $600 \div 4 = 150$ times. Light 2: $600 \div 6 = 100$ times. Light 3: $600 \div 10 = 60$ times.

Step 2. Two and three lights together – use the LCM

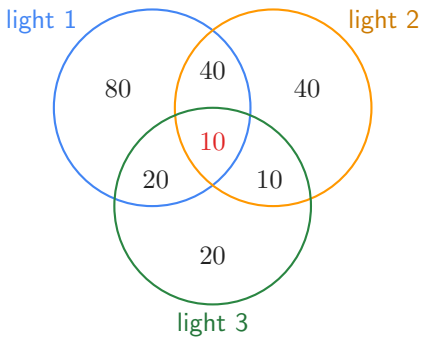
Two lights flash together when the number of seconds is a common multiple of the two periods, that is a multiple of their LCM.

- $\text{lcm}(4, 6) = 12$: $600 \div 12 = 50$ times lights 1 and 2 flash together.
- $\text{lcm}(4, 10) = 20$: $600 \div 20 = 30$ times.
- $\text{lcm}(6, 10) = 30$: $600 \div 30 = 20$ times.
- $\text{lcm}(4, 6, 10) = 60$: $600 \div 60 = 10$ times all three flash together. This answers part a.

Common mistake. $\text{lcm}(4, 6)$ is 12, not $4 \times 6 = 24$ (because 4 and 6 have the common divisor 2). Similarly $\text{lcm}(4, 6, 10) = 60$, not 240.

Step 3. Part b: exactly two lights
Of the 50 times lights 1 and 2 flash together, light 3 also flashes 10 times (all three), so only lights 1 and 2 flash $50 - 10 = 40$ times. Similarly: only lights 1 and 3: $30 - 10 = 20$; only lights 2 and 3: $20 - 10 = 10$. Total number of moments with exactly two lights: $40 + 20 + 10 = 70$.

Step 4. Part c: draw a Venn diagram and fill it in from the inside out
Only light 1: $150 - 40 - 20 - 10 = 80$; only light 2: $100 - 40 - 10 - 10 = 40$; only light 3: $60 - 20 - 10 - 10 = 20$.



The number of moments when at least one light flashes is the sum of the seven numbers in the diagram:

$$80 + 40 + 20 + 40 + 20 + 10 + 10 = 220.$$

Recall. *Inclusion–exclusion* for three groups: number of elements in at least one group = (sum of the sizes of the three groups) − (sum of the sizes of the overlaps of each pair of groups) + (size of the overlap of all three groups). The overlap of all three is added 3 times and subtracted 3 times, so it must be added back once.

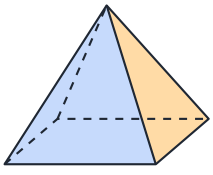
Check: By inclusion–exclusion: $150 + 100 + 60 - 50 - 30 - 20 + 10 = 220$. It matches the Venn diagram.

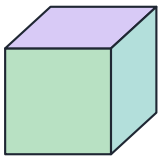
Answer: a) 10 times; b) 70 moments; c) 220 moments

Problem 46 ⋆⋆

a) A square pyramid has 5 faces: a square base and four triangular side faces. What is the smallest number of colours needed to paint the 5 faces (one colour per face) so that any two faces sharing an edge have different colours?

b) There are 6 different colours. We paint the six faces of a cube, one colour per face, using all 6 colours. Two cubes are considered the same if one can be rotated to look exactly like the other. How many different cubes are there?


square pyramid


cube with 6 colours

Detailed solution

Understanding the problem. Part a is like problem 21: find a colouring and prove that fewer colours are impossible. Part b is a counting problem "up to rotation": we must be careful because a cube can be placed in many positions.

Step 1. Part a: at least 3 colours are needed

Why is it true? The base shares an edge with all four side faces, so its colour differs from the colour of every side face. Two neighbouring side faces also share an edge (a slanted edge going up to the apex), so they must differ too. Take the base and two neighbouring side faces: any two of these three faces share an edge, so at least 3 colours are needed. □

Three colours are enough: paint the base colour 1; going around, paint the four side faces alternately colour 2, colour 3, colour 2, colour 3. The fourth side face (colour 3) is next to the first (colour 2): different. So part a needs at least 3 colours.

ThinkAlgo Tip. If the base were a pentagon (a pyramid with 5 side faces), the 5 side faces would form an odd ring and could not alternate between two colours, so 4 colours would be needed. Whether the number of side faces is even or odd decides the answer!

Step 2. Part b: put the cube in a "standard position"

Call the colours $1, 2, 3, 4, 5, 6$. Rotate each cube in two steps:

- rotate it so that the face of colour 1 is *at the bottom*;
- keeping the bottom fixed, turn it about the vertical axis so that, among the four side faces, the one with the *smallest colour number* is at the front.

Once placed like this, no rotation keeps both the bottom face and the front face in place. Therefore two cubes are the same exactly when, in standard position, they have the same colour in every position.

Step 3. Count the standard positions

- Top face (opposite colour 1): one of the 5 remaining colours: 5 ways.
- The four remaining colours go on the four side faces; the smallest of them is already placed at the front (no choice).
- The other three colours go on the three faces left, back, right: $3 \times 2 \times 1 = 6$ ways.

By the multiplication principle: $5 \times 6 = 30$ different cubes.

Step 4. Check with a second method

If rotations were *not* allowed (the six positions top, bottom, front, back, left, right fixed), the number of paintings would be $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$. Each cube can be placed in $6 \times 4 = 24$ positions (6 choices for the face at the bottom, then 4 turns about the vertical axis), and with 6 different colours these 24 positions give 24 different fixed paintings. So the number of different cubes is $720 \div 24 = 30$, matching the count in step 3.

Common mistake. Answering 720 forgets that cubes which can be rotated into each other are the same. Answering $6 \times 5 = 30$ "by luck" without explanation is not enough either: you must argue why each cube is counted exactly once.

Answer: a) 3 colours; b) 30 cubes

Problem 47 ★★

Consider the integers $1, 2, 3, \ldots, 20$.

a) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 3?

b) How many pairs (m, n) with $m < n$ are there such that $m + n$ is divisible by 4?

c) How many triples (m, n, p) with $m < n < p$ are there such that $m + n + p$ is divisible by 3?

Detailed solution

Understanding the problem. In problem 22 we sorted the numbers by parity (the remainder on division by 2). Here we sort them by *the remainder on division by* 3 (or by 4). Because $m < n$ (and $m < n < p$), each pair or triple is counted once, regardless of order.

Recall. The remainder of a sum on division by 3 equals the remainder of the sum of the remainders. For example $7 + 11 = 18$: 7 leaves remainder 1, 11 leaves remainder 2, $1 + 2 = 3$ is divisible by 3, so 18 is divisible by 3.

Step 1. Sort the 20 numbers by remainder on division by 3

Remainder	Numbers	How many
0	$3, 6, 9, 12, 15, 18$	6
1	$1, 4, 7, 10, 13, 16, 19$	7
2	$2, 5, 8, 11, 14, 17, 20$	7

Step 2. Part a

The sum of two remainders is divisible by 3 in only two cases: $0 + 0$ or $1 + 2$.

- Both numbers with remainder 0: choose 2 of the 6 numbers: $\dfrac{6 \times 5}{2} = 15$ pairs.
- One with remainder 1, one with remainder 2: $7 \times 7 = 49$ pairs (the numbers are in different groups, so automatically different; the smaller one is m).

(Two numbers with remainder 1 give a sum with remainder 2; two numbers with remainder 2 give 4, remainder 1: not allowed.) Total: $15 + 49 = 64$ pairs.

Step 3. Part b: sort by remainder on division by 4

Each remainder $0, 1, 2, 3$ has exactly 5 numbers (because $20 = 4 \times 5$). The sum of two remainders is divisible by 4 for: $0 + 0$, $2 + 2$ or $1 + 3$.

$$\frac{5 \times 4}{2} + \frac{5 \times 4}{2} + 5 \times 5 = 10 + 10 + 25 = 45 \text{ pairs.}$$

Step 4. Part c: the remainder patterns of a triple

List every pattern of remainders (order not important) of three numbers and its sum:

Pattern	000	111	222	012	001	002	011	022	112	122
Sum	0	3	6	3	1	2	2	4	4	5
Divisible by 3?	yes	yes	yes	yes	no	no	no	no	no	no

Count the four good patterns:

- Pattern 000: choose 3 of 6 numbers: $\dfrac{6\times5\times4}{3\times2\times1}=20$.
- Pattern 111: choose 3 of 7 numbers: $\dfrac{7\times6\times5}{3\times2\times1}=35$. Pattern 222: also 35.
- Pattern 012: one number from each group: $6\times7\times7=294$.

Total: $20+35+35+294=384$ triples.

Recall. The number of ways to choose 3 objects (order not important) from k objects is $\dfrac{k\times(k-1)\times(k-2)}{6}$: there are $k(k-1)(k-2)$ ordered choices, and each triple is counted $3\times2\times1=6$ times.

Common mistake. For pattern 012 we do *not* divide by 6: the three numbers come from three different groups, so each choice (one number with remainder 0, one with remainder 1, one with remainder 2) gives exactly one triple, which we then write in increasing order.

Check: The total number of triples is $\dfrac{20\times19\times18}{6}=1140$. There are 384 triples with sum divisible by 3, close to $\dfrac{1}{3}$ of 1140 (380), which is reasonable since the sums are spread "fairly evenly" among the three remainders.

Answer: a) 64 pairs; b) 45 pairs; c) 384 triples

Problem 48 ⋆

For a two-digit number $n=\overline{ab}$, call $\overline{ba}$ the *reversal* of n (as in problem 23).

a) Find all two-digit numbers n such that n plus its reversal is a perfect square.

b) Find all two-digit numbers with both digits non-zero such that n minus its reversal is a positive perfect square.

c) For a three-digit number $\overline{abc}$ ($a\neq0$, $c\neq0$), prove that the difference $\overline{abc}-\overline{cba}$ is always divisible by 99.

Detailed solution

Understanding the problem. Write the numbers in terms of their digits: $\overline{ab}=10a+b$ and $\overline{ba}=10b+a$, where a is the tens digit ($1\le a\le9$) and b is the units digit. A perfect square is the square of a natural number: $1,4,9,16,25,\ldots$

Step 1. Part a: compute the sum

$\overline{ab}+\overline{ba}=(10a+b)+(10b+a)=11a+11b=11(a+b)$.

Why is it true? If $11(a+b)=s\times s$, then $s\times s$ is divisible by the prime 11, so s is divisible by 11 (if a product of two numbers is divisible by a prime p, one of the two numbers is divisible by p). Then $s\times s$ is divisible by 121, so $11(a+b)$ is divisible by 121, and $a+b$ is divisible by 11. □

For a two-digit number (the reversal may start with 0, e.g. $\overline{ba}=b$ when $b=0$) we have

$1 \le a+b \le 18$, so $a+b=11$, and then the sum is $11 \times 11 = 121 = 11^2$, which is indeed a perfect square. The numbers are those with $a+b=11$, a from 2 to 9:

$$29,\ 38,\ 47,\ 56,\ 65,\ 74,\ 83,\ 92 \quad (8 \text{ numbers}).$$

Step 2. Part b: compute the difference

$\overline{ab} - \overline{ba} = (10a+b) - (10b+a) = 9a - 9b = 9(a-b)$. The difference is positive, so $a > b \ge 1$. Since $9 = 3 \times 3$ is a perfect square, $9(a-b)$ is a perfect square exactly when $a-b$ is a perfect square.

Why is it true? If $a-b = t \times t$, then $9(a-b) = (3t) \times (3t)$. Conversely, if $9(a-b) = s \times s$, then $s \times s$ is divisible by 3, so s is divisible by 3; writing $s = 3t$ gives $9(a-b) = 9t \times t$, that is $a - b = t \times t$. □

Since $1 \le b < a \le 9$, we have $1 \le a-b \le 8$; the perfect squares in this range are 1 and 4.

- $a-b=1$: $21, 32, 43, 54, 65, 76, 87, 98$ (8 numbers).
- $a-b=4$: $51, 62, 73, 84, 95$ (5 numbers).

There are 13 numbers in all.

Step 3. Part c: split by place value

Write both numbers using place values: $\overline{abc} = 100a + 10b + c$ and $\overline{cba} = 100c + 10b + a$. Then

$$\overline{abc} - \overline{cba} = (100a + 10b + c) - (100c + 10b + a) = 100a - a + 10b - 10b + c - 100c,$$

$$\text{that is} \quad \overline{abc} - \overline{cba} = 99a - 99c = 99(a-c).$$

Since $a-c$ is an integer, $99(a-c)$ is divisible by 99. (The tens digit b cancels out completely!)

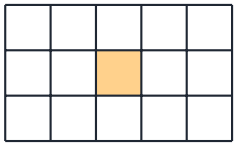
Common mistake. $\overline{ab}$ denotes the number with digits a and b, *not* the product $a \times b$. Always rewrite $\overline{ab}$ as $10a+b$ before calculating.

Check: a) $47 + 74 = 121 = 11^2$. b) $73 - 37 = 36 = 6^2$, $98 - 89 = 9 = 3^2$. c) $731 - 137 = 594 = 99 \times 6$; $205 - 502 = -297 = 99 \times (-3)$.

Answer: a) 29, 38, 47, 56, 65, 74, 83, 92; b) 13 numbers; c) $\overline{abc} - \overline{cba} = 99(a-c)$.

Problem 49 ★★

A grid of 3 rows and 5 columns of squares (15 small squares) is shown; the square in the very centre is shaded.



a) How many rectangles (including squares) with sides on the grid lines are there in the figure?

b) How many of them are squares?

c) One of the rectangles in part a is chosen at random. Find the probability that it contains the shaded square.

Detailed solution

Understanding the problem. As in problem 24, each rectangle is determined by two horizontal grid lines (top and bottom sides) and two vertical grid lines (left and right sides). A grid with 3 rows has 4 horizontal lines, numbered $0, 1, 2, 3$ from top to bottom; a grid with 5 columns has 6 vertical lines, numbered $0, 1, \ldots, 5$ from left to right.

Step 1. Part a: count the rectangles

Choose 2 of the 4 horizontal lines: $\dfrac{4 \times 3}{2} = 6$ ways. Choose 2 of the 6 vertical lines: $\dfrac{6 \times 5}{2} = 15$ ways. By the multiplication principle: $6 \times 15 = 90$ rectangles.

Step 2. Part b: count the squares by size

Recall. In a 3×5 grid, a square of side k has $(3-k+1)$ positions vertically and $(5-k+1)$ positions horizontally.

- side 1: $3 \times 5 = 15$ squares;
- side 2: $2 \times 4 = 8$ squares;
- side 3: $1 \times 3 = 3$ squares;
- side 4 or more: none (the grid is only 3 high).

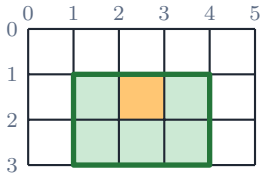
Total: $15 + 8 + 3 = 26$ squares.

Common mistake. Do not use "choose 2 horizontal and 2 vertical lines" to count squares: most of those choices give rectangles that are not squares. Squares must be counted by side length.

Step 3. Part c: rectangles containing the centre square

The centre square lies between horizontal lines 1 and 2, and between vertical lines 2 and 3. A rectangle contains it exactly when:

- its top side is horizontal line 0 or 1 (above the square): 2 ways; its bottom side is horizontal line 2 or 3: 2 ways;
- its left side is vertical line 0, 1 or 2: 3 ways; its right side is vertical line 3, 4 or 5: 3 ways.



(For example, the green rectangle: top side horizontal line 1, bottom side horizontal line 3, left side vertical line 1, right side vertical line 4.) Number of rectangles containing the centre square: $2 \times 2 \times 3 \times 3 = 36$.

$$P = \frac{36}{90} = \frac{2}{5}.$$

Why is it true? Why is this condition right? A rectangle contains the centre square exactly when it "covers" that square in both directions: its top side must be above (or on) the top side of the square, its bottom side below (or on) the bottom side of the square; similarly for the left and right sides. These four choices are independent, so we multiply. □

Check: Split by direction: of the 15 choices of two vertical lines, $3 \times 3 = 9$ "cover" the middle column; of the 6 choices of two horizontal lines, $2 \times 2 = 4$ cover the middle row. Probability $= \frac{9}{15} \times \frac{4}{6} = \frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$. It matches.

Answer: a) 90; **b)** 26; **c)** $\frac{2}{5}$

Problem 50 ★★

A box contains 2026 red marbles and 2027 blue marbles. Outside the box there is a very large pile of blue marbles. Felix repeats the following step: take 2 marbles from the box at random.

- If the two marbles have the *same colour* (both red or both blue): throw both away, then take 1 blue marble from the pile outside and put it into the box.
- If the two marbles have *different colours*: throw away the blue one and put the red one back into the box.

Felix stops when only 1 marble is left in the box.

a) How many steps did Felix carry out?

b) Find the probability that the last marble is red.

c) If the box had started with 2025 red marbles and 2027 blue marbles, what would the probability in part b be?

Detailed solution

Understanding the problem. There are thousands of marbles, so we cannot draw a probability tree as in problem 25. Instead we look for a quantity that *does not change* in any step (called an *invariant*). In problem 25 the invariant was "there is always a blue marble"; here we look at *the parity of the number of red marbles.*

Step 1. Part a: each step reduces the number of marbles by exactly 1

- Same colour: 2 taken out, 1 put in ⇒ the number of marbles in the box decreases by 1.

- Different colours: 2 taken out, 1 put back ⇒ the number decreases by 1.

At the start there are $2026 + 2027 = 4053$ marbles, and at the end 1 marble is left. The number of steps is $4053 - 1 = 4052$.

Step 2. Follow the number of red marbles in each case

Drawn	Change in red marbles	Change in blue marbles
red, red	decreases by 2	increases by 1
blue, blue	no change	down 2, up 1: decreases by 1
red, blue	no change (the red is put back)	decreases by 1

Why is it true? The number of red marbles either stays the same or decreases by exactly 2. Subtracting 2 does not change parity (an even number minus 2 is still even, an odd number minus 2 is still odd). So **the parity of the number of red marbles never changes**, no matter which marbles Felix draws. □

Step 3. Part b: 2026 red marbles at the start (an even number)
The number of red marbles is always even. When only 1 marble is left: if it were red, the number of red marbles would be 1 (odd) – impossible. So the last marble is certainly blue. The event "the last marble is red" is an impossible event:

$$P(\text{last marble red}) = 0.$$

Step 4. Part c: 2025 red marbles at the start (an odd number)
The number of red marbles is always odd, so it is never 0. When 1 marble is left, the number of red marbles is an odd number not exceeding 1, that is 1: the last marble is certainly red. This is a certain event:

$$P(\text{last marble red}) = 1.$$

ThinkAlgo Tip. The initial number of blue marbles has no effect on the result! Only the parity of the number of red marbles decides the colour of the last marble.

Note for parents. Play with your child using a few marbles (for example 2 red and 3 blue, or 3 red and 2 blue), or with coins of two colours. Seeing the same result every time makes children curious about "why"; then give a hint: "Try counting the red marbles after each turn: is the number even or odd?"

Common mistake. Seeing the word "random" and thinking the answer must be a "nice" fraction such as $\frac{1}{2}$ is a mistake. Some random processes have a final result that is completely determined.

Check: Try 2 red, 1 blue: drawing red–red → (0 red, 2 blue) → (0, 1): blue; drawing red–blue → (2, 0) → draw red–red → (0, 1): blue. Every way ends with a blue marble, consistent with an even initial number of red marbles.

Answer: a) 4052 steps; b) 0; c) 1

4 Number Sense

Big idea. "Number sense" is the ability to *understand* numbers, not just to *compute* with them: knowing which number is larger, whether a result is reasonable, how to round for a quick estimate, and seeing the hidden structure inside a number (its digits, its divisors, its multiples). In this topic you will review and connect:

- **decimals and negative numbers**: comparing and ordering them on the number line;
- the **order of operations** and **estimating** results;
- **fractions, ratios, percentages**: "a fraction of a number", equivalent fractions, discounts;
- **speed** – distance – time and unit conversion (km/h, m/s);
- **divisibility, common divisors, last digits, place value** (solving digit puzzles);
- the **mean, the median** and **number patterns** (Fibonacci, arithmetic sequences).

The general secret: *estimate before you calculate; check after you calculate.* A student with good number sense can often eliminate several unreasonable options in a multiple-choice question without any exact calculation.

① Decimals: reading, comparing, ordering, adding and subtracting

1. Place value. In a decimal, each digit sits in a *place*. After the decimal point come the tenths, the hundredths, the thousandths, ...

Tens	Ones	.	Tenths	Hundredths	Thousandths
	2	.	2	3	
	2	.	3	0	

Example: $2.23 = 2 + \frac{2}{10} + \frac{3}{100}$, while $2.3 = 2 + \frac{3}{10} = 2.30$.

2. Adding or removing zeros at the end of the decimal part does not change the value: $2.3 = 2.30 = 2.300$. Reason: $\frac{3}{10} = \frac{30}{100} = \frac{300}{1000}$ (multiply numerator and denominator by 10).

3. Rule for comparing two positive decimals.

- Compare the *whole-number parts* first: the one with the larger whole part is larger.
- If the whole parts are equal, *add zeros* so that both decimal parts have the same number of digits, then compare place by place from left to right: tenths, then hundredths, ...

Example: compare 0.304 and 0.34. Write $0.34 = 0.340$. Tenths: $3 = 3$; hundredths: $0 < 4$. So $0.304 < 0.340 = 0.34$.

4. Adding and subtracting decimals: write the numbers in a column *with the decimal points lined up* (places lined up), then add/subtract as with whole numbers.

5. Rounding decimals. To round to the nearest hundredth (two digits after the point), look at the digit just to the right of that place (the thousandths digit): if it is less than 5, keep the hundredths digit; if it is 5 or more, add 1 to the hundredths digit; then drop all digits after it.

Example: $4.6875 \approx 4.69$ (the thousandths digit is $7 \geq 5$), while $3.1416 \approx 3.14$ (because $1 < 5$). The sign "$\approx$" reads "is approximately".

Why is it true? Why can't we compare "by length" (the number with more digits is larger)? Write both in thousandths: $0.304 = \frac{304}{1000}$ and $0.34 = \frac{340}{1000}$. The fractions have the same denominator and $304 < 340$, so $0.304 < 0.34$, even though 0.304 "looks longer". Adding zeros is exactly bringing both to the same denominator. □

Common mistake. Many students think $2.23 > 2.3$ because "$23 > 3$". Wrong! Compare place by place: 2.23 has 2 tenths, while 2.3 has 3 tenths, so $2.23 < 2.3$.

ThinkAlgo Tip. To find the smallest (largest) number in a set of decimals, write all of them in *one column* with the decimal points lined up, adding zeros to make them equally long. Then you just compare them like whole numbers.

② Negative numbers and the number line

1. The number line. The integers are placed on a straight line: 0 in the middle, positive numbers to the right, negative numbers to the left.



2. Comparison rule. On a (horizontal) number line, *a number to the left is smaller than a number to the right.* Therefore:

- every negative number is less than 0, and 0 is less than every positive number;
- every negative number is less than every positive number;
- of two negative numbers, the one *farther from* 0 is *smaller*: $-8 < -2$ because $8 > 2$.

3. Everyday pictures. A temperature of $-8°$C is colder than $-2°$C; owing 8 dollars leaves you "poorer" than owing 2 dollars (an amount of -8 compared with -2).

4. The coordinate plane. A point is given by a pair of numbers $(x; y)$: x is the horizontal coordinate (moving right/left), y is the vertical coordinate (moving up/down). A horizontal move changes only x; a vertical move changes only y.

Why is it true? Why is $-8 < -2$? We have $-2 = -8 + 6$: to go from -8 to -2 on the number line we must move 6 units *to the right*. Moving right means increasing, so -2 is larger than -8. □

Common mistake. Do not compare negative numbers "by the feel of positive numbers". -8 is *less* than -2, even though 8 is greater than 2. When in doubt, sketch a quick number line.

ThinkAlgo Tip. **Parity (even – odd)** is a powerful "number sense" tool: even + even = even, odd + odd = even, even + odd = odd; a product of odd numbers is always odd. For instance, the sum of 5 odd numbers is odd, and the sum of 4 odd numbers is even. When a point moves on a

coordinate grid, looking at the parity of its coordinates often tells you at once whether a position can be reached.

③ Order of operations and estimation

1. Order of operations.

1. *Brackets* first: ( ), then [ ], then { }.
2. Next, *powers*.
3. Then *multiplication and division*, from left to right.
4. Finally *addition and subtraction*, from left to right.

Example: $6\times2-3=12-3=9$ (multiply first, then subtract), whereas $6\times(2-3)=6\times(-1)=-6$.
2. Rounding to estimate. To know quickly "roughly how much" a result is, round the factors to the nearest ten or hundred and compute mentally. Example: $99\times9\approx100\times9=900$.
3. A quick-calculation trick: $99\times9=(100-1)\times9=900-9=891$ (multiplication distributes over subtraction).

Why is it true? Why is $(100-1)\times9=100\times9-1\times9$? Imagine 9 rows of 100 sweets each, but 1 sweet is taken from every row. The sweets left are 9 rows × 99 sweets, which also equals (9 × 100 sweets) minus (the 9 sweets taken). This is the *distributive property*: $a\times(b-c)=a\times b-a\times c$. □

Common mistake. Working "from left to right regardless of the operation" is wrong: $2+3\times4\neq5\times4$. The correct value is $2+3\times4=2+12=14$.

ThinkAlgo Tip. In a multiple-choice question asking "which number is closest to…", the options often differ by factors of 10 (10; 100; 1000; …). Then a rough estimate is enough to choose correctly.

④ Fractions: a fraction of a number, equivalent fractions

1. A fraction is a part of a whole. $\frac{a}{b}$ means: split the whole into b equal parts and take a of them.
2. Finding a fraction of a number: to find $\frac{m}{n}$ of a, compute $a\times\frac{m}{n}$ (split a into n parts and take m of them). Example: $\frac{1}{3}$ of 396 is $396\div3=132$.
3. The part that is left. If $\frac{m}{n}$ of a job is done, the part left is $1-\frac{m}{n}=\frac{n-m}{n}$.
4. Equivalent fractions (the basic property). Multiplying (or dividing) the numerator and the denominator by the same non-zero number gives an equal fraction:

$$\frac{3}{7}=\frac{3\times9}{7\times9}=\frac{27}{63},\qquad\frac{12}{36}=\frac{12\div12}{36\div12}=\frac{1}{3}.$$

5. Simplifying a fraction: divide the numerator and the denominator by their greatest common divisor to get a fraction in lowest terms.

6. Two numbers with a known ratio and sum (or difference). If $\frac{x}{y} = \frac{5}{8}$, then there is a number k with $x = 5k$ and $y = 8k$ (because $\frac{5k}{8k} = \frac{5}{8}$). Knowing the sum $x + y$ (or the difference $y - x$), we form an equation in k and solve it. Example: $x + y = 91$ gives $5k + 8k = 91$, i.e. $13k = 91$, $k = 7$; hence $x = 35$, $y = 56$.

7. A fraction of a fraction. "Half of a quarter" is $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$: multiply numerator by numerator, denominator by denominator. This happens often when a shape is halved several times in a row: each halving multiplies the area fraction by $\frac{1}{2}$ again.

8. Comparing fractions. Bring them to a common denominator and compare numerators, or convert them to decimals. Examples: $\frac{2}{3} = \frac{8}{12} > \frac{7}{12}$; $\frac{3}{5} = 0.6 < \frac{3}{4} = 0.75$.

Why is it true? Why does multiplying the numerator and denominator by the same number leave the fraction unchanged? $\frac{3}{7}$ is "cut the cake into 7 parts, take 3". If we cut *each* part into 9 smaller pieces, the cake has $7 \times 9 = 63$ pieces and our share has $3 \times 9 = 27$ pieces. The amount of cake has not changed, so $\frac{3}{7} = \frac{27}{63}$. □

Common mistake. Read the question carefully: does it ask for the part *done* or the part *left*? Many students correctly find $\frac{1}{3}$ of 396 to be 132 and circle it straight away, while the question asks for the pages *left*, $396 - 132 = 264$.

Note for parents. In Grade 7, encourage your child to *use an unknown*: let x stand for the quantity we do not know, write the conditions of the problem as an equation, solve it and check. For example, "read $\frac{1}{3}$ of a 396-page book, x pages left" becomes $132 + x = 396$. This works for every problem, even when the numbers are "awkward" and a picture would not help.

⑤ Percentages, unit price and discounts

1. Percent means "out of one hundred". $80\% = \frac{80}{100} = 0.8$.

2. Finding $p\%$ of a number a: compute $a \times \frac{p}{100}$. Example: 80% of 50 is $50 \times \frac{80}{100} = 40$.

3. Expressing a as a percentage of b: compute $\frac{a}{b} \times 100\%$. Example: 32 correct answers out of 40 questions is $\frac{32}{40} \times 100\% = 80\%$.

4. Unit price. The unit price is the price of *one* unit (one kilogram, one book, ...):

$$\text{Total cost} = \text{unit price} \times \text{quantity}, \qquad \text{unit price} = \text{total cost} \div \text{quantity}.$$

5. A discount of $p\%$. New price = old price $\times \left(1 - \frac{p}{100}\right)$. A 10% discount leaves 90%: multiply by 0.9.

6. Successive discounts. Each time the price is reduced by 10% *of the current price*, we multiply by 0.9 once more. Example: 100 dollars after two reductions: $100 \to 90 \to 81$ (dollars), *not* 80.

Why is it true? Why are two 10% reductions not the same as one 20% reduction? The second reduction is 10% of 90 (that is 9), not 10% of 100 (that is 10). Since the price has already dropped, the next reduction is smaller too: in total the price drops by only $10 + 9 = 19$ dollars, i.e. 19%. □

ThinkAlgo Tip. Finding 10% very fast: just divide by 10 (move the decimal point one place to the left). From 10% you get 5% (halve it), 20% (double it), $80\% = 100\% - 20\%$, ...

⑥ Speed, distance, time and unit conversion

1. Three linked formulas. For motion at a constant speed:

$$s = v \times t, \qquad v = s \div t, \qquad t = s \div v,$$

where s is the distance, v the speed and t the time.

2. Who is faster? The faster person covers a *longer* distance in the *same amount of time*. The safest way: compute each person's speed in the same unit (for example m/s) and compare.

3. Running laps. Running n laps of a track L metres long means a distance of $n \times L$ metres.

4. Converting km/h to m/s. 1 km = 1000 m and 1 hour = 3600 seconds, so

$$1 \text{ km/h} = \frac{1000 \text{ m}}{3600 \text{ s}} = \frac{5}{18} \text{ m/s}, \qquad 18 \text{ km/h} = 18 \times \frac{5}{18} = 5 \text{ m/s}, \qquad 1 \text{ m/s} = 3.6 \text{ km/h}.$$

Why is it true? Why $v = s \div t$ and $t = s \div v$? It is just finding an unknown factor in the equation $s = v \times t$: dividing both sides by t (non-zero) gives $v = \frac{s}{t}$; dividing both sides by v (non-zero) gives $t = \frac{s}{v}$. □

Common mistake. The quantities must use *matching units*: if the distance is in metres and the time in seconds, the speed comes out in m/s; do not divide metres by hours. Before comparing two speeds, convert them to the same unit.

⑦ Divisibility, common divisors, last digits and place value

1. Multiples and divisors. a is a *multiple* of b (or b is a *divisor* of a) if $a = b \times k$ for some whole number k. Example: $45 = 9 \times 5$, so 45 is a multiple of 9.

2. Divisibility rules.

- by 2: the last digit is $0, 2, 4, 6, 8$; by 5: the last digit is 0 or 5; by 10: the last digit is 0;
- by 3: the digit sum is divisible by 3; by 9: *the digit sum is divisible by* 9.

3. Common divisors. A number that divides both a and b is a common divisor of a and b. Every common divisor is a divisor of the *greatest common divisor* $\gcd(a, b)$. Example: $\gcd(40, 72) = 8$, so the common divisors of 40 and 72 are $1, 2, 4, 8$.

4. The last digit of a product depends only on the last digits of the factors: $\ldots 3 \times \ldots 7$ ends in the same digit as $3 \times 7 = 21$, namely 1.

5. Place value. A number written with digits can be expanded by place:

$$\overline{abc} = 100a + 10b + c, \qquad \overline{PQQ} = 100P + 10Q + Q = 100P + 11Q.$$

(The bar on top shows that this is the number with digits a, b, c, not the product $a \times b \times c$.)

Why is it true? Why does the divisibility rule for 9 work? Take the number $\overline{abc} = 100a+10b+c$ and split it:

$$100a + 10b + c = (99a + 9b) + (a + b + c) = 9 \times (11a + b) + (a + b + c).$$

The part $9 \times (11a + b)$ is always divisible by 9. So $\overline{abc}$ is divisible by 9 exactly when the remaining part $a + b + c$ (the digit sum) is divisible by 9. For numbers with more digits the argument is the same, because $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$, ... □

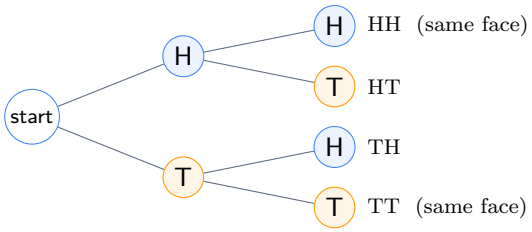
ThinkAlgo Tip. For a "digit puzzle" (each letter stands for a digit) such as $\overline{PQQ} + \overline{PPQ} + \overline{QQQ} = 876$, work *from the ones column to the left*, watching out for carries. The ones column usually has the fewest possibilities, so the first digit is found very quickly. You can also use place value to turn the puzzle into one equation. When you have to count or find all possibilities (sets of digits, outcomes of tossing coins, ...), *list them systematically* (in increasing order, or with a table/tree diagram) so that nothing is missed or repeated.

Systematic listing and chance.

- To count the three-digit numbers whose digits have a given *product*, find the sets of three digits (written in increasing order so as not to repeat), then count the arrangements of each set: three different digits can be arranged in $3 \times 2 \times 1 = 6$ ways; two equal digits and one other in 3 ways. Example: product 12 with the set $\{1; 3; 4\}$ gives 6 numbers $134, 143, 314, 341, 413, 431$; the set $\{2; 2; 3\}$ gives 3 numbers $223, 232, 322$.
- When all outcomes are *equally likely*, the probability of an event is

$$\frac{\text{number of favourable outcomes}}{\text{total number of possible outcomes}}.$$

Example: tossing 2 coins has 4 equally likely outcomes: HH, HT, TH, TT (H: heads, T: tails). The probability that both coins show the same face is $\frac{2}{4} = \frac{1}{2}$. Use a tree diagram to list them:



Common mistake. "The digit sum is divisible by 9" is the rule for 9; it does *not* work for 4, 6 or 8. And a number divisible by 3 is not necessarily divisible by 9: 12 is divisible by 3 but not by 9.

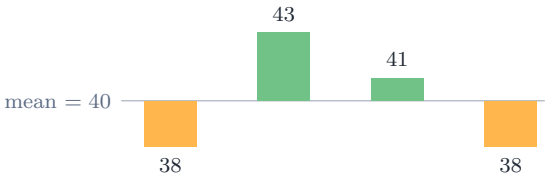
⑧ Number patterns, the mean and the median

1. The Fibonacci sequence. $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$ Each number (from the third on) is *the sum of the two numbers just before it*: $2 = 1 + 1$, $3 = 1 + 2$, $5 = 2 + 3$, ...
2. Arithmetic sequences (equally spaced). Each number (from the second on) equals the previous number plus a fixed number d (the common difference). Example: $1, 4, 7, 10$ (with $d = 3$). The n-th number is $a_1 + (n - 1) \times d$.
3. The mean (average). The mean of n numbers = (their sum) $\div n$. Hence

$$\text{Sum} = \text{mean} \times \text{how many numbers.}$$

4. The "deviation" method. The deviations from the mean always add up to 0: the "excess" of the numbers above the mean exactly equals the "shortfall" of the numbers below the mean.
5. The median. Arrange the numbers in increasing order; the number in the middle is the median (when there is an odd number of numbers). Example: the median of $42, 43, 46, 49, 50$ is 46. With an even number of numbers, the median is the mean of the two middle numbers: the median of $6, 7, 9, 10$ is $(7 + 9) \div 2 = 8$. Note: when a number is added or changed, the "middle" position can move, so re-order the list.
6. Consecutive integers. For an odd number of consecutive integers, the mean is the middle number: the mean of $19, 20, 21, 22, 23$ is 21.

Why is it true? Why do the deviations add up to 0? Suppose 4 numbers have mean 80, so their sum is $4 \times 80 = 320$. Write each number as 80+ (its deviation). Adding up: 320+ (sum of deviations) $= 320$, so the sum of the deviations must be 0. □



The four numbers $38, 43, 41, 38$ have mean 40. Deviations from 40: -2, $+3$, $+1$, -2; sum $= 0$.

Note for parents. You can illustrate the mean with stacks of coins: "level out" the tall stacks into the short ones until all stacks are the same height. That common height is the mean. The coins moved out of the tall stacks exactly equal the coins added to the short ones – this is the idea "the deviations add up to 0".

Worked examples

Example 1

Consider the set $\{0.5;\ 0.45;\ 0.405;\ 0.054;\ 0.54\}$.

a) Arrange the numbers in increasing order.

b) Find the sum of the largest and the smallest number.

Guided solution.

Step 1. Write the numbers in a column, adding zeros to get three digits after the point

Adding zeros at the end of the decimal part does not change the value, so:

$$0.5 = 0.500; \quad 0.45 = 0.450; \quad 0.405; \quad 0.054; \quad 0.54 = 0.540.$$

Step 2. Compare them like whole numbers

Dropping "0.", we compare 500; 450; 405; 54; 540:

$$54 < 405 < 450 < 500 < 540.$$

So the increasing order is: $0.054 < 0.405 < 0.45 < 0.5 < 0.54$.

Step 3. Add the largest and the smallest (decimal points lined up)

$$\begin{array}{r} 0.540 \\ +\ 0.054 \\ \hline 0.594 \end{array}$$

Check: $0.594 - 0.54 = 0.054$. Correct.

Answer: a) $0.054 < 0.405 < 0.45 < 0.5 < 0.54$; **b)** 0.594

Example 2

Emma cycles along a 45 km route. In the morning she covers $\frac{2}{5}$ of the route. How many kilometres does Emma still have to ride in the afternoon, and what fraction of the whole route is that?

Guided solution.

Step 1. Distance covered in the morning

Recall. To find $\frac{m}{n}$ of a number a, compute $a \times \frac{m}{n}$.

Morning: $45 \times \frac{2}{5} = \frac{45 \times 2}{5} = \frac{90}{5} = 18$ (km).

Step 2. Introduce an unknown and form an equation

Let x (km) be the distance still to ride in the afternoon ($x > 0$). The morning distance plus the afternoon distance is the whole route:

$$18 + x = 45.$$

Step 3. Solve the equation
$x = 45 - 18 = 27$ (km).

Step 4. The fraction left
Method 1: $1 - \frac{2}{5} = \frac{5}{5} - \frac{2}{5} = \frac{3}{5}$. Method 2: $\frac{27}{45} = \frac{27 \div 9}{45 \div 9} = \frac{3}{5}$. Both methods give the same result.

Check: $18 + 27 = 45$ km and $45 \times \frac{3}{5} = 27$ km. Correct.

Answer: 27 km, which is $\frac{3}{5}$ of the route

Example 3

a) Evaluate $20 - 4 \times 3 + 8 \div 2$ and $(20 - 4) \times (3 + 8) \div 2$.
b) Without calculating exactly, decide which number 101×49 is closest to: 50; 500; 5000; 50 000? Then compute it exactly with a quick method.

Guided solution.

Step 1. No brackets: multiply and divide first; add and subtract after

$$20 - \underbrace{4 \times 3}_{12} + \underbrace{8 \div 2}_{4} = 20 - 12 + 4 = 8 + 4 = 12.$$

(Subtraction and addition are done from left to right: $20 - 12 = 8$, then $8 + 4 = 12$.)

Step 2. With brackets: brackets first

$$(20 - 4) \times (3 + 8) \div 2 = 16 \times 11 \div 2 = 176 \div 2 = 88.$$

Step 3. Estimate
$101 \approx 100$ and $49 \approx 50$, so $101 \times 49 \approx 100 \times 50 = 5000$. Choose 5000.

Step 4. Exact value using the distributive property
$101 \times 49 = (100 + 1) \times 49 = 4900 + 49 = 4949$, indeed very close to 5000.

Answer: a) 12 and 88; b) closest to 5000; $101 \times 49 = 4949$

Example 4

a) Apples cost \$1.80 per kilogram. Grace buys 15 kg and gets a 10% discount on the whole bill. How much does Grace pay?
b) A shirt costs \$40. After every week the shop reduces the *current price* by 10%. After at least how many weeks is the price of the shirt below \$30?

Guided solution.

Step 1. Total before the discount
Total cost = unit price × quantity $= 1.80 \times 15 = 27$ (dollars).

Step 2. Subtract 10%
10% of 27 is $27 \div 10 = 2.70$ (dollars). Grace pays $27 - 2.70 = 24.30$ (dollars). (Or: 90% remains, i.e. $27 \times 0.9 = 24.3$.)

Step 3. Successive discounts: multiply by 0.9 every week
Let g_n be the price of the shirt (in dollars) after n weeks. Then $g_0 = 40$ and $g_n = g_{n-1} \times 0.9$.

Time	Start	After 1 week	After 2 weeks	After 3 weeks
Price (dollars)	40	36	32.40	29.16
Reduction (dollars)	–	4	3.60	3.24

After 2 weeks the price is \$32.40 > \$30; after 3 weeks it is \$29.16 < \$30.

Common mistake. Do not think "the price drops by \$4 every week" (because 10% of 40 is 4). Each week the reduction is 10% of *the price at that time*, so the reductions get smaller and smaller: 4, then 3.60, then 3.24, ...

Answer: a) \$24.30; b) 3 weeks

Example 5

a) Leo cycles 3 laps around a stadium with a perimeter of 400 m in 4 minutes. Mia cycles at 15 km/h. Who is faster?
b) A car travels at 72 km/h. How many metres does it travel in 3 seconds?

Guided solution.

Step 1. Leo's speed (m/s)
Distance: $3 \times 400 = 1200$ (m). Time: 4 minutes $= 4 \times 60 = 240$ (seconds).

$$v_{\text{Leo}} = 1200 \div 240 = 5 \text{ (m/s)}.$$

Step 2. Convert Mia's speed to m/s

Recall. 1 km/h $= \dfrac{1000 \text{ m}}{3600 \text{ s}} = \dfrac{5}{18}$ m/s.

$v_{\text{Mia}} = 15 \times \dfrac{5}{18} = \dfrac{75}{18} \approx 4.17$ (m/s). Since $5 > 4.17$, **Leo is faster**. (Another way: 5 m/s $= 5 \times 3.6 = 18$ km/h > 15 km/h.)

Step 3. The car
72 km/h $= 72 \times \dfrac{5}{18} = 20$ m/s. In 3 seconds it travels $20 \times 3 = 60$ (m).

Answer: a) Leo is faster (5 m/s against about 4.17 m/s); b) 60 m

Example 6

a) Which of the numbers 72; 84; 117; 135 are divisible by 9?
b) Find the digit x such that the number $\overline{3x5}$ is divisible by 9.
c) How many whole numbers are divisors of both 24 and 36?

Guided solution.

Step 1. Use the rule for 9: compute the digit sum
72: $7 + 2 = 9$ (divisible by 9). 84: $8 + 4 = 12$ (not divisible by 9).
117: $1 + 1 + 7 = 9$ (divisible). 135: $1 + 3 + 5 = 9$ (divisible).
So 72, 117, 135 are divisible by 9; 84 is not ($84 = 9 \times 9 + 3$).

Step 2. Find the digit x
The digit sum of $\overline{3x5}$ is $3+x+5=8+x$. Since x is a digit, $8 \le 8+x \le 17$. In this range only 9 is divisible by 9, so $8+x=9$, i.e. $x=1$. The number is $315=9\times 35$.

Step 3. Common divisors of 24 and 36
Divisors of 24: $1, 2, 3, 4, 6, 8, 12, 24$. Divisors of 36: $1, 2, 3, 4, 6, 9, 12, 18, 36$.
The numbers in both lists: $1, 2, 3, 4, 6, 12$ – these are exactly the divisors of $\gcd(24, 36) = 12$.

Answer: a) 72; 117; 135; **b)** $x=1$; **c) 6 numbers**

Example 7

In the calculation $\overline{AB}+\overline{AB}+\overline{AB}=\overline{1AB}$, each letter stands for a digit. Find A and B.

Guided solution.

Method 1 – Place value. We have $\overline{1AB}=100+\overline{AB}$. So

$$3\times\overline{AB}=100+\overline{AB} \;\Rightarrow\; 2\times\overline{AB}=100 \;\Rightarrow\; \overline{AB}=50.$$

So $A=5$, $B=0$.

Method 2 – Column by column. Ones column: $B+B+B=3B$ ends in the digit B, so $3B-B=2B$ ends in 0, hence $B=0$ or $B=5$.

- If $B=5$: $3\times 5=15$, write 5 carry 1. Tens column: $3A+1=\overline{1A}=10+A$, so $2A=9$ – no digit works. Rejected.
- If $B=0$: no carry. Tens column: $3A=10+A$, so $2A=10$, $A=5$.

Check: $50+50+50=150=\overline{1AB}$ with $A=5$, $B=0$. Correct.

Answer: $A=5$, $B=0$

Example 8

a) Noah's four test scores are 6; 9; 10 and x. The mean of the four scores is 8. Find x.
b) The mean of five consecutive whole numbers is 30. Find the smallest of them.

Guided solution.

Step 1. Form an equation from the definition of the mean

Recall. Mean = sum ÷ how many numbers, so sum = mean × how many numbers.

$$\frac{6+9+10+x}{4}=8 \;\Rightarrow\; 25+x=32 \;\Rightarrow\; x=32-25=7.$$

Step 2. Using deviations (faster)
Deviations from 8: -2, $+1$, $+2$, total $+1$. All deviations must add up to 0, so $x=8-1=7$.

Step 3. Five consecutive numbers
Let the smallest number be n; the five numbers are $n, n+1, n+2, n+3, n+4$, with sum $5n+10$. So $5n+10=5\times 30=150$, hence $5n=140$, $n=28$. The five numbers are $28, 29, 30, 31, 32$; the smallest is 28.

Check: $28+29+30+31+32=150=5\times 30$. Correct.

Answer: a) $x=7$; **b)** 28

Core practice

Problem 1

The smallest number in the set $\{3.2;\ 2.3;\ 3;\ 2.23;\ 3.22\}$ is

(A) 3.2 (B) 2.3 (C) 3 (D) 2.23 (E) 3.22

Detailed solution

Understanding the problem. There are 5 decimals. We need the *smallest* one.

> **Recall.** Comparing decimals: compare the whole parts first; if they are equal, compare the tenths, then the hundredths, ... You may add zeros at the end of the decimal part without changing the value.

Step 1. Compare the whole parts

Whole parts: $3.2 \to 3$; $2.3 \to 2$; $3 \to 3$; $2.23 \to 2$; $3.22 \to 3$.

The smallest number must have the smallest whole part, namely 2. Only two candidates remain: 2.3 and 2.23.

Step 2. Use the same number of decimal places and compare

$2.3 = 2.30$ and 2.23. Tenths: 3 against 2; since $2 < 3$, we get $2.23 < 2.30$.

Step 3. Conclusion

Ordering everything to be sure: $2.23 < 2.3 < 3 < 3.2 < 3.22$. The smallest number is 2.23.

> **Common mistake.** Do not choose 2.3 just because "3 is less than 23". The digit 3 in 2.3 means 3 tenths, while 2.23 has only 2 tenths.

Check: $2.23 = \frac{223}{100}$ and $2.3 = \frac{230}{100}$; $223 < 230$.

The number 2.23 is option **(D)**.

Answer: (D) 2.23

Problem 2

Keegan goes on a 36 km kayak trip. Before lunch he paddles the first 12 km. After lunch, what fraction of the whole trip does Keegan still have to paddle?

(A) $\frac{1}{2}$ (B) $\frac{5}{6}$ (C) $\frac{3}{4}$ (D) $\frac{2}{3}$ (E) $\frac{3}{5}$

Detailed solution

Understanding the problem. The whole trip is 36 km and 12 km are done. We are asked for the part *left* (not yet paddled), as a fraction.

Step 1. Distance left

Let x (km) be the distance still to paddle after lunch. Then $12 + x = 36$, so $x = 36 - 12 = 24$ (km).

Step 2. Form the fraction "part left over the whole"

Recall. The fraction "what part of the whole" $=\dfrac{\text{the part}}{\text{the whole}}$. Simplify by dividing numerator and denominator by their greatest common divisor.

$$\frac{24}{36}=\frac{24\div 12}{36\div 12}=\frac{2}{3}.$$

Step 3. Another way: find the part done and subtract it from 1

Part done: $\dfrac{12}{36}=\dfrac{1}{3}$. Part left: $1-\dfrac{1}{3}=\dfrac{3}{3}-\dfrac{1}{3}=\dfrac{2}{3}$. Both ways give the same result.

Common mistake. The fraction $\dfrac{12}{36}=\dfrac{1}{3}$ is the part *done*. The question asks for the part *left*, so the answer is $1-\dfrac{1}{3}=\dfrac{2}{3}$.

Check: $\dfrac{2}{3}\times 36=24$ km; $24+12=36$ km. Correct.

Note for parents. Ask your child two questions: "What part is done?" and "What part is left?". Getting into the habit of answering both helps avoid the trap of choosing $\dfrac{1}{3}$.

The fraction $\dfrac{2}{3}$ is option **(D)**.

Answer: (D) $\dfrac{2}{3}$

Problem 3

Which of the following statements is **true**?

(A) 0 is less than -5 (B) 7 is less than -1 (C) 10 is less than $\dfrac{1}{4}$

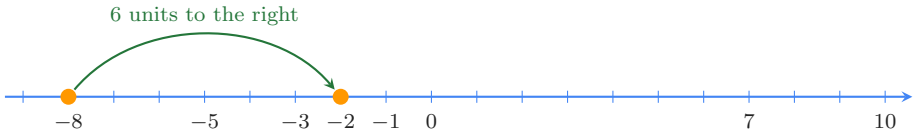
(D) -1 is less than -3 (E) -8 is less than -2

Detailed solution

Understanding the problem. Each option is a comparison. We check them one by one to find the true one.

Recall. On a horizontal number line, a number to the left is smaller than a number to the right. Every negative number is less than 0 and less than every positive number. Of two negative numbers, the one farther from 0 is smaller.

Step 1. Place the numbers on a number line



Step 2. Check each option

(A) 0 is less than −5? False: −5 is to the left of 0, so $-5 < 0$.
(B) 7 is less than −1? False: a positive number is always greater than a negative number.
(C) 10 is less than $\frac{1}{4}$? False: $\frac{1}{4} = 0.25 < 10$.
(D) −1 is less than −3? False: −3 is farther from 0, so $-3 < -1$.
(E) −8 is less than −2? **True**: −8 is to the left of −2 (farther from 0).

Common mistake. Many students choose (D), thinking "$1 < 3$ so $-1 < -3$". For negative numbers it is the other way round: $-1 > -3$ (a temperature of −1°C is warmer than −3°C).

Check: $-2 - (-8) = 6 > 0$, which means −2 is greater than −8.
The only true statement is **(E)**.

Answer: (E) −8 is less than −2

Problem 4

The value of $6 \times 2 - 3$ is

(A) 9 (B) −6 (C) 12 (D) 15 (E) 10

Detailed solution

Understanding the problem. The expression has two operations, multiplication and subtraction, and no brackets.

Recall. In an expression without brackets: multiply and divide first; add and subtract after.

Step 1. Do the multiplication first

$6 \times 2 = 12$.

Step 2. Then the subtraction

$12 - 3 = 9$.

Common mistake. Subtracting first would give $6 \times (2 - 3) = 6 \times (-1) = -6$ – option (B), a "trap" for anyone who forgets the order of operations. The expression has *no brackets*, so we must multiply first.

Check: Read the expression as a story: "6 boxes with 2 cakes each, 3 cakes eaten" leaves $12 - 3 = 9$ cakes.
The value 9 is option **(A)**.

Answer: (A) 9

Problem 5

A test is marked out of 50. Mukesh scores 80%. How many marks does Mukesh get?

(A) 40 (B) 62.5 (C) 10 (D) 45 (E) 35

Detailed solution

Understanding the problem. The maximum is 50 marks; Mukesh gets 80% of 50. We need Mukesh's marks.

Recall. To find $p\%$ of a number a, compute $a \times \frac{p}{100}$.

Step 1. Write the calculation

80% of 50 is $50 \times \frac{80}{100}$.

Step 2. Calculate

$$50 \times \frac{80}{100} = \frac{50 \times 80}{100} = \frac{4000}{100} = 40.$$

Step 3. A quick mental method

10% of 50 is 5; so $80\% = 8 \times 10\%$ is $8 \times 5 = 40$.

Common mistake. Option (B) 62.5 comes from dividing $50 \div 0.8$ – wrong, since 62.5 is even more than the maximum 50! Option (C) 10 is the number of marks Mukesh *lost* (20% of 50).

Check: $\frac{40}{50} = \frac{80}{100} = 80\%$. Correct.

The value 40 is option **(A)**.

Answer: (A) 40

Problem 6

Ground beef sells for \$5.00 per kilogram. How much do 12 kg of ground beef cost?

(A) \$5.00 (B) \$12.00 (C) \$60.00 (D) \$17.00 (E) \$2.40

Detailed solution

Understanding the problem. The unit price (price of 1 kg) is \$5.00; the quantity is 12 kg. We need the total cost.

Recall. Total cost = unit price × quantity.

Step 1. Write the calculation
Let T (dollars) be the amount to pay. Then $T = 5 \times 12$.

Step 2. Calculate
$T = 5 \times 12 = 60$ (dollars), i.e. \$60.00.

Step 3. Estimate to check
Each kilogram costs \$5 and we buy more than 10 kg, so the cost is more than $5 \times 10 = 50$ dollars. Only option (C) fits.

Common mistake. The wrong options come from using the wrong operation: (A) is the price of 1 kg; (B) is just the number of kilograms written as money; (D) is $5 + 12$ (adding instead of multiplying); (E) is $12 \div 5$ (dividing instead of multiplying).

Check: $60 \div 12 = 5$ dollars per kilogram. Correct.
The amount \$60.00 is option **(C)**.

Answer: (C) \$60.00

Problem 7

The value of $(8 \times 4) + 3$ is

(A) 96 (B) 15 (C) 56 (D) 35 (E) 28

Detailed solution

Understanding the problem. The brackets contain a multiplication; then 3 is added.

Recall. Expressions with brackets: work out the brackets first.

Step 1. Work out the brackets
$8 \times 4 = 32$.

Step 2. Add 3
$32 + 3 = 35$.

Common mistake. Misreading it as $8 \times (4 + 3) = 8 \times 7 = 56$ leads to option (C). The brackets surround 8×4, not $4 + 3$. Option (A) $96 = 8 \times 4 \times 3$ comes from mistaking the plus sign for a times sign.

Check: Without brackets the result is still $8 \times 4 + 3 = 35$, because multiplication always comes before addition; here the brackets only add emphasis.
The value 35 is option **(D)**.

Answer: (D) 35

Problem 8

Which of the following integers is closest to 99×9?

(A) 10 000 (B) 100 (C) 100 000 (D) 1000 (E) 10

Detailed solution

Understanding the problem. The options are 10; 100; 1000; 10 000; 100 000 – each 10 times the previous. An estimate of 99×9 is enough, but we also compute it exactly to be sure.

Step 1. Estimate

$99 \approx 100$, so $99 \times 9 \approx 100 \times 9 = 900$.

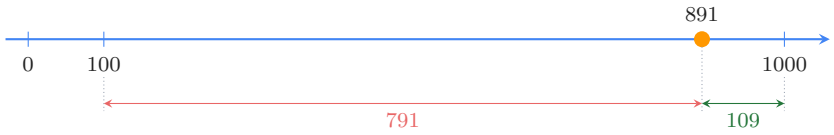
Step 2. Exact value using the distributive property

Recall. Distributive property: $(a - b) \times c = a \times c - b \times c$.

$99 \times 9 = (100 - 1) \times 9 = 900 - 9 = 891$.

Step 3. Compare the distances to the nearest options

The distance from 891 to 1000 is $1000 - 891 = 109$.
The distance from 891 to 100 is $891 - 100 = 791$.
The other options (10; 10 000; 100 000) are much farther away. So 1000 is the closest.



Check: $891 \div 9 = 99$. Correct.
The number 1000 is option **(D)**.

Answer: (D) 1000

Problem 9

Joshua is reading a 396-page book. He has read only the first $\frac{1}{3}$ of the book. How many pages does he still have to read to finish the book?

(A) 264 (B) 124 (C) 250 (D) 199 (E) 244

Detailed solution

Understanding the problem. The whole book: 396 pages. Read so far: $\frac{1}{3}$ of the book. Asked: the number of pages *left*.

Recall. To find $\frac{m}{n}$ of a number a, compute $a \times \frac{m}{n}$ (divide a by n and multiply by m).

Step 1. Pages read
$\frac{1}{3}$ of 396 is $396 \times \frac{1}{3} = 396 \div 3 = 132$ (pages).

Step 2. Introduce an unknown and form an equation
Let x be the number of pages Joshua still has to read. Pages read plus pages left make the whole book:

$$132 + x = 396 \;\Rightarrow\; x = 396 - 132 = 264.$$

Step 3. Another way: compute the part left directly
The part left is $1 - \frac{1}{3} = \frac{2}{3}$ of the book, i.e. $396 \times \frac{2}{3} = \frac{396 \times 2}{3} = \frac{792}{3} = 264$ (pages).

Common mistake. Stopping at 132 answers the wrong question (that is the number of pages *read*; and 132 is not even among the options). Always re-read the question before choosing an answer.

Check: $264 + 132 = 396$, and $132 \times 3 = 396$. Correct.
The number 264 is option **(A)**.

Answer: (A) 264

Problem 10

Which of the following numbers is a multiple of 9?

(A) 50 (B) 40 (C) 35 (D) 45 (E) 55

Detailed solution

Understanding the problem. We need the number divisible by 9 (a multiple of 9) among the five given numbers.

Recall. A number is divisible by 9 exactly when its digit sum is divisible by 9.

Step 1. Digit sum of each number

Number	50	40	35	45	55
Digit sum	5	4	8	9	10
Divisible by 9?	no	no	no	**yes**	no

Step 2. Check with the 9 times table
Multiples of 9 nearby: 36, 45, 54. Only $45 = 9 \times 5$ appears among the options.

Why is it true? Why does the rule work for a two-digit number $\overline{ab}$? We have $\overline{ab} = 10a + b = 9a + (a + b)$. The part $9a$ is divisible by 9, so $\overline{ab}$ is divisible by 9 exactly when $a + b$ is. For 45: $45 = 9 \times 4 + 9$. □

Check: $45 \div 9 = 5$, remainder 0.

The number 45 is option **(D)**.

Answer: (D) 45

Problem 11

Consider the set $\{0.34;\ 0.304;\ 0.034;\ 0.43\}$. The sum of the smallest and the largest numbers in the set is

(A) 0.77 (B) 0.734 (C) 0.077 (D) 0.464 (E) 0.338

Detailed solution

Understanding the problem. Part 1: find the smallest and the largest number. Part 2: add them.

Recall. To compare decimals, add zeros so the decimal parts have equal length, then compare place by place from left to right.

Step 1. Write the numbers with three decimal places

Original	Rewritten	Tenths	Hundredths
0.34	0.340	3	4
0.304	0.304	3	0
0.034	0.034	0	3
0.43	0.430	4	3

Step 2. Find the smallest and the largest
Comparing like whole numbers: $34 < 304 < 340 < 430$, so $0.034 < 0.304 < 0.34 < 0.43$.
Smallest: 0.034 (its tenths digit is 0). Largest: 0.43 (its tenths digit is 4).

Step 3. Add, with the decimal points lined up

$$\begin{array}{r} 0.430 \\ +\ 0.034 \\ \hline 0.464 \end{array}$$

Common mistake. Misaligning the columns (adding 43 and 34 "by the last digits") gives 0.77 or 0.077 – options (A) and (C). Keep the decimal points in line: 4 tenths plus 0 tenths, 3 hundredths plus 3 hundredths, ...

Check: $0.464 - 0.43 = 0.034$. Correct.
The sum 0.464 is option **(D)**.

Answer: (D) 0.464

Problem 12

Keiko and Leah run on a circular track 150 m long. Keiko runs 3 laps in 120 seconds, and Leah runs 5 laps in 160 seconds. Who is the faster runner, and approximately what is her speed?

(A) Keiko, 3.75 m/s (B) Keiko, 2.4 m/s (C) Leah, 3.3 m/s

(D) Leah, 4.69 m/s (E) Leah, 3.75 m/s

Detailed solution

Understanding the problem. One lap is 150 m. Keiko: 3 laps in 120 s. Leah: 5 laps in 160 s. We compute each girl's speed (m/s), compare, and choose the option.

Recall. Speed = distance ÷ time. The distance of n laps = $n\times$ the length of one lap.

Step 1. Distance run by each girl

Keiko: $3 \times 150 = 450$ (m). Leah: $5 \times 150 = 750$ (m).

Step 2. Keiko's speed

$v_{\text{Keiko}} = 450 \div 120 = 3.75$ (m/s).

Step 3. Leah's speed

$v_{\text{Leah}} = 750 \div 160 = 4.6875 \approx 4.69$ (m/s).

Step 4. Compare

$4.69 > 3.75$, so **Leah is faster**, running at about 4.69 m/s.

	Distance	Time	Speed
Keiko	450 m	120 s	3.75 m/s
Leah	750 m	160 s	$\approx$ 4.69 m/s

Common mistake. Option (E) "Leah, 3.75 m/s" pairs the right name with the slower girl's speed. Option (B) 2.4 m/s comes from the reversed calculation $120 \div 50$ (time divided by distance). Always write the units: m ÷ s = m/s.

Check: $4.6875 \times 160 = 750$ m = 5 laps; $3.75 \times 120 = 450$ m = 3 laps. Correct. The conclusion "Leah, 4.69 m/s" is option **(D)**.

Answer: (D) Leah, 4.69 m/s

Problem 13

In the addition shown, P and Q each represent a digit. The value of $P + Q$ is

$$\begin{array}{r} P\,Q\,Q \\ P\,P\,Q \\ +\,Q\,Q\,Q \\ \hline 8\;7\;6 \end{array}$$

(A) 3 (B) 5 (C) 7 (D) 6 (E) 4

Detailed solution

Understanding the problem. The three numbers $\overline{PQQ}$, $\overline{PPQ}$, $\overline{QQQ}$ add up to 876. We need the digits P, Q, then $P + Q$. We go column by column, starting with the ones.

Step 1. Ones column: $Q + Q + Q = 3Q$ ends in 6
Since Q is a digit, $3Q$ is one of $0, 3, 6, \ldots, 27$. Of these, only 6 ends in 6 (16 and 26 are not multiples of 3). So $3Q = 6$, $Q = 2$, and there is *no carry* into the tens.

Step 2. Tens column: $Q + P + Q = P + 4$ ends in 7
$P + 4$ lies between 4 and 13; ending in 7 means $P + 4 = 7$, so $P = 3$. No carry into the hundreds.

Step 3. Hundreds column: check $P + P + Q = 8$
$3 + 3 + 2 = 8$. This matches the hundreds digit of 876.

> **Recall.** Place value: $\overline{PQQ} = 100P + 11Q$, $\overline{PPQ} = 110P + Q$, $\overline{QQQ} = 111Q$.

Step 4. Another way using place value
Adding: $(100P + 11Q) + (110P + Q) + 111Q = 210P + 123Q = 876$. Try $Q = 2$: $210P = 876 - 246 = 630$, so $P = 3$. (Other values of Q do not give a digit for P.)

Check: $322 + 332 + 222 = 876$. Correct.

> **Note for parents.** A digit puzzle is like a detective game: each column is a "clue". Let your child write each column as a sentence ("three times Q ends in 6") and try the digits 0 to 9. Do not forget to check the carries.

So $P + Q = 3 + 2 = 5$, which is option **(B)**.

Answer: (B) 5

Problem 14

The mean (average) of the four integers 78; 83; 82 and x is 80. Which of the following statements is true?

(A) x is 2 greater than the mean

(B) x is 1 less than the mean

(C) x is 2 less than the mean

(D) x is 3 less than the mean

(E) x is equal to the mean

Detailed solution

Understanding the problem. We know the mean of four numbers is 80 and we know three of them. Find x and compare it with 80.

Recall. Sum of the numbers = mean × how many numbers.

Step 1. Form an equation

By the definition of the mean: $\frac{78+83+82+x}{4} = 80$, i.e. $78+83+82+x = 80\times 4 = 320$.

Step 2. Solve the equation for x

$78+83+82 = 243$, so $243+x = 320$, hence $x = 320-243 = 77$.

Step 3. Compare with the mean

$80-77 = 3$: x is 3 less than the mean.

Step 4. A faster way – the deviation method

Compared with 80: 78 is 2 short; 83 is 3 over; 82 is 2 over. Sum of these three deviations: $-2+3+2 = +3$. Since all the deviations must add up to 0, x must be 3 "short": $x = 80-3 = 77$.

Why is it true? Each number can be written as 80+(its deviation). The sum of the four numbers is 4×80+(sum of deviations). But the sum of the four numbers is exactly 4×80, so the sum of the deviations is 0. □

Check: $(78+83+82+77)\div 4 = 320\div 4 = 80$. Correct.

Note for parents. The "deviation" method is great for working out average marks mentally in everyday life. You can ask your child: "Three tests scored 7, 9, 8; to average 8, what does the fourth test need?" (Answer: 8, because $-1+1+0 = 0$.)

The statement "x is 3 less than the mean" is option **(D)**.

Answer: (D)

Problem 15

In the Fibonacci sequence $1, 1, 2, 3, 5, \ldots$, each number from the 2 onwards is the sum of the two numbers just before it. For example, the next number in the sequence is $3 + 5 = 8$. Which of the following numbers is in the sequence?

(A) 20 (B) 21 (C) 22 (D) 23 (E) 24

Detailed solution

Understanding the problem. Rule: next number = sum of the two before it. We continue the sequence until it passes 24 (the largest option), then see which option appears.

Step 1. Continue the sequence

Position	1	2	3	4	5	6	7	8	9
Number	1	1	2	3	5	8	13	**21**	34

In detail: $5 + 8 = 13$; $8 + 13 = 21$; $13 + 21 = 34$.

Step 2. Compare with the options

34 is already greater than 24, so all later numbers are greater than 24. Between 13 and 34 there is exactly one number of the sequence, 21. So 20, 22, 23, 24 are not in the sequence; only 21 is.

ThinkAlgo Tip. The Fibonacci sequence keeps increasing (from the second number on) because we keep adding positive numbers. So once the sequence has passed a number, that number can never appear later.

Check: $8 + 13 = 21$. Correct.
The number 21 is option **(B)**.

Answer: (B) 21

Problem 16

The mean (average) of five consecutive integers is 21. The smallest of the five integers is

(A) 17 (B) 21 (C) 1 (D) 18 (E) 19

Detailed solution

Understanding the problem. We have *five consecutive integers* (each one is exactly 1 more than the one before). Their mean is 21. We must find the *smallest* one.

Recall. Mean of a group of numbers = (sum of the numbers) ÷ (how many numbers there are). So: sum = mean × how many numbers.

Step 1. Find the sum of the five numbers

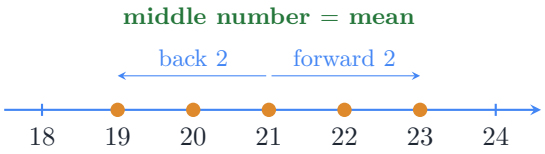
The sum of the five numbers is $21 \times 5 = 105$.

Step 2. Name the five numbers using the middle one
Call the middle (third) number n. Then the five consecutive numbers are

$$n-2, \quad n-1, \quad n, \quad n+1, \quad n+2.$$

Adding: $(n-2)+(n-1)+n+(n+1)+(n+2)=5n$ (the -2 and $+2$, and the -1 and $+1$, cancel each other).

Step 3. Find the middle number, then the smallest
We have $5n=105$, so $n=105\div 5=21$. The five numbers are 19, 20, 21, 22, 23. The smallest is $n-2=19$.



Why is it true? With an *odd* number of consecutive integers, the numbers placed symmetrically around the middle one "balance out" (one is 1 less, one is 1 more; one is 2 less, one is 2 more...). So the sum equals (middle number) × (how many numbers), which means the mean is exactly the middle number. □

Common mistake. Do not choose 21 straight away: that is the *middle* number, not the smallest. And do not take $21-4=17$ either: from the middle number to the smallest is only 2 steps back, not 4.

Check: $19+20+21+22+23=105$ and $105\div 5=21$. Correct.
The smallest number is 19, which is option **(E)**.

Answer: (E) 19

Problem 17

How many positive whole numbers, including 1, divide exactly into both 40 and 72?

(A) 9 (B) 12 (C) 4 (D) 2 (E) 5

Detailed solution

Understanding the problem. A number "divides exactly into both 40 and 72" when 40 is divisible by it *and* 72 is divisible by it. These are the *common divisors* (common factors) of 40 and 72. We must count them.

Recall. d is a divisor of a if a is divisible by d. A common divisor of two numbers is a divisor of both. The greatest common divisor (GCD) is the largest of the common divisors.

Step 1. List the divisors of 40
Find them in pairs whose product is 40: 1×40, 2×20, 4×10, 5×8. So the divisors of 40 are

$$\{1,\ 2,\ 4,\ 5,\ 8,\ 10,\ 20,\ 40\}.$$

Step 2. List the divisors of 72
Pairs with product 72: 1×72, 2×36, 3×24, 4×18, 6×12, 8×9. So the divisors of 72 are

$$\{1,\ 2,\ 3,\ 4,\ 6,\ 8,\ 9,\ 12,\ 18,\ 24,\ 36,\ 72\}.$$

Step 3. Pick the numbers that appear in both lists
Comparing the lists, the common numbers are $1,\ 2,\ 4,\ 8$. So there are 4 common divisors.

ThinkAlgo Tip. A faster way: the common divisors of two numbers are exactly the divisors of their GCD. We have $40 = 2^3 \cdot 5$ and $72 = 2^3 \cdot 3^2$, so $\text{GCD}(40, 72) = 2^3 = 8$. The divisors of 8 are $1, 2, 4, 8$: exactly 4 numbers.

Why is it true? If d divides both 40 and 72, then d also divides the difference $72 - 40 = 32$, and then $40 - 32 = 8$. Conversely, every divisor of 8 divides $40 = 8 \cdot 5$ and $72 = 8 \cdot 9$. So "common divisors of 40 and 72" and "divisors of 8" are the same set. □

Common mistake. Option 12 is the number of divisors of 72 alone; options 9 or 5 usually come from missing or miscounting some. The question asks for *common* divisors.

Check: $40 \div 1$, $40 \div 2 = 20$, $40 \div 4 = 10$, $40 \div 8 = 5$; $72 \div 1$, $72 \div 2 = 36$, $72 \div 4 = 18$, $72 \div 8 = 9$ – all exact. But $3, 6, 9$ do not divide 40, and $5, 10$ do not divide 72.
There are 4 such numbers, which is option **(C)**.

Answer: (C) 4

Problem 18

The number that goes into the box to make $\frac{3}{7} = \frac{\square}{63}$ true is

(A) 27 (B) 9 (C) 59 (D) 63 (E) 3

Detailed solution

Understanding the problem. The two fractions are equal; the denominator has changed from 7 to 63. We need the new numerator.

Recall. Basic property of fractions: multiplying both the numerator and the denominator by the same non-zero number gives an equal fraction: $\frac{a}{b} = \frac{a \cdot m}{b \cdot m}$ $(m \neq 0)$.

Step 1. What was the denominator multiplied by?
$63 \div 7 = 9$, so the denominator 7 was multiplied by 9 to become 63.

Step 2. Multiply the numerator by the same number
To keep the fraction the same, the numerator must also be multiplied by 9: $3 \times 9 = 27$. So

$$\frac{3}{7} = \frac{3 \cdot 9}{7 \cdot 9} = \frac{27}{63}.$$

Common mistake. Option 9 is the *multiplier*, not the new numerator. Option 59 comes from the "adding" mistake: $63 - 7 = 56$ and then $3 + 56 = 59$ – adding the same number to the numerator and denominator does *not* keep the fraction the same.

Check: Cross-multiply: $3 \times 63 = 189$ and $7 \times 27 = 189$. The products are equal, so the fractions are equal.
The missing number is 27, which is option **(A)**.
Answer: (A) 27

Problem 19

A fraction is equal to $\frac{5}{8}$. Its numerator and denominator add up to 91. What is the difference between the denominator and the numerator of this fraction?

(A) 21 (B) 3 (C) 33 (D) 13 (E) 19

Detailed solution

Understanding the problem. The fraction equals $\frac{5}{8}$. We know numerator + denominator = 91. We want denominator − numerator.

Recall. Every fraction equal to $\frac{5}{8}$ has the form $\frac{5k}{8k}$ for some positive integer k (multiply the numerator and denominator of $\frac{5}{8}$ by k; since $\frac{5}{8}$ is in lowest terms, there is no other form).

Step 1. Introduce an unknown
Write the fraction as $\frac{5k}{8k}$ (k a positive integer). The numerator is $5k$ and the denominator is $8k$.
Step 2. Form an equation from the sum 91

$$5k + 8k = 91 \implies 13k = 91 \implies k = 91 \div 13 = 7.$$

Step 3. Find the numerator, the denominator and the difference
Numerator $= 5 \cdot 7 = 35$; denominator $= 8 \cdot 7 = 56$. The fraction is $\frac{35}{56}$.
Denominator − numerator $= 56 - 35 = 21$. (More directly: $8k - 5k = 3k = 3 \cdot 7 = 21$.)

ThinkAlgo Tip. Another way, with two unknowns: let the numerator be a and the denominator b. Then $\frac{a}{b} = \frac{5}{8}$, so $8a = 5b$, and $a + b = 91$. Substitute $b = 91 - a$: $8a = 5(91 - a) = 455 - 5a$, so $13a = 455$, $a = 35$, $b = 56$.

Common mistake. Option 3 is the difference $8 - 5$ of the original fraction $\frac{5}{8}$, not yet multiplied by the value of one part. Option 13 is the sum $5 + 8$ (the coefficient of k in the equation), not the difference.

Check: $35 + 56 = 91$; simplifying, $\frac{35}{56} = \frac{35 \div 7}{56 \div 7} = \frac{5}{8}$. Correct.
The difference is 21, which is option **(A)**.

Answer: (A) 21

Problem 20

Sally is driving at 80 km/h along a road with equally spaced power poles. Her car passes a pole every 4 seconds. Which of the following is closest to the distance between two neighbouring poles?

(A) 50 m (B) 60 m (C) 70 m (D) 80 m (E) 90 m

Detailed solution

Understanding the problem. In 4 seconds the car travels exactly the distance between two neighbouring poles. So we need the distance travelled in 4 seconds at 80 km/h. Careful: the speed is in *km and hours*, the time is in *seconds*, and the answers are in *metres* – we must convert units.

Recall. Distance = speed × time. 1 km = 1000 m; 1 hour = 60 minutes = 3600 seconds.

Step 1. Convert the speed to metres per second
80 km/h means 80 000 m in 3600 seconds. In 1 second the car travels

$$80\,000 \div 3600 = \frac{80\,000}{3600} = \frac{200}{9} \approx 22.2 \text{ m}.$$

Step 2. Distance travelled in 4 seconds

$$\frac{200}{9} \times 4 = \frac{800}{9} \approx 88.9 \text{ m}.$$

Step 3. Compare with the options
88.9 m is about 1.1 m from 90 m and about 8.9 m from 80 m. So the closest is 90 m.

ThinkAlgo Tip. Another way to think: in 1 hour = 3600 seconds the car passes $3600 \div 4 = 900$ gaps between poles. These 900 gaps make up 80 000 m altogether, so each gap is $80\,000 \div 900 \approx 88.9$ m.

Common mistake. If you forget to convert and compute $80 \times 4 = 320$, no option matches. Converting 1 hour as 60 seconds (wrong – it is 3600 seconds) also gives an absurd result.

Check: 90 m every 4 seconds is 22.5 m/s, i.e. $22.5 \times 3600 = 81\,000$ m per hour = 81 km/h, very close to 80 km/h. Reasonable.
Choose option **(E)**.

Answer: (E) 90 m

Problem 21

Each of the integers 334 and 419 has digits whose product is 36 (since $3 \cdot 3 \cdot 4 = 36$ and $4 \cdot 1 \cdot 9 = 36$). How many 3-digit positive integers have digits whose product is 36?

(A) 21 (B) 15 (C) 18 (D) 24 (E) 12

Detailed solution

Understanding the problem. A 3-digit number $\overline{abc}$ has digit product $a \cdot b \cdot c = 36$. No digit can be 0 (a 0 would make the product 0). A tidy method: *first* find the sets of three digits (ignoring order) with product 36, *then* count how many numbers each set can make.

Step 1. Find every set of three digits (from 1 to 9) with product 36
List each set in order $a \le b \le c$ and look at the smallest digit a.

- $a = 1$: we need $b \cdot c = 36$ with $1 \le b \le c \le 9$: $36 = 4 \cdot 9 = 6 \cdot 6$ (while $1 \cdot 36$, $2 \cdot 18$, $3 \cdot 12$ use a number bigger than 9). We get $\{1, 4, 9\}$ and $\{1, 6, 6\}$.
- $a = 2$: we need $b \cdot c = 18$ with $2 \le b \le c \le 9$: $18 = 2 \cdot 9 = 3 \cdot 6$. We get $\{2, 2, 9\}$ and $\{2, 3, 6\}$.
- $a = 3$: we need $b \cdot c = 12$ with $3 \le b \le c$: $12 = 3 \cdot 4$. We get $\{3, 3, 4\}$.
- $a \ge 4$: then $a \cdot b \cdot c \ge 4 \cdot 4 \cdot 4 = 64 > 36$, no sets.

So there are exactly 5 sets: $\{1, 4, 9\}$, $\{1, 6, 6\}$, $\{2, 2, 9\}$, $\{2, 3, 6\}$, $\{3, 3, 4\}$.

Recall. Counting arrangements: three *different* digits can be arranged into $3 \cdot 2 \cdot 1 = 6$ numbers (hundreds digit: 3 choices, tens: 2 choices, units: 1 choice). Three digits with *two equal* can be arranged into only 3 numbers (we just choose the position of the odd one out).

Step 2. Count the numbers made from each set

Digit set	Type	Numbers formed	Count
$\{1, 4, 9\}$	all different	$149, 194, 419, 491, 914, 941$	6
$\{2, 3, 6\}$	all different	$236, 263, 326, 362, 623, 632$	6
$\{1, 6, 6\}$	two equal	$166, 616, 661$	3
$\{2, 2, 9\}$	two equal	$229, 292, 922$	3
$\{3, 3, 4\}$	two equal	$334, 343, 433$	3
		Total	**21**

Step 3. Add up
$6 + 6 + 3 + 3 + 3 = 21$ numbers.

Common mistake. Two common mistakes: (1) forgetting the sets containing the digit 1 (like $\{1, 4, 9\}$, $\{1, 6, 6\}$) because you only think of splitting 36 into three numbers "bigger than 1"; (2) assuming a set with two equal digits also gives 6 numbers – this double counts (for example, 334 would be counted twice).

Check: The table lists all 21 numbers, including 334 and 419 from the question. A computer check (going through every number from 100 to 999) also gives exactly 21.
Choose option **(A)**.

Answer: (A) 21

Problem 22

An *arithmetic sequence* is a sequence in which each term after the first is obtained by adding a constant to the previous term. For example, $2, 4, 6, 8$ and $1, 4, 7, 10$ are arithmetic sequences.
In the grid shown, the numbers in each row (read from left to right) must form an arithmetic sequence, and the numbers in each column (read from top to bottom) must form an arithmetic sequence. The value of x is

1			
4			25
7			x
10		36	

(A) 37 (B) 28 (C) 36 (D) 43.75 (E) 46

Detailed solution

Understanding the problem. The grid has 4 rows and 4 columns. Every row and every column is an arithmetic sequence. We need x in row 3, column 4. Idea: use the rows that already have

two known numbers to find more numbers, then use column 4 to find x.

Recall. An arithmetic sequence with 4 terms can be written $a,\ a+d,\ a+2d,\ a+3d$ (d is the constant difference). From the first to the last term there are 3 steps, so $d = (\text{last} - \text{first}) \div 3$. If we know the first and the third terms, then $d = (\text{third} - \text{first}) \div 2$.

Step 1. **Row 2:** 4, _, _, 25
From 4 to 25 there are 3 steps: $d = (25 - 4) \div 3 = 21 \div 3 = 7$. Row 2 is 4, 11, 18, 25.

Step 2. **Row 4:** 10, _, 36, _
From the first term 10 to the third term 36 there are 2 steps: $d = (36 - 10) \div 2 = 26 \div 2 = 13$. Row 4 is 10, 23, 36, 49. So the last cell of row 4 is 49.

Step 3. **Column 4:** _, 25, x, 49
In column 4, x lies between 25 and 49, the same distance from each. So x is their mean:

$$x = \frac{25 + 49}{2} = \frac{74}{2} = 37.$$

(The difference in column 4 is $37 - 25 = 12$, so the top of the column is $25 - 12 = 13$.)

Step 4. **Fill in the whole grid to check**

1	5	9	13	+4
4	11	18	25	+7
7	17	27	37	+10
10	23	36	49	+13

Column 1: $1, 4, 7, 10$ (step 3); column 2: $5, 11, 17, 23$ (step 6); column 3: $9, 18, 27, 36$ (step 9); column 4: $13, 25, 37, 49$ (step 12). Every row and every column is arithmetic.

Common mistake. You cannot find x by looking at row 3 first: row 3 has only one known number (7), not enough to find its difference. You must go "around" through row 2 and row 4 to reach column 4.

Check: Row 3: $7, 17, 27, 37$ has constant difference 10; column 4: $25, 37, 49$ has constant difference 12. Correct.
$x = 37$, which is option **(A)**.

Answer: (A) 37

Problem 23

In the six-digit number $\overline{1ABCDE}$, each letter represents a digit. Given that $\overline{1ABCDE} \times 3 = \overline{ABCDE1}$, the value of $A + B + C + D + E$ is

(A) 29 (B) 26 (C) 22 (D) 30 (E) 28

Detailed solution

Understanding the problem. When $\overline{1ABCDE}$ is multiplied by 3, the leading digit 1 "moves" to the end, and the block $ABCDE$ shifts one place to the left. We must find the digits A, B, C, D, E and add them. There are two methods; both are well worth learning.

> **Recall. Place value:** writing a digit c to the right of a number N gives $10 \cdot N + c$; writing the digit 1 to the left of a five-digit number N gives $100\,000 + N$.

Method 1: One unknown for the whole block.

Step 1. Let $N = \overline{ABCDE}$ (a five-digit number)
Then $\overline{1ABCDE} = 100\,000 + N$ and $\overline{ABCDE1} = 10N + 1$.

Step 2. Write the equation and find N

$$3 \cdot (100\,000 + N) = 10N + 1 \implies 300\,000 + 3N = 10N + 1.$$

Subtract $3N$ and 1 from both sides: $299\,999 = 7N$, so $N = 299\,999 \div 7 = 42\,857$.

Step 3. Read off the digits
$A = 4,\ B = 2,\ C = 8,\ D = 5,\ E = 7$. Sum: $4 + 2 + 8 + 5 + 7 = 26$.

Method 2: Multiply column by column, from right to left.

$$\begin{array}{r} 1\ A\ B\ C\ D\ E \\ \times \qquad\qquad 3 \\ \hline A\ B\ C\ D\ E\ 1 \end{array}$$

- Units: $3 \cdot E$ ends in 1 $\Rightarrow E = 7$ ($3 \cdot 7 = 21$, write 1 carry 2).
- Tens: $3 \cdot D + 2$ ends in $E = 7 \Rightarrow 3D$ ends in 5 $\Rightarrow D = 5$ ($15 + 2 = 17$, write 7 carry 1).
- Hundreds: $3 \cdot C + 1$ ends in $D = 5 \Rightarrow 3C$ ends in 4 $\Rightarrow C = 8$ ($24 + 1 = 25$, write 5 carry 2).
- Thousands: $3 \cdot B + 2$ ends in $C = 8 \Rightarrow 3B$ ends in 6 $\Rightarrow B = 2$ ($6 + 2 = 8$, carry 0).
- Ten thousands: $3 \cdot A + 0$ ends in $B = 2 \Rightarrow A = 4$ (12, write 2 carry 1).
- Hundred thousands: $3 \cdot 1 + 1 = 4$, exactly $A = 4$. The multiplication matches completely.

(At each step only one digit works, because $3 \cdot 0, 3 \cdot 1, \ldots, 3 \cdot 9$ all have different last digits.)

> **ThinkAlgo Tip.** 142 857 is a famous number: $142\,857 \times 2 = 285\,714$, $\times 3 = 428\,571$, $\times 4 = 571\,428$, $\times 5 = 714\,285$, $\times 6 = 857\,142$ – always the digits $1, 4, 2, 8, 5, 7$ "rotated". It is linked to $\frac{1}{7} = 0.\overline{142857}$.

Check: $142\,857 \times 3 = 428\,571$. This is indeed $\overline{ABCDE1}$ with $ABCDE = 42857$.

$A + B + C + D + E = 26$, which is option **(B)**.

Answer: (B) 26

Problem 24

On a coordinate grid, a dot starts at the point $(20, 19)$. On its first move it goes 1 unit horizontally or vertically, that is, to one of the points $(21, 19)$, $(19, 19)$, $(20, 20)$ or $(20, 18)$. On its second move it goes 2 units in a direction perpendicular to the first move (either way). Each move after that is 1 unit longer than the previous one (3 units, 4 units, 5 units,...) and must be perpendicular to the direction of the move just before it. The dot stops after 10 moves. Which of the following final positions is **not** possible?

(A) $(27, 33)$ (B) $(30, 40)$ (C) $(21, 21)$ (D) $(42, 44)$ (E) $(37, 37)$

Detailed solution

Understanding the problem. "Perpendicular to the previous move" means the moves *alternate*: horizontal, vertical, horizontal, vertical,... (or vertical, horizontal, vertical,...). The move lengths are $1, 2, 3, \ldots, 10$. We track separately the change in the x-coordinate (from horizontal moves) and in the y-coordinate (from vertical moves).

> **Recall.** For the point (x, y): x is the horizontal coordinate (moving right increases x, moving left decreases it), y is the vertical coordinate (moving up increases y, moving down decreases it). A sum of an odd number of odd numbers is odd; a sum (or difference) of even numbers is always even.

Step 1. Split the moves into two groups

Moves $1, 3, 5, 7, 9$ are all in one direction (horizontal or vertical), with lengths $1, 3, 5, 7, 9$ (all odd). Moves $2, 4, 6, 8, 10$ are in the other direction, with lengths $2, 4, 6, 8, 10$ (all even).

- If the first move is *horizontal*: x changes by $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$ and y changes by $\pm 2 \pm 4 \pm 6 \pm 8 \pm 10$.
- If the first move is *vertical*: the other way round, x changes by $\pm 2 \pm 4 \pm 6 \pm 8 \pm 10$ and y by $\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$.

(The sign $+$ or $-$ depends on which way the dot goes.)

Step 2. Odd and even

$\pm 1 \pm 3 \pm 5 \pm 7 \pm 9$ combines 5 odd numbers, so it is always *odd*, and at most 25. And $\pm 2 \pm 4 \pm 6 \pm 8 \pm 10$ is always *even*, at most 30. So of the two changes (in x and in y), exactly one is odd (from the odd group) and the other is even (from the even group).

> **Why is it true?** Start with the "all plus" total $1 + 3 + 5 + 7 + 9 = 25$. Switching the sign of a number k from $+$ to $-$ lowers the total by $2k$. So every possible value is $25 - 2s$, where s is the sum of the numbers whose sign was switched. Likewise, the even group gives the values $30 - 2s$. □

Step 3. Compute the changes for each option

Subtract $(20, 19)$ from the final point:

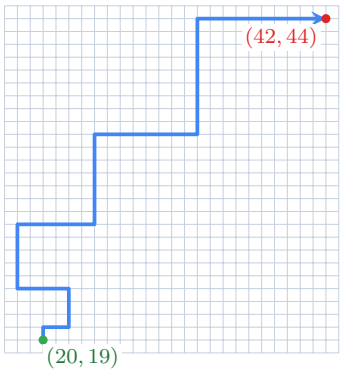
Option	Change in x	Change in y	Can it be made?
(A) $(27, 33)$	7	14	$1+3+5+7-9=7$; $2+4+6-8+10=14$
(B) $(30, 40)$	10	21	we would need $\pm1\pm3\pm5\pm7\pm9=21$ …
(C) $(21, 21)$	1	2	$1-3+5+7-9=1$; $2-4+6+8-10=2$
(D) $(42, 44)$	22	25	$2-4+6+8+10=22$; $1+3+5+7+9=25$
(E) $(37, 37)$	17	18	$-1-3+5+7+9=17$; $2+4-6+8+10=18$

Step 4. Why (B) is impossible

In (B) the odd change is 21, so we would need $\pm1\pm3\pm5\pm7\pm9=21$, i.e. $25-2s=21$, so $s=2$: we would have to switch the signs of some odd numbers adding up to 2. But a single odd number is never 2 (it can only be $1, 3, 5, \ldots$), and two different odd numbers of the group add up to at least $1+3=4$. There is no way! So the dot cannot stop at $(30, 40)$.

Step 5. An example path (option D)

First move up: up 1, right 2, up 3, left 4, up 5, right 6, up 7, right 8, up 9, right 10. The dot goes from $(20, 19)$ to $(42, 44)$.



Common mistake. Checking only "is the point too far away?" is not enough: $(30, 40)$ is no farther than $(42, 44)$. What decides it is odd/even and which totals can actually be made.

Check: Every equation in the table can be recomputed, e.g. $-1-3+5+7+9=17$. A computer search over all $2 \cdot 2^{10}=2048$ ways of moving also confirms that only $(30, 40)$ cannot be reached.

Choose option **(B)**.

Answer: (B) $(30, 40)$

Problem 25

In the "Coin Game", you toss three coins at the same time. You win only if the 3 coins all show heads or all show tails. If you play the game once only, what is the probability of winning?

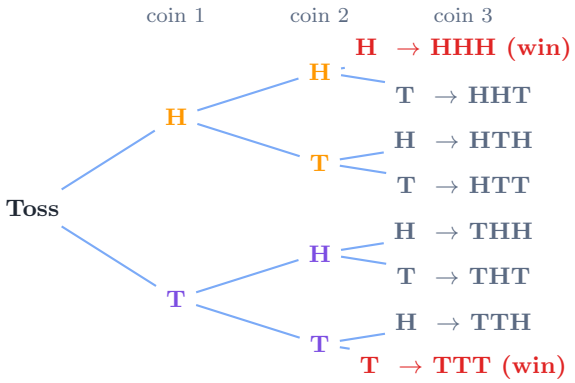
(A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{2}{27}$ (D) $\frac{2}{3}$ (E) $\frac{1}{3}$

Detailed solution

Understanding the problem. Each coin has 2 faces: heads (H) or tails (T), equally likely. You win with HHH or TTT. We list all possible outcomes and count.

Recall. When all outcomes are equally likely: the *probability* of an event = (number of favourable outcomes) ÷ (total number of possible outcomes).

Step 1. List all outcomes with a tree diagram



The first coin has 2 possibilities, for each of them the second coin has 2, and then the third coin has 2: altogether $2 \cdot 2 \cdot 2 = 8$ equally likely outcomes.

Step 2. Count the favourable outcomes

Only 2 outcomes win: HHH and TTT.

Step 3. Compute the probability

$$P(\text{win}) = \frac{2}{8} = \frac{1}{4}.$$

Common mistake. A common *wrong* argument: "The number of heads can be $0, 1, 2, 3$ – four cases, and we win in 2 of them (0 and 3), so the probability is $\frac{2}{4}$." This is wrong because these four cases are *not* equally likely: "1 head" happens in 3 ways (HTT, THT, TTH), while "3 heads" happens in only 1 way.

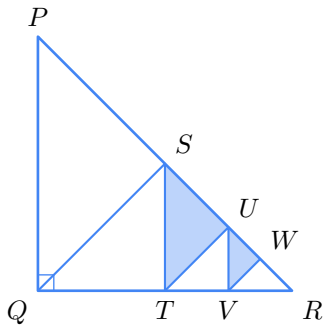
ThinkAlgo Tip. A quick way: the first coin can show anything. To win, the second coin must match the first (probability $\frac{1}{2}$) and the third must match too (probability $\frac{1}{2}$): $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

Check: The 8 outcomes in the tree are all different and cover everything; exactly 2 of them win. $\frac{2}{8} = \frac{1}{4}$.
Choose option **(B)**.

Answer: (B) $\frac{1}{4}$

Problem 26

In the right-angled triangle PQR, the right angle is at Q and $PQ = QR$. The segments QS, TU and VW are perpendicular to PR, and the segments ST and UV are perpendicular to QR, as shown. What fraction of the area of triangle PQR is shaded?



(A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{5}{16}$ (D) $\frac{5}{32}$ (E) $\frac{7}{32}$

Detailed solution

Understanding the problem. Triangle PQR is a *right isosceles* triangle with the right angle at Q. The picture is built by repeatedly "dropping a perpendicular". The shaded part consists of two triangles: STU and UVW. Key idea: *dropping a perpendicular from the right-angle vertex of a right isosceles triangle cuts it into two equal halves, and each half is again a right isosceles triangle.*

Recall. In an isosceles triangle, the altitude from the apex (to the base) is also the median, and it divides the triangle into two congruent triangles. A right isosceles triangle has two acute angles of 45°.

Why is it true? Take triangle XYZ, right isosceles at X ($XY = XZ$), and draw $XH \perp YZ$. The right triangles XHY and XHZ have equal hypotenuses $XY = XZ$ and the common leg XH, so they are congruent (hypotenuse–leg). So each has half the area of XYZ. Moreover $\widehat{Y} = 45°$ and $\widehat{XHY} = 90°$, so $\widehat{HXY} = 45°$: triangle XHY is right isosceles at H. The same holds for XHZ. □

Step 1. Keep halving
Let the area of triangle PQR be 1 (the whole triangle).

- $QS \perp PR$ cuts PQR into two halves: triangle QSR has area $\frac{1}{2}$ and is right isosceles at S.
- $ST \perp QR$ cuts QSR into two halves: triangle STR has area $\frac{1}{4}$ and is right isosceles at T.
- $TU \perp SR$ cuts STR into two halves: triangle STU **(shaded)** has area $\frac{1}{8}$; triangle TUR has area $\frac{1}{8}$ and is right isosceles at U.
- $UV \perp TR$ cuts TUR into two halves: triangle UVR has area $\frac{1}{16}$ and is right isosceles at V.
- $VW \perp UR$ cuts UVR into two halves: triangle UVW **(shaded)** has area $\frac{1}{32}$.

Step 2. Add the shaded areas

$$\frac{1}{8} + \frac{1}{32} = \frac{4}{32} + \frac{1}{32} = \frac{5}{32}.$$

Step 3. Check with a square grid

Put Q at the origin with legs of 8 units: $Q(0,0)$, $R(8,0)$, $P(0,8)$. Then $S(4,4)$, $T(4,0)$, $U(6,2)$, $V(6,0)$, $W(7,1)$ (as in the figure). Triangle PQR has area $\frac{8 \cdot 8}{2} = 32$ squares. Triangle STU has side $ST = 4$ (vertical) and U is 2 units away from ST, so its area is $\frac{4 \cdot 2}{2} = 4$ squares. Triangle UVW has $UV = 2$ and W is 1 unit from UV, area $\frac{2 \cdot 1}{2} = 1$ square. Shaded: $4 + 1 = 5$ squares out of 32, i.e. $\frac{5}{32}$.

Recall. Area of a triangle $= \frac{\text{base} \times \text{height}}{2}$.

Common mistake. Many students pick $\frac{3}{16}$ or $\frac{3}{8}$ by "estimating from the picture". Always compute with the chain of halvings: each smaller triangle is half of the triangle containing it.

Check: Both methods (halving and counting squares) give $\frac{5}{32}$.
Choose option **(D)**.

Answer: (D) $\frac{5}{32}$

Problem 27

Emily went to a second-hand market where, every 15 minutes, the price of every item was reduced by 10% of its *current* price. At 9:00 a.m., the price of a rug was \$10.00. At 9:15 a.m., the price was reduced to \$9.00. As soon as the price of the rug fell *below* \$8.00, Emily bought it. At what time did Emily buy the rug?

(A) 9:45 a.m. (B) 9:15 a.m. (C) 9:30 a.m.

(D) 10:15 a.m. (E) 10:00 a.m.

Detailed solution

Understanding the problem. Each reduction is 10% of the *current price* (not of the original \$10.00). We compute the price after each 15-minute mark until it is less than \$8.00.

Recall. Reducing a number by 10% means taking away $\frac{10}{100} = \frac{1}{10}$ of it; what remains is 90% of it. So: new price = old price $\times 0.9$ (or old price $-$ old price $\div 10$).

Step 1. Make a table of prices over time

Let g_n be the price of the rug (in dollars) after n reductions. Then $g_0 = 10$ and $g_n = g_{n-1} \times 0.9$, i.e. $g_n = 10 \times 0.9^n$.

Time	Price before	Reduction (10%)	New price
9:00 a.m.	–	–	\$10.00
9:15 a.m.	\$10.00	\$1.00	\$9.00
9:30 a.m.	\$9.00	\$0.90	\$8.10
9:45 a.m.	\$8.10	\$0.81	**\$7.29**

Step 2. Find the first time the price is below \$8.00

At 9:30 a.m. the price is \$8.10 – still *more* than \$8.00, so Emily does not buy yet. At 9:45 a.m. the price is \$7.29 $<$ \$8.00: this is the first time it falls below \$8.00. Emily buys it at 9:45 a.m.

Common mistake. (1) The reduction is 10% *of the current price*, so the amount taken off gets smaller each time (\$1.00, then \$0.90, then \$0.81...); it is not \$1.00 every time. If you thought each reduction was \$1.00, you would get \$8.00 at 9:30 a.m. (not yet *below* \$8) and \$7.00 at 9:45 a.m. – the same answer by luck, but the method is wrong. (2) At 9:30 a.m. the price \$8.10 is "very close" to \$8.00 but not yet *below* \$8.00, so it is not the moment to buy.

Check: $10 \times 0.9 \times 0.9 \times 0.9 = 7.29$ (after 3 reductions, i.e. $3 \times 15 = 45$ minutes after 9:00); after 2 reductions: $10 \times 0.81 = 8.1 > 8$.

Choose option **(A)**.

Answer: (A) 9:45 a.m.

Problem 28

The number 2018 is used to create six-digit positive integers: each of them must contain the digits $2, 0, 1, 8$ *together and in this order*. For example, 720 186 is allowed, but 209 318 and 210 893 are not. How many of these six-digit integers are divisible by 9?

(A) 28 (B) 27 (C) 31 (D) 34 (E) 22

Detailed solution

Understanding the problem. The six-digit number contains the block "2018" plus two more digits (call them a and b). We must count those divisible by 9.

> **Recall. Divisibility by 9:** a number is divisible by 9 exactly when the sum of its digits is divisible by 9. The first digit of a six-digit number cannot be 0.

Step 1. Condition on the digit sum

The digits of the block add up to $2+0+1+8=11$. The whole number has digit sum $11+a+b$. Since a, b are digits, $0 \le a+b \le 18$, so $11 \le 11+a+b \le 29$. The multiples of 9 in this range are 18 and 27. Therefore

$$a+b=7 \qquad \text{or} \qquad a+b=16.$$

The only pairs with sum 16 are $(7,9),\ (8,8),\ (9,7)$.

Step 2. Three possible positions of the block 2018

Form	Extra condition	$a+b=7$	$a+b=16$	Total
$\overline{2018ab}$	none (a may be 0)	8 pairs: $a=0,1,\ldots,7$	3	11
$\overline{a2018b}$	$a \neq 0$	7 pairs: $a=1,\ldots,7$	3	10
$\overline{ab2018}$	$a \neq 0$	7 pairs: $a=1,\ldots,7$	3	10
			Total	**31**

About the column "$a+b=7$": the pairs are $(0,7),(1,6),(2,5),(3,4),(4,3),(5,2),(6,1),(7,0)$ – that is 8 pairs. When a is the first digit of the number, we drop $(0,7)$, leaving 7 pairs. The pairs with sum 16 all have $a \ge 7$, so none is dropped.

Step 3. Is any number counted twice?

A number would be counted twice only if the block 2018 appeared in two different places in the same six-digit number. Two 4-digit blocks inside 6 digits must overlap, and the block 2018 cannot overlap itself (for example, shifting it by 1 place would force the digit 0 to coincide with the digit 2...). So no number is counted twice.

> **Common mistake.** It is easy to forget one of the two cases: many students only consider $a+b=7$ (getting $8+7+7=22$, which is exactly the distractor (E)) and forget $a+b=16$. It is also easy to forget that the first digit cannot be 0 (that would give 34).

Check: A few examples: 201 870 has digit sum 18; 720 186 has digit sum 24 – not divisible by 9; 920 187 has digit sum 27, divisible by 9. A computer search of all numbers from 100 000 to 999 999 also gives exactly 31.

Choose option **(C)**.

Answer: (C) 31

Problem 29

The number N is the product of all positive odd integers from 1 to 99 that do *not* end in the digit 5. That is,

$$N = 1 \times 3 \times 7 \times 9 \times 11 \times 13 \times 17 \times 19 \times \cdots \times 91 \times 93 \times 97 \times 99.$$

The units digit of N is

(A) 1 (B) 3 (C) 5 (D) 7 (E) 9

Detailed solution

Understanding the problem. N is a gigantic number that we cannot compute directly. But we only need its *last digit*, and the last digit of a product depends only on the last digits of the factors.

Recall. The units digit of a product equals the units digit of the product of the units digits of the factors. Example: $13 \times 17 = 221$ ends in 1, just like $3 \times 7 = 21$.

Step 1. Which units digits do the factors have?
Odd numbers that do not end in 5 can only end in $1, 3, 7, 9$. Split the factors into 10 groups, one for each ten:

$$(1 \cdot 3 \cdot 7 \cdot 9),\ (11 \cdot 13 \cdot 17 \cdot 19),\ (21 \cdot 23 \cdot 27 \cdot 29),\ \ldots,\ (91 \cdot 93 \cdot 97 \cdot 99).$$

Each group has 4 factors ending in $1, 3, 7, 9$; there are 10 groups (the tens 0 to 9), 40 factors in all.

Step 2. Units digit of each group
$1 \cdot 3 \cdot 7 \cdot 9 = 189$, which ends in 9. So the product of each group ends in 9.

Step 3. Multiply the 10 groups
The units digit of N equals the units digit of $\underbrace{9 \cdot 9 \cdots 9}_{10 \text{ factors}} = 9^{10}$. Now $9 \cdot 9 = 81$ ends in 1. Pair up the 10 nines into 5 pairs, each ending in 1, so 9^{10} ends in $1 \cdot 1 \cdot 1 \cdot 1 \cdot 1 = 1$.

ThinkAlgo Tip. Powers of 9 have alternating last digits: $9^1 \to 9$, $9^2 \to 1$, $9^3 \to 9$, $9^4 \to 1, \ldots$: odd exponents end in 9, even exponents end in 1.

Common mistake. Do not think "a product of odd numbers ends in 9", and do not choose 5 by reflex: the numbers ending in 5 have been left out. Also count the groups correctly: from 1 to 99 there are 10 groups (the first one is $1, 3, 7, 9$).

Check: Step by step: after group 1 the last digit is 9, after group 2 it is $9 \cdot 9 \to 1$, after group 3 it is $9, \ldots$, after group 10 (an even number) it is 1. A computer computing N exactly also gives last digit 1.
Choose option **(A)**.

Answer: (A) 1

Problem 30

Five students wrote a quiz with a maximum score of 50. The scores of four of the students were 42, 43, 46 and 49. The score of the fifth student was N. The mean of the five students' scores was the same as the median of the five students' scores. The number of possible values of N is

(A) 3 (B) 4 (C) 1 (D) 0 (E) 2

Detailed solution

Understanding the problem. N is a score from 0 to 50. The mean depends on N; the median also depends on where N falls when the five scores are put in order. We consider each case for the position of N.

Recall. The **median** of five numbers is the *middle* one (the third) when they are arranged from smallest to largest. The **mean** = sum ÷ how many numbers.

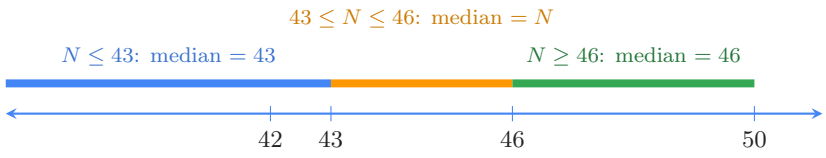
Step 1. Write the mean in terms of N

The four known scores add to $42 + 43 + 46 + 49 = 180$. The mean of the five scores is $\dfrac{180 + N}{5}$.

Step 2. What is the median? – Three cases

Arrange $42, 43, 46, 49$ together with N in increasing order:

- If $N \le 43$: the order is $\ldots, 43, 46, 49$ with N in one of the first two places; the third number is 43. Median = 43.
- If $43 \le N \le 46$: the order is $42, 43, N, 46, 49$. Median = N.
- If $N \ge 46$: the order is $42, 43, 46, \ldots$ with N in one of the last two places. Median = 46.



Step 3. Solve each case

- **Case 1** (median 43): $\dfrac{180 + N}{5} = 43 \Rightarrow 180 + N = 215 \Rightarrow N = 35$. Check: $35 \le 43$ – it is in the case being considered. **Accept.**
- **Case 2** (median N): $\dfrac{180 + N}{5} = N \Rightarrow 180 + N = 5N \Rightarrow 4N = 180 \Rightarrow N = 45$. Check: $43 \le 45 \le 46$. **Accept.**
- **Case 3** (median 46): $\dfrac{180 + N}{5} = 46 \Rightarrow 180 + N = 230 \Rightarrow N = 50$. Check: $50 \ge 46$, and 50 does not exceed the maximum score. **Accept.**

Common mistake. After solving for N in each case, you *must check* that N really belongs to that case (and lies between 0 and 50). In this problem all three values are valid, but in many other problems some values get rejected. Do not stop after finding one value!

Check: $N = 35$: scores $35, 42, 43, 46, 49$, sum 215, mean 43, median 43. $N = 45$: scores $42, 43, 45, 46, 49$, sum 225, mean 45, median 45. $N = 50$: scores $42, 43, 46, 49, 50$, sum 230, mean 46, median 46.
There are 3 values of N, which is option **(A)**.

Answer: **(A)** 3

Extension and challenge

Problem 31

Consider the numbers: -1.5; 0.25; -0.75; 1.2; -2.

a) Arrange the numbers in increasing order.

b) Which numbers lie between -1 and 1 on the number line?

c) Find the sum of the largest and the smallest number.

Detailed solution

Understanding the problem. This problem combines two skills from Problems 1 and 3: comparing decimals and comparing negative numbers.

> **Recall.** Negative numbers are always less than positive numbers. Of two negative numbers, the one farther from 0 is smaller. For two positive numbers, compare the whole parts, then the decimal places one by one.

Step 1. Split into two groups: negative and positive

Negative: -1.5; -0.75; -2. Positive: 0.25; 1.2.

Step 2. Order the negative numbers

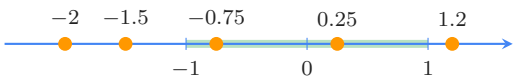
Distance from 0: -2 is 2 away; -1.5 is 1.5 away; -0.75 is 0.75 away. The one farthest from 0 is the smallest:

$$-2 < -1.5 < -0.75.$$

Step 3. Order the positive numbers, then put everything together

$0.25 < 1.2$ (whole parts $0 < 1$). Negative numbers first, positive numbers after:

$$-2 < -1.5 < -0.75 < 0.25 < 1.2.$$



Step 4. The numbers between -1 and 1

On the number line, the green segment from -1 to 1 contains -0.75 and 0.25. (-1.5 is to the left of -1; 1.2 is to the right of 1.)

Step 5. Sum of the largest and the smallest

The largest is 1.2, the smallest is -2:

$$1.2 + (-2) = -(2 - 1.2) = -0.8.$$

> **Recall.** Adding two numbers with opposite signs: subtract the smaller size from the larger (ignoring signs), then use the sign of the number farther from 0.

Common mistake. Many students put -0.75 before -1.5 because "$0.75 < 1.5$". For negative numbers it is reversed: $-1.5 < -0.75$.

Check: $-0.8 - 1.2 = -2$. Correct.

Answer: **a)** $-2 < -1.5 < -0.75 < 0.25 < 1.2$; **b)** -0.75 **and** 0.25; **c)** -0.8

Problem 32

Consider the expression $2 + 3 \times 4 - 1$.

a) Evaluate the expression.

b) Insert *one* pair of brackets ( ) into the expression (without changing the order of the numbers or the signs) to get the largest possible value. What is that largest value?

c) Same question as b), but for the smallest value.

Detailed solution

Understanding the problem. This extends Problems 4 and 7: the position of the brackets changes the order of the operations, and so changes the result.

Recall. Brackets first; then multiplication and division; finally addition and subtraction (from left to right).

Step 1. Value without brackets

$2 + 3 \times 4 - 1 = 2 + 12 - 1 = 13$.

Step 2. List systematically every way to place one pair of brackets

A pair of brackets must enclose at least two neighbouring numbers. Call the four numbers, in order, number $1, 2, 3, 4$ (2; 3; 4; 1). The brackets can start at number i and end at number j with $i < j$: 6 ways.

Brackets around	Expression	Value
numbers 1 – 2	$(2 + 3) \times 4 - 1 = 5 \times 4 - 1$	**19**
numbers 1 – 3	$(2 + 3 \times 4) - 1 = 14 - 1$	13
numbers 1 – 4	$(2 + 3 \times 4 - 1)$	13
numbers 2 – 3	$2 + (3 \times 4) - 1$	13
numbers 2 – 4	$2 + (3 \times 4 - 1) = 2 + 11$	13
numbers 3 – 4	$2 + 3 \times (4 - 1) = 2 + 9$	**11**

Step 3. Read off the results

The largest is 19, with $(2 + 3) \times 4 - 1$: the sum $2 + 3$ is "blown up" 4 times.

The smallest is 11, with $2 + 3 \times (4 - 1)$: 3 is multiplied by a smaller number (3 instead of 4).

ThinkAlgo Tip. For a large result, let the multiplication act on a large sum; for a small result, let it act on a difference. But to be sure, the best way is still to list every case, as in the table.

Check: $(2+3)\times 4-1=20-1=19$ and $2+3\times 3=11$.

Answer: a) 13; **b)** $(2+3)\times 4-1=19$; **c)** $2+3\times(4-1)=11$

Problem 33

Anna's storybook has 360 pages. On the first day Anna reads $\frac{1}{4}$ of the book. On the second day Anna reads $\frac{1}{3}$ *of the remaining pages*. After two days, how many pages has Anna not read yet, and what fraction of the book is that?

Detailed solution

Understanding the problem. Careful: the fraction $\frac{1}{3}$ on the second day is taken of the *remaining pages*, not of the whole book. This is a "two-step" version of Problem 9.

Recall. $\frac{m}{n}$ of a equals $a \div n \times m$.

Step 1. First day
Anna reads $360 \div 4 = 90$ (pages). Remaining: $360 - 90 = 270$ (pages).

Step 2. Second day (taken of 270 pages)
Anna reads $270 \div 3 = 90$ (pages). Remaining: $270 - 90 = 180$ (pages).

Step 3. Fraction of the book left
The unread pages make up $\frac{180}{360} = \frac{180\div 180}{360\div 180} = \frac{1}{2}$ of the book.

Common mistake. A common mistake: adding $\frac{1}{4}+\frac{1}{3}=\frac{7}{12}$ and saying $\frac{5}{12}$ of the book (150 pages) is left. Wrong, because $\frac{1}{3}$ is of the *remaining part* (270 pages), not of the whole book (360 pages).

Step 4. Another way: multiply the "remaining" fractions
After day 1, $\frac{3}{4}$ of the book remains; after day 2, $\frac{2}{3}$ of that remains, i.e. $\frac{2}{3}\times\frac{3}{4}=\frac{6}{12}=\frac{1}{2}$ of the book, which is $360 \div 2 = 180$ pages.

Check: $90+90+180=360$. Correct.

Answer: 180 pages, which is $\frac{1}{2}$ of the book

Problem 34

The first test has 40 questions, and Ben answers 32 of them correctly.

a) What percentage of the questions in the first test did Ben answer correctly?

b) The second test has 60 questions. At least how many questions must Ben answer correctly in the second test so that, over both tests together, his correct answers make up at least 85% of all the questions?

Detailed solution

Understanding the problem. Part a) asks for a percentage (the reverse of Problem 5). Part b) asks for a percentage of a total.

> **Recall.** a as a percentage of b is $\frac{a}{b} \times 100\%$. $p\%$ of a is $a \times \frac{p}{100}$.

Step 1. Percentage in the first test

$\frac{32}{40} \times 100\% = 0.8 \times 100\% = 80\%$.

Step 2. Total number of questions in both tests

$40 + 60 = 100$ (questions).

Step 3. Correct answers needed overall

85% of 100 questions is $100 \times \frac{85}{100} = 85$ (questions).

Step 4. Correct answers needed in the second test

Ben already has 32 correct answers, so he needs at least $85 - 32 = 53$ (questions) in the second test.

> **Common mistake.** Do not take "the average of the two percentages": solving $(80\% + x) \div 2 = 85\%$ gives $x = 90\%$ of 60 questions, i.e. 54 questions – wrong! The tests have different numbers of questions, so we must work with the *total number of questions*.

Check: With 53 correct: $(32+53) \div 100 = 85\%$ (just enough). With 52: $84 \div 100 = 84\% < 85\%$ (not enough).

Answer: a) 80%; b) at least 53 questions

Problem 35

A shop sells cashew nuts in packs: a 3 kg pack costs \$45, a 5 kg pack costs \$70 (no loose sales).

a) Which pack is cheaper per kilogram?

b) Grace needs to buy *exactly* 15 kg. How should she buy them to spend as little as possible? How much does it cost?

c) How much does it cost to buy *exactly* 13 kg?

Detailed solution

Understanding the problem. This extends Problem 6: compare unit prices, then find how to combine packs to get the exact mass.

> **Recall.** Unit price = total cost ÷ quantity.

Step 1. Unit price of each pack

3 kg pack: $45 \div 3 = 15$ (dollars/kg). 5 kg pack: $70 \div 5 = 14$ (dollars/kg).
So the 5 kg pack is cheaper ($14 < 15$).

Step 2. Ways to make exactly 15 kg

Let a be the number of 3 kg packs and b the number of 5 kg packs (a, b whole numbers). We need $3a + 5b = 15$, i.e. $3a = 15 - 5b$. Since $5b \le 15$, $b \in \{0, 1, 2, 3\}$. Try each value:

b	$15 - 5b$	a	Cost (dollars)
0	15	5	$5 \times 45 = 225$
1	10	not divisible by 3	–
2	5	not divisible by 3	–
3	0	0	$3 \times 70 = \mathbf{210}$

Buying 3 packs of 5 kg is cheapest: \$210.

Step 3. Exactly 13 kg

$3a+5b = 13$, and $5b \le 13$ gives $b \in \{0, 1, 2\}$: $b = 0 \Rightarrow 3a = 13$ (not divisible by 3); $b = 1 \Rightarrow 3a = 8$ (not divisible by 3); $b = 2 \Rightarrow 3a = 3$, $a = 1$. There is only one way: 1 pack of 3 kg and 2 packs of 5 kg.
Cost: $45 + 2 \times 70 = 45 + 140 = 185$ (dollars).

> **Common mistake.** Part c) cannot be answered by computing $13 \times 14 = 182$ dollars, because the shop does not sell loose nuts: the packs must add up to exactly 13 kg.

Check: $3+5+5 = 13$ kg and $45+70+70 = 185$; $5+5+5 = 15$ kg and $70+70+70 = 210 < 225$.

Answer: a) the 5 kg pack (\$14/kg); b) 3 packs of 5 kg, \$210; c) \$185

Problem 36

The five members of a group weigh (in kg) 38; 41; 40; 37 and x. Their mean weight is 40 kg.

a) Find x. Is the fifth member heavier or lighter than the mean, and by how many kilograms?

b) Find the median weight of the group.

c) A sixth member joins the group, and the mean weight of all six becomes 41 kg. How much does the sixth member weigh?

Detailed solution

Understanding the problem. This extends Problem 14: use the sum and the deviations to find the missing number; add the median and an "extra member" question.

Recall. Sum = mean × how many numbers. The deviations from the mean add up to 0.

Step 1. Find x using deviations

Compared with 40: $38 \to -2$; $41 \to +1$; $40 \to 0$; $37 \to -3$. Sum: $-2 + 1 + 0 - 3 = -4$.
For the deviations to add up to 0, the fifth member must be $+4$: $x = 40 + 4 = 44$ (kg), 4 kg heavier than the mean.
Check with the sum: $5 \times 40 = 200$; $200 - (38 + 41 + 40 + 37) = 200 - 156 = 44$.

Step 2. Median

In increasing order: 37; 38; **40**; 41; 44. The middle (third) number is 40. The median is 40 kg.

Step 3. The sixth member

Total weight of six members: $6 \times 41 = 246$ (kg). The first five weigh 200 kg in total. The sixth weighs $246 - 200 = 46$ (kg).

ThinkAlgo Tip. Think in deviations: the mean rises by 1 kg for all 6 people, so 6 kg extra above the level of 40 are needed; the new member must weigh $40 + 6 = 46$ kg.

Common mistake. It is not "the sixth member weighs 41 kg" – then the mean would rise only a little: $241 \div 6 \approx 40.17$ kg.

Check: $(37 + 38 + 40 + 41 + 44 + 46) \div 6 = 246 \div 6 = 41$. Correct.

Answer: a) $x = 44$ kg, 4 kg heavier than the mean; b) 40 kg; c) 46 kg

Problem 37 *

A running track around the schoolyard is 200 m long. Mia runs 6 laps in 4 minutes. Tom runs at a constant 18 km/h.

a) Who is faster?

b) Keeping the same speeds, how many minutes does each of them need to run 3 km?

c) They start at the same time from the same point of the track, but run in *opposite directions*. After how many seconds do they meet for the first time?

d) One day Tom increases his speed to 21.6 km/h while Mia runs as before. They start at the same time from the same point, running in the *same direction*. How long is it until Tom is exactly one lap ahead of Mia?

Detailed solution

Understanding the problem. This extends Problem 12: find a speed from laps run, convert km/h to m/s, then study two runners on the same circular track.

Recall. $v = s \div t$, $t = s \div v$. 1 km/h $= \frac{5}{18}$ m/s, so to convert km/h to m/s multiply by 5 and divide by 18.

Step 1. Mia's speed

Distance: $6 \times 200 = 1200$ (m). Time: 4 minutes = 240 seconds. $v_{\text{Mia}} = 1200 \div 240 = 5$ (m/s).

Step 2. Tom's speed (in m/s)

$v_{\text{Tom}} = 18 \times \frac{5}{18} = 5$ (m/s). They run **equally fast**!

Common mistake. Comparing "5" with "18" directly is meaningless, because the units differ (m/s and km/h). Convert to the same unit first.

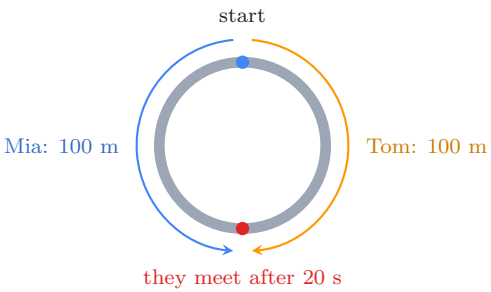
Step 3. Time to run 3 km

3 km = 3000 m. $t = 3000 \div 5 = 600$ (seconds) $= 600 \div 60 = 10$ (minutes). Both need 10 minutes.

Step 4. Opposite directions

Every second Mia runs 5 m and Tom runs 5 m the other way, so the "combined" distance they cover grows by $5 + 5 = 10$ m per second. They first meet when the *total* of their distances is exactly one lap, 200 m:

$$t = 200 \div 10 = 20 \text{ (seconds)}.$$



After 20 seconds each has run $5 \times 20 = 100$ m, exactly half a lap: they meet at the point opposite the start.

Step 5. Same direction, Tom faster

$v_{\text{Tom}} = 21.6 \times \frac{5}{18} = \frac{108}{18} = 6$ (m/s). Every second Tom runs $6 - 5 = 1$ m more than Mia. Tom is exactly one lap ahead when he has run 200 m more than Mia:

$$t = 200 \div 1 = 200 \text{ (seconds)} = 3 \text{ minutes } 20 \text{ seconds}.$$

Check: After 200 seconds: Tom has run $6 \times 200 = 1200$ m (6 laps), Mia $5 \times 200 = 1000$ m (5 laps). Tom is exactly 1 lap ahead.

Note for parents. You can act out parts c) and d) with two coins on a round plate: in opposite directions the "gap to close" shrinks by the *sum* of the speeds; in the same direction it shrinks by the *difference* of the speeds.

Answer: a) equally fast (5 m/s); b) 10 minutes; c) 20 seconds; d) 200 seconds (3 minutes 20 seconds)

Problem 38

a) Find the digits a, b such that the number $\overline{4a7b}$ is divisible by 2, 5 and 9.

b) How many four-digit numbers of the form $\overline{5a3b}$ are divisible by 9? List them.

Detailed solution

Understanding the problem. This extends Problem 10: use several divisibility rules at once and count systematically.

Recall. Divisible by 2 and 5 $\iff$ the last digit is 0. Divisible by 9 $\iff$ the digit sum is divisible by 9.

Why is it true? Why is "divisible by both 2 and 5" the same as "ends in 0"? Divisibility by 2 needs an even last digit; divisibility by 5 needs a last digit of 0 or 5. The only digit that is even and in $\{0;5\}$ is 0. □

Step 1. Part a) – find b
$\overline{4a7b}$ is divisible by 2 and 5, so $b = 0$.

Step 2. Part a) – find a
Digit sum: $4 + a + 7 + 0 = 11 + a$. Since $0 \le a \le 9$, we have $11 \le 11 + a \le 20$. The only multiple of 9 in this range is 18. So $11 + a = 18$, $a = 7$. The number is 4770.

Check: $4770 = 9 \times 530$, and it ends in 0, so it is divisible by 2 and 5.

Step 3. Part b) – the condition
The digit sum of $\overline{5a3b}$ is $5 + a + 3 + b = 8 + a + b$. Since $0 \le a + b \le 18$, we have $8 \le 8 + a + b \le 26$. The multiples of 9 in this range are 9 and 18. Hence $a + b = 1$ or $a + b = 10$.

Step 4. Part b) – count systematically
$a + b = 1$: $(a; b) = (0; 1)$ or $(1; 0)$ – 2 numbers.
$a + b = 10$: a runs from 1 to 9, $b = 10 - a$ (from 9 down to 1) – 9 numbers. ($a = 0$ would give $b = 10$, not a digit.)
In total: $2 + 9 = 11$ numbers:

$$5031,\ 5130,\ 5139,\ 5238,\ 5337,\ 5436,\ 5535,\ 5634,\ 5733,\ 5832,\ 5931.$$

Common mistake. Do not forget that a may be 0 (the first digit is 5), and do not miss the case where the digit sum is 9 (that is, $a + b = 1$).

Check: Each number above has digit sum 9 or 18, e.g. 5 436: $5 + 4 + 3 + 6 = 18$.

Answer: a) $a = 7$, $b = 0$ (the number 4770); b) 11 numbers

Problem 39

In the addition below, M and N are two different non-zero digits. Find all pairs $(M;N)$ that work.

$$\begin{array}{cccc} & M & N \\ & N & M \\ + & N & N \\ \hline 1 & 8 & 7 \end{array}$$

Detailed solution

Understanding the problem. This extends Problem 13. This time, instead of trying column by column, we use place value to get one simple equation, and then find *all* solutions.

Recall. $\overline{MN} = 10M + N$, $\overline{NM} = 10N + M$, $\overline{NN} = 10N + N = 11N$.

Step 1. Write the sum using place value
Using place value for the three addends, we get:

$$\overline{MN} + \overline{NM} + \overline{NN} = (10M + N) + (10N + M) + 11N = 11M + 22N = 11 \times (M + 2N).$$

Step 2. Set up the equation
$11 \times (M + 2N) = 187$, so $M + 2N = 187 \div 11 = 17$.

Step 3. Find every pair of digits
$M = 17 - 2N$ must be a digit from 1 to 9, so $17 - 2N \leq 9$, i.e. $N \geq 4$; and $17 - 2N \geq 1$, i.e. $N \leq 8$.

N	4	5	6	7	8
$M = 17 - 2N$	9	7	5	3	1
Addition	$94 + 49 + 44$	$75 + 57 + 55$	$56 + 65 + 66$	$37 + 73 + 77$	$18 + 81 + 88$

All five pairs have $M \neq N$, and every sum equals 187.

ThinkAlgo Tip. When letters appear "symmetrically", as in $\overline{MN}$ and $\overline{NM}$, try place value: there is often a common factor (11 here) that makes the problem much simpler.

Common mistake. The problem asks for *all* pairs. Finding one pair (say $M = 9$, $N = 4$) and stopping is not enough.

Check: $94 + 49 + 44 = 187$; $75 + 57 + 55 = 187$; $56 + 65 + 66 = 187$; $37 + 73 + 77 = 187$; $18 + 81 + 88 = 187$.

Answer: $(M;N) \in \{(9;4),\ (7;5),\ (5;6),\ (3;7),\ (1;8)\}$

Problem 40 ★★

A sequence is built by the Fibonacci rule: from the third number on, each number is the sum of the two numbers just before it.

a) The first number is 2 and the fifth number is 19. Find the second and the eighth numbers of the sequence.

b) Find the mean of the first five numbers of the sequence in part a).

c) In the familiar Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$, is the 2026th number even or odd? Explain.

Detailed solution

Understanding the problem. This extends Problem 15. In part a) the second number is unknown, so we call it b and express the following numbers in terms of b. In part c) we cannot write out 2026 numbers; we need a pattern of even and odd.

Step 1. Express the numbers in terms of b

Position	1	2	3	4	5
Number	2	b	$2+b$	$b+(2+b)=2+2b$	$(2+b)+(2+2b)=4+3b$

Step 2. Find b

$4+3b=19 \Rightarrow 3b=15 \Rightarrow b=5$. The second number is 5.

Step 3. Continue up to the eighth number

2, 5, 7, 12, 19, 31, 50, 81 $\quad(12+19=31;\ 19+31=50;\ 31+50=81)$. The eighth number is 81.

Step 4. Mean of the first five numbers

$(2+5+7+12+19)\div 5 = 45\div 5 = 9$.

Step 5. The even–odd pattern of the Fibonacci sequence

Recall. odd + odd = even; odd + even = odd; even + odd = odd.

Write O (odd) and E (even) for the numbers $1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$:

Position	1	2	3	4	5	6	7	8	9
Number	1	1	2	3	5	8	13	21	34
Even/odd	O	O	**E**	O	O	**E**	O	O	**E**

The pattern "O, O, E" repeats forever.

Why is it true? Suppose at some point we have two consecutive numbers O, O. Then the next number is O + O = E; the one after is O + E = O; the one after that is E + O = O. We again have two consecutive numbers O, O, so the cycle "O, O, E" starts again. The sequence begins with 1, 1 (O, O), so this pattern repeats all the way: the numbers in positions divisible by 3 are even, all the others are odd. □

Step 6. Look at position 2026

$2026 = 3\times 675 + 1$, i.e. 2026 leaves remainder 1 when divided by 3. This position corresponds to the first "O" in a cycle. So the 2026th number is **odd**.

Common mistake. You cannot conclude "2026 is even, so the 2026th number is even too". The parity of the *position* is different from the parity of the *number* in that position; for example, the 2nd number is 1 (odd).

Check: Part a): $2+5=7$, $5+7=12$, $7+12=19$ – indeed the fifth number is 19. Part c): test the pattern on small positions: position $4=3\times1+1$ holds 3 (odd); position $7=3\times2+1$ holds 13 (odd).

Answer: a) the second number is 5, the eighth is 81; b) 9; c) odd

Problem 41 *

Sums of consecutive numbers (extends Problem 16).

a) Find *all* the ways of writing 100 as a sum of at least two consecutive positive integers.

b) Prove that 64 cannot be written as a sum of at least two consecutive positive integers.

Detailed solution

Understanding the problem. A sum of consecutive positive integers is determined by two pieces of information: *how many numbers* (call it k, with $k\geq2$) and *the first number* (call it a, with $a\geq1$). We need a quick formula for such a sum, and then we use it to "narrow down" the possibilities.

Recall. Sum of an arithmetic sequence (Gauss's method):

$$\text{sum}=\frac{(\text{first}+\text{last})\times\text{number of terms}}{2}.$$

Why is it true? Write the sum twice, once forwards and once backwards, and add column by column: for example $S=9+10+\cdots+16$ and $S=16+15+\cdots+9$. Each column adds up to $9+16=25$ and there are 8 columns, so $2S=25\cdot8$, $S=100$. The same argument works for any arithmetic sequence. □

Step 1. Set up the key equation

With k consecutive numbers starting at a, the last number is $a+k-1$. Let $m=(\text{first}+\text{last})=2a+k-1$. By Gauss's formula: $\text{sum}=\dfrac{m\cdot k}{2}$. If the sum is 100, then

$$k\cdot m=200.$$

Step 2. Two important observations about k and m

- $m-k=2a-1$ is odd, so k **and** m **have opposite parity** (one is even, the other odd).
- $m-k=2a-1\geq1$, so $m>k$.

Step 3. List the factorisations $200 = k \cdot m$ with $2 \le k < m$
Since $k < m$ and $k \cdot m = 200$, we get $k \cdot k < 200$, so $k \le 14$. The divisors of 200 from 2 to 14 are $2, 4, 5, 8, 10$.

k	$m = 200 \div k$	Opposite parity?	Result
2	100	no (both even)	rejected
4	50	no	rejected
5	40	yes	$2a + 4 = 40 \Rightarrow a = 18$
8	25	yes	$2a + 7 = 25 \Rightarrow a = 9$
10	20	no	rejected

Step 4. Write out the sums

$$100 = 18 + 19 + 20 + 21 + 22 \quad \text{and} \quad 100 = 9 + 10 + 11 + 12 + 13 + 14 + 15 + 16.$$

There are exactly **two** ways.

ThinkAlgo Tip. Remember Problem 16: with 5 consecutive numbers (an odd number of terms), the middle number equals the mean $= 100 \div 5 = 20$, so the five numbers are $18, 19, 20, 21, 22$. The "middle number" idea gives a very quick check.

Step 5. Part b): why 64 is impossible
Suppose 64 is a sum of $k \ge 2$ consecutive positive integers. Arguing as above: $k \cdot m = 2 \cdot 64 = 128$ with $m > k \ge 2$ and k, m of opposite parity. So one of k, m is odd and divides 128. But $128 = 2^7$ has only *one* odd divisor, namely 1, while both k and m are ≥ 2. Contradiction! So 64 cannot be written in this way.

Why is it true? The same argument shows that *every power of* 2 $(2, 4, 8, 16, 32, 64, \dots)$ cannot be written as a sum of at least two consecutive positive integers. Conversely, every number with an odd divisor greater than 1 can be. □

Common mistake. Do not forget the word "positive": if 0 and negative numbers were allowed there would be extra "fake" ways, e.g. $(-17) + (-16) + \cdots + 21 + 22 = 100$ (the numbers from -17 to 17 cancel, leaving $18 + 19 + 20 + 21 + 22$). Here every number must be ≥ 1, so $a \ge 1$, which gives $m > k$.

Check: $18 + 19 + 20 + 21 + 22 = 100$; $9 + 10 + \cdots + 16 = \dfrac{(9 + 16) \cdot 8}{2} = 100$. Trying 64 by hand: 2 consecutive numbers have an odd sum; 3 consecutive numbers have a sum divisible by 3; ... no way gives 64.

Answer: a) Two ways: $18 + 19 + 20 + 21 + 22$ and $9 + 10 + \cdots + 16$. b) See the proof.

Problem 42 *

Common divisors and counting divisors (extends Problem 17).

a) How many positive integers divide both 720 and 1080?

b) Find the smallest positive integer with exactly 12 divisors.

Detailed solution

Understanding the problem. Listing the divisors of 720 and 1080 would be very long (each has dozens of divisors). We use two powerful tools: (1) the common divisors of two numbers are the divisors of their GCD; (2) the formula counting divisors from the prime factorisation.

Recall. Counting divisors. If $n = p^x \cdot q^y \cdot r^z$ (p, q, r different primes), then n has $(x+1)(y+1)(z+1)$ divisors.

Why is it true? Each divisor of n has the form $p^i \cdot q^j \cdot r^l$ with $0 \le i \le x$, $0 \le j \le y$, $0 \le l \le z$. The exponent i has $x+1$ choices $(0, 1, \ldots, x)$, j has $y+1$ choices, l has $z+1$ choices. By the multiplication rule there are $(x+1)(y+1)(z+1)$ divisors, and different choices give different divisors (because prime factorisation is unique). □

Part a).

Step 1. Prime factorisation

$720 = 2^4 \cdot 3^2 \cdot 5$ (since $720 = 16 \cdot 45 = 16 \cdot 9 \cdot 5$); $1080 = 2^3 \cdot 3^3 \cdot 5$ (since $1080 = 8 \cdot 135 = 8 \cdot 27 \cdot 5$).

Step 2. Find the GCD

Take each common prime with the *smaller* exponent: $\mathrm{GCD}(720, 1080) = 2^3 \cdot 3^2 \cdot 5 = 8 \cdot 9 \cdot 5 = 360$. (Another check: $1080 - 720 = 360$ and $720 = 2 \cdot 360$, so the GCD is 360.)

Step 3. Count the divisors of 360

$360 = 2^3 \cdot 3^2 \cdot 5^1$ has $(3+1)(2+1)(1+1) = 4 \cdot 3 \cdot 2 = 24$ divisors. So 24 numbers divide both 720 and 1080.

Part b).

Step 4. The "shapes" of a number with exactly 12 divisors

The number of divisors is a product of factors (exponent +1). We write 12 as a product of numbers ≥ 2:

$$12 = 12, \qquad 12 = 6 \cdot 2, \qquad 12 = 4 \cdot 3, \qquad 12 = 3 \cdot 2 \cdot 2.$$

These correspond to four shapes: p^{11}, $p^5 q$, $p^3 q^2$, $p^2 qr$.

Step 5. The smallest number of each shape

To make the number small, use the smallest primes $2, 3, 5$, and give the *larger* exponent to the *smaller* prime:

Shape	Smallest number	Value
p^{11}	2^{11}	2048
$p^5 q$	$2^5 \cdot 3$	96
$p^3 q^2$	$2^3 \cdot 3^2$	72
$p^2 qr$	$2^2 \cdot 3 \cdot 5$	**60**

The smallest is 60.

Common mistake. Do not give the large exponent to the large prime: for example $2 \cdot 3^5 = 486$ is much bigger than $2^5 \cdot 3 = 96$. And do not forget the shape with *three* primes (p^2qr) – it is the one that gives the answer.

Check: The divisors of 60: $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$ – exactly 12. A computer check of the numbers from 1 to 59: none has 12 divisors.

Answer: a) 24 numbers. b) 60.

Problem 43 *

How does a fraction change? (extends Problems 18 and 19).

a) When the same whole number n is added to both the numerator and the denominator of $\frac{5}{8}$, the result is a fraction equal to $\frac{4}{5}$. Find n.

b) Let $\frac{a}{b}$ be a fraction with $0 < a < b$. Prove that for every positive integer n, $\frac{a+n}{b+n} > \frac{a}{b}$.

c) How many fractions equal to $\frac{3}{7}$ have a two-digit denominator?

Detailed solution

Understanding the problem. Parts a) and b) are both about "adding the same number to the numerator and the denominator". What does *not* change when we do that is the *difference* between the denominator and the numerator – this is the key.

Part a).

Step 1. Find the quantity that stays the same

For $\frac{5}{8}$, denominator − numerator is $8 - 5 = 3$. After adding n to both: $(8+n) - (5+n) = 3$, still 3.

Step 2. Set up an equation

After adding n to both the numerator and the denominator we get the fraction $\frac{5+n}{8+n}$. By the problem:

$$\frac{5+n}{8+n} = \frac{4}{5}.$$

Recall. Property of proportions: if $\frac{a}{b} = \frac{c}{d}$ (with $b, d \neq 0$), then $a \cdot d = b \cdot c$ (the cross products are equal).

Applying it: $5 \cdot (5+n) = 4 \cdot (8+n)$.

Step 3. Solve the equation for n

$$25 + 5n = 32 + 4n \implies 5n - 4n = 32 - 25 \implies n = 7.$$

Then the new numerator is $5+7=12$ and the new denominator is $8+7=15$; the difference is still $15-12=3$ (as in Step 1).

Check: $\dfrac{5+7}{8+7}=\dfrac{12}{15}=\dfrac{12\div 3}{15\div 3}=\dfrac{4}{5}$. Correct.

Part b).

Step 4. Compare "what is missing to reach 1"

Since $a<b$, both fractions are less than 1. What is missing to make 1 is

$$1-\frac{a}{b}=\frac{b-a}{b},\qquad 1-\frac{a+n}{b+n}=\frac{(b+n)-(a+n)}{b+n}=\frac{b-a}{b+n}.$$

The fractions $\dfrac{b-a}{b}$ and $\dfrac{b-a}{b+n}$ have *the same positive numerator*, and the denominator $b+n>b$.

The fraction with the larger denominator is smaller, so $\dfrac{b-a}{b+n}<\dfrac{b-a}{b}$.

Step 5. Conclusion

The fraction $\dfrac{a+n}{b+n}$ is "missing less" to reach 1, so it is larger: $\dfrac{a+n}{b+n}>\dfrac{a}{b}$.

Recall. Two positive fractions with the same numerator: the one with the larger denominator is smaller (the same cake shared among more people gives smaller pieces).

Why is it true? Another way (cross-multiplying): with positive denominators, $\dfrac{a+n}{b+n}>\dfrac{a}{b}$ exactly when $(a+n)\cdot b>a\cdot(b+n)$, i.e. $ab+nb>ab+na$, i.e. $nb>na$. This is true because $n>0$ and $b>a$. □

Common mistake. This is exactly why in Problem 18 we may *not* "add" to the numerator and denominator: adding changes the fraction (a fraction less than 1 gets bigger). Only *multiplying* (or dividing) both by the same non-zero number keeps the fraction the same.

Part c).

Step 6. General form and condition

Fractions equal to $\dfrac{3}{7}$ have the form $\dfrac{3k}{7k}$ (k a positive integer). The denominator $7k$ has two digits: $10\le 7k\le 99$. We have $7\cdot 1=7<10$, $7\cdot 2=14\ge 10$; $7\cdot 14=98\le 99$, $7\cdot 15=105>99$. So $k=2,3,\ldots,14$: that is $14-2+1=13$ values. The fractions are $\dfrac{6}{14},\dfrac{9}{21},\ldots,\dfrac{42}{98}$.

Answer: a) $n=7$. b) See the proof. c) 13 fractions.

Problem 44 *

Poles, a bicycle and a train (extends Problem 20).

a) Leo cycles steadily at 18 km/h along a road whose power poles are equally spaced 50 m apart. How many minutes pass from the moment Leo passes the first pole to the moment he passes the 25th pole?

b) A train travelling at constant speed takes 9 seconds to pass a power pole and 30 seconds to cross a bridge 350 m long (from the moment the front of the train reaches the bridge until the end of the train has just left it). Find the length of the train and its speed in km/h.

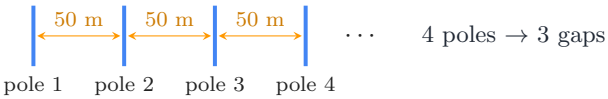
Detailed solution

Understanding the problem. Part a) hides the "fence-post trap": 25 poles make how many gaps? Part b) requires understanding how far a train (an object with length) must travel to "pass" something.

Recall. Distance = speed × time. Converting 1 km/h: 1000 m in 3600 seconds, so 1 km/h $= \frac{1000}{3600} = \frac{5}{18}$ m/s; conversely 1 m/s = 3.6 km/h.

Part a).

Step 1. Count the gaps between poles



Between the first and the 25th pole there are $25 - 1 = 24$ gaps, with total length $24 \cdot 50 = 1200$ m.

Step 2. Convert the speed and find the time

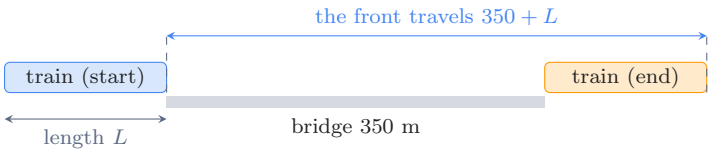
18 km/h $= 18 \cdot \frac{5}{18} = 5$ m/s. Time: $1200 \div 5 = 240$ seconds $= 240 \div 60 = 4$ minutes.

Common mistake. Counting 25 gaps ($25 \cdot 50 = 1250$ m) gives 250 seconds – wrong. The number of gaps is always one less than the number of poles (counting from the first pole to the last).

Part b).

Step 3. What "passing a pole" and "crossing a bridge" mean

- Passing a pole: from when the front of the train is level with the pole until the end of the train is level with it, the train travels exactly *one train length* L.
- Crossing the bridge: from when the front is at the start of the bridge until the end leaves the far side, the train travels *the bridge length plus the train length*: $350 + L$.



Step 4. Compare the two situations
Crossing the bridge takes 30 seconds, passing the pole takes 9 seconds. The difference $30 - 9 = 21$ seconds is exactly the time to travel the extra 350 m (the bridge length). The speed of the train is

$$v = 350 \div 21 = \frac{350}{21} = \frac{50}{3} \text{ m/s}.$$

Step 5. Find the train length and convert the speed
Train length $L = v \cdot 9 = \frac{50}{3} \cdot 9 = 150$ m. Speed: $\frac{50}{3} \cdot 3.6 = \frac{50 \cdot 3.6}{3} = 60$ km/h.

Common mistake. A common mistake: thinking the train travels only 350 m to cross the bridge (forgetting to add the train length); then the speed would be $350 \div 30 \approx 11.7$ m/s, which is wrong. Follow *one fixed point* on the train (e.g. its front) to know how far the train has travelled.

Check: With $v = \frac{50}{3}$ m/s and $L = 150$ m: passing the pole takes $150 \div \frac{50}{3} = 9$ seconds; crossing the bridge takes $(350 + 150) \div \frac{50}{3} = 500 \cdot \frac{3}{50} = 30$ seconds. Both facts check out.

Answer: a) 4 minutes. b) The train is 150 m long; its speed is 60 km/h.

Problem 45 ★★

Products of digits (extends Problem 21).

a) How many four-digit numbers have digits whose product is 36?

b) Find the smallest positive integer whose digits have product 360.

c) Find the largest positive integer with *all different* digits whose product is 360.

Detailed solution

Understanding the problem. All three parts start by *factorising into primes*, then "sharing" the prime factors among the digits (each digit from 1 to 9). Note: no digit can be 0 (otherwise the product is 0).

Recall. $36 = 2^2 \cdot 3^2$; $360 = 2^3 \cdot 3^2 \cdot 5$. The digits and their factorisations: 1; 2; 3; $4 = 2^2$; 5; $6 = 2 \cdot 3$; 7; $8 = 2^3$; $9 = 3^2$.

Part a).

Step 1. Find the sets of four digits (ignoring order) with product 36

As in Problem 21, list $a \le b \le c \le d$:

- Two digits equal to 1: the other two digits have product 36: $\{1, 1, 4, 9\}$, $\{1, 1, 6, 6\}$.
- Exactly one digit 1: the other three digits (≥ 2) have product 36: $\{1, 2, 2, 9\}$, $\{1, 2, 3, 6\}$, $\{1, 3, 3, 4\}$.
- No digit 1: four digits ≥ 2 with product 36: only $\{2, 2, 3, 3\}$ (since $36 = 2 \cdot 2 \cdot 3 \cdot 3$ cannot be split further).
- Three digits equal to 1: the remaining digit would be 36 – impossible.

Recall. Arranging 4 digits: all four different: $4 \cdot 3 \cdot 2 \cdot 1 = 24$ ways. Exactly two equal (the other two different from each other and from them): choose 2 of the 4 positions for the equal pair (6 ways), then place the other two digits in the 2 remaining spots (2 ways): 12 ways. Two equal pairs (like $1, 1, 6, 6$): just choose 2 positions for one pair: 6 ways.

Step 2. Count

Digit set	Type	Arrangements
$\{1, 1, 4, 9\}$	one equal pair	12
$\{1, 1, 6, 6\}$	two equal pairs	6
$\{1, 2, 2, 9\}$	one equal pair	12
$\{1, 2, 3, 6\}$	all different	24
$\{1, 3, 3, 4\}$	one equal pair	12
$\{2, 2, 3, 3\}$	two equal pairs	6
	Total	**72**

Part b).

Step 3. Use as few digits as possible

A number with fewer digits is smaller. One or two digits give a product of at most $9 \cdot 9 = 81 < 360$, so we need at least 3 digits.

Step 4. Find three digits with product 360

360 has the prime factor 5, and the only digit containing the factor 5 is 5 itself. So one digit is 5, and the other two have product $360 \div 5 = 72 = 8 \cdot 9$ (the only way with digits ≤ 9). Digit set: $\{5, 8, 9\}$.

Step 5. Arrange to get the smallest number

Arrange the digits from smallest to largest: 589.

Part c).

Step 6. For the largest number: as many digits as possible

The digits are different, non-zero, and one of them must be 5. Six different digits have product at least $1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720 > 360$, so there are at most 5 digits.

Step 7. Find the sets of 5 different digits with product 360

One digit is 5; the other four are different, not 5, with product $72 = 2^3 \cdot 3^2$. Four different digits with a product as small as 72 must include 1 (since $2 \cdot 3 \cdot 4 \cdot 6 = 144 > 72$); the remaining three different digits have product 72: $72 = 2 \cdot 4 \cdot 9 = 3 \cdot 4 \cdot 6$ (while $2 \cdot 6 \cdot 6$ and $3 \cdot 3 \cdot 8$ repeat a digit, and $1 \cdot 8 \cdot 9$ repeats the digit 1). Two sets: $\{1, 2, 4, 5, 9\}$ and $\{1, 3, 4, 5, 6\}$.

Step 8. Arrange to get the largest number

Arrange from largest to smallest: 95421 and 65431. The larger is 95421.

Common mistake. In part c), do not jump to a number with a big first digit such as 985 (product 360): a 5-digit number is always larger than a 3-digit number. Maximise the *number of digits* first, and only then the leading digit.

Check: $5 \cdot 8 \cdot 9 = 360$; $9 \cdot 5 \cdot 4 \cdot 2 \cdot 1 = 360$. A computer search over all numbers with up to 6 digits confirms: there are 72 four-digit numbers with digit product 36; the smallest number with digit product 360 is 589; the largest with all different digits and digit product 360 is 95 421.

Answer: a) 72 numbers. b) 589. c) 95 421.

Problem 46 *

Arithmetic grids (extends Problem 22). In a square grid, the numbers in every row and in every column form arithmetic sequences.

a) A 4×4 grid has these four corner numbers: top left 1, top right 10, bottom left 19, bottom right 37. Fill in the whole grid.

b) Find the sum of all 16 numbers in the grid of part a). Compare it with the sum of the four corners: what do you notice? Explain.

c) Prove that in a 3×3 grid of this kind, the number in the centre always equals the mean of the four corner numbers.

1			10
19			37

Detailed solution

Understanding the problem. Only the four corners are given. The first and last columns (and the first and last rows) each have a known first and last term, so they can be found immediately. Then we use the middle rows or columns.

Recall. Arithmetic sequence with 4 terms: difference $d = (\text{last} - \text{first}) \div 3$. Sum of the 4 terms $= 2 \times (\text{first} + \text{last})$ (pair term 1 with term 4, term 2 with term 3: both pairs have the same sum). With 3 terms: middle term = mean of the two outer terms.

Part a).

Step 1. First and last rows

Row 1: from 1 to 10, $d = (10 - 1) \div 3 = 3$: $1, 4, 7, 10$.
Row 4: from 19 to 37, $d = (37 - 19) \div 3 = 6$: $19, 25, 31, 37$.

Step 2. The four columns (first and last terms of each are known)

Column 1: $1 \to 19$, $d = 6$: $1, 7, 13, 19$. Column 2: $4 \to 25$, $d = 7$: $4, 11, 18, 25$.
Column 3: $7 \to 31$, $d = 8$: $7, 15, 23, 31$. Column 4: $10 \to 37$, $d = 9$: $10, 19, 28, 37$.

Step 3. Check the middle rows

Row 2: $7, 11, 15, 19$ (difference 4); row 3: $13, 18, 23, 28$ (difference 5). The grid satisfies every condition:

1	4	7	10	row sum: 22
7	11	15	19	row sum: 52
13	18	23	28	row sum: 82
19	25	31	37	row sum: 112

Part b).

Step 4. Compute the sum

Sum of the rows: $22 + 52 + 82 + 112 = 268$. Sum of the corners: $1 + 10 + 19 + 37 = 67$. We see that $268 = 4 \cdot 67$: **the sum of the whole grid is 4 times the sum of the corners.**

Step 5. Explanation (true for every such 4×4 grid)

Each row is an arithmetic sequence of 4 terms, so each row sum $= 2 \times$(first term + last term of the row). Adding the 4 rows:

$$\text{Grid sum} = 2 \times (\text{sum of column 1} + \text{sum of column 4}).$$

Columns 1 and 4 are also arithmetic sequences of 4 terms: sum of column $1 = 2 \times (1 + 19)$, sum of column $4 = 2 \times (10 + 37)$. So

$$\text{Grid sum} = 2 \times \big(2 \cdot (1 + 19) + 2 \cdot (10 + 37)\big) = 4 \times (1 + 10 + 19 + 37) = 4 \times 67 = 268.$$

Part c).

Step 6. Name the cells of the 3×3 grid

a	$\cdot$	b
u	x	v
c	$\cdot$	d

The corners are a, b, c, d; the centre is x; u and v are the middle cells of the left and right columns.

Step 7. Use "middle term = mean of the outer terms"

The left column a, u, c is arithmetic, so $u = \dfrac{a+c}{2}$. Right column: $v = \dfrac{b+d}{2}$. The middle row u, x, v is arithmetic, so

$$x = \frac{u+v}{2} = \frac{\frac{a+c}{2} + \frac{b+d}{2}}{2} = \frac{a+b+c+d}{4}.$$

This is exactly the mean of the four corner numbers.

Why is it true? Why does an arithmetic sequence p, q, r satisfy $q = \dfrac{p+r}{2}$? Because $q - p = r - q$ (both equal the difference d), so $2q = p + r$. □

Check: Apply part c) to the top-left 3×3 block of the grid in part a): corners $1, 7, 13, 23$, mean $\dfrac{1+7+13+23}{4} = 11$, exactly the centre 11.

Answer: a) See the grid. b) Sum $268 = 4 \times 67$ (4 times the sum of the corners). c) See the proof.

Problem 47 **

"Rotating" numbers (extends Problem 23). In each part, every letter represents a digit, and $A \neq 0$.

a) Find the six-digit number $\overline{2ABCDE}$ such that $\overline{2ABCDE} \times 3 = \overline{ABCDE2}$.

b) Find all digits d (from 1 to 9) for which there is a six-digit number $\overline{dABCDE}$ with $\overline{dABCDE} \times 3 = \overline{ABCDEd}$.

c) Prove that there is no six-digit number $\overline{1ABCDE}$ with $\overline{1ABCDE} \times 2 = \overline{ABCDE1}$.

Detailed solution

Understanding the problem. All three parts are about "moving the first digit to the end". We reuse Method 1 of Problem 23: treat the block $\overline{ABCDE}$ as one number N and set up an equation. This time we do it once for every leading digit d.

Recall. With $N = \overline{ABCDE}$: $\overline{dABCDE} = d \cdot 100\,000 + N$ and $\overline{ABCDEd} = 10 \cdot N + d$. Since $A \neq 0$, N has exactly five digits: $10\,000 \leq N \leq 99\,999$.

Step 1. The general equation for parts a) and b)

$$3 \cdot (d \cdot 100\,000 + N) = 10N + d \implies 300\,000\,d + 3N = 10N + d \implies 299\,999\,d = 7N.$$

Since $299\,999 = 7 \cdot 42\,857$, dividing both sides by 7 gives

$$N = 42\,857 \cdot d.$$

Step 2. Part a): $d = 2$
$N = 42\,857 \cdot 2 = 85\,714$. The number is 285 714.

Step 3. Part b): when does N have five digits?
$N = 42\,857\,d \leq 99\,999$. For $d = 1$: $N = 42\,857$ (five digits, fine). For $d = 2$: $N = 85\,714$ (fine). For $d = 3$: $N = 128\,571$ has six digits – too big to be the block $\overline{ABCDE}$. For $d \geq 3$, N is even bigger. So only $d = 1$ and $d = 2$ work, giving the two numbers 142 857 and 285 714.

ThinkAlgo Tip. These two numbers use the same digits $1, 4, 2, 8, 5, 7$ arranged in a circle. That is no accident: $\frac{1}{7} = 0.\overline{142857}$ and $\frac{2}{7} = 0.\overline{285714}$. Multiplying by 3 corresponds to "rotating" the circle of digits.

Step 4. Part c): set up the equation
With $N = \overline{ABCDE}$:

$$2 \cdot (100\,000 + N) = 10N + 1 \implies 200\,000 + 2N = 10N + 1 \implies 199\,999 = 8N.$$

Step 5. Part c): an odd/even contradiction
The right-hand side $8N$ is even (since 8 is even). The left-hand side 199 999 is odd. An odd number cannot equal an even number. So no such number exists.

Why is it true? You can see it immediately without an equation: $\overline{1ABCDE} \times 2$ is even, so its last digit must be even; but $\overline{ABCDE1}$ ends in 1 (odd). Contradiction. □

Common mistake. In part b), many students stop after finding $d = 1$ (the answer of Problem 23) and forget $d = 2$; or they accept $d = 3, 4, \ldots$ without checking whether N has more than five digits. Always check the condition "N has exactly five digits".

Check: $142\,857 \times 3 = 428\,571$; $285\,714 \times 3 = 857\,142$. In both cases the first digit has indeed moved to the end.

Answer: a) 285 714. **b)** $d = 1$ **or** $d = 2$ **(the numbers** 142 857 **and** 285 714**). c) See the proof.**

Problem 48 ★★

A frog on the number line (extends Problem 24). A frog sits at the point 0 on the number line. On its k-th jump ($k = 1, 2, 3, \dots$), the frog jumps exactly k units, to the left or to the right as it likes.

a) After exactly 10 jumps, can the frog be back at 0? Why?

b) After exactly 10 jumps, which points can the frog be at? How many such points are there?

c) What is the smallest number of jumps the frog needs to reach the point 20?

Detailed solution

Understanding the problem. The frog's position after n jumps is $\pm 1 \pm 2 \pm 3 \cdots \pm n$ (a $+$ sign for a jump to the right, a $-$ sign for a jump to the left). The idea is the same as in Problem 24: start from "all plus" and see how switching signs changes the result.

> **Recall.** $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$ (Gauss's formula). For example $1 + 2 + \cdots + 10 = \dfrac{10 \cdot 11}{2} = 55$.

Step 1. The "switching" rule

If the frog always jumps right, after 10 jumps it is at 55. Switching one jump k from right to left lowers the position by $2k$ (it loses k and goes back another k). So if the jumps to the left add up to s, the final position is

$$55 - 2s.$$

Part a).

Step 2. Odd or even

$55 - 2s$ is always *odd* (odd minus even). The point 0 is even, so the frog **cannot** be back at 0 after 10 jumps.

Part b).

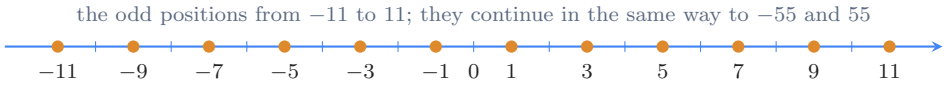
Step 3. Which values can s take?

s is the sum of a group (possibly empty) of different numbers from $\{1, 2, \dots, 10\}$, so $0 \le s \le 55$. We show that *every* integer s from 0 to 55 can be made.

> **Why is it true?** Take the largest numbers one by one: $10, 9, 8, \dots$ The totals 0, 10, $10 + 9 = 19$, $19 + 8 = 27$, $27 + 7 = 34$, $34 + 6 = 40$, $40 + 5 = 45$, $45 + 4 = 49$, $49 + 3 = 52$, $52 + 2 = 54$, $54 + 1 = 55$ cut the interval $[0, 55]$ into pieces. A number s between two consecutive marks, e.g. $27 < s \le 34$, is written as $s = 10 + 9 + 8 + (s - 27)$ with $s - 27$ from 1 to 7 – a number not yet used. The same works for every other piece (for $0 < s \le 10$ just take the number s). So every s from 0 to 55 can be made. □

Step 4. The final positions

For $s = 0, 1, 2, \dots, 55$, the position $55 - 2s$ takes the values $55, 53, 51, \dots, -53, -55$: that is, **every odd number from -55 to 55**. Counting: from -55 to 55 there are 28 positive odd numbers $(1, 3, \dots, 55)$ and 28 negative ones, 56 points altogether (the same as the number of values of s: 56).



Part c).

Step 5. Two necessary conditions

After n jumps, let $T = 1 + 2 + \cdots + n$. The position is $T - 2s$, so: (1) $T \geq 20$ (the frog cannot get farther than T); (2) $T - 20$ must be even, i.e. T must be even like 20.

Step 6. Try n one at a time

n	1	2	3	4	5	6	7
T	1	3	6	10	15	21	28
Reach 20?	no ($T < 20$)					no (T odd)	**possible**

Step 7. Show a way with $n = 7$

We need $28 - 2s = 20$, i.e. $s = 4$: just jump left on the 4th jump:

$$1 + 2 + 3 - 4 + 5 + 6 + 7 = 20.$$

So the smallest number of jumps is 7.

Common mistake. Checking only "$T \geq 20$" is not enough: for $n = 6$, $T = 21 \geq 20$, but the position is always odd and can never be 20. This is exactly the odd/even idea of Problem 24.

Check: $1 + 2 + 3 - 4 + 5 + 6 + 7 = 20$. For $n = 6$ the position is always $21 - 2s$, an odd number, never 20. A computer search of all jump patterns also confirms: after 10 jumps there are exactly 56 positions, not including 0; the smallest number of jumps to reach 20 is 7.

Answer: a) No (the position is always odd). b) Every odd number from −55 to 55: 56 points. c) 7 jumps.

Problem 49 ⋆

Numbers containing the block 2025 (extends Problem 28). Consider the six-digit numbers that contain the four digits $2, 0, 2, 5$ *together and in this order* (for example: 720 251, 202 513, 132 025). How many of these numbers are divisible by 45?

Detailed solution

Understanding the problem. The number consists of the block 2025 plus two more digits a, b in different positions. "Divisible by 45" should be split into two conditions that are easier to check.

Recall. $45 = 5 \cdot 9$, and 5 and 9 have no common divisor other than 1. So a number is divisible by 45 exactly when it is divisible by both 5 and 9. **Divisible by 5:** the last digit is 0 or 5. **Divisible by 9:** the digit sum is divisible by 9.

Step 1. The condition for divisibility by 9
The digits of the block add up to $2+0+2+5=9$. The whole number has digit sum $9+a+b$, which is divisible by 9 when $a+b$ is divisible by 9, i.e. $a+b \in \{0,\ 9,\ 18\}$.

Step 2. The three positions of the block, case by case

- **Form** $\overline{2025ab}$ (the last digit is b, so $b \in \{0,5\}$; a is any digit): $b=0$: $a \in \{0,9\} \to 202\,500$, $202\,590$. $b=5$: $a+5 \in \{9,18\} \to a=4 \to 202\,545$. **3 numbers.**
- **Form** $\overline{a2025b}$ ($a \neq 0$, $b \in \{0,5\}$): $b=0$: $a=9 \to 920\,250$. $b=5$: $a=4 \to 420\,255$. **2 numbers.**
- **Form** $\overline{ab2025}$ (ends in 5 – always divisible by 5; $a \neq 0$): $a+b=9$: $a=1,2,\ldots,9$ (with $b=9-a$) $\to$ 9 numbers: $182\,025$, $272\,025$, …, $902\,025$. $a+b=18$: $a=b=9 \to 992\,025$. ($a+b=0$ is impossible because $a \neq 0$.) **10 numbers.**

Step 3. Check for double counting
A number would be counted twice if the block 2025 appeared twice among its six digits, i.e. two overlapping blocks. Shifting the block by 1 place: "025" would have to match "202" – false; shifting by 2 places: "25" would have to match "20" – false. So no number is counted twice.

Step 4. Add up
$3+2+10=15$ numbers.

Common mistake. (1) Do not forget the case $a+b=0$ in the form $\overline{2025ab}$ (the number $202\,500$). (2) In the forms $\overline{a2025b}$ and $\overline{ab2025}$, the first digit a cannot be 0. (3) Check *both* conditions, divisibility by 5 and by 9, not just one.

Check: Example $992\,025$: it ends in 5, and its digit sum $9+9+2+0+2+5=27$ is divisible by 9, so it is divisible by 45 ($992\,025 = 45 \cdot 22\,045$). A computer search over all six-digit numbers also gives exactly 15.

Answer: 15 numbers.

Problem 50 **

Mean and median (extends Problem 30). A list of five positive integers (repeats allowed) has mean 12 and median 10.

a) What is the largest possible value of the largest number in the list?

b) What is the smallest possible value of the largest number in the list?

c) Answer both questions again if the five numbers must all be different.

Detailed solution

Understanding the problem. Arrange the five numbers in increasing order: $a \le b \le c \le d \le e$. The median is the middle number, so $c = 10$. The mean is 12, so the sum is $5 \cdot 12 = 60$. We want to make e as large or as small as possible. This is an "optimisation" problem: we must both *prove* that a limit cannot be passed and *give an example* that reaches exactly that limit.

Recall. Mean = sum ÷ how many numbers, so sum = $12 \cdot 5 = 60$. The median of five numbers (in increasing order) is the third one.

Step 1. The key equation

$$a + b + 10 + d + e = 60 \quad \Longrightarrow \quad a + b + d + e = 50, \qquad a \le b \le 10 \le d \le e.$$

Part a): make e as large as possible.

Step 2. Upper limit

For e to be large, the other numbers must be as small as possible: $a \ge 1$, $b \ge 1$, $d \ge 10$. Hence $e = 50 - a - b - d \le 50 - 1 - 1 - 10 = 38$.

Step 3. An example that reaches it

The list $1, 1, 10, 10, 38$: sum 60, median 10. So the largest possible e is 38.

Part b): make e as small as possible.

Step 4. Lower limit

For e to be small, a and b must be as large as possible, but $a \le b \le 10$, so $a + b \le 20$. Hence $d + e = 50 - (a + b) \ge 30$. Since $d \le e$, $2e \ge d + e \ge 30$, so $e \ge 15$.

Step 5. An example that reaches it

The list $10, 10, 10, 15, 15$: sum 60, median 10. So the smallest possible e is 15.

Part c): five different numbers ($a < b < 10 < d < e$).

Step 6. Largest e

As small as possible: $a \ge 1$, $b \ge 2$, $d \ge 11$. So $e \le 50 - 1 - 2 - 11 = 36$. Example: $1, 2, 10, 11, 36$ (sum 60). So the largest e is 36.

Step 7. Smallest e

As large as possible: $b \le 9$, $a \le 8$, so $a + b \le 17$ and $d + e \ge 50 - 17 = 33$. Since $d < e$, $d \le e - 1$, so $2e - 1 \ge d + e \ge 33$, i.e. $e \ge 17$. Example: $8, 9, 10, 16, 17$ (sum 60). So the smallest e is 17.

Common mistake. A "largest/smallest" problem needs *two* parts: (1) a proof that the limit cannot be passed, and (2) a concrete example reaching exactly that limit. An example alone may not be the best; a limit alone may not be reachable.

Note for parents. Let your child share 60 sweets among 5 "boxes" kept in order, with the middle box always holding 10. Ask: to make the last box as full as possible, should the other boxes hold few or many? And to make the last box as small as possible? This game helps children "feel" optimisation reasoning.

Check: $1 + 1 + 10 + 10 + 38 = 60$; $10 + 10 + 10 + 15 + 15 = 60$; $1 + 2 + 10 + 11 + 36 = 60$; $8 + 9 + 10 + 16 + 17 = 60$; all have median 10. A computer search over all lists also confirms the values 38, 15, 36, 17.

Answer: a) 38. b) 15. c) Largest 36, smallest 17.

5 Others: Logic and Puzzles

Big idea. Logical thinking is *the art of drawing certain conclusions from what we already know.* A logic puzzle usually needs no hard calculation: the difficulty lies in *reading carefully*, *recording information systematically* and *not missing any case.* In this chapter you will get to know the following "tools":

- **truth-tellers and liars**, "half-true" statements, and how to negate a statement;
- **case analysis** and **proof by contradiction** (assume, then look for a contradiction);
- **ordering** with arrow diagrams and number lines; telling "*must be true*" apart from "*could be true*";
- **shift ciphers** (the Caesar cipher) and **remainders** on division by 26 (or by 7 for the days of the week);
- **magic squares**, **balance scales** (which are really equations), **sums on figures** and **dice**;
- **Latin squares** and 4×4 **sudoku**.

Four golden habits: *(1) write it down – use symbols; (2) draw it – diagrams, tables; (3) consider every case; (4) check the answer against* ***all*** *the conditions of the problem.*

① True and false statements: truth-tellers and liars

1. Statements. A sentence that is either *true* or *false* (it cannot be both) is called a statement. Examples: "7 is odd" (true), "Paris is the capital of Italy" (false).

2. Truth-tellers and liars. In puzzles, everything a *truth-teller* says is true, and everything a *liar* says is false. Therefore:

- if we know who the speaker is, we know whether the statement is true or false;
- conversely, if we know whether the statement is true or false, we know who the speaker is.

3. Negating a statement (saying "the opposite").

Statement	Negation (true exactly when the statement is false)
Anna *and* Ben are both liars	*At least one* of the two tells the truth
Every apple is red	*There is at least one* apple that is not red
The flower is a red rose	The flower is not red, *or* it is not a rose (or both)

4. "Half-true" statements. If a statement has two parts and we know that *exactly one* part is true, then: first part true ⇒ second part false; first part false ⇒ second part true. Decide true/false for *one* part, and all the other parts follow like a row of dominoes.

5. Lying on certain days. Someone who "lies on Tuesdays" says only false things on Tuesday. To find on which days they *could* make a given statement, make a table: for each day, decide whether the statement is true or false, and compare with whether the person tells the truth or lies on that day.

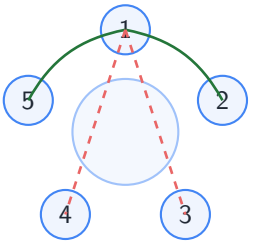
Why is it true? Why can nobody say "I am a liar"? If the speaker were a truth-teller, the statement would be true, so they would be a liar – impossible. If the speaker were a liar, the statement would be false, so they would *not* be a liar – impossible too. Both cases lead to a contradiction, so the sentence can never be spoken (on an "island" with only truth-tellers and liars). □

Common mistake. The negation of "Anna and Ben are both liars" is *not* "Anna and Ben both tell the truth". As soon as *one* of them tells the truth, the original statement is already false.

ThinkAlgo Tip. For a truth-and-lies puzzle, draw a small table: each row is one possibility (who tells the truth, who lies), each column is one statement. Fill each cell with T/F; the row with no contradiction is the answer.

② Case analysis and proof by contradiction

1. Case analysis. When something is unknown, we split into *all* the possibilities (missing none) and examine them one by one. A possibility that leads to something impossible (a contradiction with the problem) is ruled out. The possibility that remains is the answer.
2. Proof by contradiction. To show that a statement P is true, we *assume* that P is false and keep reasoning. If we arrive at something impossible, the assumption was wrong, so P is true.
3. Two contradictory statements. If two statements cannot both be true and cannot both be false (for example Leo says "Helen is lying" about Helen's statement), then *exactly one* of them is true. This is the "key" to very many puzzles: we immediately know that one statement of the pair is true and the other false.
4. Seating around a round table. Rotating the whole table does not change who sits next to whom. So we may *fix* the seat of one person and only consider the seats of the others. Around a round table with 5 seats, each person has exactly 2 neighbours, and exactly 2 people who are *not* next to them.



solid: seat 1 is next to seats 2 and 5

dashed: seat 1 is not next to seats 3, 4

Why is it true? Why is proof by contradiction valid reasoning? A statement can only be true or false. If "P is false" leads to something impossible, then the possibility "P is false" cannot happen. Only the possibility "P is true" remains. This is just case analysis with two cases: P true or P false. □

ThinkAlgo Tip. Start with the "tightest" piece of information – the one that allows the fewest possibilities. For instance, "Jack and Kate do *not* sit next to each other" at a 5-seat table means there is *exactly one* person between them on one side. Fewer possibilities means fewer cases to check!

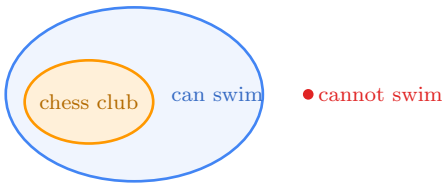
③ Ordering; "must be true" and "could be true"

1. Use symbols. Translate each sentence into an inequality: "Ryan runs faster than Henry" becomes $R > H$; "Kayla ate less than Max" becomes $K < M$ (or $M > K$).
2. Transitivity. If $A > B$ and $B > C$, then $A > C$. This lets us chain inequalities together.
3. Diagrams. Draw each person as a point on a *vertical axis*: whoever is "more" goes higher; or draw an arrow from the "lesser" person to the "greater" one. Two people who are not joined by a chain of arrows in the same direction *cannot be compared* yet.
4. Position in a line. If someone is k-th from the front and m-th from the back of a line, the line has $k + m - 1$ people.
5. "Must be true" and "could be true".

- A statement that **must be true** (is always true): true in *every* situation that satisfies the problem.
- A statement that **could be true**: true in *at least one* situation that satisfies the problem.
- To refute a "must be true" claim, *one counterexample* is enough: a situation that satisfies the problem but in which the claim is false.

6. "Every A is a B". From this we may conclude: "whoever is *not* a B is *not* an A". But we may *not* conclude "whoever is a B is an A".





Why is it true? Why does being k-th from the front and m-th from the back give $k + m - 1$ people? Counting from the front up to that person gives k people, counting from that person to the back gives m people. Adding gives $k + m$, but that person has been counted *twice*, so we subtract 1. Why does "every A is a B" give "not a B means not an A"? Look at the right-hand picture: the "chess club" oval lies *entirely* inside the "can swim" oval. A point outside the big oval is certainly outside the small one. A point inside the big oval, however, may be inside or outside the small one. □

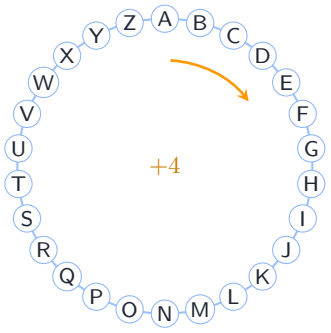
Common mistake. When you see "$A > B$" and "$A > C$", do not jump to conclusions about B and C: they *cannot be compared* yet. A claim like "$B > C$" then only *could* be true; it is not *certainly* true.

④ Shift ciphers and remainders

1. The Caesar cipher. Arrange the 26 letters A, B, C, …, Z of the English alphabet in a circle (after Z comes A again). To encrypt with *shift* k: replace each letter by the letter k places after it, clockwise. To decrypt: shift *back* k places.
2. Numbering the letters. Number them A $= 0$, B $= 1$, …, Z $= 25$. To encrypt letter number x with shift k: compute $x + k$; if the result is ≥ 26, subtract 26 (that is, take the *remainder* of $x + k$ on division by 26).

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

For example, with shift 4: Y $= 24$, $24 + 4 = 28 \geq 26$, $28 - 26 = 2 =$ C. So Y $\to$ C.
3. Shifting several times. Shifting by a and then by b is the same as shifting once by $a + b$. Shifting by 26 brings every letter back to itself, so only the *remainder* of the number of steps on division by 26 matters.
4. Days of the week "go round" too. After Sunday comes Monday again: the days repeat in a cycle of 7. Which day it will be "in n days" depends only on the remainder of n on division by 7.



Shift 4:
A $\to$ E, B $\to$ F, …
W $\to$ A, X $\to$ B,
Y $\to$ C, Z $\to$ D.

Why is it true? Why does a shift of 26 bring us back to the start? The circle has exactly 26 letters, so 26 steps is one full turn. Likewise, a shift of $30 = 26 + 4$ is one full turn followed by 4 more steps: exactly the same as a shift of 4. And 4 is precisely the remainder when 30 is divided by 26. □

Common mistake. The most common mistake is forgetting to "wrap around" at the end of the alphabet: W shifted by 4 does *not* give some strange symbol after Z, but A (W $\to$ X $\to$ Y $\to$ Z $\to$ A). The second mistake: decrypting by shifting *forwards* instead of *backwards*.

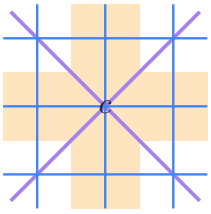
⑤ Magic squares

1. Definition. A 3×3 magic square is a grid of 9 cells filled with 9 different numbers so that the sums of the numbers in every row, every column and both diagonals are equal. This common sum is called the **magic sum**, written S.
2. The magic sum. The three rows cover the whole grid, each cell exactly once, so

$$3S = \text{sum of all 9 numbers}, \qquad S = \frac{\text{sum of all 9 numbers}}{3}.$$

Example: for the numbers $1, 2, \ldots, 9$ the total is 45, so $S = 15$.
3. The centre number. In a 3×3 magic square: **centre number** $= S \div 3$ (the average of the 9 numbers). For the numbers $1, \ldots, 9$ the centre is 5.
4. Quick filling trick. Once you know S and a row (column, diagonal) already has two numbers, the missing number is $S-$ (sum of the two known numbers).

2	7	6
9	5	1
4	3	8



4 lines through the centre

Why is it true? Why is the centre number equal to $S \div 3$? Exactly 4 lines pass through the centre: the middle row, the middle column and the two diagonals (right-hand picture). Adding the sums of these 4 lines gives $4S$. In that sum the centre c is counted 4 times, while every other cell is counted exactly once (check on the picture: each outer cell lies on exactly one of the four lines). Hence

$$4S = (\text{sum of the 9 numbers}) + 3c = 3S + 3c.$$

Subtract $3S$ from both sides: $S = 3c$, that is, $c = S \div 3$. □

ThinkAlgo Tip. Magic squares may use negative numbers too. With the numbers $-6, -5, \ldots, 2$: the total is -18, so $S = -6$ and the centre is -2. The formula works just the same!

⑥ A balance scale is an equation

1. Balanced scale = equality. If a scale balances, the total masses on the two pans are equal. Giving each kind of object a letter for its mass, we can write an equality (an equation).
2. Removing identical objects. Removing identical objects from *both* pans keeps the scale balanced (subtract the same number from both sides).
3. Substitution. If a group of objects A balances a group B, then on any other scale we may replace A by B and the scale stays balanced.
4. Reduce to one unknown. Let the mass of *one* kind of object be x, then use the equalities to express the mass of every other object in terms of x. Comparing two groups of objects then

becomes comparing two expressions in x.



$4O = 3T$. Let $O = x$: $3T = 4x$, $T = \frac{4x}{3}$.

Why is it true? Why may we do this? A balanced scale means that the two sides of the equation $4O = 3T$ are equal. Dividing both sides by the same non-zero number still gives a true equation: $T = 4O \div 3$. So however many grams a circle really weighs, a triangle always weighs 4/3 of a circle. For instance, if $x = 30$ g then $T = 40$ g; check: $4 \cdot 30 = 120 = 3 \cdot 40$. □

Common mistake. Never remove an object from *one* pan without removing an identical object from the other. And only remove objects that are *identical* (have the same mass).

⑦ Clever counting: sums on figures, dice, fair games

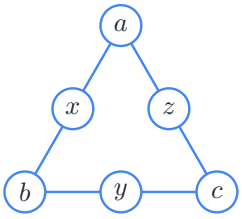
1. How many times is each number counted? When numbers are written in circles along the sides of a figure and every side has the same sum S, add up the sums of *all the sides*. A number at a *corner* belongs to two sides, so it is counted twice; a number in the *middle of a side* is counted once. For a triangle:

$$3S = (\text{sum of all the numbers}) + (\text{sum of the three corner numbers}).$$

2. Lower and upper bounds. The sum of the three corners cannot be smaller than the sum of the three smallest numbers, nor larger than the sum of the three largest. This pins down S; then check each possible value.

3. Dice. On a standard die, *opposite faces add up to* 7: 1 – 6, 2 – 5, 3 – 4. The six faces add up to $1+2+3+4+5+6 = 21$. Therefore: sum of the *hidden* faces $= 21 -$ sum of the visible faces.

4. Fair games. Rolling a die has 6 equally likely outcomes. A game "you win if ..." is *fair* when the chance of winning equals the chance of losing, that is, there are exactly 3 winning and 3 losing outcomes.



left side: $a + x + b$
base: $b + y + c$
right side: $c + z + a$
a, b, c appear twice.



visible $3 + 6 + 2 = 11$
hidden $21 - 11 = 10$

Why is it true? Why is $3S = (\text{sum of all the numbers}) + (a + b + c)$? Add the three sides: $(a + x + b) + (b + y + c) + (c + z + a) = (a + b + c + x + y + z) + (a + b + c)$. Each middle number x, y, z appears once; each corner number a, b, c appears twice, so $a + b + c$ must be added once more. □

Common mistake. For a die standing on a table, the *bottom* face is hidden as well. Do not forget it! The safest method is to use "sum of the 6 faces = 21" and subtract the visible faces.

⑧ Latin squares and sudoku

1. 4×4 Latin square. Write the digits $1, 2, 3, 4$ in a 4×4 grid so that every row and every column contains all four digits (no repeats). A 4×4 **sudoku** adds one more condition: each 2×2 box (thick border) also contains all four digits.

2. Three "regions". Each cell belongs to exactly three regions: one row, one column, one 2×2 box. The digit in the cell must be *different* from every digit already present in all three regions.

3. Two ways of reasoning.

- *Elimination in one cell:* cross out every digit already in the cell's row, column and box; if exactly one digit is left, write it in.
- *Finding a place for a digit:* if, in some region, digit d can go in only one cell, then d must go there.

4. Write "candidates". If a cell still has two possibilities, write both in small (e.g. $\{1, 4\}$) and move on; information elsewhere will settle it.

1		3	
	2		
			4

Cell in row 1, column 2: row 1 already has $1, 3$;
the 2×2 box already has $1, 2$.
The only digit left is 4.

Common mistake. Do not "guess" and write the digit straight in. Every digit you write must have a reason (all other possibilities are ruled out). If you really must try something, write "suppose" clearly and be ready to go back when you meet a contradiction.

Note for parents. Logic puzzles are great to do together with your child: read the problem aloud, let your child retell it in their own words, then draw a diagram or a table together. When your child gives an answer, do not say at once whether it is right or wrong, but ask: "Have you checked it against *each* condition of the problem?". The habit of checking is worth more than any trick.

Example 1 (Truth-tellers and liars)

On an island, every person either always tells the truth or always lies. You meet two people, Peter and Quinn. Peter says: "Quinn is a liar." Quinn says: "Peter and I are the same type of person." Who tells the truth and who lies?

Guided solution.

Step 1. Case: Peter tells the truth

Then Peter's statement is true: Quinn is a liar. So Quinn's statement must be false. Quinn says "we are the same type"; in fact Peter is truthful and Quinn lies – *different* types, so Quinn's

statement is indeed false. **No contradiction.**

Step 2. Case: Peter lies

Then Peter's statement is false: Quinn is *not* a liar, so Quinn tells the truth. Then Quinn's statement must be true: "Peter and Quinn are the same type". But Peter lies and Quinn tells the truth – different types. So Quinn's statement is false after all. **Contradiction!** This case is ruled out.

Step 3. Summary in a table

Peter	Quinn	Peter's statement	Quinn's statement	Conclusion
truthful	liar	true (fine)	false (fine)	**OK**
liar	truthful	false (fine)	false (impossible!)	ruled out

(The cases "both truthful" and "both liars" are ruled out straight away by Peter's statement: if Peter is truthful then Quinn lies; if Peter lies then Quinn is truthful.)

Answer: Peter tells the truth, Quinn lies.

Example 2 (Each person is half right)

Three teams, Red, Blue and Yellow, play a tug-of-war; no two teams share a place. Before the match, three fans predict:

- Grace: "Red first, Blue second." Jack: "Blue first, Yellow second." Lucy: "Blue third, Yellow first."

In the end, each person got exactly one part right. Find the places of the three teams.

Guided solution.

Step 1. Pick one part to split into cases

Each prediction has exactly one true part. Look at Grace's: either "Red first" is true or "Blue second" is true.

Step 2. Case 1: "Blue second" is true (and "Red first" is false)

If Blue is second, Blue is not first, so Jack's part "Blue first" is false; hence his other part "Yellow second" must be true. But Blue is already second, so Yellow cannot be second as well. **Contradiction**, ruled out.

Step 3. Case 2: "Red first" is true (and "Blue second" is false)

Blue is not first (Red is) and not second, so **Blue is third**. That leaves Yellow second.

Step 4. Check all three people

Person	First part	Second part
Grace	Red first: true	Blue second: false
Jack	Blue first: false	Yellow second: true
Lucy	Blue third: true	Yellow first: false

Each person has exactly one part right.

Answer: Red first, Yellow second, Blue third.

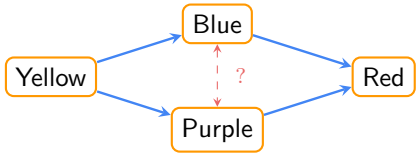
Example 3 (Ordering – must or could?)

There are four coloured pencils, red, blue, yellow and purple, all of different lengths. We know: the red pencil is longer than the blue one; the yellow pencil is shorter than the blue one; the purple pencil is longer than the yellow one but shorter than the red one.
a) Which pencil is certainly the longest? Which is certainly the shortest?
b) Is the statement "The purple pencil is the second longest" certainly true, possibly true, or certainly false?

Guided solution.

Step 1. **Use symbols**
Call the lengths R, B, Y, P. We are told: $R > B$; $B > Y$; $P > Y$; $R > P$.

Step 2. **Draw an arrow diagram (arrows point from the shorter pencil to the longer one)**



Step 3. **Answer part a**
From $R > B > Y$ and $R > P > Y$: the red pencil is longer than all three others and the yellow pencil is shorter than all three others. So **red is certainly the longest** and **yellow is certainly the shortest**.

Step 4. **Answer part b: find two situations**
No condition compares Blue with Purple. Try two sets of lengths (cm), both satisfying every condition:

- Red 15, Purple 13, Blue 12, Yellow 8: purple is second longest – the statement is *true*;
- Red 15, Blue 12, Purple 10, Yellow 8: purple is third longest – the statement is *false*.

So "The purple pencil is the second longest" is only **possibly true**, not certainly true.

Answer: a) Red is the longest, yellow the shortest (certainly). b) Possibly true (not certain).

Example 4 (Shift cipher – finding the shift)

Tom encrypts a word with a Caesar cipher and gets **JVKL**. He tells us that with this encryption the letter A becomes H.
a) Find the original word. b) After this shift, what is the smallest number of further steps (in the same direction) that brings every letter back to its original letter?

Guided solution.

Step 1. **Find the shift**
A = 0, H = 7. A becomes H, so the shift is $k = 7$.

Step 2. **Decrypt: shift back 7 (subtract 7; if the result is negative, add 26)**

Cipher letter	J	V	K	L
Number	9	21	10	11
Subtract 7	2	14	3	4
Original letter	**C**	**O**	**D**	**E**

The original word is **CODE**.

Common mistake. Subtracting 7 does not always give a negative number; we add 26 (wrap around past A) only when the number of the cipher letter is less than 7. Here all four letters have numbers ≥ 7, so no wrapping is needed.

Step 3. Part b

A total shift of 26 steps goes once round the circle. We have shifted 7 steps, so we need $26 - 7 = 19$ more steps.

Check: Encrypt CODE again with shift 7: C(2) $\to 9 =$ J, O(14) $\to 21 =$ V, D(3) $\to 10 =$ K, E(4) $\to 11 =$ L. We get JVKL.

Answer: a) CODE. b) 19 steps.

Example 5 (Completing a magic square)

Place the even numbers $2, 4, 6, \ldots, 18$ (each once) in the grid so that every row, column and diagonal has the same sum.

4		12
	6	

Guided solution.

Step 1. Find the magic sum S

Sum of the nine numbers: $2+4+\cdots+18 = 2\times(1+2+\cdots+9) = 2\times 45 = 90$. So $S = 90 \div 3 = 30$.

Step 2. The centre number

Recall. In a 3×3 magic square the centre number equals $S \div 3$.

Centre $= 30 \div 3 = 10$.

Step 3. Fill in the lines that already have two numbers

- Top row: middle cell $= 30 - 4 - 12 = 14$.
- Diagonal 4 – 10 – ?: bottom-right corner $= 30 - 4 - 10 = 16$.
- Diagonal 12 – 10 – ?: bottom-left corner $= 30 - 12 - 10 = 8$.
- Left column: $30 - 4 - 8 = 18$; right column: $30 - 12 - 16 = 2$.

4	14	12
18	10	2
8	6	16

Check: Bottom row: $8+6+16=30$; middle row: $18+10+2=30$; middle column: $14+10+6=30$. Every even number from 2 to 18 is used exactly once.

Answer: Top row $4, 14, 12$; middle row $18, 10, 2$; bottom row $8, 6, 16$.

Example 6 (A balance-scale problem)

Both scales below are balanced; identical shapes have equal masses. How many circles does one square weigh?



Guided solution.

Step 1. Write equalities

Let the masses of a square, a triangle and a circle be V, T, O. Scale 1: $2V = 3T$. Scale 2: $T = 2O$.

Step 2. Substitute

On scale 1, replace each triangle by two circles: $2V = 3 \times 2O = 6O$.

Step 3. Halve both pans

$2V = 6O$, so $V = 3O$.

Check: Take $O = 1$, $T = 2$, $V = 3$: scale 1: $2 \times 3 = 6 = 3 \times 2$; scale 2: $2 = 2 \times 1$. Correct.

Answer: One square weighs as much as 3 circles.

Example 7 (Sums along the sides of a triangle)

Place the numbers $1, 2, 3, 4, 5, 6$ in the six circles (three at the corners, three in the middles of the sides) so that the three numbers on each side add up to the same number S. Find all possible values of S.

Guided solution.

Step 1. Count how many times each number is counted

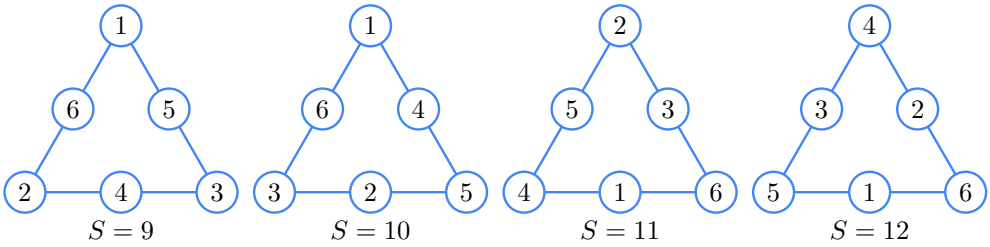
> **Recall.** When adding the three side sums, corner numbers are counted twice and middle numbers once.

$3S = (1 + 2 + \cdots + 6) + (\text{sum of the three corners}) = 21 + (\text{sum of the three corners})$.

Step 2. Pin it down

The smallest possible corner sum is $1+2+3=6$, the largest is $4+5+6=15$. So $27 \le 3S \le 36$, that is, $9 \le S \le 12$.

Step 3. Check each value with an actual arrangement



For instance, with $S = 10$: $1 + 6 + 3 = 10$, $3 + 2 + 5 = 10$, $5 + 4 + 1 = 10$.

Answer: S **can be** 9, 10, 11 **or** 12.

Example 8 (4×4 sudoku)

Complete the sudoku: every row, every column and every thick-bordered 2×2 box must contain the digits $1, 2, 3, 4$.

2	1		
		1	
	4		
			3

Guided solution.

Step 1. **Top-left box**

It already has $2, 1$ in row 1, so row 2, columns 1 and 2, are $\{3, 4\}$. Column 2 already has a 4 (in row 3), so row 2 column 2 is 3 and row 2 column 1 is 4.

Step 2. **Row 2 and the top-right box**

Row 2: $4, 3, 1, _ \Rightarrow$ the last cell is 2. Row 1, columns 3 and 4, are $\{3, 4\}$; column 4 already has a 3 (row 4), so row 1 column 4 is 4 and row 1 column 3 is 3.

Step 3. **Column 2 and the bottom-left box**

Column 2: $1, 3, 4, _ \Rightarrow$ row 4 column 2 is 2. Column 1 is still missing $\{1, 3\}$; row 4 already has a 3, so row 4 column 1 is 1 and row 3 column 1 is 3.

Step 4. **The last two rows**

Row 3: $3, 4, _, _$ needs $\{1, 2\}$; column 3 already has a 1 (row 2), so row 3 column 3 is 2 and column 4 is 1. Row 4: $1, 2, _, 3 \Rightarrow$ the empty cell is 4.

2	1	3	4
4	3	1	2
3	4	2	1
1	2	4	3

Check: Every row, column and 2×2 box contains all of $1, 2, 3, 4$.

Answer: **Row** 1: 2134; **row** 2: 4312; **row** 3: 3421; **row** 4: 1243.

Core practice

Problem 1

Mum has just bought a new pot of flowers. Three children guess what flower it is. Anna guesses: "a red rose". Ben guesses: "a purple daisy". Chloe guesses: "a red dahlia". Mum says that each child got *exactly one* of the two things right – either the colour or the kind of flower. The flower Mum bought is a

(A) purple dahlia (B) purple rose (C) red dahlia

(D) yellow rose (E) red daisy

Detailed solution

Analysing the problem. Each guess has two parts: a *colour* and a *kind of flower*. Each child has exactly one part right: if one part is right, the other is wrong. We must find the real colour and kind.

Recall. "Half-true" statement: if we know one part is true, the other part is false, and vice versa.

Step 1. The colour: is the flower red?

Suppose the flower is not red. Then Anna's part "red" is wrong, so her other part "rose" must be right. Likewise Chloe's part "red" is wrong, so "dahlia" must be right. The flower would be both a rose and a dahlia – **impossible**. So **the flower is red**.

Step 2. Deduce the kind of flower

The flower is red, so Anna has the colour right ⇒ Anna's "rose" is wrong: it is not a rose.
Chloe also has the colour right ⇒ Chloe's "dahlia" is wrong: it is not a dahlia.
Ben's "purple" is wrong (the flower is red), so Ben's "daisy" must be right: **a daisy**.

Step 3. Check with a table

Child	Colour guessed	Kind guessed	Parts right (real flower: red daisy)
Anna	red (right)	rose (wrong)	1
Ben	purple (wrong)	daisy (right)	1
Chloe	red (right)	dahlia (wrong)	1

Common mistake. Try option (B) "purple rose": Anna has the kind right (rose), Ben has the colour right (purple), but Chloe has *nothing* right – against the conditions. You must check *all three* children, not just one or two.

Check: With "red daisy" each child has exactly one part right. Every other option gives some child 0 or 2 right parts: (A) purple dahlia – Anna has 0; (C) red dahlia – Chloe has 2; (D) yellow rose – Ben has 0.
So the answer is **(E)**.

Answer: (E) red daisy

Problem 2

The 26 letters of the English alphabet are written in order A, B, C, ..., Z clockwise around a circle. To *encrypt* a message, each letter is replaced by the letter 4 places after it, clockwise (this is called a *Caesar cipher*). For example, the word ZAP is encrypted as DET. The word WIN is encrypted as

(A) ALN (B) ZLN (C) AMR (D) AMQ (E) ZMQ

Detailed solution

Analysing the problem. The shift is 4, forwards (clockwise). We must encrypt the letters W, I, N one by one.

Recall. Number A $= 0, \ldots,$ Z $= 25$. Encrypt: add 4; if the result is ≥ 26, subtract 26 (going past Z brings you back to A).

Step 1. Check our understanding with ZAP $\to$ DET

$Z(25) + 4 = 29$, $29 - 26 = 3 =$ D; $A(0) + 4 = 4 =$ E; $P(15) + 4 = 19 =$ T. That is indeed DET, and we see Z "wrap around" to D.

Step 2. Encrypt each letter of WIN

Original letter	W	I	N
Number	22	8	13
Add 4	$26 \to 26 - 26 = 0$	12	17
Cipher letter	**A**	**M**	**R**

W $\to$ X $\to$ Y $\to$ Z $\to$ **A**; I $\to$ J $\to$ K $\to$ L $\to$ **M**; N $\to$ O $\to$ P $\to$ Q $\to$ **R**.

Common mistake. Each wrong option has at least one letter shifted by the wrong number of steps: (B) ZLN and (E) ZMQ shift W by only 3 (stopping at Z and forgetting that after Z we carry on to A); (D) AMQ and (E) ZMQ shift N by only 3; (A) ALN and (B) ZLN shift I by 3 and leave N unchanged. Count carefully: "4 places after" means taking exactly 4 steps, not counting the starting letter.

Check: Decrypt AMR by going back 4 steps: A $\to$ Z $\to$ Y $\to$ X $\to$ W; M $\to$ I; R $\to$ N. We get WIN back.

So the answer is **(C)**.

Answer: (C) AMR

Problem 3

Five friends, Ryan, Henry, Felix, Tom and Oscar, run a race; no two finish at the same time. Ryan runs faster than Henry and Felix. Henry runs slower than Felix. Tom runs faster than Ryan but slower than Oscar. Who finishes fourth?

(A) Felix (B) Henry (C) Oscar (D) Ryan (E) Tom

Detailed solution

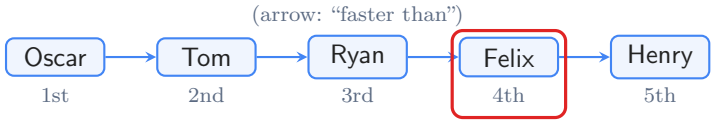
Analysing the problem. Five runners and several speed comparisons. We must find the finishing order and the fourth runner. Faster runners finish earlier.

Step 1. Use symbols (> means "faster than")

- "Ryan is faster than Henry and Felix": $R > H$ and $R > F$.
- "Henry is slower than Felix": $F > H$.
- "Tom is faster than Ryan but slower than Oscar": $O > T > R$.

Step 2. Chain them together (transitivity)

From $O > T > R$, $R > F$ and $F > H$ we get: $O > T > R > F > H$.



Step 3. Read off the result

Finishing order: Oscar (1st), Tom (2nd), Ryan (3rd), **Felix (4th)**, Henry (5th).

Common mistake. "Ryan is faster than Henry and Felix" does not tell us which of Henry and Felix is faster – we need "Henry is slower than Felix". Do not skip any sentence!

Check: Ryan (3rd) is faster than Henry (5th) and Felix (4th) ✓; Henry is slower than Felix ✓; Tom (2nd) is faster than Ryan and slower than Oscar ✓.
So the answer is **(A)**.

Answer: (A) Felix

Problem 4

In a magic square, the numbers in every row, every column and both diagonals have the same sum. The magic square shown uses each of the integers from -6 to 2 exactly once. The value of Y is

(A) -1 (B) 0 (C) -6
(D) 2 (E) -2

1		Y
-4		
-3		-5

Detailed solution

Analysing the problem. The nine numbers are $-6, -5, -4, -3, -2, -1, 0, 1, 2$. Four cells are known; we need the cell Y in the top-right corner.

Step 1. Find the magic sum S

Sum of the nine numbers: $(-6)+(-5)+(-4)+(-3)+(-2)+(-1)+0+1+2$. Pair $(-2)+2=0$, $(-1)+1=0$; what is left is $(-6)+(-5)+(-4)+(-3)=-18$. So the total is -18, and

$$S = (-18) \div 3 = -6.$$

Check: The left column is already complete: $1+(-4)+(-3)=-6$. This agrees with $S=-6$.

Step 2. Find the centre number

> **Recall.** In a 3×3 magic square the centre number equals $S \div 3$.

Centre $= (-6) \div 3 = -2$. (Another way: the main diagonal gives $1 + c + (-5) = -6$, so $c = -6 - 1 + 5 = -2$.)

Step 3. Find Y using the other diagonal

The diagonal through Y, the centre and the cell -3: $Y+(-2)+(-3)=-6$, so $Y=-6+2+3=-1$.

Step 4. Complete the grid to check

Top row: middle cell $= -6-1-(-1) = -6$. Middle row: right cell $= -6-(-4)-(-2) = 0$. Bottom row: middle cell $= -6-(-3)-(-5) = 2$.

1	−6	−1
−4	−2	0
−3	2	−5

Check: Middle column: $-6-2+2=-6$; right column: $-1+0-5=-6$; middle row: $-4-2+0=-6$. Each number from -6 to 2 is used exactly once.

> **Common mistake.** Be careful with minus signs: $-6-(-4) = -6+4 = -2$. Subtracting a negative number means adding its opposite.

So the answer is **(A)**.

Answer: (A) -1

Problem 5

A game is called *fair* if your chance of winning equals your chance of losing. You roll a fair six-sided die. How many of the following four games are fair?

- You win if you roll a 2.
- You win if you roll an even number.
- You win if you roll a number less than 4.
- You win if you roll a number divisible by 3.

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

Detailed solution

Analysing the problem. There are 6 equally likely outcomes: $1, 2, 3, 4, 5, 6$. A game is fair when the number of winning outcomes equals the number of losing outcomes, i.e. 3 winning and 3 losing outcomes.

Recall. With a fair die all six faces are equally likely. Chance of winning = (number of winning outcomes) $\div$ 6.

Step 1. List the winning outcomes of each game

Game	Winning outcomes	Wins	Losses	Fair?
Roll a 2	2	1	5	no
Even number	$2, 4, 6$	3	3	**yes**
Less than 4	$1, 2, 3$	3	3	**yes**
Divisible by 3	$3, 6$	2	4	no

Step 2. Count

There are 2 fair games: "even number" and "less than 4".

Common mistake. "Less than 4" does *not* include 4: only $1, 2, 3$. Counting 4 by mistake gives 4 winning outcomes – not fair. And "divisible by 3" includes both 3 and 6 (do not forget 6).

Check: Game "even number": chance of winning $\frac{3}{6} = \frac{1}{2}$, chance of losing also $\frac{1}{2}$. The same holds for "less than 4".
So the answer is **(C)**.

Answer: (C) 2

Problem 6

Five friends, Jack, Kate, Leo, Mia and Noah, sit on the 5 chairs around a round table. Leo and Mia sit next to each other. Jack and Kate do *not* sit next to each other. The two people sitting on either side of Noah are

(A) Jack and Leo (B) Jack and Kate (C) Kate and Mia

(D) Leo and Mia (E) Mia and Jack

Detailed solution

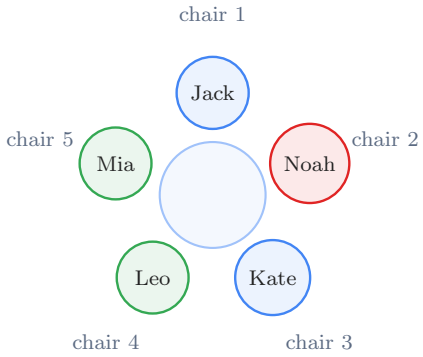
Analysing the problem. Five people around a round table with 5 chairs. Two conditions: Leo – Mia are neighbours; Jack – Kate are not. We must find Noah's two neighbours.

Recall. Around a round table we may fix one person's seat (rotating the table does not change who sits next to whom). With 5 chairs, each person is not next to exactly 2 people – the two who are "one seat away".

Step 1. Start with the tightest condition: Jack and Kate are not neighbours
Number the chairs 1 to 5 around the table. Put Jack on chair 1. The chairs next to chair 1 are chairs 2 and 5, so Kate sits on chair 3 or chair 4. These two cases are mirror images of each other, so we consider Kate on chair 3.

Step 2. Seat Leo and Mia
Chairs 2, 4, 5 are free. Leo and Mia need two chairs *next to each other*. Of these three chairs, only chairs 4 and 5 are neighbours (chair 2 is next to chairs 1 and 3, both taken). So Leo and Mia sit on chairs 4 and 5.

Step 3. Noah takes the remaining chair
Noah sits on chair 2, between chair 1 (Jack) and chair 3 (Kate).



(If Kate is on chair 4, it is similar: Leo and Mia are on chairs 2, 3 and Noah is on chair 5 – still between Jack and Kate.)

Why is it true? Why is there no other arrangement? Jack and Kate are not neighbours, so between them there is exactly one chair on one side and two chairs on the other side. Leo – Mia need two adjacent chairs, so they must take the side with two chairs. The single chair between Jack and Kate belongs to Noah. □

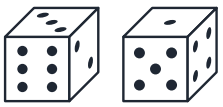
Check: Leo – Mia are neighbours ✓; Jack (chair 1) and Kate (chair 3) are one chair apart ✓.

So the answer is **(B)**.

Answer: (B) Jack and Kate

Problem 7

On each die, the numbers of pips on opposite faces always add up to 7. Two dice stand on a table as shown. The total number of pips on all the faces of the two dice that *cannot be seen* is



(A) 14 (B) 20 (C) 21

(D) 24 (E) 30

Detailed solution

Analysing the problem. Each die has 6 faces; we see 3 of them (top, front, right) and 3 are hidden (bottom, back, left). We need the total number of pips on the 6 hidden faces of the two dice.

Recall. The six faces of a die carry $1+2+3+4+5+6=21$ pips in total.

Step 1. Read the visible faces

First die: top 3, front 6, right 2 $\Rightarrow$ total $3+6+2=11$.

Second die: top 1, front 5, right 4 $\Rightarrow$ total $1+5+4=10$.

Step 2. Compute the hidden faces

First die: $21-11=10$. Second die: $21-10=11$. Total: $10+11=21$.

Step 3. Second method: use opposite faces

Each hidden face is opposite a visible face, so it has $7-$(visible face) pips. First die: $(7-3)+(7-6)+(7-2)=4+1+5=10$. Second die: $(7-1)+(7-5)+(7-4)=6+2+3=11$.

Why is it true? Why is each hidden face opposite exactly one visible face? The three visible faces (top, front, right) are pairwise adjacent, so no two of them are opposite. The three faces opposite them (bottom, back, left) are exactly the three hidden faces. □

Common mistake. Some students count only the two hidden faces "at the back" and forget the bottom face resting on the table. The bottom face cannot be seen either!

Check: All faces of both dice total $21+21=42$; visible faces $11+10=21$; hidden faces $42-21=21$.

So the answer is **(C)**.

Answer: (C) 21

Problem 8

Chris lies on Fridays, Saturdays and Sundays and tells the truth on the other days. Mark lies on Tuesdays, Wednesdays and Thursdays and tells the truth on the other days. On which day of the week could both of them say: "Tomorrow I will lie"?

(A) Monday (B) Thursday (C) Friday (D) Sunday (E) Tuesday

Detailed solution

Analysing the problem. A person can say "Tomorrow I will lie" in only two situations:

- they tell the *truth* today and the statement is *true*: truthful today, lying tomorrow;
- they *lie* today and the statement is *false*: lying today, truthful tomorrow.

In short: the person can say it when **today and tomorrow are of different types** (one truthful day, one lying day).

Recall. A truth-teller only says true things; a liar only says false things. After Sunday comes Monday (the week goes round).

Step 1. Table for Chris (L = lies, T = truthful)

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Chris today	T	T	T	T	L	L	L
Chris tomorrow	T	T	T	L	L	L	T
Can say it?	–	–	–	**yes**	–	–	**yes**

Chris can say it on **Thursday** and **Sunday** (the two "switching" days).

Step 2. Table for Mark

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Mark today	T	L	L	L	T	T	T
Mark tomorrow	L	L	L	T	T	T	T
Can say it?	**yes**	–	–	**yes**	–	–	–

Mark can say it on **Monday** and **Thursday**.

Step 3. Find the common day

The day in both lists: **Thursday**.

Common mistake. Many students choose Friday, thinking "Chris lies on Friday". But on Friday Chris lies and tomorrow (Saturday) he lies too, so "Tomorrow I will lie" is a *true* statement – and a person who is lying cannot say something true.

Check: Thursday: Chris tells the truth, and tomorrow (Friday) he lies – the statement is true, so he can say it. Mark lies, and tomorrow (Friday) he tells the truth – the statement is false, which fits Mark lying today.

So the answer is **(B)**.

Answer: (B) Thursday

Problem 9

Both scales below are balanced; identical shapes have equal masses.



Which of the following groups has the same mass as the group ●▲▲▲ (one circle and three triangles)?

(A) ■■■ (B) ▲▲■■ (C) ●●●●

(D) ●●●■ (E) ●●●

Detailed solution

Analysing the problem. Let the masses of one circle, one triangle and one square be O, T and V (in the same unit). The two scales give two equations. We need the group whose mass equals $O + 3T$.

> **Recall.** A balanced scale means the two pans have equal mass. Removing identical objects from *both* pans (subtracting the same number from both sides) keeps the scale balanced.

Step 1. Write the two equations

First scale: $3O = 2T$. Second scale: $V + O + T = 2V$.

Step 2. Simplify the equation of the second scale

Remove one square from each pan (subtract V from both sides): $O + T = V$. *One square weighs as much as one circle and one triangle.*

Step 3. Introduce an unknown and express every mass in terms of it

Let $O = x$ $(x > 0)$. From $2T = 3O = 3x$ we get $T = \dfrac{3x}{2}$. Then

$$V = O + T = x + \frac{3x}{2} = \frac{5x}{2}.$$

Step 4. The mass of the given group

$O + 3T = x + 3 \cdot \dfrac{3x}{2} = \dfrac{2x}{2} + \dfrac{9x}{2} = \dfrac{11x}{2}$.

Step 5. The mass of each option (all written over the denominator 2 for easy comparison)

Option	Mass	Result
(A)	$3V = 3 \cdot \frac{5x}{2}$	$\frac{15x}{2}$
(B)	$2T + 2V = 3x + 5x = 8x$	$\frac{16x}{2}$
(C)	$4O = 4x$	$\frac{8x}{2}$
(D)	$3O + V = 3x + \frac{5x}{2}$	$\mathbf{\frac{11x}{2}}$ ✓
(E)	$3O = 3x$	$\frac{6x}{2}$

Only option (D) has mass $\frac{11x}{2}$.

Step 6. Another way: substitution only

$O + 3T = (O + T) + 2T = V + 3O$ (replace $O + T$ by V from Step 2, and $2T$ by $3O$ from the first scale). That is exactly three circles and one square.

Common mistake. The first scale gives $3O = 2T$: we need *more* circles to balance *fewer* triangles, so a circle is *lighter* than a triangle. Writing $2O = 3T$ by mistake ruins everything from the start.

Check: Take $x = 20$ g: circle 20 g, triangle 30 g, square 50 g. Scale 1: $3 \cdot 20 = 60 = 2 \cdot 30$; scale 2: $50 + 20 + 30 = 100 = 2 \cdot 50$. Given group: $20 + 3 \cdot 30 = 110$ g; option (D): $3 \cdot 20 + 50 = 110$ g. So the answer is **(D)**.

Answer: (D) three circles and one square

Problem 10

Five friends, Chris, Max, Brandon, Kayla and Tanya, go strawberry picking; no two of them pick the same number of strawberries. Chris picks more than Max. Brandon picks fewer than Kayla. Kayla picks fewer than Max but more than Tanya. Who picks the second most?

(A) Brandon (B) Chris (C) Kayla (D) Max (E) Tanya

Detailed solution

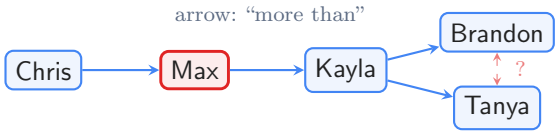
Analysing the problem. Five friends, four comparisons of numbers of strawberries. We must find the person in second place (behind only one other person).

Step 1. Use symbols ("$>$" means "picked more than")

$C > M$; $K > B$; $M > K$; $K > T$.

Step 2. Chain the inequalities

$C > M > K$, and K is more than both B and T.



Step 3. Read off the result

Chris picks the most (more than Max, who picks more than everyone else). Max is beaten only by Chris and beats Kayla, Brandon, Tanya. So **Max picks the second most**. We do not know whether Brandon or Tanya picked more, but that does not affect the question.

Common mistake. The sentence "Kayla picks fewer than Max but more than Tanya" contains *two* pieces of information: $M > K$ and $K > T$. Reading too fast, it is easy to miss one.

Check: For example Chris 30, Max 25, Kayla 20, Brandon 12, Tanya 10 strawberries satisfies every condition; Max is second.

So the answer is **(D)**.

Answer: (D) Max

Problem 11

Three friends, Amy, Helen and Tom, count the comics and the science books they read over the summer (no two of them read the same number of comics, and no two read the same number of science books). Amy read more comics than Helen but fewer science books than Helen. Tom read more comics than Helen but fewer science books than Amy. Which of the following statements **must** be true?

(A) Tom read the most comics.

(B) Amy read the most comics.

(C) Helen read the fewest comics.

(D) Amy read the fewest science books.

(E) Tom read the most science books.

Detailed solution

Analysing the problem. There are two completely separate comparisons: *comics* and *science books*. We split the information into two groups, order each group, then examine each statement.

> **Recall.** "Must be true" = true in *every* situation satisfying the problem. A *single* situation that satisfies the problem but makes a statement false is enough to rule that statement out.

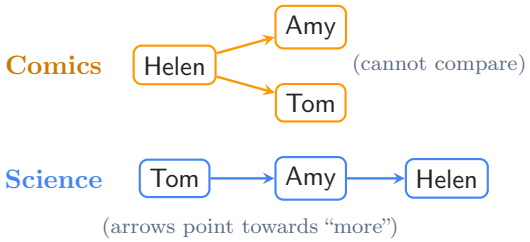
Step 1. Comics

"Amy read more comics than Helen": $A > H$. "Tom read more comics than Helen": $T > H$. So Helen read the fewest comics. As for Amy and Tom: *no sentence compares them.*

Step 2. Science books

"Amy read fewer science books than Helen": $H > A$. "Tom read fewer science books than Amy": $A > T$.

Chained: $H > A > T$. Helen read the most science books, Tom the fewest.



Step 3. Examine each statement

- (A) "Tom read the most comics": not certain. For example Amy 5, Tom 3, Helen 1 comics satisfies the problem, but Amy read the most.
- (B) "Amy read the most comics": not certain. For example Amy 3, Tom 5, Helen 1 comics satisfies the problem, but Tom read the most.

- (C) "Helen read the fewest comics": **must be true** (Step 1: Helen read fewer than both Amy and Tom).
- (D) "Amy read the fewest science books": **false**, since Tom read fewer than Amy.
- (E) "Tom read the most science books": **false**, Tom read the fewest.

Common mistake. Do not mix the two kinds of books into one chain! "Amy read more comics than Helen" says nothing about science books. Draw *two separate diagrams*.

So the answer is **(C)**.

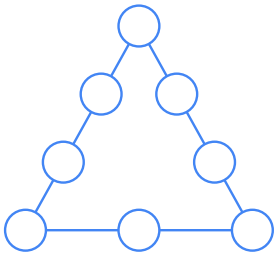
Answer: (C) Helen read the fewest comics

Problem 12

Each of the numbers $1, 2, 3, 4, 5, 6, 7, 8$ is placed in a different circle of the triangle shown, so that the sums of the numbers on the three sides of the triangle (the two slanted sides have 4 circles each, the base has 3 circles) are all equal to the same number S. The sum of all the different possible values of S is

(A) 85 (B) 99 (C) 66

(D) 81 (E) 67



Detailed solution

Analysing the problem. Call the number at the top corner a, the bottom-left corner b, the bottom-right corner c; the two middle numbers of the left side x_1, x_2; the two middle numbers of the right side y_1, y_2; the middle number of the base z. The three sums:

$$\text{left: } a + x_1 + x_2 + b = S, \qquad \text{right: } a + y_1 + y_2 + c = S, \qquad \text{base: } b + z + c = S.$$

We must find *every* possible value of S and add them up.

Step 1. Add the three sides – how many times is each number counted?

Recall. A corner number belongs to two sides, so it is counted twice; a middle number is counted once.

$3S = (1 + 2 + \cdots + 8) + (a + b + c) = 36 + (a + b + c)$, so $a + b + c = 3S - 36$.

Step 2. Pin down S

The corner sum is at least $1 + 2 + 3 = 6$ and at most $6 + 7 + 8 = 21$. Hence $6 \le 3S - 36 \le 21$, i.e. $42 \le 3S \le 57$, or $14 \le S \le 19$. Candidates: $S = 14, 15, 16, 17, 18, 19$.

Step 3. Use the base to rule some out

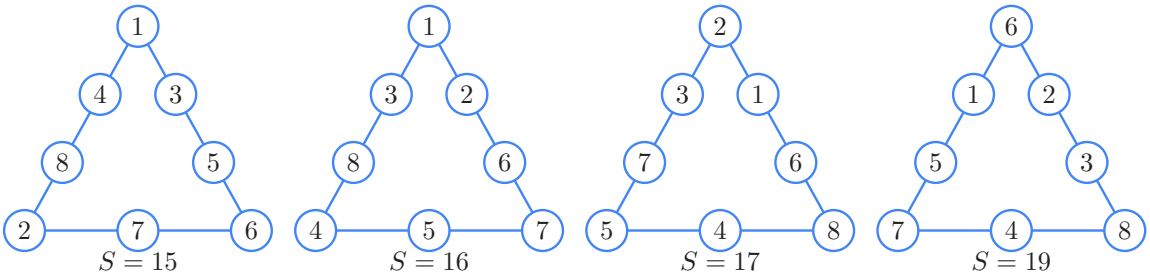
The base has only 3 numbers: $z = S - (b + c)$. And $b + c = (a + b + c) - a = 3S - 36 - a$, so

$$z = S - (3S - 36 - a) = 36 - 2S + a.$$

- $S = 14$: $z = 8 + a \ge 9$ – impossible (the largest number is 8). **Ruled out.**
- $S = 18$: $z = 36 - 36 + a = a$ – two circles with the same number, not allowed. **Ruled out.**

- $S = 15, 16, 17, 19$: not ruled out; we look for actual arrangements.

Step 4. Show arrangements for $S = 15, 16, 17, 19$



Check, e.g. $S = 15$: left $1 + 4 + 8 + 2 = 15$; right $1 + 3 + 5 + 6 = 15$; base $2 + 7 + 6 = 15$. $S = 19$: left $6 + 1 + 5 + 7 = 19$; right $6 + 2 + 3 + 8 = 19$; base $7 + 4 + 8 = 19$.

Step 5. Add the possible values

$15 + 16 + 17 + 19 = 67$.

Common mistake. If you only pin down $14 \le S \le 19$ and add all six values, you get 99 – option (B), a "trap". Pinning down only gives *candidates*; each candidate must be checked: either show an arrangement or prove that none exists.

Note for parents. Let your child cut out 8 slips of paper numbered 1 to 8 and try placing them on the figure. After a few attempts, ask: "Why can we never manage $S = 18$?" – that is when your child sees the value of the argument in Step 3.

So the answer is **(E)**.

Answer: (E) 67

Problem 13

The 4×4 grid shown is to be filled so that each row and each column contains the digits $1, 2, 3, 4$. The grid is divided into four 2×2 squares (thick borders), and each of these squares must also contain the digits $1, 2, 3, 4$. Which digit goes in cell P?

1		3	
	2		
	P		
			4

(A) 1

(B) 2

(C) 3

(D) 4

(E) The digit cannot be determined

Detailed solution

Analysing the problem. This is a 4×4 sudoku. Cell P is in row 3, column 2. We reason step by step; every digit we write must have a reason.

> **Recall.** Each cell belongs to one row, one column and one 2×2 box; its digit must differ from every digit already in those three regions.

Step 1. Top-left box
This box already has 1 (row 1, column 1) and 2 (row 2, column 2); it still needs 3 and 4 in two cells: row 1 column 2 and row 2 column 1. Row 1 already has a 3, so row 1 column 2 is 4, and row 2 column 1 is 3.

Step 2. Column 2 leaves only two possibilities for P
Column 2 already has 4 and 2, so $P \in \{1, 3\}$.

Step 3. Rows 1 and 2 on the right
Row 1: $1, 4, 3, _ \Rightarrow$ row 1 column 4 is 2. Row 2: $3, 2, _, _$ still needs $\{1, 4\}$. Column 4 already has a 4 (in row 4), so row 2 column 4 is 1 and row 2 column 3 is 4.

Step 4. Column 4 and row 3
Column 4: $2, 1, _, 4 \Rightarrow$ row 3 column 4 is 3. So row 3 already contains a 3, hence $P \neq 3$. With Step 2: $\mathbf{P = 1}$.

1	4	3	2
3	2	4	1
	1		3
			4

cells just deduced

1	4	3	2
3	2	4	1
4	1	2	3
2	3	1	4

completed grid

Step 5. Complete the grid to be sure there is no contradiction
Column 1 still needs $\{2, 4\}$; row 4 has a 4, so row 4 column 1 is 2 and row 3 column 1 is 4. Row 3: $4, 1, _, 3 \Rightarrow 2$. Row 4: $2, _, _, 4$ needs $\{1, 3\}$; column 2 still needs a 3, so row 4 column 2 is 3 and column 3 is 1.

Check: Every row, column and 2×2 box of the completed grid contains all of $1, 2, 3, 4$. The grid can be filled in only one way, so P is determined.

> **Common mistake.** Option (E) is a trap for those who look only at column 2 (seeing two possibilities, 1 and 3) and stop there. Keep going in other regions!

So the answer is **(A)**.

Answer: (A) 1

Problem 14

On an island, every person either always tells the truth or always lies. You meet two people, Alice and Ben. Alice says: "We are both liars." Which of the following is correct?

(A) Both tell the truth.

(B) Both are liars.

(C) Alice tells the truth, Ben lies.

(D) Alice lies, Ben tells the truth.

(E) It cannot be determined.

Detailed solution

Analysing the problem. There is only one statement, Alice's. We consider two cases, Alice truthful or Alice lying, and see which case gives no contradiction.

Recall. A truth-teller only says true things; a liar only says false things. The negation of "both are liars" is "at least one of them tells the truth".

Step 1. Case: Alice tells the truth
Then Alice's statement is true: both are liars – so Alice is a liar too. But we assumed Alice tells the truth. **Contradiction**, ruled out.

Step 2. Case: Alice lies
Then Alice's statement is false: it is *not* the case that both are liars, so at least one of them tells the truth. Alice lies, so the truth-teller must be **Ben**. No contradiction.

Step 3. Conclusion
Alice lies, Ben tells the truth.

Why is it true? This is a version of "I am a liar": a truth-teller never calls himself a liar. So as soon as we hear Alice say "we are both liars" (which includes Alice), we know Alice must be a liar. □

Common mistake. Many students choose (B), thinking "Alice lies, so the opposite is true: both tell the truth". Wrong! The opposite of "both are liars" is only "at least one tells the truth".

Check: Alice lies, Ben is truthful: the statement "we are both liars" is false (because Ben tells the truth) – indeed a lie, which fits Alice.
So the answer is **(D)**.

Answer: (D) Alice lies, Ben tells the truth

Problem 15

A vase has been broken. Exactly one of the four friends David, Helen, Leo and Mia broke it. When asked, they answer:

- David: "Helen broke it." Helen: "Leo broke it."
- Leo: "Helen is lying." Mia: "I did not break it."

We know that **exactly one** of them is telling the truth. Who broke the vase?

(A) David (B) Helen (C) Leo (D) Mia (E) Cannot tell

Detailed solution

Analysing the problem. Four statements, exactly one true, three false. We must find who broke the vase.

> **Recall.** For two contradictory statements (they cannot both be true and cannot both be false), exactly one of them is true.

Step 1. Find a contradictory pair
Leo says "Helen is lying". If Helen's statement is true, Leo's is false; if Helen's statement is false, Leo's is true. So among the statements of Helen and Leo, **exactly one** is true.

Step 2. So David and Mia are both lying
The whole group has only one true statement, and it lies in the pair Helen – Leo. Therefore the statements of David and Mia are both **false**.

Step 3. Use Mia's false statement
Mia's "I did not break it" is false, so **Mia broke the vase**.

Step 4. Check everything

Friend	Statement	True / false (if Mia broke it)
David	"Helen broke it"	false
Helen	"Leo broke it"	false
Leo	"Helen is lying"	**true** (Helen indeed said something false)
Mia	"I did not"	false

Exactly one true statement (Leo's). This matches the problem.

> **Common mistake.** Do not test people at random and stop as soon as something "looks fine". The sure way is to test all 4 possibilities: if David broke it, Leo's and Mia's statements are both true (2 true statements); if Helen broke it, David's, Leo's and Mia's are all true; if Leo broke it, Helen's and Mia's are true. Only "Mia broke it" gives exactly one true statement.

So the answer is **(D)**.

Answer: (D) Mia

Problem 16

Kate encrypts a short wish with a Caesar cipher with shift 5 (each letter is replaced by the letter 5 places after it on the circle of letters A–Z, where A follows Z). She gets: **MFAJ KZS**. The original wish is

(A) HAVE FUN (B) RKFO PEX (C) IBWF GVO

(D) HAVE FUM (E) GZUD ETM

Detailed solution

Analysing the problem. We have the ciphertext and know the shift is 5. To find the original we must *decrypt*: shift *back* 5 places.

Recall. Decrypt: take the letter's number (A = 0, . . . , Z = 25) and subtract 5; if the result is negative, add 26.

Step 1. Decrypt each letter

Cipher letter	M	F	A	J	K	Z	S
Number	12	5	0	9	10	25	18
Subtract 5	7	0	$-5 \to 21$	4	5	20	13
Original	**H**	**A**	**V**	**E**	**F**	**U**	**N**

The letter A has to "wrap around": A → Z → Y → X → W → V (going back 5 steps), which is exactly $-5 + 26 = 21 =$ V.

Step 2. Read the result

The wish is HAVE FUN.

Common mistake. (B) RKFO PEX is what you get by shifting *forwards* another 5 – encrypting once more instead of decrypting. (C) and (E) are shifts back by 4 and by 6 (miscounting). (D) is wrong in the last letter: S shifted back 5 is N, not M.

Check: Encrypt HAVE FUN with shift 5: H → M, A → F, V(21) + 5 = 26 − 26 = 0 = A, E → J, F → K, U → Z, N → S. We get MFAJ KZS.

So the answer is **(A)**.

Answer: (A) HAVE FUN

Problem 17

Nora encrypts the letter M with a Caesar cipher three times in a row: first with shift 9, then with shift 11 (applied to the result of the first step), and finally with shift 9 again. After the three steps, which letter has M become?

(A) V (B) G (C) J (D) P (E) O

Detailed solution

Analysing the problem. Three shifts in the same direction. We can do them one at a time, or combine them into a single shift.

Recall. Shifting by a and then by b is the same as shifting by $a + b$. A shift of 26 returns to the start, so only the remainder of the number of steps on division by 26 matters.

Step 1. Combine the three shifts
Total steps: $9 + 11 + 9 = 29$. Since $29 = 26 + 3$, a shift of 29 is exactly the same as a shift of 3.

Step 2. Shift M by 3
M $= 12$; $12 + 3 = 15 =$ P. (M $\to$ N $\to$ O $\to$ P.)

Step 3. Check by doing one shift at a time

Start	After step 1 (+9)	After step 2 (+11)	After step 3 (+9)
M (12)	$21 =$ V	$32 - 26 = 6 =$ G	$15 =$ P

Common mistake. Options (A) V and (B) G are *intermediate* results (after steps 1 and 2). Read carefully: the question asks for the result after *all three* steps. (C) J is a shift *backwards* by 3.

Check: Going back 29 steps from P, i.e. back 3 steps (since going back 26 returns to the same letter): P $\to$ O $\to$ N $\to$ M. Correct.
So the answer is **(D)**.

Answer: (D) P

Problem 18

Five friends, Nina, Olivia, Peter, Quinn and Sam, all have different heights. Nina is taller than Olivia. Peter is shorter than Quinn but taller than Nina. Sam is shorter than Olivia. When the five friends line up by height, who stands *exactly in the middle*?

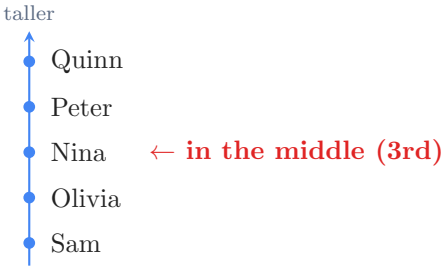
(A) Nina (B) Olivia (C) Peter (D) Quinn (E) Sam

Detailed solution

Analysing the problem. There are five friends; "in the middle" is the third tallest (two taller and two shorter).

Step 1. **Use symbols (“>” means “taller than”)**
$N > O$; $Q > P > N$; $O > S$.

Step 2. **Chain them**
$Q > P > N$ and $N > O > S$ give: $Q > P > N > O > S$.



Step 3. **Read off the result**
From tallest to shortest: Quinn, Peter, **Nina**, Olivia, Sam. The person in the middle is Nina.

Common mistake. “Peter is shorter than Quinn but taller than Nina” contains *two* facts. Many students write only $Q > P$ and forget $P > N$; then the chain cannot be completed.

Check: Nina (3rd) is taller than Olivia (4th) ✓; Peter (2nd) is shorter than Quinn and taller than Nina ✓; Sam (5th) is shorter than Olivia ✓.
So the answer is **(A)**.

Answer: (A) Nina

Problem 19

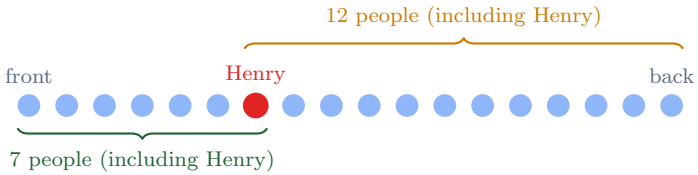
In a queue for tickets, Henry is 7th from the front and 12th from the back. How many people are in the queue?

(A) 17 (B) 18 (C) 19 (D) 20 (E) 5

Detailed solution

Analysing the problem. We know Henry’s position counted from both ends. We must find the total number of people in the queue.

Step 1. **Draw a diagram**



Step 2. **Count**
Counting from the front to Henry: 7 people. Counting from Henry to the back: 12 people. Adding gives $7 + 12 = 19$, but Henry has been counted **twice**. So the queue has $19 - 1 = 18$ people.

Recall. Being k-th from the front and m-th from the back means the line has $k + m - 1$ people.

Step 3. Another way

In front of Henry there are $7 - 1 = 6$ people, behind him $12 - 1 = 11$ people. The whole queue: $6 + 1 + 11 = 18$ people.

Common mistake. Option (C) $19 = 7 + 12$ counts Henry twice; option (E) $5 = 12 - 7$ takes the difference – which means nothing here.

Check: In a queue of 18 people, if Henry is 7th from the front, then from the back he is $18 - 7 + 1 = 12$th. Correct.

So the answer is **(B)**.

Answer: (B) 18

Problem 20

Place the numbers $1, 2, 3, \ldots, 9$ (each once) in the grid shown to make a magic square (the numbers in every row, every column and both diagonals have the same sum). Three numbers are already filled in. Complete the grid. What is the number in the bottom-left corner?

2	7	
		1

Detailed solution

Analysing the problem. A 3×3 magic square with the numbers 1 to 9; we know the top-left cell (2), the middle of the top row (7) and the right cell of the middle row (1). We must fill the 6 remaining cells.

Step 1. Magic sum and centre

Recall. S = (sum of the 9 numbers) $\div$ 3; centre = $S \div 3$.

$1 + 2 + \cdots + 9 = 45$, so $S = 45 \div 3 = 15$; centre $= 15 \div 3 = 5$.

Step 2. Fill in the lines that already have two numbers

- Top row: $2 + 7 + \square = 15 \Rightarrow$ top-right corner $= 6$.
- Middle row: $\square + 5 + 1 = 15 \Rightarrow$ left cell of the middle row $= 9$.
- Diagonal 2 – 5 – ?: bottom-right corner $= 15 - 2 - 5 = 8$.
- Diagonal 6 – 5 – ?: bottom-left corner $= 15 - 6 - 5 = \mathbf{4}$.
- Middle column: $7 + 5 + \square = 15 \Rightarrow$ middle of the bottom row $= 3$.

2	7	6
9	5	1
4	3	8

Check: Rows: $2+7+6=15$, $9+5+1=15$, $4+3+8=15$. Columns: $2+9+4=15$, $7+5+3=15$, $6+1+8=15$. Diagonals: $2+5+8=15$, $6+5+4=15$. The numbers 1 to 9 are each used exactly once.

Common mistake. You must check *all* eight lines (three rows, three columns, two diagonals) and make sure no number is repeated. A grid that works for rows and columns but not for a diagonal is not a magic square.

Answer: Grid: $2,7,6$ / $9,5,1$ / $4,3,8$. The bottom-left number is 4.

Problem 21

The numbers $3, 4, 5, \ldots, 11$ (nine consecutive whole numbers) are placed in the grid shown, each once, to make a magic square. The number x in the bottom-right corner is

(A) 6 (B) 8 (C) 10

(D) 3 (E) 5

4	9	
11		
		x

Detailed solution

Analysing the problem. Nine numbers from 3 to 11. Three cells are known; the cell x lies on the main diagonal together with the cell 4 and the centre.

Step 1. Find the magic sum

Sum of the nine numbers: $3+4+\cdots+11$. Pair first with last: $(3+11)+(4+10)+(5+9)+(6+8)+7 = 4\times 14+7=63$. So $S = 63 \div 3 = 21$.

Step 2. The centre

Recall. In a 3×3 magic square, centre $= S \div 3$.

Centre $= 21 \div 3 = 7$ (which is also the middle number of the list $3, 4, \ldots, 11$).

Step 3. Find x on the main diagonal

$4+7+x=21 \Rightarrow x = 21-11 = 10$.

Step 4. Complete the grid to check

Top row: $21-4-9=8$. Middle row: $21-11-7=3$. Left column: $21-4-11=6$. Bottom row: $21-6-10=5$.

4	9	8
11	7	3
6	5	**10**

Check: Middle column $9+7+5=21$; right column $8+3+10=21$; other diagonal $8+7+6=21$. All nine numbers 3 to 11 are used, no repeats.

Common mistake. If you mistakenly take the magic sum to be 15 (as for the numbers 1 to 9), you get $x=4$ – a number that is already used! The magic sum depends on which numbers are used.

So the answer is **(C)**.

Answer: (C) 10

Problem 22

Both scales below are balanced. The circles all have the same mass.



How many grams does the square ■ weigh?

Detailed solution

Analysing the problem. There are two unknown quantities: the mass of one circle and the mass of the square. The two scales give two equalities.

Recall. If a group of objects A balances a group B, then on another scale we may replace A by B (substitution).

Step 1. Choose the unknown

Let the mass of one circle be x grams, $x>0$. The first scale tells us that the square weighs as much as three circles, so the square weighs $3x$ grams.

Step 2. Form an equation from the second scale

The left pan holds the square and one circle: $3x+x$ grams; the right pan holds the 800 g weight. So

$$3x+x=800, \qquad \text{that is,} \qquad 4x=800.$$

Step 3. Solve the equation

$x=800\div 4=200$. One circle weighs 200 g.

Step 4. Answer the question

The square weighs $3x=3\cdot 200=600$ g.

Common mistake. The question asks for the mass of the *square*, not the circle. Stopping at $x = 200$ answers the wrong question.

Check: Scale 1: $3 \cdot 200 = 600$ g, the mass of the square ✓. Scale 2: $600 + 200 = 800$ g ✓.

Answer: The square weighs 600 g.

Problem 23

There are four kinds of boxes: red, yellow, blue, purple; boxes of the same colour have equal masses. The following three scales are all balanced:



How many yellow boxes does one purple box weigh?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

Detailed solution

Analysing the problem. Let the masses of the red, yellow, blue and purple boxes be R, Y, B, P. The three scales give three equalities:

$$R + Y = B, \qquad B + Y = P, \qquad R + R = B.$$

We must express P in terms of Y.

Step 1. Compare scales 1 and 3

Both have a single blue box on the right pan, so their left pans weigh the same: $R + Y = R + R$. Remove one red box from each side: $Y = R$. *A red box weighs the same as a yellow box.*

Step 2. The blue box

$B = R + Y = Y + Y = 2Y$: a blue box weighs as much as 2 yellow boxes.

Step 3. The purple box

$P = B + Y = 2Y + Y = 3Y$: a purple box weighs as much as 3 yellow boxes.



Common mistake. Do not assume a red box is heavier than a yellow one just because "red" appears more often. Use the scales to *prove* the relationships.

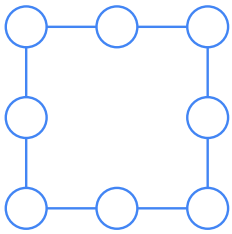
Check: Take $Y = R = 1$ kg, $B = 2$ kg, $P = 3$ kg: scale 1: $1 + 1 = 2$; scale 2: $2 + 1 = 3$; scale 3: $1 + 1 = 2$. All three balance.
So the answer is **(B)**.

Answer: (B) 3

Problem 24

The numbers $1, 2, 3, 4, 5, 6, 7, 8$ are placed in the eight circles of the square shown (four at the corners, four in the middles of the sides), each once, so that the three numbers on each side add up to 13. The sum of the four corner numbers is

(A) 12 (B) 13 (C) 16 (D) 20 (E) 36



Detailed solution

Analysing the problem. The square has 4 sides, each made of two corners and one middle number. We must find the sum of the four corners.

Recall. When we add the sums of all the sides, a corner number is counted twice (it belongs to two sides) and a middle number once.

Step 1. Add the sums of the four sides

Each side sums to 13, so the four sides together give $4 \times 13 = 52$.

Step 2. Count how many times each number is counted

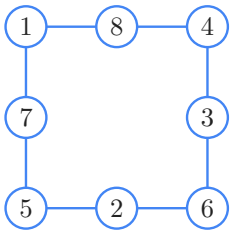
In those 52, each number from 1 to 8 is counted at least once, and the four corner numbers are counted once more:

$$52 = (1 + 2 + \cdots + 8) + (\text{sum of the four corners}) = 36 + (\text{sum of the four corners}).$$

Step 3. Compute

Sum of the four corners $= 52 - 36 = 16$.

Step 4. An actual arrangement (to be sure the puzzle has a solution)



top: $1 + 8 + 4 = 13$
right: $4 + 3 + 6 = 13$
bottom: $6 + 2 + 5 = 13$
left: $5 + 7 + 1 = 13$
corners: $1 + 4 + 6 + 5 = 16$

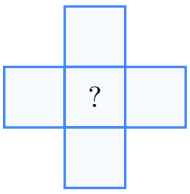
Common mistake. Option (E) 36 is the sum of *all* eight numbers; option (B) 13 is the sum of one side. The question asks for the sum of the *corner* numbers.

So the answer is **(C)**.

Answer: (C) 16

Problem 25

Place the numbers $1, 2, 3, 4, 5$ in the five cells of the plus-shaped figure shown, each once, so that the sum of the three numbers in the horizontal row equals the sum of the three numbers in the vertical column. How many different values can the centre cell take?



(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

Detailed solution

Analysing the problem. The centre cell (call it c) belongs to both the row and the column. The other four cells form two pairs: the left – right pair (row) and the top – bottom pair (column).

Step 1. The condition for equal sums

Row: (left + right) + c; column: (top + bottom) + c. The two sums are equal exactly when

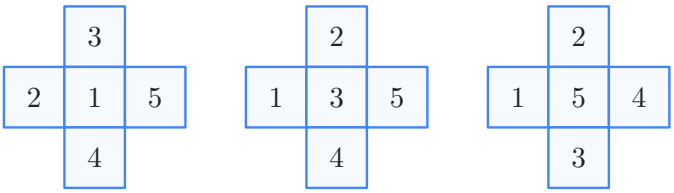
$$\text{left} + \text{right} = \text{top} + \text{bottom}.$$

Step 2. The sum of the other four numbers must be even

The four non-centre numbers add up to $15 - c$ (since $1 + 2 + 3 + 4 + 5 = 15$). The two pairs have equal sums, so $15 - c$ must split into two equal halves, i.e. $15 - c$ is even. Since 15 is odd, c must be **odd**: $c \in \{1, 3, 5\}$.

Step 3. Check each odd value

- $c = 1$: $2, 3, 4, 5$ remain, each pair must sum to $14 \div 2 = 7$: $2 + 5 = 3 + 4 = 7$. ✓
- $c = 3$: $1, 2, 4, 5$ remain, each pair must sum to $12 \div 2 = 6$: $1 + 5 = 2 + 4 = 6$. ✓
- $c = 5$: $1, 2, 3, 4$ remain, each pair must sum to $10 \div 2 = 5$: $1 + 4 = 2 + 3 = 5$. ✓



$c = 1$ $c = 3$ $c = 5$

Step 4. Why does an even c not work?

If $c = 2$: the remaining numbers $1, 3, 4, 5$ add up to 13 – odd, so they cannot form two pairs with equal sums. If $c = 4$: $1, 2, 3, 5$ remain, with sum 11 – odd, also impossible.

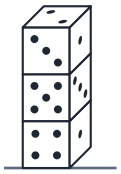
> **Common mistake.** Do not think "the centre must be the middle number (3)". Here there are three possibilities!

So the centre can be 1, 3 or 5: 3 values. The answer is **(C)**.

Answer: (C) 3

Problem 26

Three standard dice (opposite faces add up to 7) are stacked on a table as shown. The top face shows 2 pips. The total number of pips on the five hidden *horizontal* faces (the faces pressed together between two dice and the bottom face resting on the table) is



(A) 14 (B) 17 (C) 19 (D) 21 (E) 23

Detailed solution

Analysing the problem. The five hidden horizontal faces are: the bottom face of the top die; the top and bottom faces of the middle die; the top and bottom faces of the bottom die. We do not need to know each face separately – we only use "opposite faces add up to 7".

> **Recall.** On a standard die the top and bottom faces are opposite, so together they carry 7 pips.

Step 1. Bottom die and middle die
For each of them the top and bottom faces are both hidden, and each pair totals 7. Two dice: $7 + 7 = 14$.

Step 2. Top die
The visible top face has 2 pips, so the (opposite) bottom face has $7 - 2 = 5$ pips, and it is hidden.

Step 3. Add up
$14 + 5 = 19$.

> **Common mistake.** (D) $21 = 3 \times 7$ forgets that the very top face (2 pips) *can be seen*; it is not hidden. (A) 14 forgets the bottom face of the top die (also hidden, with 5 pips).

Check: The six horizontal faces of the three dice total $3 \times 7 = 21$; of these only the very top face (2 pips) is visible, so the five hidden ones total $21 - 2 = 19$.
So the answer is **(C)**.

Answer: (C) 19

Problem 27

The 4×4 grid shown is to be filled so that every row, every column and every thick-bordered 2×2 box contains the digits $1, 2, 3, 4$. The digit in cell Q is

	3		2
			1
4			
		2	Q

(A) 1
(B) 2
(C) 3
(D) 4
(E) It cannot be determined

Detailed solution

Analysing the problem. Cell Q is in row 4, column 4 (bottom-right corner), in the bottom-right 2×2 box.

> **Recall.** The digit in a cell must differ from every digit already in its row, its column and its 2×2 box.

Step 1. Column 4
Column 4 already has 2 (row 1) and 1 (row 2). The two remaining cells – row 3 column 4 and cell Q – must be 3 and 4 in some order.

Step 2. Row 3 decides
Row 3 already has a 4 (in column 1), so row 3 column 4 *cannot* be 4. Hence row 3 column 4 is 3, and $Q = 4$.

Step 3. Check with the bottom-right box and row 4
The bottom-right box contains 3 (row 3 column 4), 2 (row 4 column 3), $Q = 4$, and the remaining cell (row 3 column 3) is 1 – all of $1, 2, 3, 4$. Row 4 has 2 and 4, no repeat.

Step 4. Complete the grid (to be sure there is no contradiction)

1	3	4	2
2	4	3	1
4	2	1	3
3	1	2	4

Row 1: 1 3 4 2
Row 2: 2 4 3 1
Row 3: 4 2 1 3
Row 4: 3 1 2 4

Check: Every row, column and 2×2 box contains all of $1, 2, 3, 4$; a computer check confirms this is the only way to fill the grid.

> **Common mistake.** Looking only at column 4, Q could be 3 or 4, and it is tempting to choose (E). Also look at row 3 – a different region containing a "neighbour" of Q.

So the answer is **(D)**.

Answer: (D) 4

Problem 28

In class 7A, *all* members of the chess club can swim. Leo cannot swim. Nora can swim. Which of the following statements **must** be true?

(A) Leo is in the chess club.

(B) Leo is not in the chess club.

(C) Nora is in the chess club.

(D) Nora is not in the chess club.

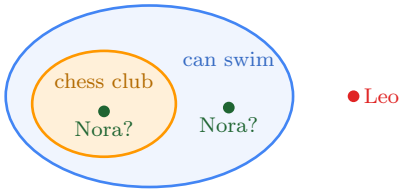
(E) Everyone in the class who can swim is in the chess club.

Detailed solution

Analysing the problem. "All members of the chess club can swim" has the form "every A is a B" (A: in the chess club, B: can swim).

> **Recall.** From "every A is a B" we may conclude "not a B means not an A"; but we may *not* conclude "a B means an A".

Step 1. Draw a diagram



The "chess club" oval lies entirely inside the "can swim" oval.

Step 2. Leo

Leo cannot swim, so Leo is *outside* the big oval, and therefore certainly outside the small one: **Leo is not in the chess club**. If Leo were in the club, Leo would have to be able to swim – contradicting the problem (proof by contradiction).

Step 3. Nora

Nora can swim: Nora is inside the big oval, but may be inside or outside the small one. So (C) and (D) are each only *possibly* true, not certain.

Step 4. Statement (E)

(E) reverses the given statement. The problem says nothing about pupils who can swim but do not play chess, so (E) is not certainly true.

> **Common mistake.** A very common mistake: from "chess players can all swim" concluding "swimmers all play chess". Think of an example: "every cat has four legs" does not mean "every four-legged animal is a cat"!

So the answer is **(B)**.

Answer: (B) Leo is not in the chess club

Problem 29

Mr Fox lies on Mondays, Tuesdays and Wednesdays and tells the truth on the other days of the week. One day, Fox says: "Yesterday I lied." Which day could it be?

(A) Only Monday.

(B) Only Thursday.

(C) Monday or Thursday.

(D) Wednesday or Thursday.

(E) Sunday or Monday.

Detailed solution

Analysing the problem. Fox can say "Yesterday I lied" when:

- he tells the truth today and the statement is true: truthful today, lied yesterday; or
- he lies today and the statement is false: lying today, truthful yesterday.

That is: **yesterday and today are of different types**.

Recall. The day before Monday is Sunday (the week goes round).

Step 1. Make a table

Today	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Fox today	L	L	L	T	T	T	T
Fox yesterday	T (Sun)	L	L	L	T	T	T
Different?	**yes**	–	–	**yes**	–	–	–

Step 2. Check the two days found

Monday: Fox lies; yesterday (Sunday) he told the truth, so "yesterday I lied" is false – indeed a lie. ✓

Thursday: Fox tells the truth; yesterday (Wednesday) he lied, so the statement is true – indeed the truth. ✓

Common mistake. Wednesday: Fox lies, and yesterday (Tuesday) he lied too, so "yesterday I lied" is a *true* statement. Someone who is lying does not say true things, so Wednesday is ruled out – this is the trap in option (D).

So the answer is **(C)**.

Answer: (C) Monday or Thursday

Problem 30

Four friends, Amy, Ben, Chloe and David, run a race; no two finish at the same time. Three spectators make predictions:

- Henry: "Amy comes first, Ben comes second."
- Kate: "Chloe comes first, David comes fourth."
- Lucy: "Ben comes third, David comes second."

In the end, each spectator got *exactly one* part right. Find the finishing order of the four friends.

Detailed solution

Analysing the problem. Each prediction has two parts, exactly one of them right. We split into cases using Henry's prediction.

Recall. "Half-true" statement: if one part is true the other is false, and if one part is false the other is true.

Step 1. **Case 1: "Ben second" is true (and "Amy first" is false)**
Look at Lucy's prediction. Ben is second, so "Ben third" is false; hence the other part "David second" must be true. But Ben is already second, so David cannot be second too. **Contradiction** – this case is ruled out.

Step 2. **Case 2: "Amy first" is true (and "Ben second" is false)**

- Kate: Amy is first, so "Chloe first" is false ⇒ "David fourth" is true.
- Lucy: David is fourth, so "David second" is false ⇒ "Ben third" is true.
- Second place is left for Chloe.

Order: Amy (1st), Chloe (2nd), Ben (3rd), David (4th).

Step 3. **Check**

Spectator	First part	Second part
Henry	Amy first: true	Ben second: false
Kate	Chloe first: false	David fourth: true
Lucy	Ben third: true	David second: false

ThinkAlgo Tip. Choose the prediction to split on so that each case "triggers" as many other parts as possible. Here "Ben second" clashes immediately with Lucy's "David second", so case 1 is ruled out very quickly.

Note for parents. You can use four slips of paper with the names, line them up as "1st – 2nd – 3rd – 4th" and swap them around to test. Then ask your child: "Have we tried every case? How can we be sure there is no other answer?" – the answer is Step 1.

Answer: Amy 1st, Chloe 2nd, Ben 3rd, David 4th.

Extension and challenge

Problem 31

On an island, every inhabitant is either a *truth-teller* (always tells the truth) or a *liar* (always lies).

a) You meet two inhabitants, Kate and Leo. Kate says: "Both of us are liars." What type is each of them?

b) You meet three inhabitants, Anna, Ben and Chloe.
Anna says: "Ben is a liar."
Ben says: "Chloe is a liar."
Chloe says: "Anna and Ben are both liars."
What type is each of them?

Detailed solution

Understanding the problem. Each person has only two possibilities. The safest method is to *assume* one possibility and check whether the statements fit that assumption. If we reach a contradiction, we rule that possibility out.

> **Recall.** Whatever a truth-teller says is **true**. Whatever a liar says is **false**. So each statement gives us an "equation": *the statement is true* $\iff$ *the speaker is a truth-teller*.

Part a).

Step 1. Suppose Kate is a truth-teller
Then her statement is true, so Kate is a liar – contradicting the assumption. So Kate **cannot** be a truth-teller.

Step 2. Kate is a liar
The statement "Both of us are liars" is false. It is false when *at least one* of them is not a liar. Kate is a liar, so the other one – Leo – must be a truth-teller.

> **Common mistake.** The opposite of "both are liars" is **not** "both tell the truth", but "at least one of them tells the truth".

Part b).

Step 3. Suppose Anna is a truth-teller
Anna's statement is true ⇒ Ben is a liar ⇒ Ben's statement is false ⇒ Chloe is a truth-teller ⇒ Chloe's statement is true ⇒ Anna is a liar. This contradicts the assumption! Ruled out.

Step 4. So Anna is a liar
Anna's statement is false ⇒ Ben is *not* a liar, i.e. Ben is a truth-teller ⇒ Ben's statement is true ⇒ Chloe is a liar.

Step 5. Check Chloe's statement
Chloe is a liar, so the statement "Anna and Ben are both liars" must be false. Indeed it is false, because Ben is a truth-teller. There is no contradiction.

Person	Type	Statement	True/false
Anna	liar	Ben is a liar	false
Ben	truth-teller	Chloe is a liar	true
Chloe	liar	Anna and Ben are both liars	false

Check: In the table, every truth-teller makes a true statement and every liar a false one. We have also seen that "Anna is a truth-teller" is impossible, so this is the only solution.

Answer: a) Kate is a liar, Leo a truth-teller; b) Anna and Chloe are liars, Ben is a truth-teller

Problem 32

Mia lies on Mondays, Tuesdays and Wednesdays and tells the truth on the other days of the week. Henry lies on Thursdays, Fridays and Saturdays and tells the truth on the other days. One day, both of them say: "Yesterday was a day on which I lie." Which day of the week is it?

Detailed solution

Understanding the problem. The statement "Yesterday was a day on which I lie" may be true or false depending on the day. For each person we find the days on which they *could* say it, then take the day common to both.

Recall. On a truth day the statement must be true; on a lying day the statement must be false.

Step 1. Table for Mia (lies Mon, Tue, Wed)

Today	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Mia today	lies	lies	lies	truth	truth	truth	truth
Mia yesterday	truth (Sun)	lies	lies	lies	truth	truth	truth
Statement true?	no	yes	yes	yes	no	no	no
Can Mia say it?	**yes**	no	no	**yes**	no	no	no

Mia can say it when she lies today and the statement is false (Monday), or when she tells the truth today and the statement is true (Thursday).

Step 2. Table for Henry (lies Thu, Fri, Sat)

Today	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Henry today	truth	truth	truth	lies	lies	lies	truth
Henry yesterday	truth	truth	truth	truth	lies	lies	lies
Statement true?	no	no	no	no	yes	yes	yes
Can Henry say it?	no	no	no	**yes**	no	no	**yes**

Henry can say it on Thursday or Sunday.

Step 3. The common day
Mia: {Monday, Thursday}; Henry: {Thursday, Sunday}. The only common day is **Thursday**.

ThinkAlgo Tip. Quick rule: a person can say "Yesterday I lied" only on a *switch day* (yesterday and today are of different types): the first lying day, or the first day back to telling the truth.

Check: Thursday: Mia tells the truth, and yesterday (Wednesday) she lied – the statement is true, consistent. Henry lies, and yesterday (Wednesday) he told the truth – the statement is false, consistent with Henry lying today.

Answer: Thursday

Problem 33

Six friends Lucy, Max, Noah, Olivia, Peter and Quinn run a race. No two of them finish at the same time. We know that:

- Max finishes after Lucy but before Noah;
- Olivia finishes after Noah;
- Peter finishes *immediately after* Lucy;
- Quinn is neither first nor last, and Quinn finishes before Max.

Find the finishing order of the six friends. Who finishes third?

Detailed solution

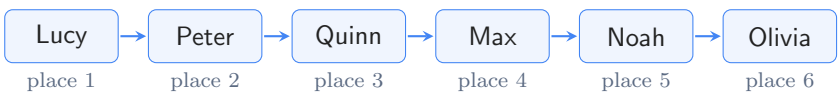
Understanding the problem. We have "before – after" conditions. A good method is to join them into a *chain* and then insert the remaining friends one by one. Write $X \to Y$ for "X finishes before Y".

Step 1. Join conditions 1 and 2
Lucy → Max → Noah and Noah → Olivia, so: Lucy → Max → Noah → Olivia.

Step 2. Insert Peter
Peter finishes *immediately after* Lucy, i.e. nobody comes between Lucy and Peter: Lucy, Peter → Max → Noah → Olivia.

Step 3. Insert Quinn
Quinn finishes before Max, so Quinn is in one of these places: before Lucy, or between Peter and Max (he may not come between Lucy and Peter). But Quinn is not first, so he is not before Lucy. Hence:

$$\text{Lucy} \to \text{Peter} \to \text{Quinn} \to \text{Max} \to \text{Noah} \to \text{Olivia}.$$



Step 4. Check every condition again
Max (4th) is after Lucy (1st) and before Noah (5th): true. Olivia (6th) is after Noah: true. Peter (2nd) is immediately after Lucy: true. Quinn is 3rd – not first, not last, before Max: true.

Common mistake. If you forget the word "immediately", Quinn could come between Lucy and Peter and the problem would have two answers. Read every word of each condition carefully!

Check: We tried every possible position for Quinn; only one order satisfies all four conditions.

Answer: Lucy, Peter, Quinn, Max, Noah, Olivia; Quinn finishes third

Problem 34

Five friends Hannah, Kevin, Lily, Maya and Tom sit in a row of 5 chairs numbered $1, 2, 3, 4, 5$ from left to right. We know that: Hannah and Kevin sit next to each other; Lily and Maya do *not* sit next to each other; Tom sits at one end of the row (chair 1 or chair 5).

a) How many seating arrangements satisfy these conditions?

b) Which of the following statements is *certainly true*?
(1) Lily sits at one end of the row.
(2) Hannah does not sit at an end of the row.
(3) Kevin sits in chair 3.

c) If we also know that Lily sits in chair 1 and Hannah sits to the left of Kevin, find each friend's seat.

Detailed solution

Understanding the problem. Start with the "tightest" condition (Tom has only 2 places), then look at the 4 remaining chairs.

Step 1. Case: Tom sits in chair 5

The other four sit in chairs $1, 2, 3, 4$. The pair Hannah – Kevin takes two adjacent chairs: $(1, 2)$, $(2, 3)$ or $(3, 4)$.

- Hannah – Kevin in chairs $(1, 2)$: Lily and Maya must sit in chairs $3, 4$ – next to each other, **impossible**.
- Hannah – Kevin in chairs $(3, 4)$: Lily and Maya sit in chairs $1, 2$ – next to each other, **impossible**.
- Hannah – Kevin in chairs $(2, 3)$: Lily and Maya sit in chairs 1 and 4 – not next to each other, **possible**.

In the possible case, Hannah and Kevin can swap (2 ways) and Lily and Maya can swap (2 ways): $2 \times 2 = 4$ ways.

Step 2. Case: Tom sits in chair 1

Exactly the same (turn the row around): Hannah – Kevin in chairs $(3, 4)$, Lily and Maya in chairs 2 and 5: also 4 ways.

Step 3. Total number of ways

$4 + 4 = 8$ ways.

Recall. Multiplication rule: if a task consists of consecutive steps, step 1 in m ways and step 2 in n ways, then the task can be done in $m \times n$ ways. **Addition rule:** separate cases are added.

Step 4. Part b): check each statement

Chair 1	Chair 2	Chair 3	Chair 4	Chair 5
Lily/Maya	Hannah/Kevin	Hannah/Kevin	Maya/Lily	Tom
Tom	Lily/Maya	Hannah/Kevin	Hannah/Kevin	Maya/Lily

(1) False: when Tom is in chair 1, Lily may sit in chair 2. (2) **True**: in both rows, Hannah and Kevin are always in middle chairs (2, 3 or 3, 4). (3) False: Kevin may sit in chair 2 or chair 4.

Step 5. Part c)

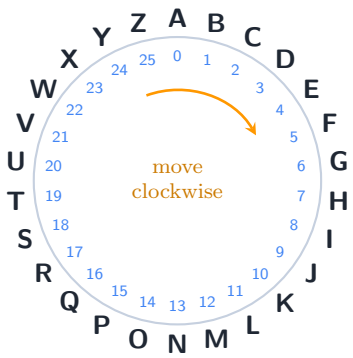
Lily is in chair 1, so Tom is not in chair 1; Tom is in chair 5 (first row of the table). Then Maya is in chair 4 and Hannah – Kevin in chairs 2, 3; Hannah is to the left of Kevin, so Hannah is in chair 2 and Kevin in chair 3.

Check: Lily(1), Hannah(2), Kevin(3), Maya(4), Tom(5): Hannah and Kevin are adjacent; Lily in chair 1 and Maya in chair 4 are not adjacent; Tom is at an end. Correct.

Answer: a) 8 ways; b) statement (2); c) Lily – Hannah – Kevin – Maya – Tom (chairs 1 to 5)

Problem 35

Number the letters $A, B, C, \ldots, Z$ (the 26-letter English alphabet) as $0, 1, 2, \ldots, 25$. The *Caesar cipher with key* k replaces each letter with number x by the letter whose number is the *remainder* of $x + k$ when divided by 26 (spaces are kept).



a) Encrypt the text MATH CLUB with key $k = 5$.

b) Decrypt the text FVB DPU, knowing that the key is $k = 7$.

c) In a message, the word CAT was encrypted as RPI. Find the key k (with $0 \le k \le 25$), then decrypt the word SDV from the same message.

d) A message is encrypted with key 10, and the result is encrypted once more with key 20. These two encryptions together are the same as *one* encryption with which key?

e) Sam encrypts a word with key 4, then encrypts the result with key 4 again, and so on. After how many encryptions, at least, does Sam get back exactly the original word?

Detailed solution

Understanding the problem. When adding the key takes us past 25, we "wrap around" to the start of the alphabet – that is exactly taking the *remainder on division by* 26. Decrypting is the reverse: *subtract* the key (if the result is negative, add 26).

Recall. The remainder of a on division by 26 is the number r with $0 \le r \le 25$ such that $a - r$ is divisible by 26. Examples: 30 leaves remainder 4; -6 "leaves remainder" 20, because $-6 + 26 = 20$.

Step 1. Part a): encrypt MATH CLUB with $k = 5$

Plain letter	M	A	T	H	C	L	U	B
Number x	12	0	19	7	2	11	20	1
$x + 5$	17	5	24	12	7	16	25	6
Cipher letter	R	F	Y	M	H	Q	Z	G

Result: RFYM HQZG. (No number exceeds 25 here, so no wrapping is needed yet.)

Step 2. Part b): decrypt FVB DPU with $k = 7$ (subtract 7)

Cipher letter	F	V	B	D	P	U
Number	5	21	1	3	15	20
Number -7	$-2 \to 24$	14	$-6 \to 20$	$-4 \to 22$	8	13
Plain letter	Y	O	U	W	I	N

Original text: YOU WIN. Note: the letters F, B, D have numbers smaller than 7, so subtracting 7 gives a negative number and we must add 26; for example $5 - 7 = -2$, $-2 + 26 = 24$ (the letter Y).

Step 3. Part c): find the key from the pair CAT → RPI

C (2) becomes R (17), so $k = 17 - 2 = 15$. Check: A (0) $\to 0 + 15 = 15$ (P), correct; T (19) $\to 19 + 15 = 34$, remainder 8 on division by 26 (I), correct. So $k = 15$.

Decrypt SDV (subtract 15): S $= 18$, $18 - 15 = 3$ (D); D $= 3$, $3 - 15 = -12 \to 14$ (O); V $= 21$, $21 - 15 = 6$ (G). The original word is DOG.

Step 4. Part d): two encryptions in a row

Each letter gets 10 added and then 20 added, i.e. 30 added. Since 30 leaves remainder 4 on division by 26, "add 30 and take the remainder" is exactly the same as "add 4 and take the remainder". The two encryptions equal **one encryption with key** 4.

Step 5. Part e): repeated encryption with key 4

After n encryptions, every letter has moved $4n$ places. To get back to the original letter we must go round a whole number of times, i.e. $4n$ must be divisible by 26. Try $n = 1, 2, 3, \ldots$: $4n = 4, 8, 12, \ldots$ The smallest (non-zero) $4n$ divisible by 26 is $\text{LCM}(4, 26) = 52$, which gives $n = 52 \div 4 = 13$.

Recall. $4n$ is divisible by $26 = 2 \times 13 \Leftrightarrow 2n$ is divisible by $13 \Leftrightarrow n$ is divisible by 13 (because 13 is a prime that does not divide 2).

ThinkAlgo Tip. With key 13 you only need 2 encryptions to get back ($2 \times 13 = 26$). So with key 13, decrypting is the same as encrypting once more!

Check: Part a): decrypting RFYM HQZG by subtracting 5 gives back MATH CLUB. Part c): DOG encrypted with key 15: D $3 + 15 = 18$ (S), O $14 + 15 = 29 \to 3$ (D), G $6 + 15 = 21$ (V). Indeed SDV.

Answer: a) RFYM HQZG; b) YOU WIN; c) $k = 15$, DOG; d) key 4; e) 13 times

Problem 36

Again number $A = 0, B = 1, \ldots, Z = 25$. A *multiplicative cipher* replaces the letter with number x by the letter whose number is the remainder of $3x + 2$ on division by 26.

a) Encrypt the word MAP.

b) Decrypt the word EGW.

c) Nathan suggests using $2x + 2$ instead of $3x + 2$. Find two different letters with the same cipher letter. Why does that make decryption impossible?

d) Explain why, with $3x + 2$, two different letters always get two different cipher letters.

Detailed solution

Understanding the problem. This cipher does not just "shift": it also multiplies and then takes the remainder. Encrypting is easy (compute directly); to decrypt we need x when we know the remainder of $3x + 2$ – we use the trick "add 26 until the number is divisible by 3".

Step 1. Part a): encrypt MAP

M $= 12$: $3 \cdot 12 + 2 = 38$, remainder 12 on division by 26 $\to$ M.
A $= 0$: $3 \cdot 0 + 2 = 2 \to$ C.
P $= 15$: $3 \cdot 15 + 2 = 47$, remainder 21 $\to$ V.
Result: MCV. (Fun fact: M is encrypted as itself.)

Step 2. Part b): decrypt EGW

For a cipher letter with number y we need x ($0 \le x \le 25$) such that $3x + 2$ leaves remainder y, i.e. $3x = y - 2$, or $y - 2 + 26$, or $y - 2 + 52$ (try them in turn and take the one divisible by 3).

Cipher letter	y	Try $y-2,\ y+24,\ y+50$	$3x$	Plain letter
E	4	2 (no), 28 (no), 54 (divisible)	$54 \Rightarrow x = 18$	S
G	6	4 (no), 30 (divisible)	$30 \Rightarrow x = 10$	K
W	22	20 (no), 46 (no), 72 (divisible)	$72 \Rightarrow x = 24$	Y

The original word is SKY.

Step 3. Part c): why $2x + 2$ does not work

A $= 0$: $2 \cdot 0 + 2 = 2 \to$ C. N $= 13$: $2 \cdot 13 + 2 = 28$, remainder 2 $\to$ C.
Both A and N become C. When the receiver sees a C, they cannot tell whether the original letter was A or N, so decryption is not certain. (In fact every number $2x + 2$ is even, so there are only 13 cipher letters for 26 plain letters – some pairs must coincide.)

Step 4. Part d): why $3x + 2$ never collides

Why is it true? Suppose two letters x and z (with $0 \le z < x \le 25$) have the same cipher letter. Then $3x + 2$ and $3z + 2$ leave the same remainder on division by 26, so their difference $3(x - z)$ is divisible by 26. But 3 and 26 have no common divisor greater than 1, so $x - z$ must be divisible by 26. However $0 < x - z \le 25$ – not divisible by 26. Contradiction. So different letters always get different cipher letters. □

Common mistake. With $2x + 2$, the argument breaks down here: $2(x - z)$ is divisible by 26 as soon as $x - z$ is divisible by 13, for example $x - z = 13$ (A and N).

Check: Encrypt SKY again: S $3 \cdot 18 + 2 = 56 \to 56 - 52 = 4$ (E); K $3 \cdot 10 + 2 = 32 \to 6$ (G); Y $3 \cdot 24 + 2 = 74 \to 74 - 52 = 22$ (W). We get EGW, correct.

Answer: a) MCV; b) SKY; c) A and N both become C; d) see the proof

Problem 37

In a 3×3 *magic square*, the numbers in every row, every column and both diagonals have the same sum (called the *magic sum* M).

a) Let c be the number in the centre cell. Prove that $M = 3c$.

b) Fill in the empty cells of the magic square below.

9	**19**	
		13

Detailed solution

Understanding the problem. In part b) only three numbers are known; we know neither the centre nor M. Part a) is the "key": it links the centre number to M.

Step 1. Part a): add the four lines through the centre

Name the cells:

a	b	d
e	c	f
g	h	i

Four lines pass through the centre: the middle row $e + c + f$, the middle column $b + c + h$, and the two diagonals $a + c + i$ and $d + c + g$. Adding these four sums:

$$(e + c + f) + (b + c + h) + (a + c + i) + (d + c + g) = 4M.$$

On the left, each outer cell appears exactly *once*, while the centre c appears *four* times. So the left side equals (sum of all 9 cells) $+ 3c$.

Why is it true? The sum of all 9 cells is the sum of the three rows, i.e. $3M$. Therefore $3M + 3c = 4M$, which gives $3c = M$. □

Step 2. Part b): find the centre

The main diagonal passes through 9, the centre c and 13: $9 + c + 13 = M = 3c$. So $22 = 2c$, $c = 11$ and $M = 33$.

Step 3. Fill the cells one by one (each time choose a line with only one cell missing)

- Top row: $33 - 9 - 19 = 5$ (top right corner).
- Other diagonal: $33 - 5 - 11 = 17$ (bottom left corner).
- Left column: $33 - 9 - 17 = 7$; middle row: $33 - 7 - 11 = 15$.
- Bottom row: $33 - 17 - 13 = 3$.

9	**19**	5
7	11	15
17	3	**13**

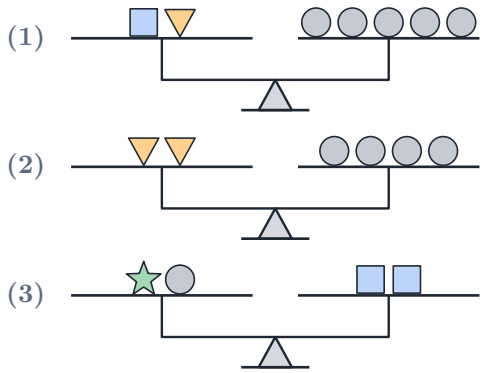
Check: Middle column $19 + 11 + 3 = 33$; right column $5 + 15 + 13 = 33$; middle row $7 + 11 + 15 = 33$; the diagonals $9 + 11 + 13$ and $5 + 11 + 17$ both equal 33.

ThinkAlgo Tip. The result $M = 3c$ holds for *every* 3×3 magic square, even with negative numbers or fractions.

Answer: a) see the proof; b) top row $9, 19, 5$; middle row $7, 11, 15$; bottom row $17, 3, 13$

Problem 38

The three equal-arm balances below are all balanced. Objects of the same shape have the same mass.



a) How many ○ does each of ▽, □, ☆ weigh?

b) How many ▽ are needed to balance ☆☆□□?

c) Each ○ weighs 12 g. How many grams do the three objects ☆▽□ weigh altogether?

Detailed solution

Understanding the problem. Each balanced scale gives us an *equation*. We introduce a letter for the mass of each kind of object, write the three equations, and solve them one by one, starting with the simplest (the one with only two kinds of object).

> **Recall.** A balanced scale is like an equation: mass on the left pan = mass on the right pan. We may add (or subtract) the same amount on both sides, or divide both sides by the same non-zero number – the equation stays true (just as the scale stays balanced).

Step 1. Introduce letters and write the equations

Let the masses of one ○ (circle), □ (square), ▽ (triangle), ☆ (star) be o, q, t, s. Reading the three scales:

$$(1)\ \ q + t = 5o; \qquad (2)\ \ 2t = 4o; \qquad (3)\ \ s + o = 2q.$$

Step 2. From equation (2): find t

Divide both sides of $2t = 4o$ by 2: $t = 2o$. So each ▽ weighs as much as 2 objects ○.

Step 3. Substitute into equation (1): find q

Substitute $t = 2o$ into (1): $q + 2o = 5o$. Subtract $2o$ from both sides: $q = 5o - 2o = 3o$. Each □ weighs as much as 3 objects ○.

Step 4. Substitute into equation (3): find s

Substitute $q = 3o$ into (3): $s + o = 2 \cdot 3o = 6o$. Subtract o from both sides: $s = 6o - o = 5o$. Each ☆ weighs as much as 5 objects ○.

Step 5. Part b)

The four objects ☆☆□□ weigh $2s + 2q = 2 \cdot 5o + 2 \cdot 3o = 10o + 6o = 16o$.

Let n be the number of objects ▽ needed. We need $n \cdot t = 16o$, i.e. $n \cdot 2o = 16o$, so $n = 16 \div 2 = 8$.

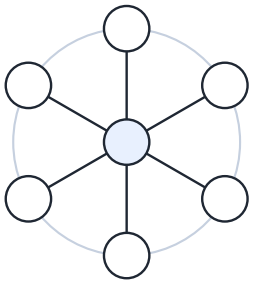
Step 6. Part c)
With $o = 12$ g: $s + t + q = 5o + 2o + 3o = 10o = 10 \cdot 12 = 120$ g.

Check: With $o = 1$, $t = 2$, $q = 3$, $s = 5$: balance (1) $3 + 2 = 5$; balance (2) $2 \cdot 2 = 4$; balance (3) $5 + 1 = 6 = 2 \cdot 3$. All three balances are balanced. Part b): 8 triangles weigh $8 \cdot 2 = 16$, exactly $5 + 5 + 3 + 3$.

Answer: a) ▽ $= 2$, □ $= 3$, ☆ $= 5$ **objects** ○; **b)** 8 **objects** ▽; **c)** 120 **g**

Problem 39

The "wheel" below has 7 circles: one at the hub and 6 on the rim, forming 3 straight "spokes" of 3 circles each (two opposite circles on the rim and the hub). Place the numbers $1, 2, 3, 4, 5, 6, 7$, one in each circle, so that the three numbers on each spoke have the same sum S.



a) Let c be the number at the hub. Prove that $3S = 28 + 2c$.

b) Deduce that the hub can only be 1, 4 or 7. In each case find S and give one arrangement.

Detailed solution

Understanding the problem. This is the "count twice" technique: add all the spokes and see how many times each circle is counted.

Step 1. Part a): add the three spokes
The three spokes total $S + S + S = 3S$. In this sum every rim circle is counted *once* (each rim circle lies on only one spoke), while the hub is counted *three* times. So

$$3S = (1 + 2 + \cdots + 7) + 2c = 28 + 2c.$$

Recall. $1 + 2 + 3 + 4 + 5 + 6 + 7 = 28$. The hub already appears once in the total 28, so we only add $2c$ (the other two times).

Step 2. Part b): which values can c take?
$3S$ is divisible by 3, so $28 + 2c$ must be divisible by 3. Try $c = 1, 2, \ldots, 7$:

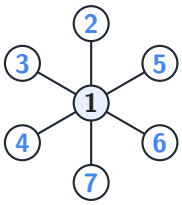
c	1	2	3	4	5	6	7
$28 + 2c$	**30**	32	34	**36**	38	40	**42**
Divisible by 3?	yes	no	no	yes	no	no	yes

So $c \in \{1; 4; 7\}$, giving $S = 10$; 12; 14 respectively.

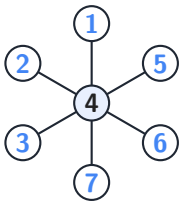
Step 3. Concrete arrangements

On each spoke the two opposite circles must add up to $S - c$, so we split the remaining 6 numbers into 3 pairs with equal sums.

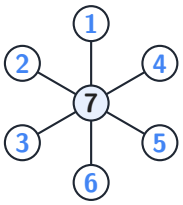
- $c = 1$, $S = 10$: pairs with sum 9: $(2, 7), (3, 6), (4, 5)$.
- $c = 4$, $S = 12$: pairs with sum 8: $(1, 7), (2, 6), (3, 5)$.
- $c = 7$, $S = 14$: pairs with sum 7: $(1, 6), (2, 5), (3, 4)$.



$S = 10$



$S = 12$



$S = 14$

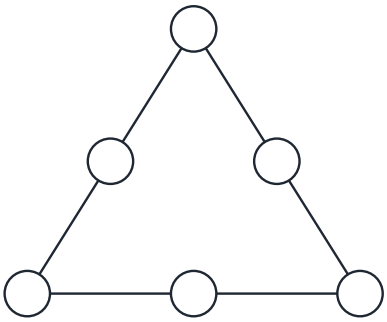
Common mistake. Do not assume "the hub must be the middle number of the list (4)". Both 1 and 7 can also go at the hub!

Check: $S = 10$: $1+2+7 = 1+3+6 = 1+4+5 = 10$. $S = 12$: $4+1+7 = 4+2+6 = 4+3+5 = 12$. $S = 14$: $7 + 1 + 6 = 7 + 2 + 5 = 7 + 3 + 4 = 14$.

Answer: a) see the proof; b) $c = 1, S = 10$; $c = 4, S = 12$; $c = 7, S = 14$ (arrangements as in the figure)

Problem 40

Place the numbers $1, 2, 3, 4, 5, 6$, one in each circle on the sides of the triangle (three circles at the vertices, three at the midpoints of the sides), so that the three numbers on each side have the same sum S.



a) Let V be the sum of the three numbers at the vertices. Prove that $3S = 21 + V$. Hence find all possible values of S.

b) For each value of S, give one arrangement.

c) When $S = 10$, which three numbers are at the vertices? Explain why there is no other possibility.

Detailed solution

Understanding the problem. As in Problem 39, we "count twice": each vertex circle lies on *two* sides, each midpoint circle on *one* side.

Step 1. Part a): add the three sides

The three sides total $3S$. In this sum each midpoint number is counted once and each vertex number twice. So

$$3S = (1+2+3+4+5+6) + V = 21 + V.$$

Step 2. Bounds on V

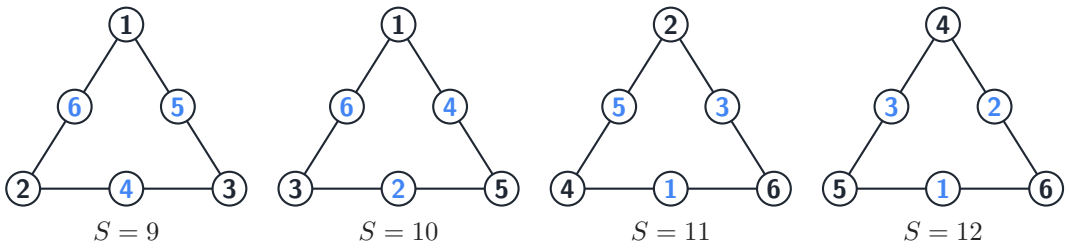
V is the sum of three different numbers from $1, \ldots, 6$: at least $1+2+3=6$, at most $4+5+6=15$. Also $21+V$ is divisible by 3, and 21 is divisible by 3, so V is divisible by 3:

$$V \in \{6;\ 9;\ 12;\ 15\} \quad \Rightarrow \quad S = \frac{21+V}{3} \in \{9;\ 10;\ 11;\ 12\}.$$

Recall. If a is divisible by 3 and $a+b$ is divisible by 3, then b is also divisible by 3.

Step 3. Part b): one arrangement for each S

Choose three vertex numbers with sum V; then subtract the sum of the two end numbers of each side from S to get the number at the middle of that side.



Example for $S=9$: vertices $1, 2, 3$; side 1–2 needs $9-3=6$; side 1–3 needs $9-4=5$; side 2–3 needs $9-5=4$.

Step 4. Part c): $S=10$ gives $V = 3 \cdot 10 - 21 = 9$

The triples of different numbers with sum 9 are $\{1,2,6\}$, $\{1,3,5\}$, $\{2,3,4\}$. Try each one:

- Vertices $\{1,2,6\}$: the side joining 1 and 2 needs a middle number $10-3=7$ – there is no 7. **Ruled out.**
- Vertices $\{2,3,4\}$: the side joining 3 and 4 needs $10-7=3$ – but 3 is already at a vertex. **Ruled out.**
- Vertices $\{1,3,5\}$: the middle numbers are $10-4=6$, $10-6=4$, $10-8=2$ – exactly the remaining numbers $2, 4, 6$. **Works.**

So when $S=10$, the vertices must be $1, 3, 5$.

Common mistake. The condition "V is divisible by 3" is only a *necessary* condition: it tells us what S *could* be, but we still have to show an actual arrangement (part b) and check each triple (part c).

Note for parents. Let your child cut out 6 slips of paper numbered 1 to 6 and try placing them on the drawing. After a few tries, ask: "How many times is a vertex number added?" – that is the main idea of the solution.

Check: $S = 10$: $1+6+3 = 3+4+5 = 5+2+1 = 10$. $S = 12$: $4+3+5 = 5+1+6 = 6+2+4 = 12$.

Answer: a) $S \in \{9; 10; 11; 12\}$; **b) as in the figure; c) the vertices are** $1, 3, 5$

Problem 41 ★★

Magic squares with the numbers 1 to 9. Place the numbers $1, 2, \ldots, 9$ (each exactly once) in a 3×3 grid so that the three numbers in every row, every column and both diagonals have the same sum.

a) Find the magic sum M and prove that the centre cell must contain 5.

b) Prove that the number 9 cannot be in a corner of the grid.

c) How many such magic squares are there altogether? Explain.

Detailed solution

Understanding the problem. We do not try all $9! = 362\,880$ arrangements; instead we use reasoning to "force" the cells one after another: first the centre, then the position of 9, finally the remaining cells.

Step 1. Part a): the magic sum

The three rows contain each number from 1 to 9 exactly once, so $3M = 1 + 2 + \cdots + 9 = 45$, hence $M = 15$.

Step 2. Part a): the centre

Recall. Problem 37: in every 3×3 magic square the centre number equals $M \div 3$ (add the four lines through the centre: $4M = 3M + 3c$).

So the centre is $c = 15 \div 3 = 5$.

Step 3. Part b): 9 is not in a corner

Why is it true? Suppose 9 is in a corner, say the top left. Three lines pass through this corner: the top row, the left column and a diagonal.

- Diagonal: $9 + 5 + x = 15$, so the opposite corner is $x = 1$.
- Top row: the other two numbers add up to $15 - 9 = 6$. Left column: the other two numbers also add up to 6.

The numbers still available are $2, 3, 4, 6, 7, 8$ (since $1, 5, 9$ are placed). Among them there is only *one* pair with sum 6, namely $2 + 4$ ($1 + 5$ cannot be used, and $3 + 3$ would use a number twice). But the top row and the left column need *two* different pairs with sum 6. Contradiction! So 9 is not in a corner. □

Therefore 9 is in the middle of a side (one of the 4 cells above, below, left or right of the 5).

Step 4. Part c): completing the square when 9 is in the middle of the top side

- Middle column: $9+5+y=15$, so the middle of the bottom side is $y=1$.
- Top row: the two top corners add up to $15-9=6$, chosen from $2,3,4,6,7,8$: they must be 2 and 4. There are 2 ways to place them (2 on the left or 4 on the left).
- With top left 2 and top right 4: one diagonal gives bottom right $15-2-5=8$; the other diagonal gives bottom left $15-4-5=6$; middle row: left $15-2-6=7$, right $15-4-8=3$.

2	**9**	4
7	**5**	3
6	1	8

4	**9**	2
3	**5**	7
8	1	6

Check: bottom row $6+1+8=15$, columns $2+7+6=15$, $4+3+8=15$ – correct. So when 9 is in the middle of the top side, there are exactly 2 magic squares (the two above, mirror images in the middle column).

Step 5. Part c): counting all of them

The number 9 has 4 possible positions (the middles of the four sides); for each position there are exactly 2 ways (the same argument, just with the grid rotated). In total $4\times 2=8$ magic squares. All of them are obtained from one square by rotations and reflections.

ThinkAlgo Tip. Look at the picture: the even numbers $2,4,6,8$ are in the corners, the odd numbers $1,3,7,9$ in the middles of the sides, and two numbers symmetric about the centre always add up to 10. This is the famous Lo Shu square of ancient China.

Common mistake. You may not conclude "there are 8" just by "seeing 8 rotations/reflections". Two things must be proved: every magic square has the form above (part b and step 4), and each case really gives a correct magic square (the check).

Check: A computer search through all 362 880 arrangements also finds exactly 8 magic squares, and all 8 have 5 in the centre.

Answer: a) $M=15$, the centre is 5; b) see the proof; c) 8 magic squares

Problem 42

4×4 **Sudoku.** Each row, each column and each 2×2 square (thick border) must contain all the digits $1, 2, 3, 4$.

a) Complete grid (I) below (only 4 given digits!). Explain each step.

b) Grid (II) has first row $1, 2, 3, 4$ and first column $1, 3, 2, 4$. In how many ways can grid (II) be completed?

	2		
		1	
			3
4			

(I)

1	2	3	4
3			
2			
4			

(II)

Detailed solution

Understanding the problem. Call the cell in row i, column j cell (i, j) (rows counted from the top, columns from the left). Each cell "sees" its row, its column and its 2×2 block. If these already contain three different digits, the cell has only *one* possibility left.

> **Recall. Single candidate:** an empty cell can only take a digit that does not yet appear in its row, column and block. If only one such digit is left, write it in straight away.

Part a).

Step 1. Row 1

Cell $(1, 4)$: row 1 has 2; column 4 has 3; the top right block has 1. What is left: **4**.

Cell $(1, 3)$: row 1 has $2, 4$; column 3 has 1. What is left: **3**. Cell $(1, 1)$: row 1 is missing only **1**.

Step 2. Row 2

Cell $(2, 1)$: column 1 has $1, 4$; row 2 has 1; the top left block has $1, 2$. Digits seen: $1, 2, 4$. What is left: **3**.

Cell $(2, 2)$: the top left block already has $1, 2, 3 \Rightarrow$ **4**. Cell $(2, 4)$: row 2 already has $3, 4, 1 \Rightarrow$ **2**.

Step 3. Row 3

Cell $(3, 1)$: column 1 already has $1, 3, 4 \Rightarrow$ **2**. Cell $(3, 2)$: row 3 has $2, 3$; column 2 has $2, 4 \Rightarrow$ **1**.

Cell $(3, 3)$: row 3 is missing only **4**.

Step 4. Row 4

Cell $(4, 2)$: column 2 already has $2, 4, 1 \Rightarrow$ **3**. Cell $(4, 3)$: column 3 already has $3, 1, 4 \Rightarrow$ **2**. Cell $(4, 4)$: **1**.

1	**2**	3	4
3	4	**1**	2
2	1	4	**3**
4	3	2	1

Part b).

Step 5. The top left and bottom left blocks

Cell $(2,1)=3$, so cell $(2,2)$ must be 4 (the top left block needs all of $1,2,3,4$). Cells $(2,3),(2,4)$ are $1,2$ in some order: 2 possibilities.

Column 2 already has $2,4$, so cells $(3,2),(4,2)$ are $1,3$ in some order: 2 possibilities.

Step 6. Look at the $2\times 2=4$ cases; the bottom right block is then determined

Row 2	Column 2 (bottom)	Each row/column still needs (bottom right block)	Ways
$3,4,1,2$	$1,3$	R3: $\{3,4\}$, R4: $\{1,2\}$, C3: $\{2,4\}$, C4: $\{1,3\}$	1
$3,4,1,2$	$3,1$	R3: $\{1,4\}$, R4: $\{2,3\}$, C3: $\{2,4\}$, C4: $\{1,3\}$	1
$3,4,2,1$	$1,3$	R3: $\{3,4\}$, R4: $\{1,2\}$, C3: $\{1,4\}$, C4: $\{2,3\}$	1
$3,4,2,1$	$3,1$	R3: $\{1,4\}$, R4: $\{2,3\}$, C3: $\{1,4\}$, C4: $\{2,3\}$	0

(R3 means row 3, C3 means column 3.) Example, last line: cell $(4,3)$ must be in $\{2,3\}$ (because of row 4) and in $\{1,4\}$ (because of column 3) – no digit works. In the first three lines, each cell of the bottom right block has exactly one digit that suits both its row and its column; e.g. first line: $(3,3)=4$, $(3,4)=3$, $(4,3)=2$, $(4,4)=1$.

Step 7. Conclusion

There are $1+1+1+0=3$ ways to complete grid (II).

1	2	3	4
3	4	1	2
2	1	4	3
4	3	2	1

1	2	3	4
3	4	1	2
2	3	4	1
4	1	2	3

1	2	3	4
3	4	2	1
2	1	4	3
4	3	1	2

Common mistake. Do not rush to multiply $2\times 2=4$! A choice made in an earlier step can lead to a dead end later. Always check each case to the end.

Check: A computer lists all 288 4×4 Sudoku grids; exactly 3 of them have first row $1,2,3,4$ and first column $1,3,2,4$, and exactly 1 contains the 4 given digits of grid (I).

Answer: a) Row 1: 1234; row 2: 3412; row 3: 2143; row 4: 4321; b) 3 ways

Problem 43 ✷

6×6 **Sudoku.** Fill in the digits 1 to 6 so that every row, every column and every 2×3 block (thick border) contains all six digits $1, 2, 3, 4, 5, 6$. The grid has only 9 given digits.

a) Fill in the first row and explain why each cell can be filled in only one way.

b) Complete the whole grid.

3				2	6
				6	
		4	1		
		1			
6				3	

Detailed solution

Understanding the problem. We reuse the "single candidate" method of Problem 42, but now each cell must rule out 5 digits. We always look for the cell that "sees" the most digits and fill it first. Cell (i, j) is the cell in row i, column j.

> **Recall.** An empty cell can only take a digit that is not yet in its row, column and block. If five digits are ruled out, the remaining digit is the answer for that cell.

Step 1. **Part a): cell** $(1, 3)$
Row 1 has $3, 2, 6$; column 3 has $4, 1$. Ruled out: $1, 2, 3, 4, 6 \Rightarrow$ cell $(1, 3) = 5$.

Step 2. **Cell** $(1, 4)$
Row 1 now has $3, 5, 2, 6$; column 4 has 1. Ruled out: $1, 2, 3, 5, 6 \Rightarrow$ cell $(1, 4) = 4$.

Step 3. **Cell** $(1, 2)$
Row 1 already has $3, 5, 4, 2, 6$; only 1 is missing $\Rightarrow$ cell $(1, 2) = 1$. Row 1 is: $3, 1, 5, 4, 2, 6$.

> **ThinkAlgo Tip.** At first, cell $(1, 2)$ "saw" only 3 digits. But after its two neighbours were filled, it became a single candidate. In Sudoku, every cell you fill opens up new cells – like a chain of dominoes.

Step 4. **Part b): carry on in the same way**
The table records the filling order. The columns "row/column/block" list the digits already in the row, column and block of the cell *at the moment it is filled*; the single missing digit is the answer.

Move	Cell	Row has	Col. has	Block has	Fill	Move	Cell	Row has	Col. has	Block has	Fill
4	(4,5)	1,4	2,3,6	1,6	**5**	16	(4,6)	1,2,4,5,6	4,6	1,2,4,5,6	**3**
5	(2,5)	–	2,3,5,6	2,4,6	**1**	17	(2,6)	1,2,4,6	3,4,6	1,2,4,6	**5**
6	(4,1)	1,4,5	3,6	4	**2**	18	(2,4)	1,2,4,5,6	1,2,4	1,2,4,5,6	**3**
7	(2,1)	1	2,3,6	1,3,5	**4**	19	(5,1)	1	1,2,3,4,6	1,6	**5**
8	(3,3)	6	1,4,5	2,4	**3**	20	(5,4)	1,5	1,2,3,4	3	**6**
9	(3,2)	3,6	1	2,3,4	**5**	21	(5,5)	1,5,6	1,2,3,5,6	3,6	**4**
10	(3,1)	3,5,6	2,3,4,6	2,3,4,5	**1**	22	(5,2)	1,4,5,6	1,2,5,6	1,5,6	**3**
11	(3,4)	1,3,5,6	1,4	1,5,6	**2**	23	(5,6)	1,3,4,5,6	3,4,5,6	3,4,6	**2**
12	(3,6)	1,2,3,5,6	6	1,2,5,6	**4**	24	(6,2)	3,6	1,2,3,5,6	1,3,5,6	**4**
13	(4,2)	1,2,4,5	1,5	1,2,3,4,5	**6**	25	(6,3)	3,4,6	1,3,4,5,6	1,3,4,5,6	**2**
14	(2,2)	1,4	1,5,6	1,3,4,5	**2**	26	(6,4)	2,3,4,6	1,2,3,4,6	2,3,4,6	**5**
15	(2,3)	1,2,4	1,3,4,5	1,2,3,4,5	**6**	27	(6,6)	2,3,4,5,6	2,3,4,5,6	2,3,4,5,6	**1**

Example, move 4, cell $(4,5)$: row 4 has $1,4$; column 5 has $2,3,6$; the middle right block has $1,6$. Together $1,2,3,4,6$ are present, so this cell is 5. Example, move 5, cell $(2,5)$: row 2 is still empty, but column 5 has $2,3,5,6$ and the top right block has $2,4,6$; together only 1 is missing.

3	1	5	4	**2**	**6**
4	2	6	3	1	5
1	5	3	2	**6**	4
2	6	**4**	**1**	5	3
5	3	**1**	6	4	2
6	4	2	5	**3**	1

Common mistake. When a cell has two possibilities, do not guess and carry on. Look for another cell that is certain; in this puzzle there is always at least one cell with only one possibility.

Check: Every row of the final grid (e.g. row 4: $2,6,4,1,5,3$), every column (e.g. column 1: $3,4,1,2,5,6$) and every 2×3 block contains 1 to 6; the 9 given digits are still in place. A computer also confirms that the puzzle has a *unique* solution.

Answer: Row 1: 315426**; row 2:** 426315**; row 3:** 153264**; row 4:** 264153**; row 5:** 531642**; row 6:** 642531

Problem 44 *

Finding the fake (lighter) coin. There are 9 identical-looking coins; exactly one of them is fake and *lighter* than the others (the genuine coins all have the same mass). We only have an equal-arm balance (no weights); each weighing only tells us which side is heavier or that both sides are equal.

a) Show how to find the fake coin with only 2 weighings.

b) Explain why 1 weighing is not enough.

c) With 3 weighings, among at most how many coins can we find the (lighter) fake coin? Describe the weighings.

Detailed solution

Understanding the problem. Each weighing has **three** outcomes: left heavier, right heavier, balanced. The secret is to split the coins into *three* equal groups, not two.

Step 1. Part a): the first weighing

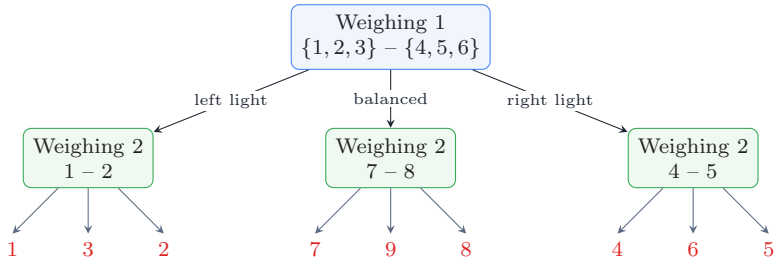
Number the coins 1 to 9. Put group $\{1, 2, 3\}$ on the left pan, group $\{4, 5, 6\}$ on the right pan, and keep $\{7, 8, 9\}$ aside.

- Left pan lighter: the fake coin is in $\{1, 2, 3\}$.
- Right pan lighter: the fake coin is in $\{4, 5, 6\}$.
- Balanced: all 6 coins on the scale are genuine; the fake coin is in $\{7, 8, 9\}$.

Whatever happens, only 3 "suspect" coins remain.

Step 2. Part a): the second weighing

Call the three suspects x, y, z. Put x on the left pan and y on the right pan. The lighter side holds the fake coin; if they balance, the fake coin is z.



(In the diagram, at weighing 2: left arrow – left pan lighter; middle – balanced; right – right pan lighter.)

Step 3. Part b): why 1 weighing is not enough

Why is it true? There are 9 possibilities (the fake is coin 1, coin 2, ..., coin 9). One weighing gives only 3 different outcomes. By the pigeonhole principle, at least one outcome corresponds to ≥ 3 possibilities (because $9 > 3$), and when that outcome occurs we cannot tell which coin is fake. So 1 weighing cannot always find the fake coin. □

Step 4. Part c): 3 weighings

Arguing as in part b: 3 weighings give at most $3 \times 3 \times 3 = 27$ different sequences of outcomes, so we cannot distinguish more than 27 possibilities. Conversely, 27 coins can be handled: split them into 3 groups of 9, weigh the first group against the second; this tells us which group of 9 contains the fake; then use the 2 weighings of part a) on that group.

ThinkAlgo Tip. Rule: with n weighings we can find a lighter coin among at most 3^n coins: 1 weighing – 3 coins, 2 weighings – 9 coins, 3 weighings – 27 coins.

Common mistake. The "halving" method ($9 \to 4 + 4$, then $2 + 2$, then $1 + 1$) needs 3 weighings, because it wastes the "balanced" outcome.

Check: Try all 9 cases in turn (the fake is $1, 2, \ldots, 9$): the method of part a) always finds the right coin after 2 weighings.

Answer: a) Weigh 3 against 3 to find the group with the fake, then 1 against 1 inside that group; b) 9 possibilities $>$ 3 outcomes; c) 27 coins

Problem 45 ⋆⋆

The famous 12-coin problem. There are 12 identical-looking coins, exactly one of which is fake. The fake coin has a different mass from the genuine ones, but we do *not* know whether it is heavier or lighter. We may only use an equal-arm balance (no weights).

a) How many "possibilities" must be distinguished? Why are 2 weighings certainly not enough?

b) Show how to use exactly 3 weighings to find the fake coin *and* tell whether it is heavier or lighter.

Detailed solution

Understanding the problem. Unlike Problem 44, we do not know whether the fake is heavy or light, so every unbalanced result has *two* explanations (the heavy side contains a heavy coin, *or* the light side contains a light coin). We must keep careful notes of which "suspects" could be heavy and which could be light.

Step 1. Part a): counting possibilities

The fake can be any of the 12 coins, and for each coin there are 2 cases (heavier / lighter): $12 \times 2 = 24$ possibilities in total. Two weighings give at most $3 \times 3 = 9$ sequences of outcomes. Since $24 > 9$, different possibilities would share the same sequence of outcomes, so 2 weighings are not enough. With 3 weighings there are $27 \geq 24$ sequences – it *might* be enough, if we weigh cleverly.

Recall. Notation: $\uparrow k$ means "coin k might be the fake and heavier", $\downarrow k$ means "coin k might be the fake and lighter". A coin known to be genuine can be used as a "reference coin".

Step 2. Weighing 1: $\{1, 2, 3, 4\}$ against $\{5, 6, 7, 8\}$

Case A – balanced. Coins 1–8 are all genuine. The fake is among $9, 10, 11, 12$.

Case B – left pan heavier. Suspects: $\uparrow 1, \uparrow 2, \uparrow 3, \uparrow 4, \downarrow 5, \downarrow 6, \downarrow 7, \downarrow 8$ (8 possibilities); coins 9–12 are genuine.

Case C – right pan heavier. Like B with the roles of the two groups swapped.

Step 3. Case A (fake among 9–12)

Weighing 2: $\{9, 10, 11\}$ against $\{1, 2, 3\}$ (three genuine reference coins).

- Balanced: the fake is 12. Weighing 3: 12 against coin 1; if 12 is heavier the fake is heavy, if lighter the fake is light (it cannot balance).
- Left pan heavier: the fake is one of $9, 10, 11$ and *heavier*. Weighing 3: 9 against 10; the heavier side is the fake; if balanced it is 11.
- Left pan lighter: similarly the fake is among $9, 10, 11$ and *lighter*; weigh 9 against 10, the lighter side is the fake, and if balanced it is 11.

Step 4. Case B (suspects $\uparrow 1, \uparrow 2, \uparrow 3, \uparrow 4, \downarrow 5, \downarrow 6, \downarrow 7, \downarrow 8$)
The key trick: weighing 2 *swaps* some coins, *removes* some coins, and adds the reference coin 9:

Weighing 2: $\{1, 2, 5\}$ against $\{3, 6, 9\}$.

Result of weighing 2	Remaining suspects	Weighing 3 and how to read it
Balanced	$\uparrow 4$, $\downarrow 7$, $\downarrow 8$ (the coins taken off)	7 against 8: the lighter side is the fake (light); balanced $\Rightarrow$ 4 heavy
Left heavier	$\uparrow 1$, $\uparrow 2$, $\downarrow 6$	1 against 2: the heavier side is the fake (heavy); balanced $\Rightarrow$ 6 light
Right heavier	$\uparrow 3$, $\downarrow 5$	3 against 9: if 3 is heavier $\Rightarrow$ 3 heavy; balanced $\Rightarrow$ 5 light

Why is it true? Why does the line "Left heavier" leave only $\uparrow 1, \uparrow 2, \downarrow 6$? The left pan $\{1, 2, 5\}$ being heavier means: either there is a heavy coin on the left – it can only be 1 or 2 (coin 5 can only be *light*); or there is a light coin on the right – it can only be 6 (coin 3 can only be *heavy*, coin 9 is genuine). The other lines are argued the same way. □

Step 5. Case C
Exactly like case B with the roles swapped: the suspects are $\uparrow 5, \uparrow 6, \uparrow 7, \uparrow 8, \downarrow 1, \downarrow 2, \downarrow 3, \downarrow 4$; weighing 2 is $\{5, 6, 1\}$ against $\{7, 2, 9\}$, and the result is read from the table above (replace $1, 2, 3, 4$ by $5, 6, 7, 8$ and vice versa).

ThinkAlgo Tip. After each weighing the number of possibilities drops at best to one third: $24 \to 8 \to 3 \to 1$ (or fewer). A "good" weighing is one whose three outcomes split the possibilities into three nearly equal parts. In weighing 2 of case B, the 8 possibilities are split as $3 + 3 + 2$.

Common mistake. A common mistake in weighing 2: weighing $\{1, 2\}$ against $\{3, 4\}$. If it balances, 4 possibilities $\downarrow 5, \downarrow 6, \downarrow 7, \downarrow 8$ remain, with only one weighing (3 outcomes) left – not enough!

Check: All 24 possibilities (each coin 1 to 12, heavy or light) were run through the scheme above by computer: every time the fake coin and "heavy/light" were identified correctly after 3 weighings.

Answer: a) 24 possibilities $>$ 9 sequences of outcomes; b) weigh $\{1, 2, 3, 4\} - \{5, 6, 7, 8\}$, then follow the scheme in steps 3–5

Problem 46 *

A mini football tournament. Four teams A, B, C, D play a single round robin (each team plays every other team exactly once). A win gives 3 points, a draw 1 point, a loss 0 points. At the end, team A has 7 points, B has 5, C has 3 and D has 1.

a) How many matches were drawn?

b) Determine the result (win/draw/loss) of every match.

Detailed solution

Understanding the problem. We do not know any scores, only the point totals. We will (1) count the points of the whole tournament to find the number of draws; (2) analyse each team's points; (3) "pair up" the draws.

Step 1. Number of matches

Each pair of teams plays once: AB, AC, AD, BC, BD, CD – 6 matches in total. Each team plays 3 matches.

Step 2. Part a): total points and number of draws

A match with a winner gives the two teams $3+0=3$ points in total; a draw gives $1+1=2$ points. The total number of points in the tournament is $7+5+3+1=16$.

Let x be the number of draws (x is a whole number, $0 \le x \le 6$). Then $6-x$ matches had a winner, and the total number of points is

$$2x + 3(6-x) = 16.$$

Solve the equation: $2x+18-3x=16$, $18-x=16$, $x=2$. So there were 2 **draws** (and $6-2=4$ matches with a winner).

Check: 2 draws give $2\cdot 2=4$ points, 4 matches with a winner give $4\cdot 3=12$ points; $4+12=16$, correct.

Step 3. Analyse each team's points (3 matches, each worth 3, 1 or 0 points)

Team	Points	Possible ways
A	7	$3+3+1$: 2 wins, 1 draw
B	5	$3+1+1$: 1 win, 2 draws
C	3	$3+0+0$ (1 win, 2 losses) or $1+1+1$ (3 draws)
D	1	$1+0+0$: 1 draw, 2 losses

(For example, with 7 points: $3+3+3=9$ is too many, $3+3+0=6$ too few, $3+1+1=5$ too few; only $3+3+1=7$.)

Step 4. Count the "draw entries"

Each draw counts for *two* teams, so the four teams have $2\times 2=4$ draw entries in total. A has 1, B has 2, D has 1: that already makes 4. So C has *no draws*, hence C has 1 win and 2 losses.

Step 5. Which matches were drawn?

The two draws involve only teams A, B, D. B drew 2 matches against two different opponents, while A and D each drew exactly 1. So the two draws are **A – B** and **B – D** (if A had drawn with D, then A or D would have 2 draws once B's draws are added).

Step 6. The remaining matches

A wins its other 2 matches: A beats C, A beats D. B wins its other match: B beats C. D loses its other 2 matches: to A and to C, so C beats D.

	A	B	C	D	Points
A	–	draw	win	win	$1+3+3=7$
B	draw	–	win	draw	$1+3+1=5$
C	loss	loss	–	win	$0+0+3=3$
D	loss	draw	loss	–	$0+1+0=1$

Common mistake. C has 3 points, so many students hastily think "C drew all 3 matches". We need the extra information "there are only 2 draws" to rule this out.

Check: There are exactly 2 draws (A–B, B–D). Points: A $1+3+3=7$, B $1+3+1=5$, C $0+0+3=3$, D $0+1+0=1$. A computer check of all $3^6=729$ possible sets of results also finds exactly one giving the table $7,5,3,1$.

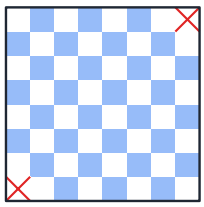
Answer: a) 2 draws; b) A drew with B, B drew with D; A beat C, A beat D, B beat C, C beat D

Problem 47 ⋆

Invariants.

a) The numbers $1,2,3,\ldots,10$ are written on the board. In each turn, Oscar erases any two numbers a and b and writes the number $|a-b|$ (the larger minus the smaller). After 9 turns, only one number is left on the board. Can it be 0? Can it be 1?

b) Two opposite corner squares are cut off an 8×8 chessboard (see the figure). Can the remaining 62 squares be covered exactly by 31 dominoes, each covering two adjacent squares?





one domino

Detailed solution

Understanding the problem. Both parts ask "can it be done?". For "it cannot be done", trying again and again proves nothing – we need an *invariant*: a property that does not change, whatever we do.

Recall. An **invariant** is a quantity (or property) that stays the same after every step. If the starting position and the target position have different invariants, we cannot get from one to the other.

Part a).

Step 1. Look at the parity of the sum

When a,b are replaced by $|a-b|$, the sum of the numbers on the board decreases by $a+b-|a-b|$. If $a\geq b$, that is $a+b-(a-b)=2b$ – an even number. So the sum changes by an even number, and **the parity of the sum never changes**.

Step 2. Conclusion about 0

The initial sum $1+2+\cdots+10=55$ is odd. So the sum is always odd, and the last number (which is the sum at that moment) is odd. 0 is even, so the last number **cannot** be 0.

Step 3. 1 **is possible: show a way**
Erase $(10, 9)$, $(8, 7)$, $(6, 5)$, $(4, 3)$, $(2, 1)$: this gives five 1s. Erase two 1s to get 0; erase two other 1s to get 0. The board now shows $0, 0, 1$. Erase $(0, 0)$ to get 0; finally erase $(0, 1)$ to get 1. Count: $5 + 2 + 1 + 1 = 9$ turns.

Common mistake. An invariant only tells us what is *impossible*. To claim "it is possible" we must show a concrete way, as in step 3.

Part b).

Step 4. **Colour and count**
A chessboard has 32 dark and 32 white squares. Two opposite corner squares always have the *same colour* (dark in the figure). After cutting, 30 dark and 32 white squares remain.

Step 5. **Each domino covers one dark and one white square**
Two adjacent squares on a chessboard always have different colours. So 31 dominoes (however they are placed) cover exactly 31 dark and 31 white squares. But the board has only 30 dark squares left. Contradiction! So it is **impossible**.

Why is it true? The invariant here is: "number of dark squares covered = number of white squares covered". It holds after each domino is placed, but the part of the board to be covered has $30 \neq 32$. □

Note for parents. Try a small 4×4 board with two opposite corners removed (14 squares, 7 dominoes needed): let your child try it to feel that "it just never works", and only then explain the reason using the colours.

Check: Part a): every final result is odd (because 55 is odd), and the method of step 3 gives 1. Part b): $30 + 32 = 62$ squares; 31 dominoes need 31 squares of each colour.

Answer: a) It cannot be 0; it can be 1; b) Impossible

Problem 48 *

An Einstein-style puzzle. Four houses are numbered $1, 2, 3, 4$ from left to right. Each house is painted a different colour (red, yellow, green, white), has one inhabitant (Anna, Ben, Chloe, David), and each inhabitant keeps a different animal (dog, cat, fish, rabbit). We know that:

(1) Anna lives in the red house.

(2) The green house is *immediately to the right* of the white house.

(3) Ben keeps the dog.

(4) The fish owner lives in house 1.

(5) David lives right next to the cat owner.

(6) House 2 is white.

(7) Anna keeps the rabbit.

Who keeps the cat? Whose house is yellow?

Detailed solution

Understanding the problem. There are three kinds of information (colour, person, animal) for four houses. We draw a 3×4 table and fill it in step by step, starting with the "definite" clues (those giving a specific position), then the "pairing" clues and the "next to" clues.

Step 1. House colours: clues (6) and (2)
House 2 is white (6), so the green house immediately to its right is house 3 (2). The colours red and yellow go to houses 1 and 4.

Step 2. Anna: clues (1), (7), (4)
Anna lives in the red house (1) and keeps the rabbit (7). The red house is house 1 or house 4. But house 1 has the fish (4), while Anna has the rabbit, so Anna is not in house 1. So Anna lives in house 4, which is red and has the rabbit; house 1 is yellow and has the fish.

Step 3. Dog and cat
The remaining animals, dog and cat, are in houses 2 and 3. Ben keeps the dog (3), so Ben lives in house 2 or house 3.

Step 4. Where is David? Clue (5)
David lives in one of houses $1, 2, 3$. Check each possibility:

- David in house 2: David does not keep the dog (that is Ben's), so David keeps the cat. David's neighbours are house 1 (fish) and house 3 (dog) – nobody keeps a cat, contradicting (5). **Ruled out.**
- David in house 3: similarly David keeps the cat; the neighbours are house 2 (dog) and house 4 (rabbit) – contradicting (5). **Ruled out.**
- David in house 1 (fish): his only neighbour is house 2, so house 2 has the cat and house 3 the dog. **Works.**

Step 5. Complete the table
House 3 has the dog, so Ben lives in house 3; the remaining person, Chloe, lives in house 2.

	House 1	House 2	House 3	House 4
Colour	yellow	white	green	red
Person	David	Chloe	Ben	Anna
Animal	fish	cat	dog	rabbit

ThinkAlgo Tip. Whenever a clue says "right next to", try each position in turn and eliminate – usually only one possibility survives.

Check: Check each clue: (1) Anna – red house ✓; (2) green (house 3) immediately right of white (house 2) ✓; (3) Ben – dog ✓; (4) house 1 – fish ✓; (5) David (house 1) next to house 2 with the cat ✓; (6) house 2 white ✓; (7) Anna – rabbit ✓. A computer check of all $24^3 = 13\,824$ arrangements also finds only this solution.

Answer: Chloe keeps the cat; the yellow house is David's

Problem 49 ⋆

Clocks and calendars.

a) 1 January 2026 is a Thursday. On which day of the week is 2 September 2026? On which day is 1 January 2027?

b) At 3:40, what angle do the hour hand and the minute hand make (the angle not exceeding 180°)?

c) After exactly 3 o'clock, how many minutes later does the minute hand first catch up with the hour hand?

d) ⋆ Prove that every year has at least one Friday the 13th.

Detailed solution

Understanding the problem. The days of the week repeat every 7 days, so we only need the *remainder on division by* 7 of the number of days that pass. For the clock, we convert the position of each hand into "degrees measured from the 12".

Recall. A common year has 365 days, a leap year 366 days (February has 29 days). The year 2026 is not a leap year. Month lengths divided by 7: a 31-day month leaves remainder 3, a 30-day month 2, a 28-day month 0, a 29-day month 1.

Step 1. Part a): 2 September 2026

Days from 1 January to 2 September: January to August have $31+28+31+30+31+30+31+31=243$ days, plus 1 more day (from 1 September to 2 September): 244 days in total. Now $244=7\times34+6$. Moving 6 days on from Thursday: Friday, Saturday, Sunday, Monday, Tuesday, **Wednesday**. 2 September 2026 is a **Wednesday**.

Step 2. Part a): 1 January 2027

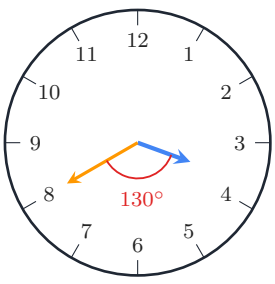
The year 2026 has $365=7\times52+1$ days, so the weekday of 1 January 2027 is one day later than that of 1 January 2026: **Friday**.

Common mistake. Do not count 243 instead of 244: from 1 January to 2 September is 243 days (up to 1 September) plus 1 more day. Check with a small example: from 1 January to 2 January is 1 day, not 2.

Step 3. Part b): the angle at 3:40

The minute hand turns 360° in 60 minutes, i.e. 6° per minute: at 40 minutes it is at $40\times6=240°$. The hour hand turns 30° per hour, i.e. 0.5° per minute: at 3:40 it is at $3\times30+40\times0.5=90+20=110°$.

The angle between the hands: $240-110=130°$.



Step 4. Part c): the minute hand catches up with the hour hand
At 3 o'clock the hour hand is 90° ahead of the minute hand. Every minute the minute hand gains $6 - 0.5 = 5.5°$ on the hour hand. Time to catch up: $90 \div 5.5 = \frac{180}{11} = 16\frac{4}{11}$ minutes (about 16 minutes 22 seconds).

Recall. "Catch-up" motion problems: catch-up time = initial gap ÷ difference of speeds.

Step 5. Part d): the 13th of each month
Call the weekday of 13 January "offset 0". The 13th of the next month is shifted by exactly the number of days of the previous month (remainder on division by 7). Adding up (and taking remainders on division by 7):

Month	1	2	3	4	5	6	7	8	9	10	11	12
Common year	0	3	3	6	1	4	6	2	5	0	3	5
Leap year	0	3	4	0	2	5	0	3	6	1	4	6

Example for a common year: January has 31 days (remainder 3), so 13 February is offset 3; February has 28 days (remainder 0), so 13 March is still offset 3; March has 31 days, so 13 April is offset $3 + 3 = 6$; April has 30 days, so 13 May is offset $6 + 2 = 8$, remainder 1; ...

Why is it true? In both rows the offsets take all seven values $0, 1, 2, 3, 4, 5, 6$. That means the 13ths of the year fall on all seven days of the week. In particular, at least one 13th always falls on a Friday. □

Check: By the calendar: 2 September 2026 is a Wednesday, 1 January 2027 a Friday. In 2026, Friday the 13th occurs in February, March and November.

Answer: a) Wednesday; Friday; b) 130°; c) $16\frac{4}{11}$ minutes; d) see the proof

Problem 50 **

The pigeonhole principle.

a) Class 7A has 40 students. Prove that at least 4 of them were born in the same month.

b) Prove that among any 5 integers there are always two whose difference is divisible by 4.

c) Choose any 6 different numbers from $1, 2, \ldots, 10$. Prove that two of the chosen numbers always add up to 11. If only 5 numbers are chosen, is this still certain?

d) At a meeting there are 10 friends; some pairs of friends shake hands (each pair at most once). Prove that two friends have shaken the same number of hands.

Detailed solution

Understanding the problem. All four parts say "prove that there are always..." – we do not need to find out who exactly, only to prove that it *must exist*. The tool is the pigeonhole principle.

Recall. The pigeonhole principle. If $n+1$ pigeons are put into n holes, some hole contains 2 or more pigeons. In general: if more than $k \times n$ pigeons are put into n holes, some hole contains at least $k+1$ pigeons.

Why is it true? If every hole contained at most k pigeons, there would be at most $k \times n$ pigeons altogether – contradicting the assumption that there are more than $k \times n$. □

Step 1. Part a)
The "holes" are the 12 months, the "pigeons" are the 40 students. If each month had at most 3 students, the class would have at most $12 \times 3 = 36 < 40$ students – impossible. So some month has at least 4 birthdays.

Step 2. Part b)
On division by 4, every integer leaves one of 4 remainders: $0, 1, 2, 3$ ("4 holes"). There are 5 numbers ("5 pigeons"), so two of them leave the same remainder, say $a = 4p + r$ and $b = 4q + r$. Then $a - b = 4(p - q)$ is divisible by 4.

Step 3. Part c)
Split the 10 numbers into 5 "holes", each hole a pair with sum 11:

$$\{1, 10\}, \quad \{2, 9\}, \quad \{3, 8\}, \quad \{4, 7\}, \quad \{5, 6\}.$$

We choose 6 numbers but there are only 5 pairs, so two chosen numbers fall into the same pair – their sum is 11.
With 5 numbers it is **not certain**: choose $1, 2, 3, 4, 5$; the two largest add up to only $4+5 = 9 < 11$.

ThinkAlgo Tip. The hardest part of the pigeonhole principle is *choosing the holes*. Look at what you need to prove ("sum equal to 11") and build holes such that two pigeons in the same hole immediately have that property.

Step 4. Part d): the "holes" are the numbers of handshakes
Each friend shakes hands with at most 9 others, so each friend's number of handshakes is one of the 10 values $0, 1, 2, \ldots, 9$. 10 friends and 10 values – not enough for the pigeonhole principle yet! We need one more observation.

Why is it true? The values 0 and 9 cannot both occur. Indeed, if someone shook hands 9 times, they shook hands with *everybody*, so nobody has 0 handshakes. So in fact the 10 friends take at most 9 different values (either $0, 1, \ldots, 8$ or $1, 2, \ldots, 9$). 10 pigeons, 9 holes: two friends have the same number of handshakes. □

Common mistake. In part d), if you stop at "10 friends, 10 values", you cannot conclude anything. The observation "0 and 9 cannot both occur" is the key.

Note for parents. You can illustrate with 5 cups and 6 marbles: however the marbles are dropped in, some cup holds 2. Then ask your child: "In this problem, what are the cups and what are the marbles?"

Check: Part a): $12 \times 3 = 36 < 40 \leq 12 \times 4$. Part c): a computer checks all 210 ways of choosing 6 numbers from 1 to 10 – every one contains two numbers with sum 11.

Answer: a), b), c), d): see the proofs; in part c) with 5 numbers it is not certain (e.g. $1, 2, 3, 4, 5$)